

GE Healthcare

Discovery MR450 & MR750 System Installation

Technical Publication



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DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



ПРЕДУПРЕЖДЕНИЕ

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.
- V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKladU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ Vlivem ELEKTRICKÉHO PROUDU, RESPEKTIVE Vlivem MECHANICKÝCH ČI JINÝCH RIZIK.

ADVARSEL

(DA)

- DENNE SERVICE MANUAL FINDES KUN PÅ ENGELSK.
- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FORSØG IKKE AT SERVICERE Udstyret medmindre denne service manual har været konsulteret og er forstået.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING

(NL)

- DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.
- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.
- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.

WARNING

(EN)

- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR, OR PATIENT FROM ELECTRIC SHOCK, OR FROM MECHANICAL OR OTHER HAZARDS.

HOIATUS

(ET)

- KÄESOLEV TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.
- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.

VAROITUS

(FI)

- KÄESOLEVA HOIATUSE EIRAMINE VÖIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.
- TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.
- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

ATTENTION

(FR)

- CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

(DE)

- DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.
- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

FIGYELMEZTETÉS

(HU)

- EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHEŐ EL.
- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÚTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

AÐVÖRUN

(IS)

- ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU.
- EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLDA VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAPJÓNUSTU.
- REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓK HEFUR VERIÐ SKOÐUÐ OG SKILIN.
- BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.

AVVERTENZA

(IT)

- IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

 경고

(KO)

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BRĪDINĀJUMS

(LV)

- ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGLŪ VALODĀ.
- JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGLŪ, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.
- NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.
- ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.

ISPĒJIMAS

(LT)

- ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŪ KALBA.
- JEI KLIENTO PASLAUGŪ TIEKĒJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŪ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĒ.
- NEMĒGINKITE ATLIKTI ĮRANGOS TECHINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.
- JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO,
- MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.

ADVARSEL

(NO)

- DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.
- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

OSTRZEŻENIE

(PL)

- NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIĘ W JĘZYKU ANGIELSKIM.
- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BADŹ INNEGO.

ATENÇÃO

(PT)

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENȚIE

(RO)

- ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.
- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIEȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ȘI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPARANATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

ОСТОРОЖНО!

(RU)

- ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.
- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСООБЛЮЖДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

UPOZORNENIE

(SK)

- TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.
- AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
- NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBSLUHU A NEPOROZUMIETE MU.
- ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYUSTÍŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRUDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

ATENCION

(ES)

- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.
- SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION.
- NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

VARNING

(SV)

- DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.
- OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.
- FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.
- OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KÄN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.

DİKKAT

(TR)

- BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.
- EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.
- SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.
- BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.

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REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
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1	Apr 11, 2008	Initial verification release.
2	May 5, 2008	Replaced <i>3.0T DVMR</i> with <i>MR750 3.0T</i> throughout the entire manual. Section 1: Updated the tools list and System Options section. Section 2: Revised the cabinet moving and positioning procedures with more specific details. Section 4: Revised cable maps to correct cable designations. Updated gradient cable trimming and routing at the gradient filter on the penetration panel. Added screen room ground cable connections to SPW and penetration panel. Revised ground connection at HEC. Updated part numbers for air hoses and clamps. Updated instructions for connecting rear pedestal to enclosure. Revised positioning of Dock mounting channels. Added instructions for installing shunt connector in the rear pedestal. Added OCP installation instructions and bore light cabling. Section 5: Added installation of chimney cover and rear pedestal covers. Section 6: Revised entire section with more details for installing GOC.
3	Jul 7, 2008	Updates for MRIhc38592. Section 1: Added subsection 1–3, <i>MR System Shipping and Storage Requirements</i> (MRIhc38679). Section 2: Added note for disposing dollies outside of North America. Section 3: Corrected run # to E0009. Section 4: Added 16 Channel cable map. Updated all cable maps to correct minor changes, including routing of M1006 & M1015. Section 4–2, added cables for pulling thru overhead trays and corrected run numbers. Section 4–7, revised length of cutting gradient cables from bracket. Added note about not showing cable tray dividers for clarity reasons. Section 4–8, added steps for installing body RF cable to SPW. Section 4–11, added connection of Run E3027. Section 4–16, added <i>Hose and Fitting Installation Notes</i> for installing air and water hoses. Updated procedures for installing ball valves and fittings (MRIh36779, MRIhc38225). Corrected hose connections between blower box and magnet. Section 4–17, Added LPCA rear bridge travel adjustment. Section 4–20, Revised cable connections to dock brackets. Section 4–21, added callout for 32 channel only RCVR #2. Added note indicating area on magnet where cables cannot be routed.
4	Jul 23, 2008	Updates for MRIhc39003: Section 1: Updated Table 1–1, Transportation and Storage Environmental Conditions and added table 1–2 for Magnet and XRM B Transportation and Storage Environmental Conditions.
5	Aug 18, 2008	Renamed to <i>Discovery</i> and replaced <i>Signa</i> throughout entire manual. Section 4: Section 4–1, revised MagMon ethernet cable E3034 to connect to site ethernet. Section 4–8–2, added note for not using flat washer for connecting RF body coil to pen panel (MRIhc37467). Sections 4–11–1 and 4–11–3, added power cable restraining brackets installation on PGR and PEN cabinets (ECO2056794). Section 4–16–7, Added views of blower box hose labeling and updated hose installation instructions (MRIhc38739). Section 4–17–1, added bridge height measurement, added instructions for routing I and Q cable Runs M1322 and M1323 for proper spacing distances, revised drive belt routing and attaching procedure.
6	Nov 14, 2008	Section 1: Updated installation flowchart to reflect changes in other sections of manual. Section 3: Removed E3003 and E3027 as GE supplied cables. Section 4: Corrected Patent Alert tube routing and removed errant line on map (PQR 13207687). Added temp/humidity sensor and cable (PQR 13210222). Added Section 4–5, installation of rear cable cover to check for cable clearance (PQR 13210525). Section 4–6, updated cable information in overhead cable trays (PQR 13207746). Section 4–12, Added power cable restraint bracket to PEN cabinet (PQR 13210222). Added Section 4–13, installing temp/humidity sensor on PEN cabinet. Section 4–16, Updated RF Door Switch connection information. Section 4–18, added section for installation cork/rubber wrap on hose fittings. Updated Section 4–20 for InRoom Display installation. Section 4–22–4, Revised cable routing for M3300 and M3301 (PQR 13207635). Added Sections 4–24 and 4–25 for installing rear covers and foam insulation. Updated instructions on installing ball valves. Section 4–28, Revised information to include the possible reuse of existing Runs 296 and 297 for upgraded sites. Removed Section 5–5 (PQR 13208970). Added Section 8, Appendix. This section includes information on system configuration and cable identification.
7	Mar 5, 2009	Revised title and page headers to reflect addition of MR450 product. Added new procedures specific to MR450 throughout manual. Section 1–4: Removed reference to color coded labels on boxes delivered to site (PQR223407). Section 1–12: Revised flowchart with addition of MR450. Added Section 2–3 for removing components from top of PGR cabinet for ease of moving thru doorways (PQR 13244029). Section 3–3: Revised illustration to show power connection to HEC (PQR 13218037). Section 4–1: Added MR450 cable maps, updated the MNS cable map, and added remote MRU, Oxygen Monitor, and IRD option cable maps. Corrected 1.5T Hybrid Splitter connections (SPR MRIhc43057). Removed Run E3003 (PQR 13218037) and E3027 (ECO 2067970). Revised 32–Channel Dock cables with alternative Dock Harness to represent new installations that will not use MAG–DCK_R (ECO 2066118). Sections 4–6 & 4–7: Added MR450 cables and revised phrase “overhead cable trays” to “overhead cable routing mechanisms”. Added Section 4–8 and 4–9 for removing rear magnet cover and for providing instructions on routing cryogen lines on top of magnet (PQR 13222580). Section 4–10: Revised gradient cable installation procedures to accommodate new gradient filter covers (PQR 13228694). Section 4–10–2–C: Added steps for removing and attaching safety shield (SPR MRIhc42946). Section 4–12–2: Added equipment room side to ground cable install instructions (SPR MRIhc42947). Section 4–14–2: Corrected run # from E3014 to E3013 (SPR MRIhc42948). Section 4–14–4: For step 3, added note that it's for 3T only (SPR MRIhc429520). On panel illustration, switched J16 and J17 (PQR 13252966 & ECO 2082206). Section 4–20: Revised instructions for installing ball valves and hose barbs (PQR's 13210526, 13234878, 13236543, & ECO2077212). Removed ball valves from HEC end of return/supply lines (PQR 13238186). Revised 4 inch hose

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REVISION HISTORY (continued)

REV	DATE	PRIMARY REASON FOR CHANGE
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7 (continued)		part number from P15 to Pxx to reflect multiple lengths of hoe offerings (ECO 2067150). Added Section 4-22: Installing the Bridge and Rear pedestal using the new Bridge front support bracket. Section 4-24: Add step to make sure connectors are seated properly in LPCA (PQR 13235774). Section 4-25: Remove control panel screw and washer part numbers (PQR 13218394). Section 4-25-4: Added step 4 for connecting Emergency off cables. (SPR MRIhc42953). Added Section 4-26: Installing the Dock using the new Bridge front support bracket. Section 4-27: Added installation of rubber pad under dock (PQR 13224977). Added section for installing dock at an upgrade site using the floor bolt (PQR 13209814). Revised 32-channel cable installation for new dock that will not be using the cable bracket on the side of the right leg (ECO 2066118). Section 4-28: Corrected routing connections of cables at hybrid and corrected 3T cable maps (PQR 13223005). Section 4-28-2: Corrected J number connections for runs M1341 and M1342 (SPR MRIhc42950). Added installation steps in Section 4-28-7 for routing and connecting Runs M1325 and M1326 (PQR 13209846 & 13229304). Section 4-29: Added rear view of CAM module to show connection of fiber optic wire of E2020 to J18/19 (PQR 13210022). Section 4-29-1: Added step for installing double fiber optic cable to back of ICN's (SPR MRIhc42954). Separated out the cable connections to the SPW and PEN Panels by creating Sections 4-30 and 4-31. Section 5: Added section 5-5 which provides instructions for installing the cable curtain at top of the magnet (PQR 13224796). Section 6-8: Added missing section reference number in step 4 (PQR 13221882). Section 8: Added section 8-2. This section provided information on MR450 catalog offerings. Updated section 8-3 by adding MR450 1.5T specific cables.
8	... Nov 4, 2009	.. Section 1-5: Removed last row in Table 1-2 (MRIhc38291). Section 1-12-2: Removed "Option" from top of In-Room Display box in flowchart (DOC0593027). Section 2-3: Updated removal step for coolant hose covers (ECO 2088746). Section 4-1: Revised Cable Interconnect TOC by removing 3T 16-Channel w/o IRD cable maps. These systems are not offered anymore (DOC0593027). Section 4-14-4: Corrected typing error in step 7 with inclusion of F2 and F4. Section 4-20: Updated valve assembly part number information and added installation notes to clarify valve and hose connection procedures (ECO 2092926). Section 4-20-6 & 7: Revised 4 inch hose and Y-fitting part numbers (ECO 2082392). Section 4-25-3 & 4: Updated photos and instruction for connecting E-Stop, M3366, and M3337 cables to left and right Operator Control Panels (PQR 13274947). Section 4-28-7: Updated step 3 with note and added step 12 to provide guidance on installing Run M3307 and/or terminator (ECO 2088407). Section 8-1: Updated MR750 catalog offerings by adding 16-Channel configuration (DOC0593027).
9	.. Aug 22, 2010	.. Section 1-5: Removed Table 1-2 and moved magnet Temperature Range information to Table 1-1 (MRIhc46642). Section 1-11-1: Revised installation flow chart to reflect changes to respective sections in manual. Inserted Section 2-2 to expand on LOTO for PGR PDU. Section 2-6: Added note to save cabinet shipping brackets for seismic anchoring (MRIhc49004). Section 3: Added view of new PDU that does not require DIP switch setting (ECO 2090162). Section 4-1: Revised cable maps to reflect new I/O bracket, with cables bypassing bracket to connect directly to rear pedestal (PCN 290378). Updated cable map for removal of Run M3307 and other cable corrections (MRIhc52352) and adding fiber optics installation information (MRIhc51560). Inserted Section 4-5 for alternate stacking of overhead cable trays (MRIhc52726). Section 4-6 & 4-7: Added 32 channel designation to P2004 (MRIhc50338). Section 4-10-1-A & 4-10-2-B: Added instruction for gradient cables to be in a single layer (MRIhc48840). Section 4-10-2-C: Added note for correct placement of gradient cable lockwasher (MRIhc48789). section 4-12-2: Added step for connecting PGR ground to RF common ground (MRIhc48937). Section 4-14-2: Added note for connecting E4007 to alternate ground stud (MRIhc 48936 & MRIhc48938). Section 4-19: Added note for alternate method of installing Blower Box (MRIhc48792). Section 4-20: Updated illustrations to show new HEC cabinet (ECO 2100354). Added instruction to refer to Direction 500097 for installing cryo compressor cooling hoses (MRIhc49003). Added Important Note to not apply cork/rubber wrap until after leak check (MRIhc48856). Section 4-22: Revised and rearranged section pertaining to Bridge Alignment (MRIhc49005, MRIhc47416, & ECO 2093536). Added Section 4-22-13, Reed Switch installation (MRIhc48850). Removed Section 4-23 (PCN 290378). Section 4-25: Removed Dock installation procedures pertaining to obsolete I/O bracket. (ECO 2071893). Section 4-26: Updated cable connecting instructions (PCN 290378). Section 4-26-7: Added step for correct routing of 8w8 cable at rear pedestal (MRIhc52329). Section 4-27: Added note for fiber optic cable minimum bend radius (MRIhc49402). Updated cable map and illustrations to show new ICN, 5337894 (ECO 2090162). Section 4-34: Updated E-Off installation and wire connections (MRIhc47230). Section 6-10: Added view of New Z400 computer for cable connections (G60002DA). Also added information that SCSI tower is not supported for Z400 computer. Section 6-11: Corrected connection of cable connection to J1 of F/O converter box (MRIhc49338). Section 8: Sections 8-1 & 8-2: Removed catalog information from both sections to align information to that of MR450w installation manual. Section 8-3: Updated information in table 8-2 for cables removed by VCP project (PCN 290378).

LIST OF EFFECTIVE PAGES

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i	7	TAB 2		4-3	8	TAB 6		2164904-9	—
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SECTION 1 – GETTING STARTED

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1-1 SYSTEM CONFIGURATION

This manual provides instructions to complete the installation of an MR450 1.5T system and an MR750 3.0T System. The installation procedures of these systems are very similar. When different procedures are required, they will be noted in the flow chart or in the applicable section within the manual.

Additional options are installed according to the installation manual delivered with the option.

1-2 SITE READY CHECK FOR MECHANICAL INSTALLATION

Before equipment is delivered, the following must be complete to avoid delays and confusion.

- ☐ Pre-installation work must be complete. Refer to either:
 - Direction 5500101, *Discovery MR750 3.0T Preinstallation*
 - Direction 5500109, *Discovery MR450 1.5T Preinstallation*
- ☐ Magnet Delivery and Installation complete. Refer to either:
 - Direction 5500098, *GE 3.0T LCC300 Magnet Delivery and Installation*.
 - Direction 5500108, *GE 1.5T and 1.0T LCC Magnet Delivery and Installation*
- ☐ Magnet Subsystem should be installed. Refer to either:
 - Direction 5500099, *GE MR750 LCC300 Active Shield Magnet and Cryogen Subsystem*, Section 1.
 - Direction 2192624, *GE 1.5T & 1.0T LCC Active Shield Magnet and Cryogen Subsystem*, Section 1.

1-3 PRODUCT DELIVERY INSTRUCTIONS

Note

Two or more people are required to accept delivery of equipment due to weight and size of cabinets and other components.

The “Shipping Document” lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Note

INSTALLERS: Make sure the right combination of short and/or long cables have been delivered to site.

Labels that are attached to the outside of packing boxes summarize box contents.

“Product Delivery Instructions” (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M7000BL is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1-4 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-5 MR SYSTEM SHIPPING AND STORAGE REQUIREMENTS

1. The customer must provide a shipping dock or other staging area for unloading system components.
2. The customer is responsible for unloading and moving all system components from the delivery truck or van to final locations in the MR suite. This includes:
 - Door and other openings are sufficiently wide to allow passage
 - Sufficient room is provided for any required dollies or rigging
3. The MR system must be shipped and stored within the environmental requirements listed below:
 - Some equipment is liquid cooled. After coolant is added, the equipment must be kept from freezing.

TABLE 1-2
TRANSPORTATION AND STORAGE ENVIRONMENTAL CONDITIONS

System Equipment	Temperature Range	Temperature Change	Relative Humidity	Humidity Change	Atmospheric Pressure
Cabinets and System	–22 to 158°F (–30 to 70°C)	68°F/Hr (20°C/Hr)	5–95% Non–condensing	30%/Hr	1060 to 525 hPa
Host Computer ¹	–22 to 140°F (–30 to 60°C)		8–90% Non–condensing		1060 to 500 hPa
LCD Monitor	–22 to 158°F (–30 to 70°C)		10–80% Non–condensing		1060 to 525 hPa
8–Port Ethernet Hub	–22 to 158°F (–30 to 70°C)		10–90% Non–condensing		1060 to 525 hPa
Magnet	–22 to 122°F (–30 to 50°C)		n/a		n/a

1-6 NON-ENGLISH LABEL INSTALLATION

Components and accessories delivered to this site may have English labels attached to them. For non-English sites, optional multi-language labels have also been supplied.

If required, non-English labels should be attached over the English labels.

- For Surface Coils and Coil Cables – Apply the appropriate label over the English label. The label is included with the coil component.
- For all other labels, follow the label update procedures in the manual or on the applicable multi-language label sheet.

Verify at the end of the installation that the original English labels are covered by new labels representing the native language of the customer.

1-7 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://supportcentral.ge.com/15563> or <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-8 REQUIRED TOOLS

WARNING!

FERROUS MATERIAL HAZARD! IF MAGNET IS ENERGIZED, THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

Tools to install the MR450 or MR750 system are listed in Table 1–3 below:

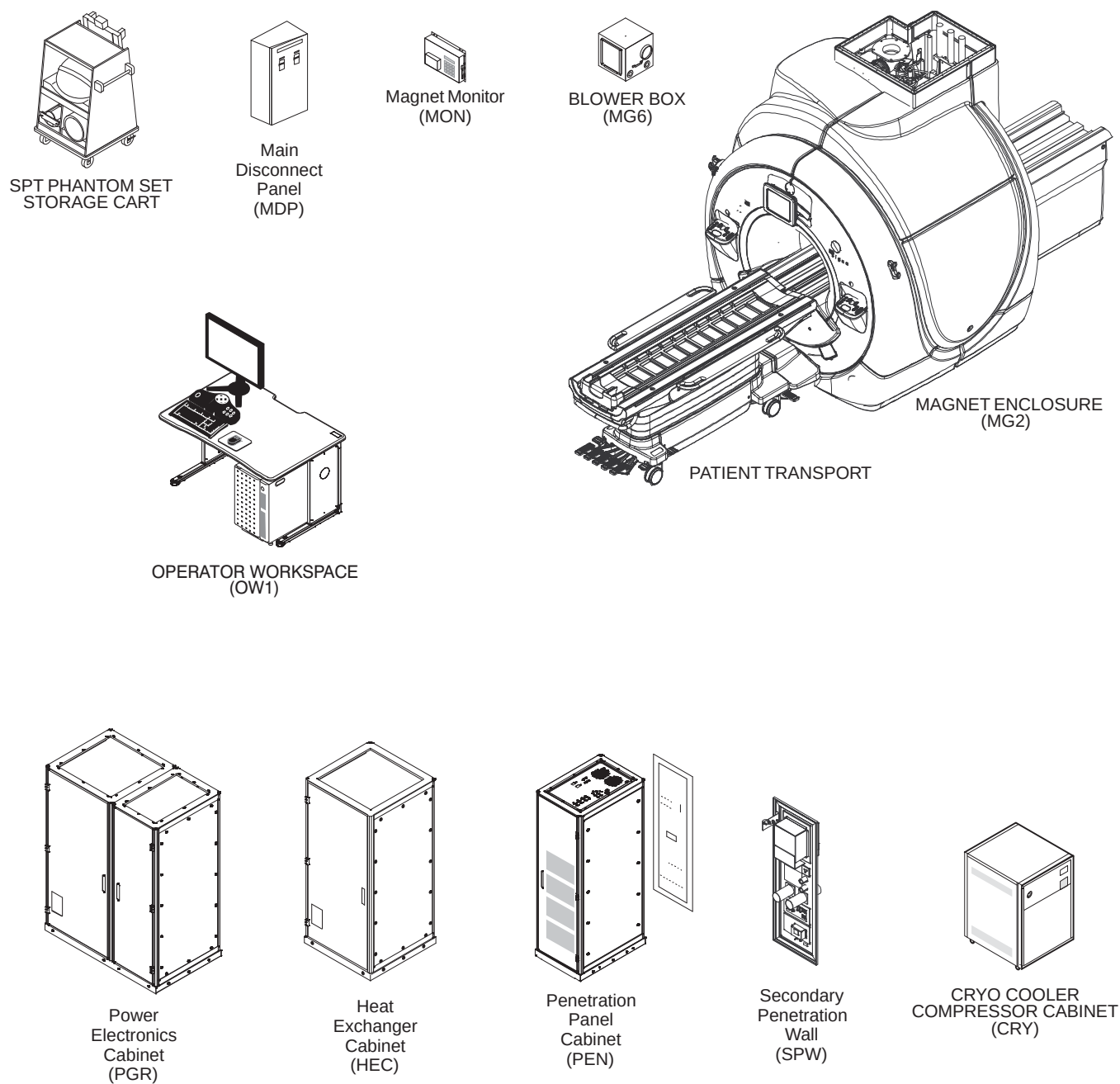
TABLE 1–3
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	—	Wrecking bar
3	—	Claw hammer, 3/4 lb
4	46–271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5		Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
6		Gradient Cable Crimper, Thomas & Betts TBM5 Gradient Cable Cutter, Greenlee 45207
7	46–320273G3 or G4	Non–Magnetic Tool Kit – Universal (See Note 2)
8	46–301450G1	Fiber optic connector repair kit (See Note 2)
9	46–198094P1	Wrist grounding strap
10	—	Volt Meter
11	—	Extension cords, with ground conductor
12	—	Power strip, grounded type, with minimum of five outlets
13	—	Plastic or aluminum flashlight
14	—	Assorted crimp tools.
15	—	Non–magnetic level, 2 foot length minimum
16	—	Non–magnetic tape rule, 12 ft
17	—	Assorted drill bits
18	—	Inspection mirror
19	—	Hobby and utility knives
Note		
1	Supplied as part of Discovery.	
2	Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.	

1-9 BASIC SYSTEM

The basic system, Illustration 1-1, consists of the following major equipment:

- The Magnet with Magnet Enclosure and Magnet Accessories consists of either:
 - 3.0T (30 kilogauss) LCC300
 - 1.5T (15 kilogauss) LCC
- Shield/Cryo Cooler Compressor Cabinet
- Gradient (XRMB) and RF body coils and Head Coil
- Main Disconnect Panel
- System electronics cabinets:
 - PGR Cabinet containing HFD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480 Volt, 50/60 Hz with power filter, CAM Chassis with boards for either 16 or 32 channels, components for the Volume Reconstruction Engine (VRE), Narrow Band RF Amplifier
 - HEC Cabinet containing the Cryo Coolers, Cooling Loops, and Gradient Blower.
 - PEN Cabinet containing the PHPS2, Scan Room power supply, CAN Fiber I/F, T/R DD Driver module, Penetration Panel, and High Order Shim Filter (optional)
 - SPW (Second Penetration Wall) containing Gradient Filter, Scan Room Filters & I/F, and Ramp/Shim access
- Operator Workspace equipment: GOC Computer Cabinet (containing Linux PC and GOCAA), LCD Widescreen monitor, keyboard, SCIM, mouse and mouse pad, and chair
- Pneumatic Patient Alert System
- Patient Transport Table and cradle
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows

1-9 BASIC SYSTEM (Continued)

DISCOVERY MR450 or MR750 SYSTEM
ILLUSTRATION 1-1

1-10 SYSTEM OPTIONS

This section lists options for the Discovery MR450 or MR750 system which have site preparation impact.

- Advantage Workstation (AW)

Note

LCD monitor is recommended for a AW to be located near the system Operator Workspace due to gauss field proximity limits.

- Bar Code Reader for use in conjunction with HIS/RIS software option.
- Additional Patient Transport table with cradle.
- Multi-Nuclear Spectroscopy (MNS) options.
- BrainWave option
- Other hard copy devices and patient accessories.

1-11 INSTALLATION PROCEDURE

1-11-1 Installation Flow

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

The flow chart on the next page should be followed for an orderly and efficient installation of the Discovery MR450 or MR750 system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

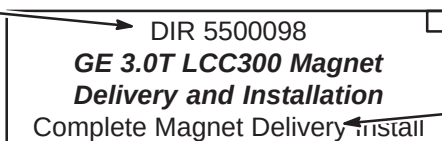
The flow chart references the Tab number and/or Section number for the respective installation procedures. These procedures are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The Procedures must be returned to the binder for future reference upon completion of the installation.

This installation flowchart has been developed assuming that all MR450 or MR750 equipment has been delivered together. Make sure that every part required is available before starting.

1-11-2 Installation Flowchart Explanation

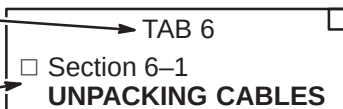
Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.

Use the correct Magnet Manual that applies to the System being installed to locate the installation procedure listed below



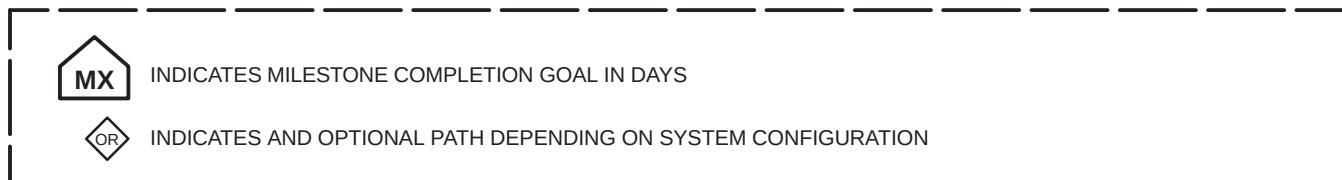
Section within manual to be completed.

Tab within this manual where the Section listed below is located



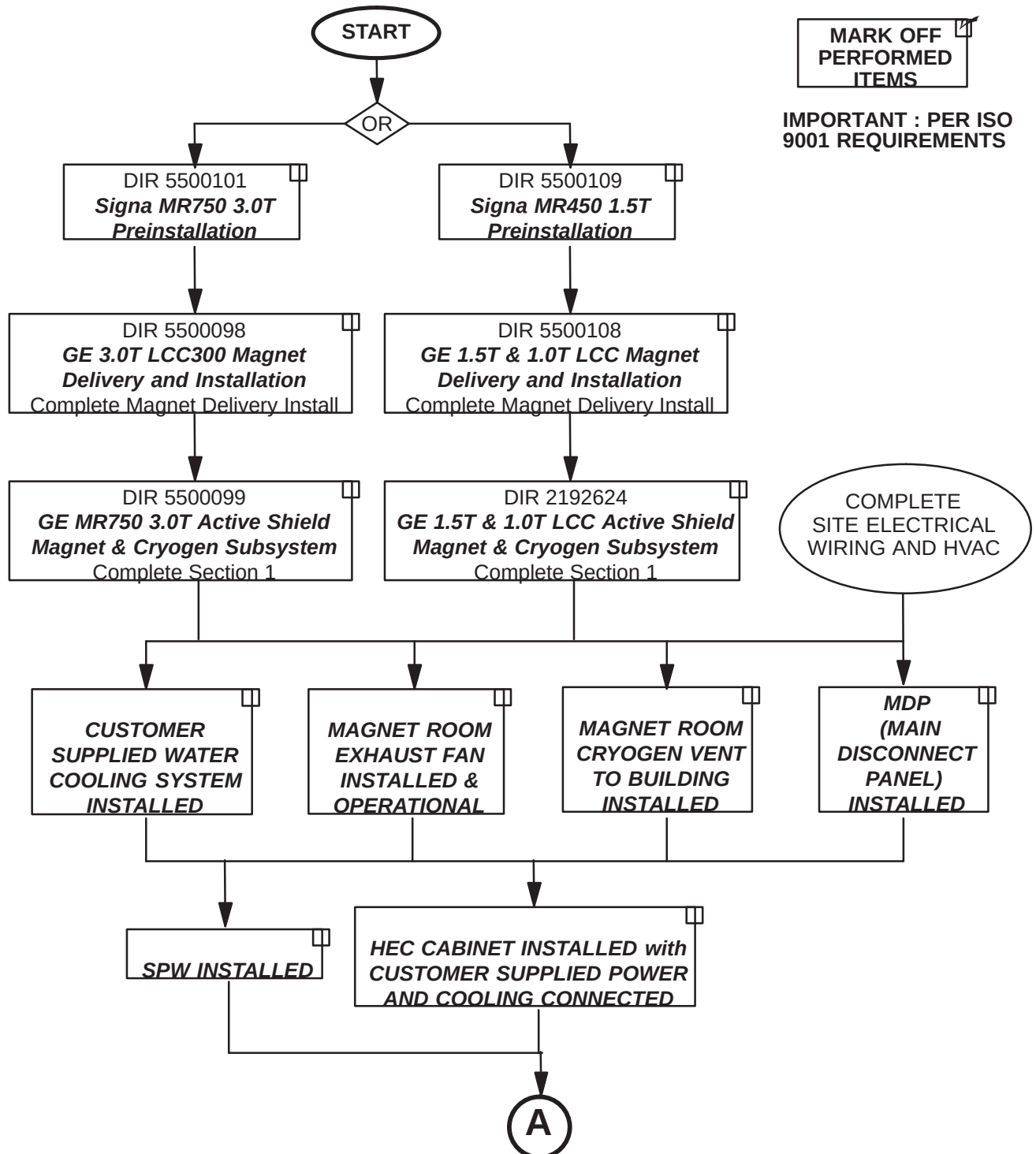
Section within the Tab where the Installation Procedure is located

LEGEND FOR FOLLOWING INSTALLATION FLOWCHART



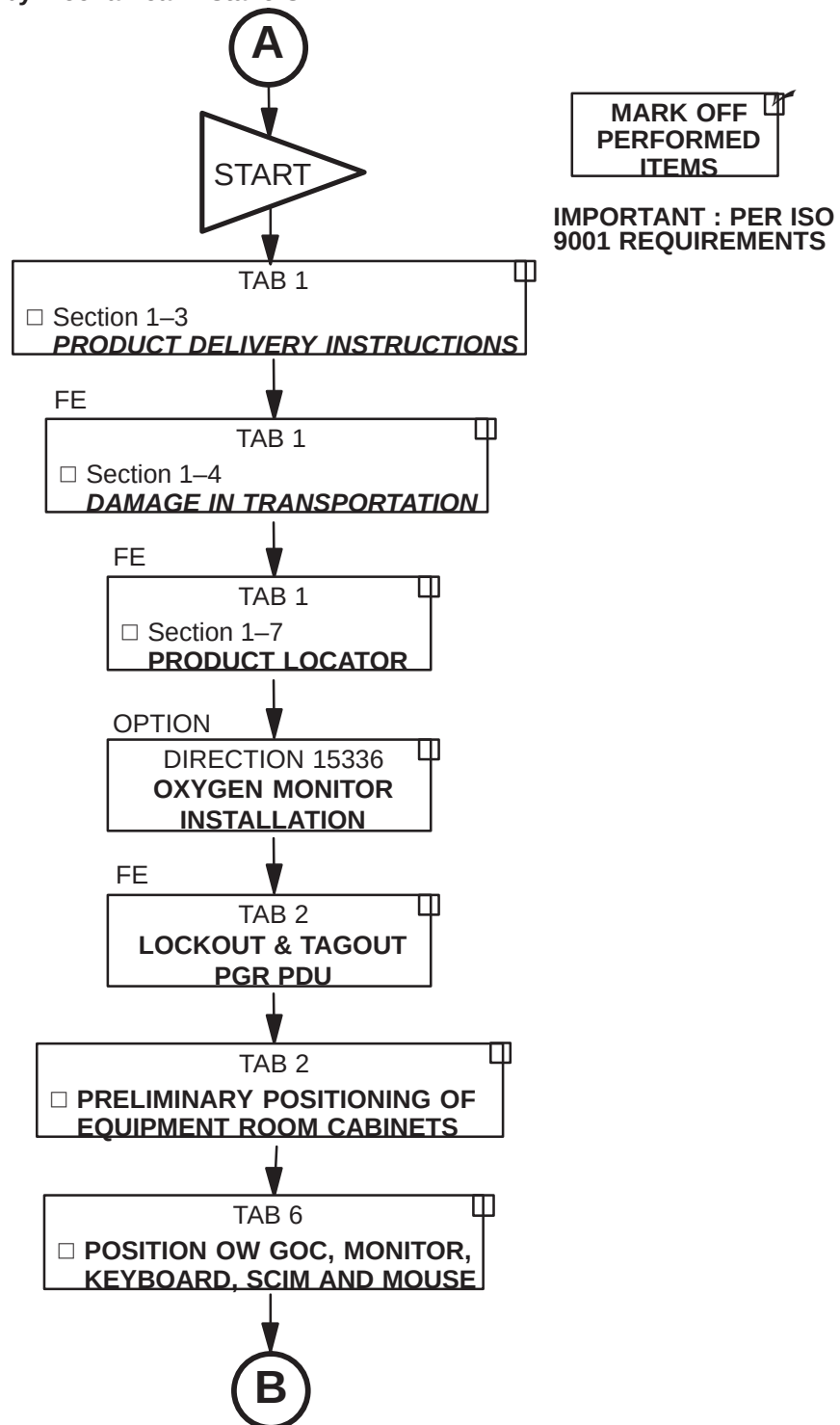
1-12 INSTALLATION FLOWCHART

1-12-1 PREREQUISITES FOR SYSTEM INSTALLATION

Site Installations Required Prior to System Mechanical Installation

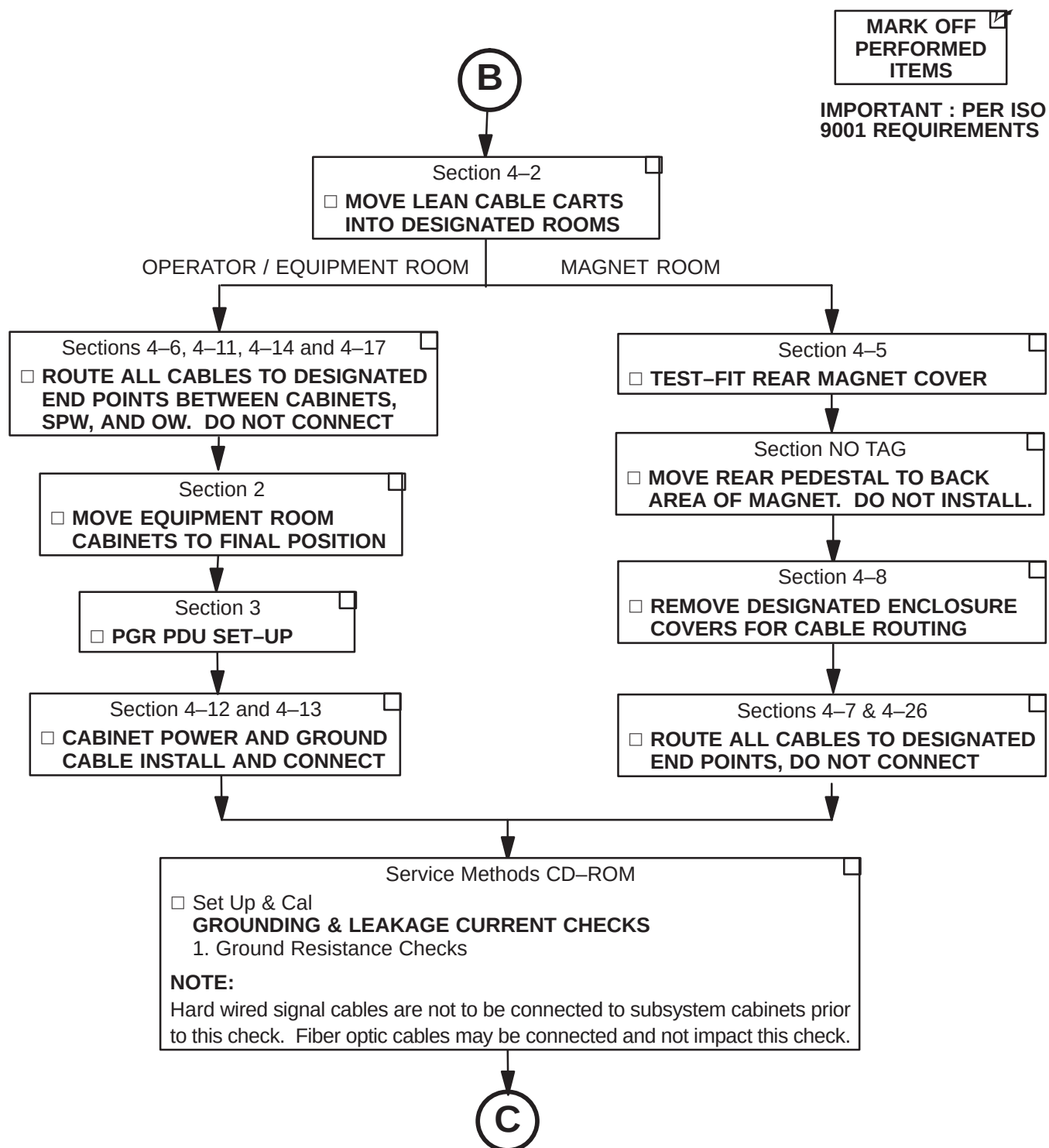
1-12-2 SYSTEM MECHANICAL INSTALLATION

Procedures to be Completed by Mechanical Installers

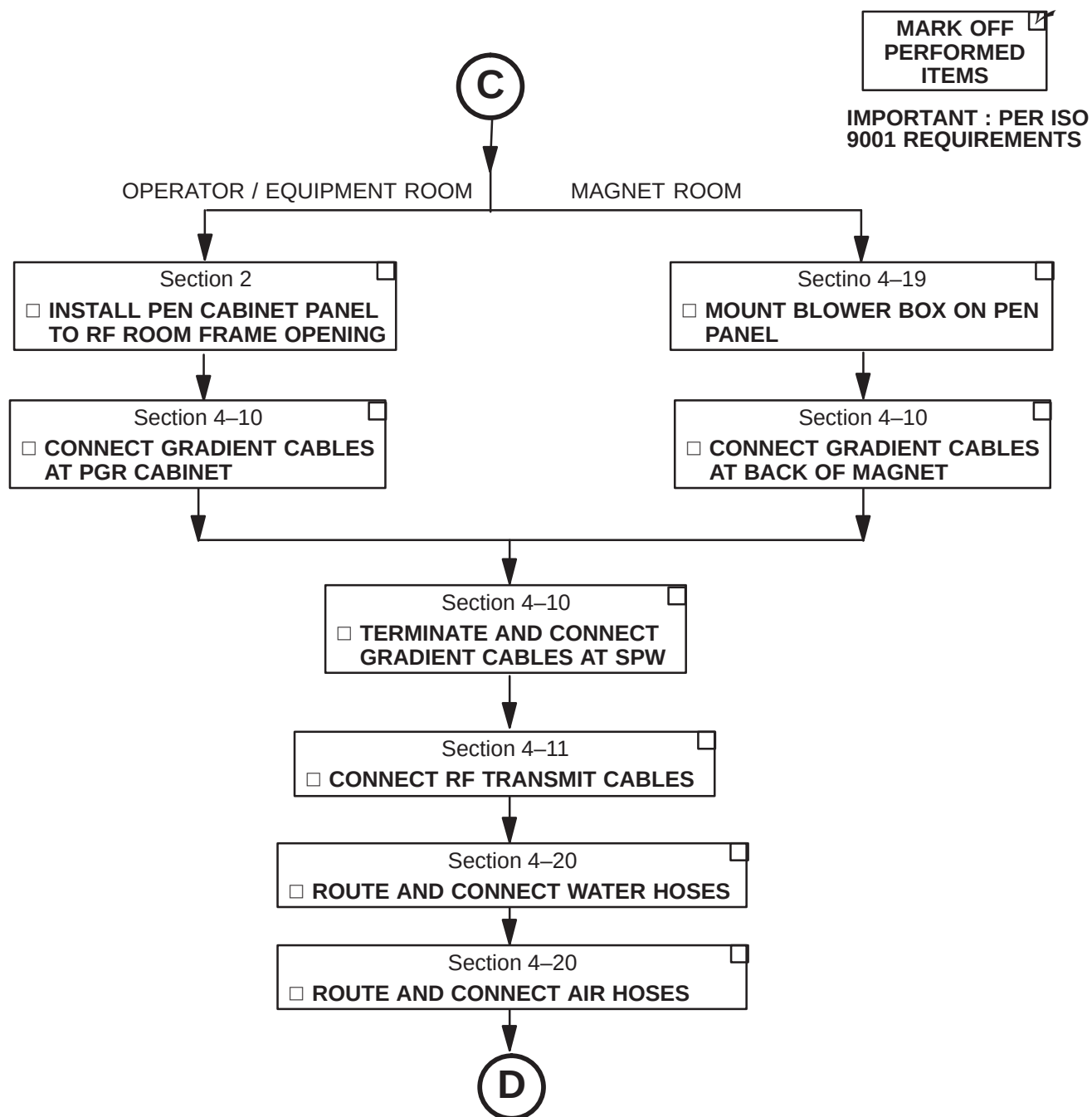


1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Mechanical Installers



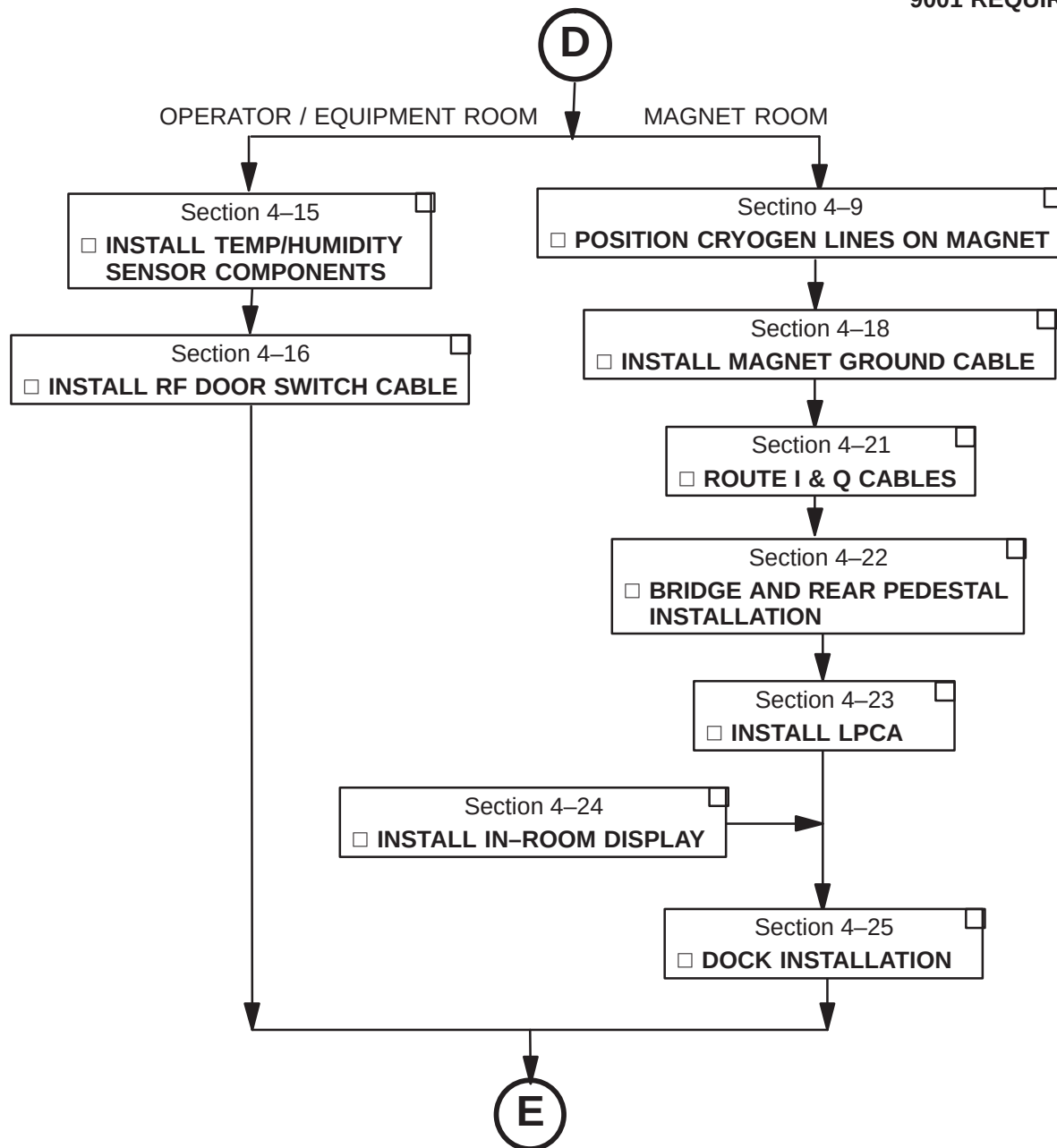
1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Mechanical Installers

Procedures to be Completed by Mechanical Installers

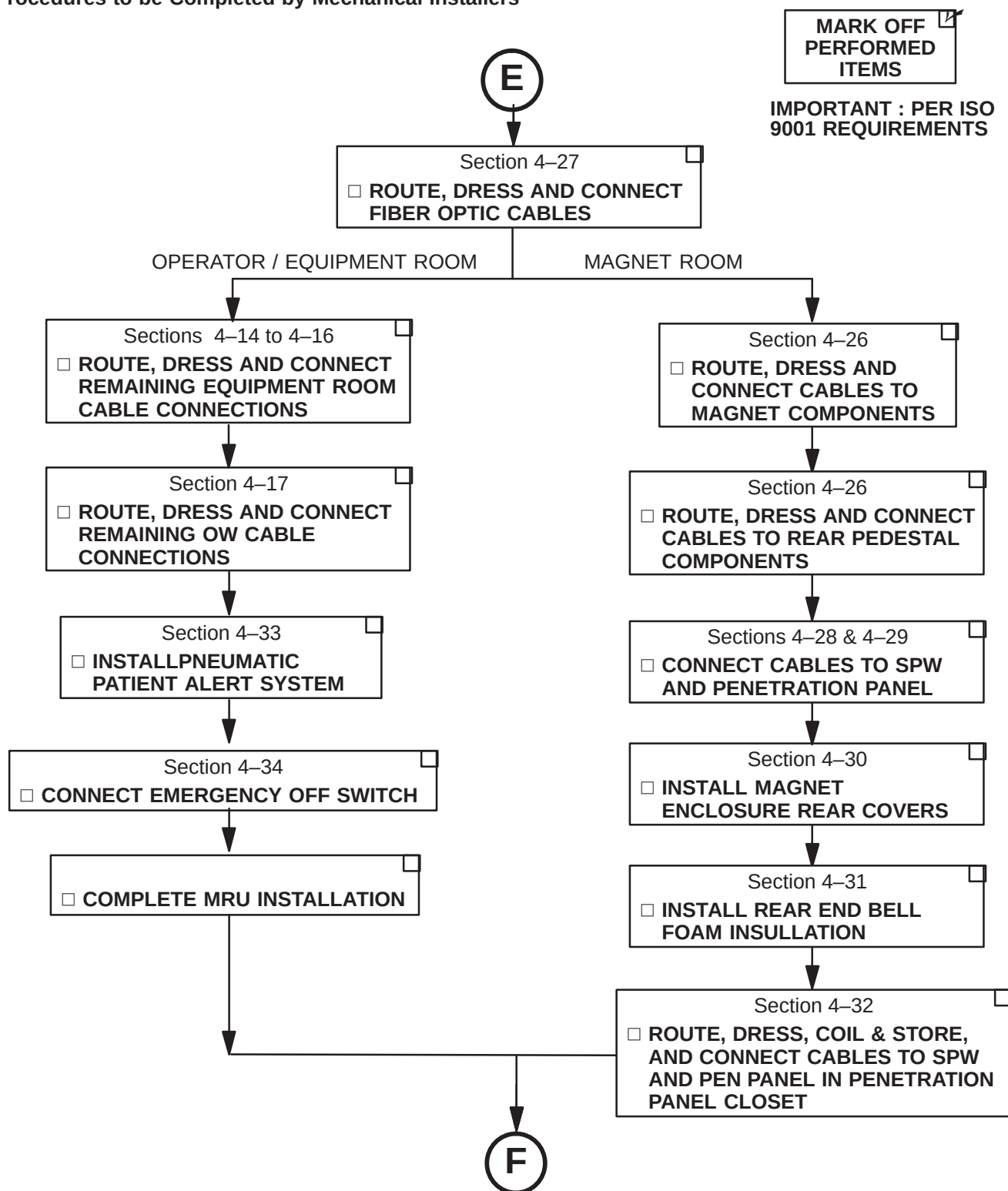
**MARK OFF
PERFORMED
ITEMS**

**IMPORTANT : PER ISO
9001 REQUIREMENTS**



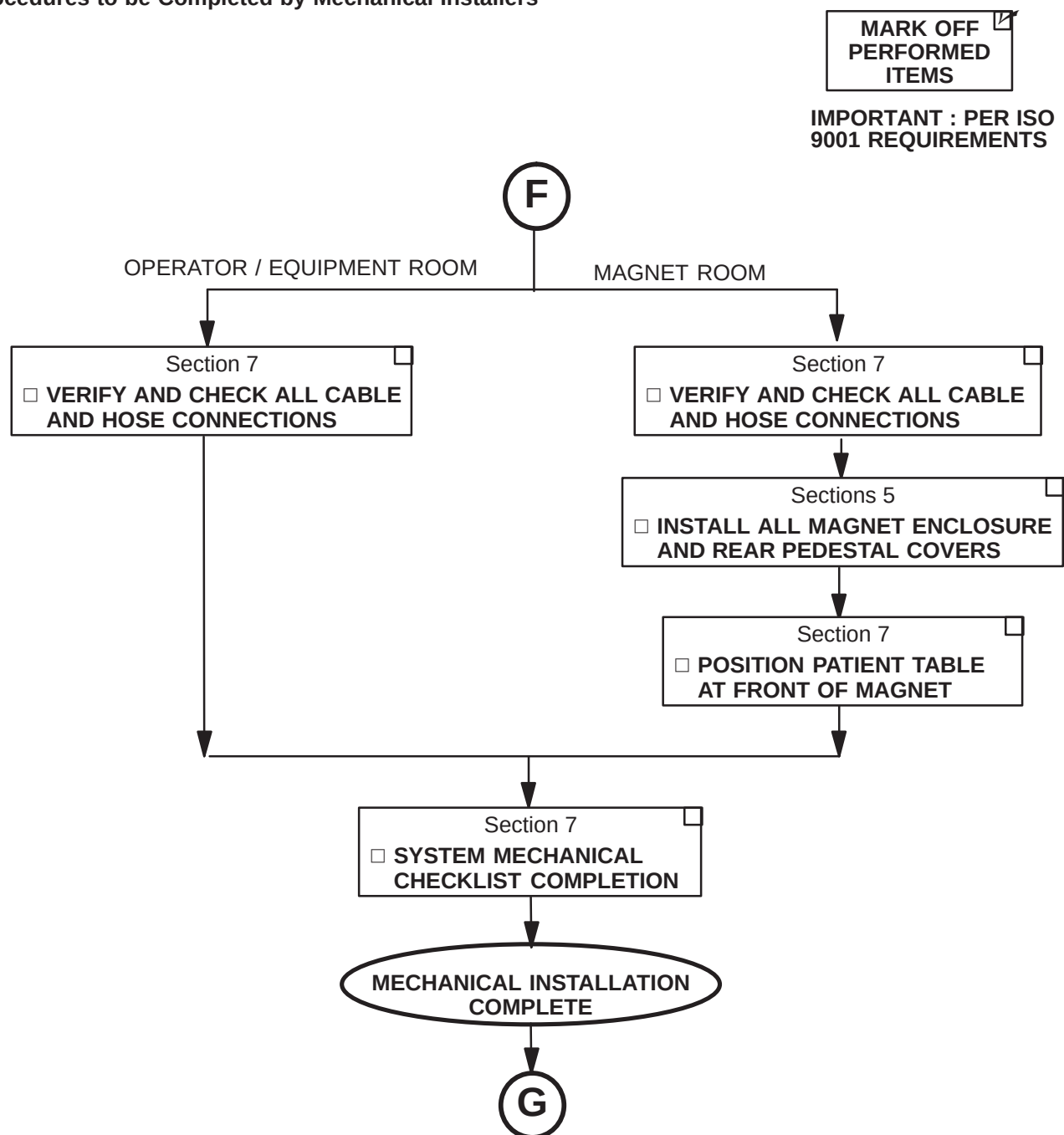
1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Mechanical Installers



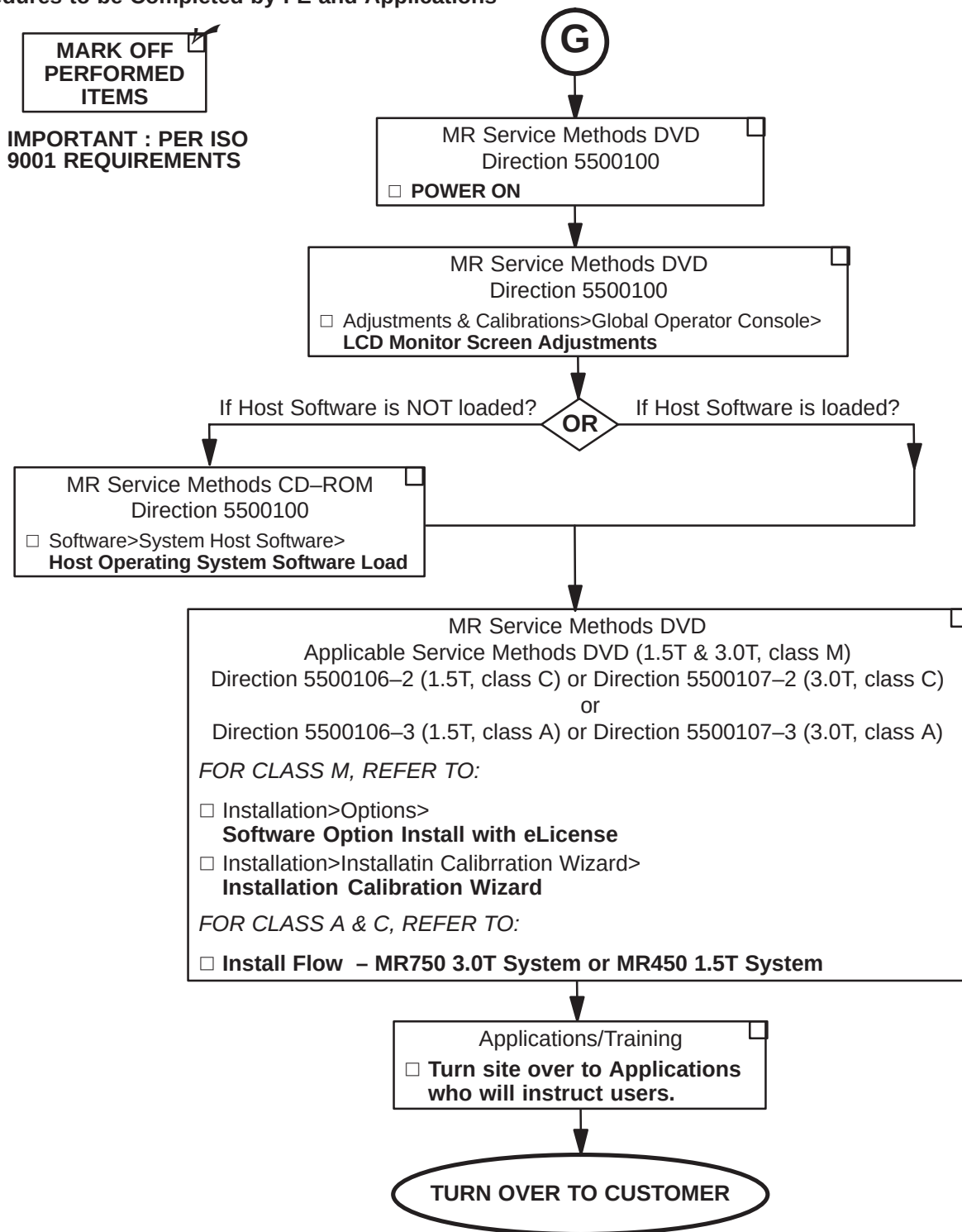
1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Mechanical Installers



1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by FE and Applications



SECTION 2 – CABINET POSITIONING

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2-9 FINAL POSITION OF PGR CABINET	2-11
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2-1 CABINET INSTALLATION REQUIREMENTS

Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)
- For crane rigging, contact your local field service engineer or Project Manager Installation, PMI.



Equipment room cabinets must not be installed in an area with magnetic fields higher than 50 gauss. Refer to site layout plans for positioning of cabinets.



DUE TO WEIGHT OF CABINETS, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE ANY CABINET

2-2 Lock Out/Tag Out (LOTO) for PGR PDU

The Main Disconnect Panel (MDP) has been installed and is providing power to the HEC Cabinet and Cryogen Cooler Compressor. Before installing the PGR Cabinet, the PDU input switch on the front of the MDP must be locked out and tagged.

The steps in this Lock Out/Tag Out (LOTO) procedure must be performed by a LOTO authorized GE Field Engineer. Completing all the steps in this procedure ensures a safe environment and avoids equipment damage when working on these parts.

TABLE 2-1
AFFECTED BY LOTO

Name of Equipment	Number of Locks	Titles of Employees Authorized to perform LOTO	Titles of Affected Employees	How to Notify
Heat Exchanger Cabinet (HEC) Power, Gradient, RF Cabinet (PGR) with Power Distribution Unit (PDU)	2 per GE Field Engineer (FE)	GE Field Engineers	Hospital Personnel	Verbal, Posted signs

2-2-1 Preliminary Requirements

TABLE 2-2
TOOLS AND TEST EQUIPMENT

Item	Part #
Personal Lock	46-194427P320
Multi-locking Device (If multiple service people)	46-194427P313
Red Warning LOTO Tag	46-194427P322
Digital Voltmeter	46-194427P284

2-2-2 Safety



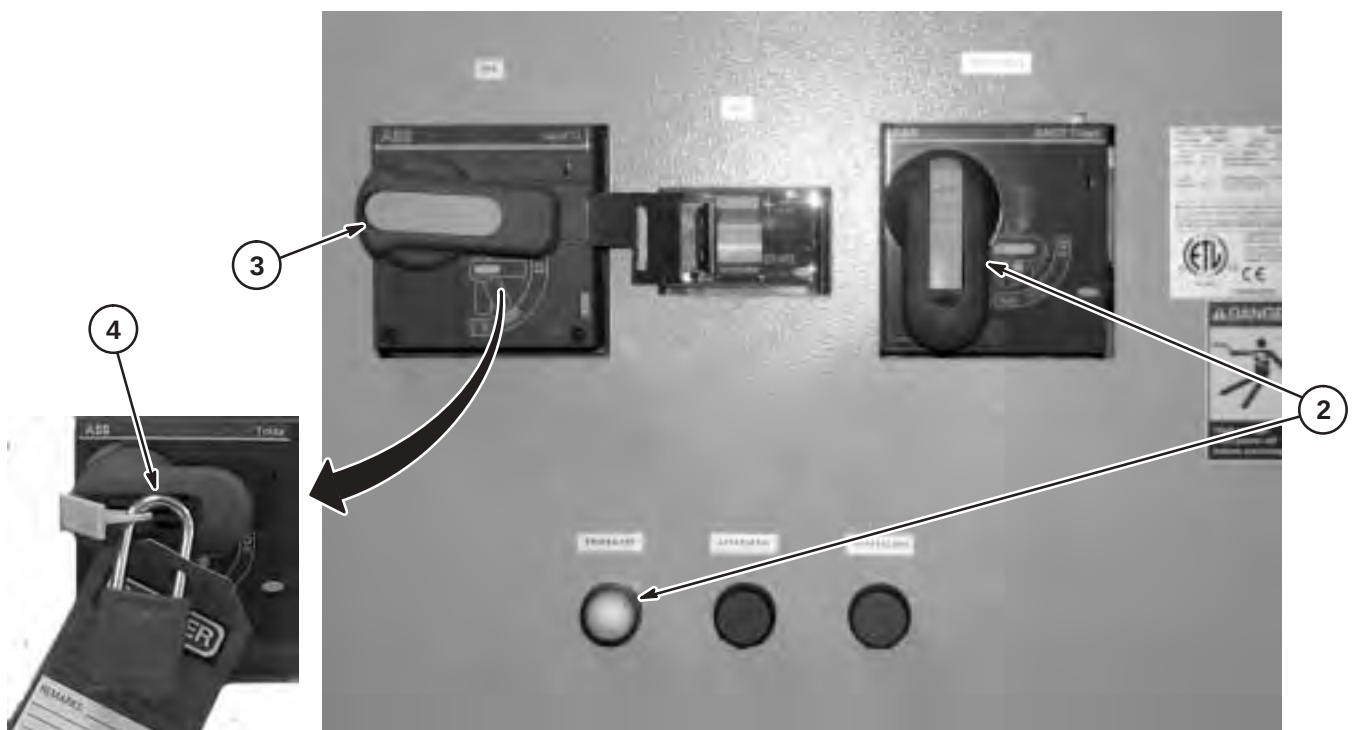
**ELECTROCUTION HAZARD!
HIGH VOLTAGE PRESENT.**

**USE PROPER LOCK-OUT / TAG-OUT PROCEDURES BEFORE INSTALLING, SERVICING
AND/OR MAINTAINING.**

2-2-3 Applying LOTO to PGR Cabinet PDU on MDP

CAUTION

The following process only removes power from the PGR Cabinet. The HEC will continue to receive power so that the cryogen compressor can continue operation uninterrupted. Powering off the HEC circuit breaker will cause the Cryogen Cooler Compressor to also lose power. If this is done make sure that the HEC downtime is minimized to avoid boiling off Cryogenes.



1. To prepare for shutdown of equipment, notify affected personnel working in the area the LOTO is being performed.
2. Observe that the POWER ON light is on. It signifies the main breaker is closed and control power is on.
3. Make sure the PDU input switch is in the OFF position
4. Lock Out/Tag Out the PDU input switch on the MDP

2-2-4 Removing LOTO

Refer to MR450/MR750 Service Methods DVD for procedures to remove LOTO on PDU input switch.

2-3 CABINET POSITIONING PREPARATION

The illustration below shows a typical layout for the equipment room. For ease of cable installation in overhead cable trays, it is recommended that cabinets NOT be positioned in final location until after cables have been routed.

5. The HEC cabinet has already been positioned and installed prior to delivery of PEN and PGR cabinets.
6. Refer to site layout drawings to determine location of PEN and PGR cabinets.
 - If available, use the cabinet floor layout templates for positioning cabinets
 - If templates are not available, using the minimum spacing distances as noted below, apply tape on floor to mark final location of front corners of cabinets.

Note: The procedures for removing the dollies from the HEC cabinet and attaching them to the PEN and PGR cabinets are provided in Sections 2-5 thru 2-10.

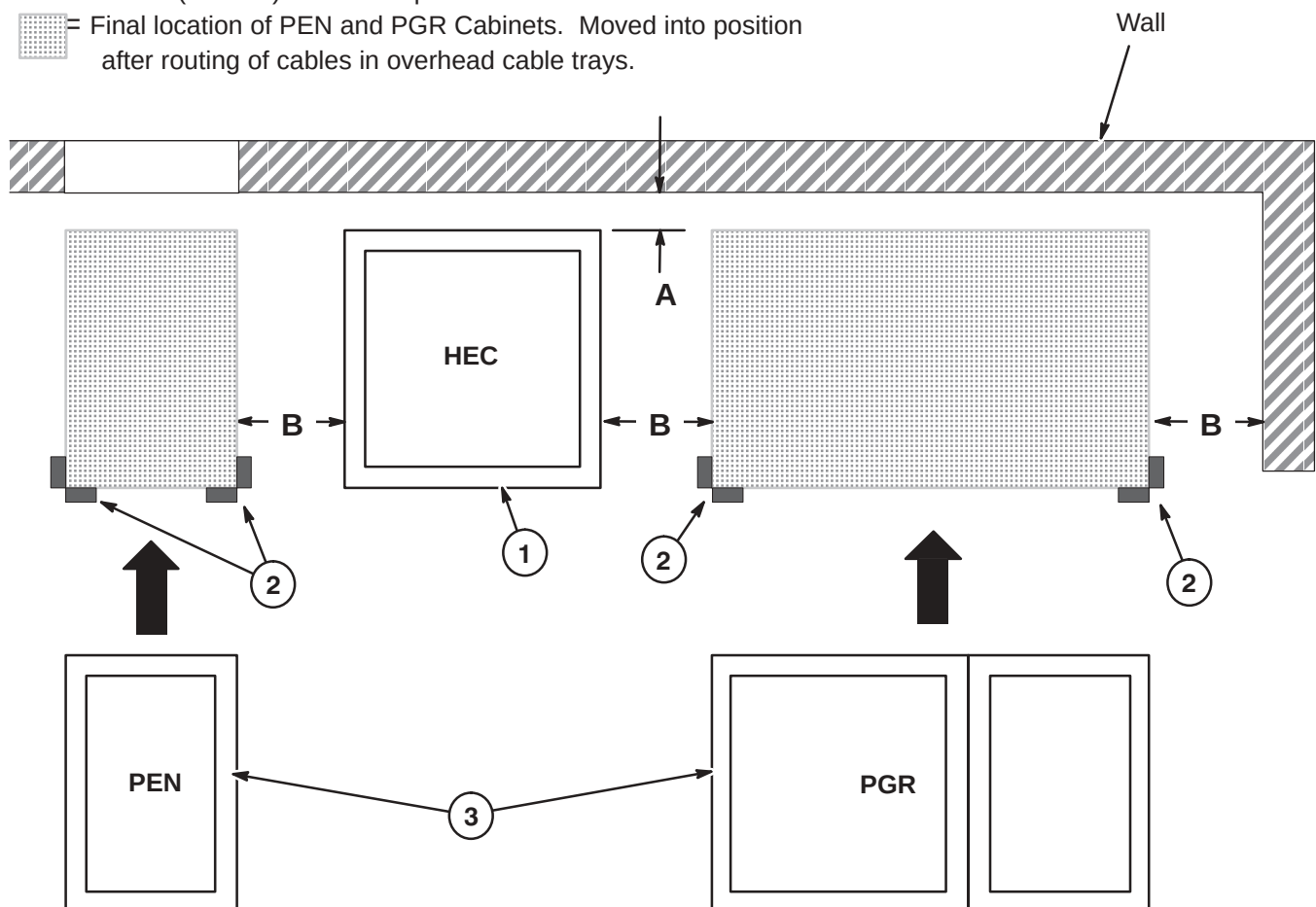
7. Using the dollies, move PEN and PGR cabinets into position in front of final location.

Legend:

A = 4 inch (102mm) minimum space distance from wall

B = 15 inch (380mm) minimum space distance at side of cabinet

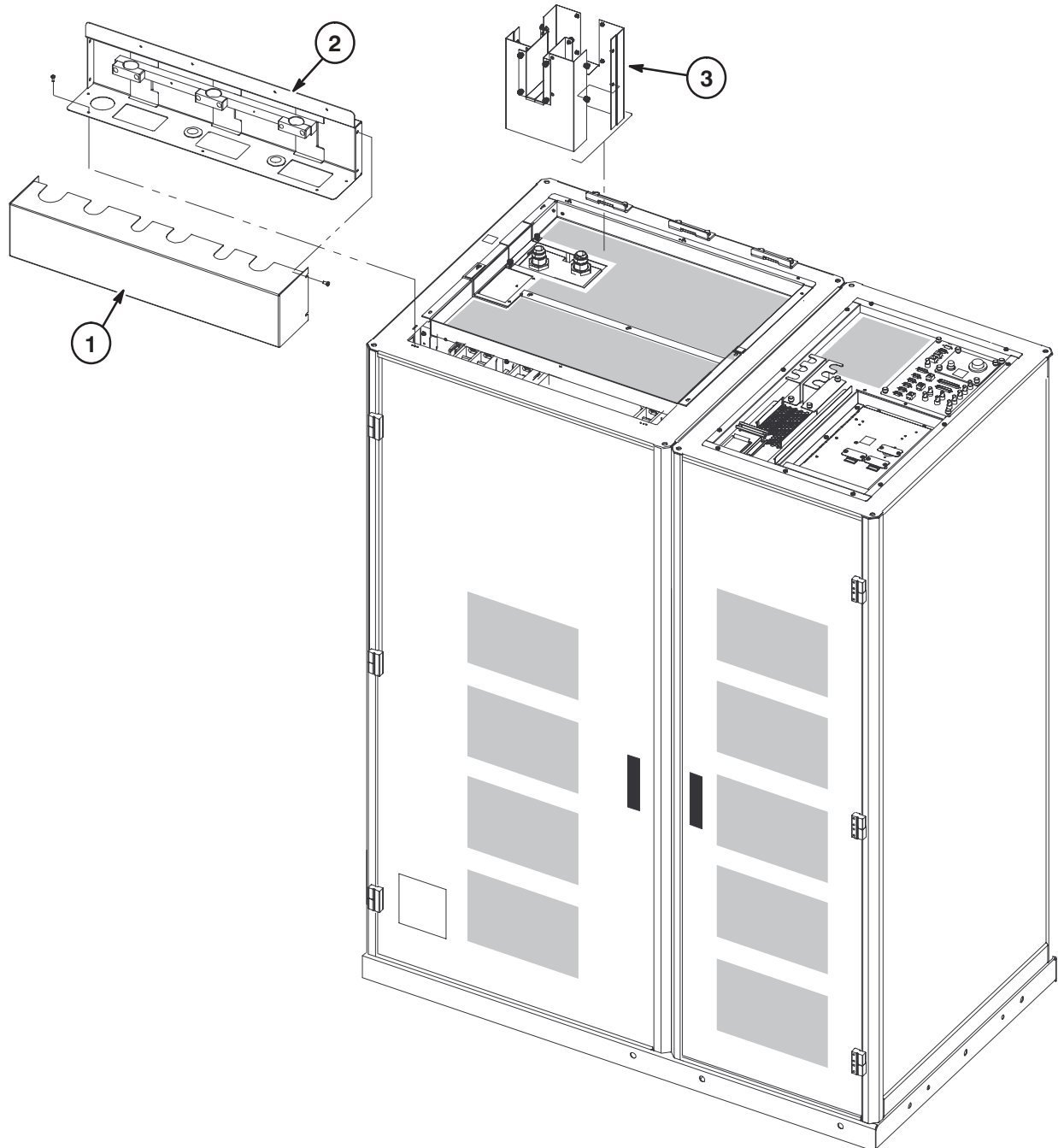
 = Final location of PEN and PGR Cabinets. Moved into position after routing of cables in overhead cable trays.



2-4 REMOVING COMPONENTS FROM TOP OF PGR CABINET

If the PGR cabinet, with the brackets and covers attached to the top of cabinet, is too tall to go thru doorways, then these components will have to be removed. They will be re-installed after cabinet is placed in final position.

1. Unfasten the four screws that attach the front cover to the cable bracket and remove cover.
2. Unfasten the five screws that attach the cable bracket to the top of the cabinet and remove bracket.
3. If provided on cabinet, lift and remove the protective coolant hose covers. These will be re-installed after the coolant hoses are connected.



2-5 CABINET MOVING DOLLIES

The HEC cabinet has already been delivered to site and installed. If dollies are not on site, they can be ordered online at www.umi-dollyshop.com. For technical questions, contact: UMI, 1-800-638-7457. The moving dollies that are required for positioning the cabinets should have remained attached to the HEC cabinet.

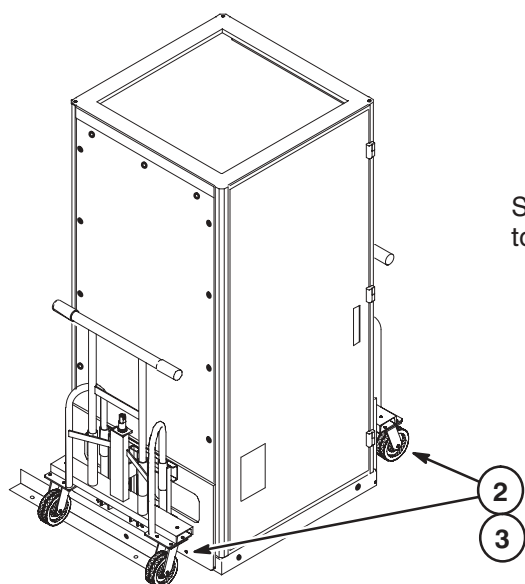
Note: The dollies have the following features to aid in their use:

- Wheel Locking Pin – Keeps the wheels locked into a certain direction
- Parking Brakes – For holding the cabinet in position
- Slide and Lock – The dollies can be removed in limited space (15 inches or 380mm)
- Twinning Hardware – hardware to connect each dolly to the other dolly during transport without a cabinet

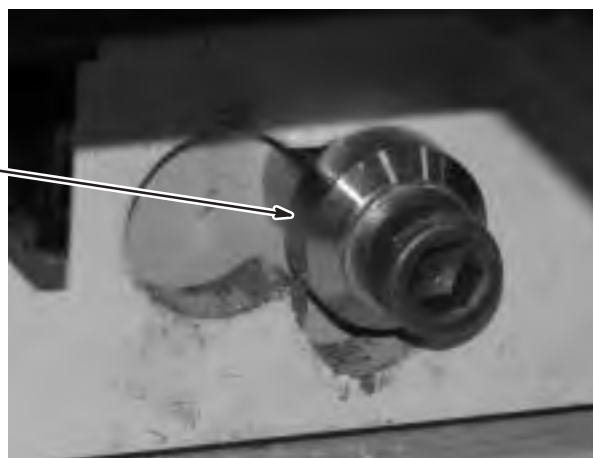
1. The dollies are designed with a slide and lock attachment plate. Slide the dolly until the cabinet bushing lines up with the large opening on the attachment plate.
2. Remove each dolly.

Note: Dolly height may have to be adjusted to aid in the removal process.

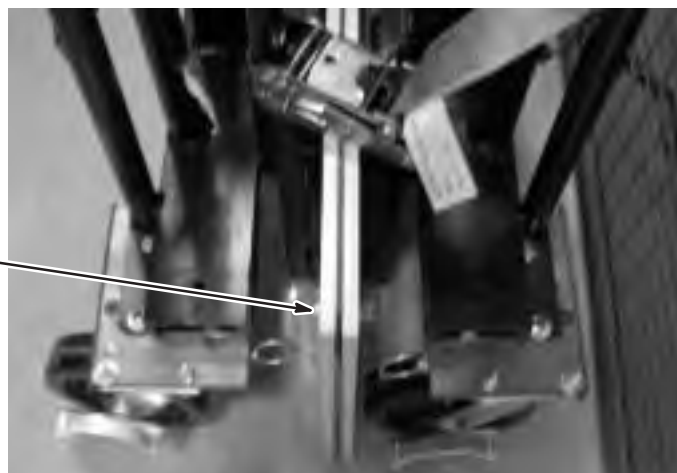
3. Remove each bushing and place in attached dolly bag.
4. Attach the two dollies together using the twinning hardware, if desired.
5. Move dollies to next cabinet to be positioned.



1
Slide Dolly
to big hole

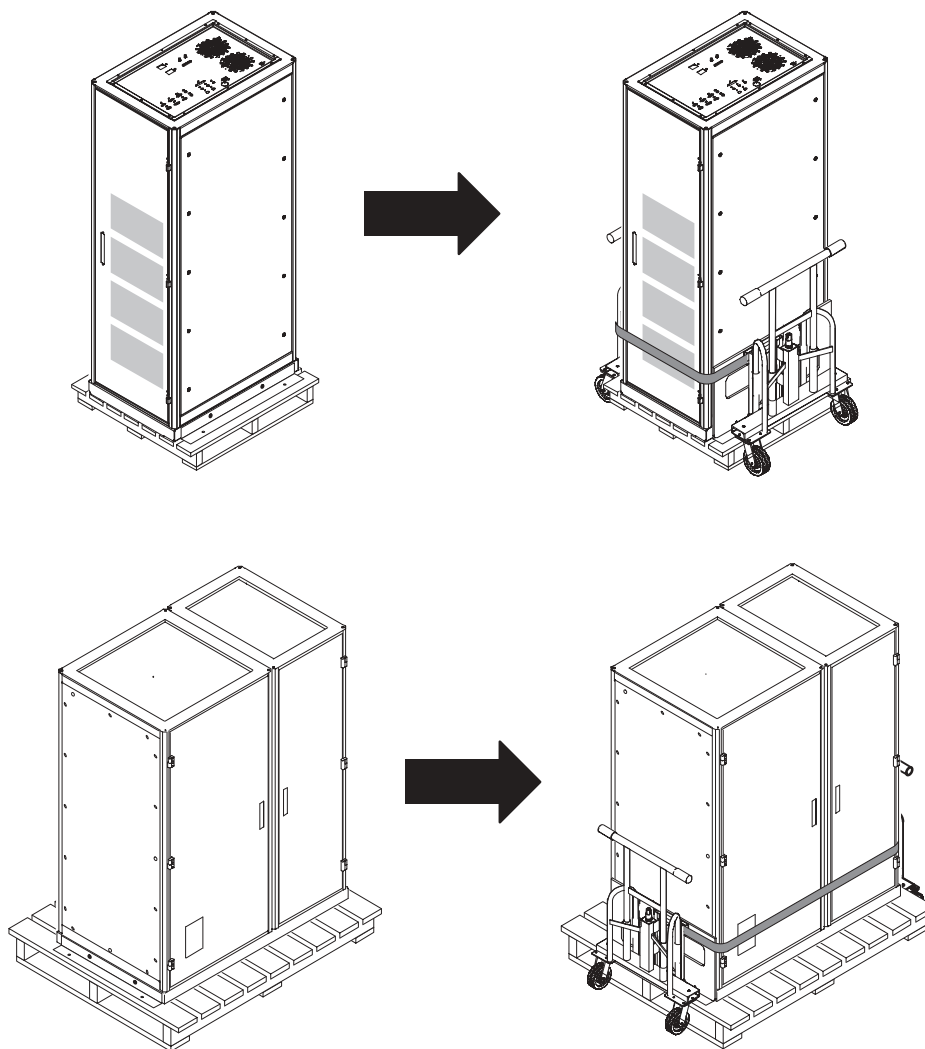


4



2-6 REMOVE CABINETS FROM SHIPPING PALLETS

The PEN and PGR cabinets are delivered to site secured to pallets. The cabinets must be removed from the pallets and positioned in Equipment Room according to site layout plans. The cabinet moving dollies that were attached to the HEC cabinet, which was delivered and installed prior to magnet delivery, are required to move the PEN and PGR cabinets off the pallets.



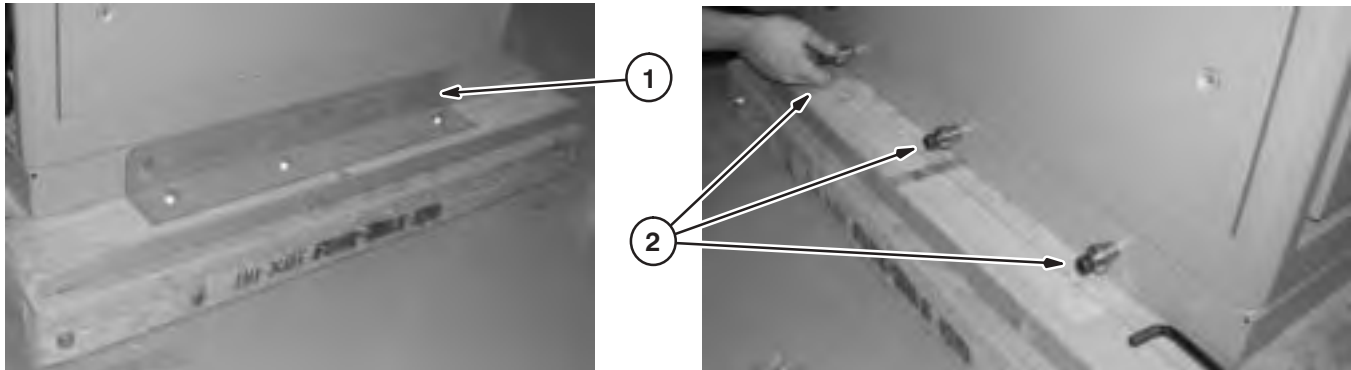
2-7 MOVE PGR CABINET INTO EQUIPMENT ROOM

1. Unfasten screws for each angle shipping bracket and remove bracket from each side of cabinet.

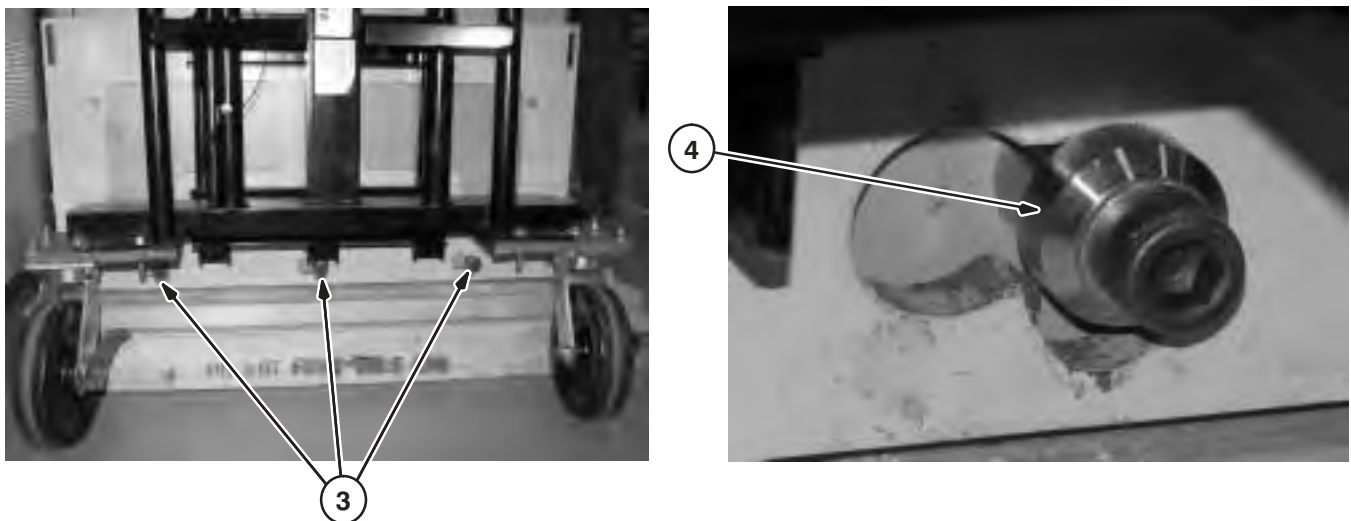
NOTE:

The screws and brackets used to secure the cabinets to the pallets during shipment of cabinets are also used to secure the cabinet to the floor at sites requiring seismic support. **DO NOT DISCARD BRACKETS AND FASTENING SCREWS.** Retain and save for customer.

2. Install the six bushings, three on each side, to the bottom of the cabinet.

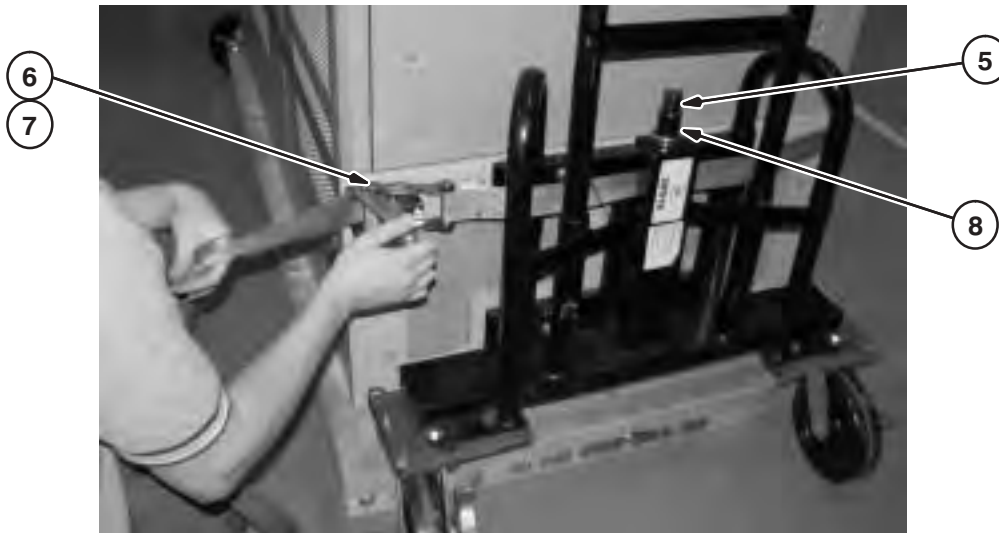


3. Maneuver the dolly so that all three bushings slip into the slide lock on the attachment plate.
4. Slide the dolly sideways to position the bushings in the lifting part of the attachment plate.

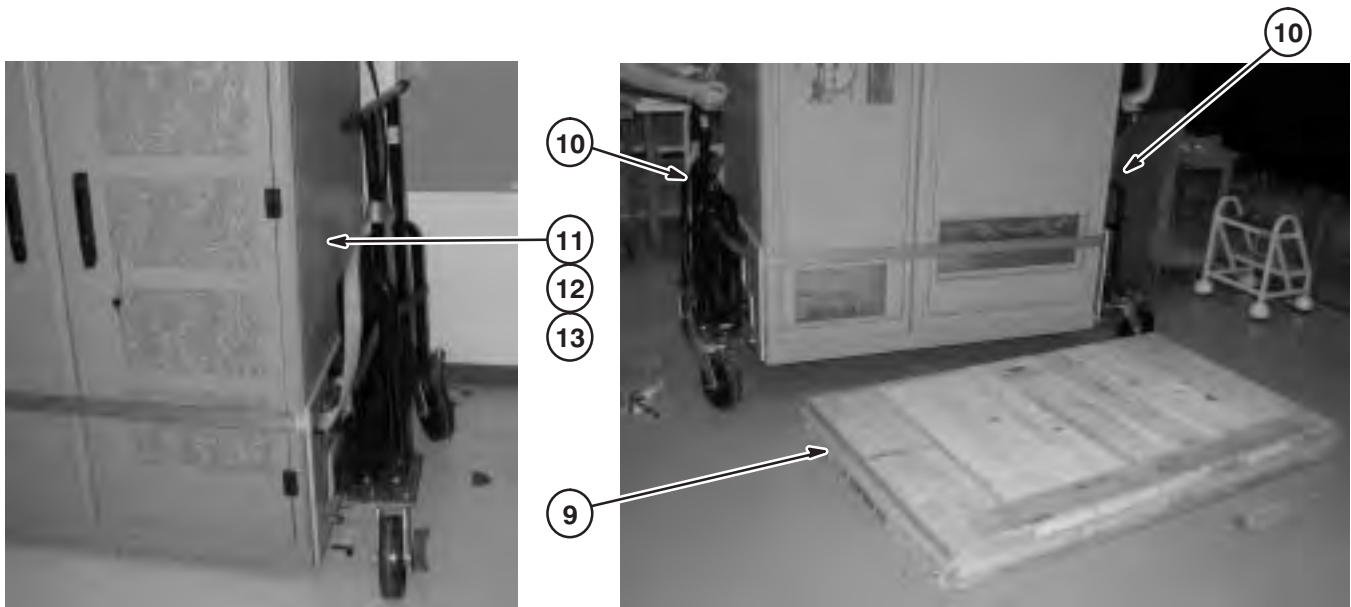


2-7 MOVE PGR CABINET INTO EQUIPMENT ROOM (continued)

5. Raise the dolly by turning the jack screw clockwise. This will lock the dolly into position.
6. Secure to dollies with strap. Make sure the strap goes through the slot of the attachment plate and the clamp is on flat surface of cabinet.
7. Tighten strap and close clamp.
8. Lift cabinet off of pallet by turning the jack screw clockwise until the cabinet surface is free from the pallet surface.
Note: Alternate raising each dolly in 1/4 inch (6mm) increments until the cabinet is above the pallet.



9. Slide the pallet out from under the cabinet.
10. Lower the cabinet by turning the jack screw counter-clockwise until the cabinet is about 1 inch (25mm) above the floor. **Note:** Alternate lowering each side of cabinet.
11. Move the cabinet into the equipment room.
12. Refer to Section 2-3 for placement of cabinet and lower cabinet into position.
13. Remove dollies for attachment to PEN cabinet.



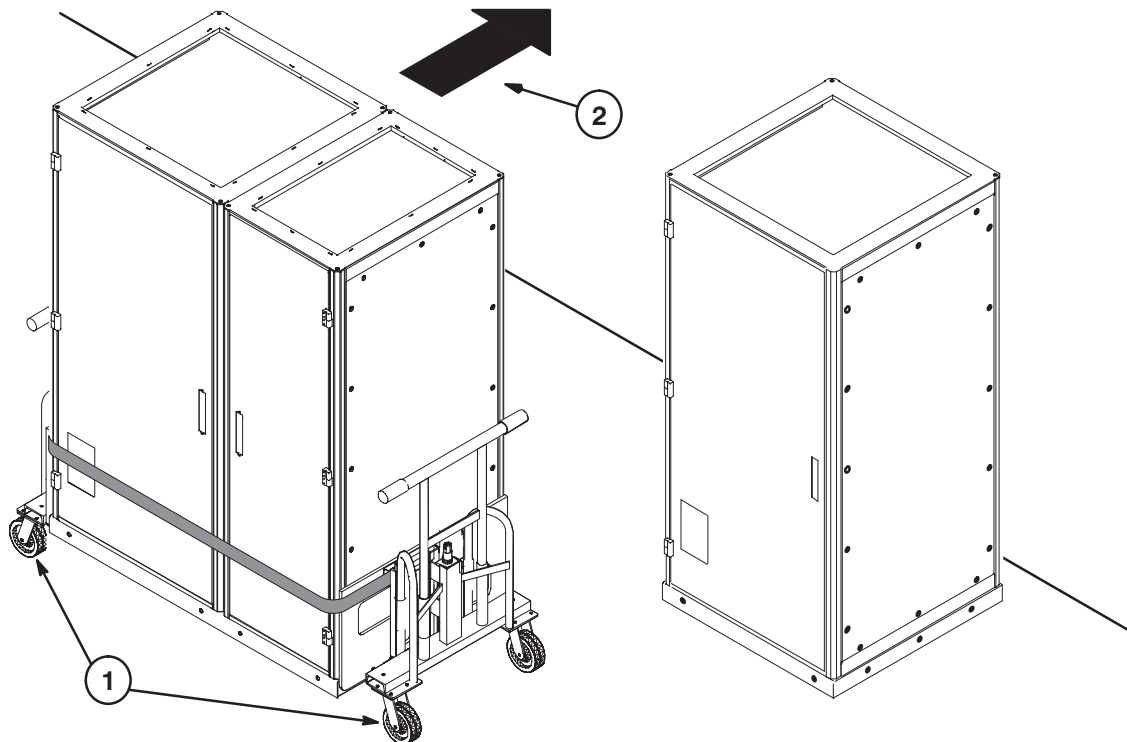
2-8 MOVE PEN CABINET INTO EQUIPMENT ROOM

Repeat the steps in Section 2-7 for removing the PEN cabinet from the shipping pallet and moving it into the Equipment Room.

2-9 FINAL POSITION OF PGR CABINET

The final positioning of the cabinet is recommended after all the cables have been routed in the overhead cable trays.

1. If dollies are not attached to cabinet, re-attach moving dollies to each side of cabinet
2. Lift and move cabinet to final position as indicated by site layout drawings.



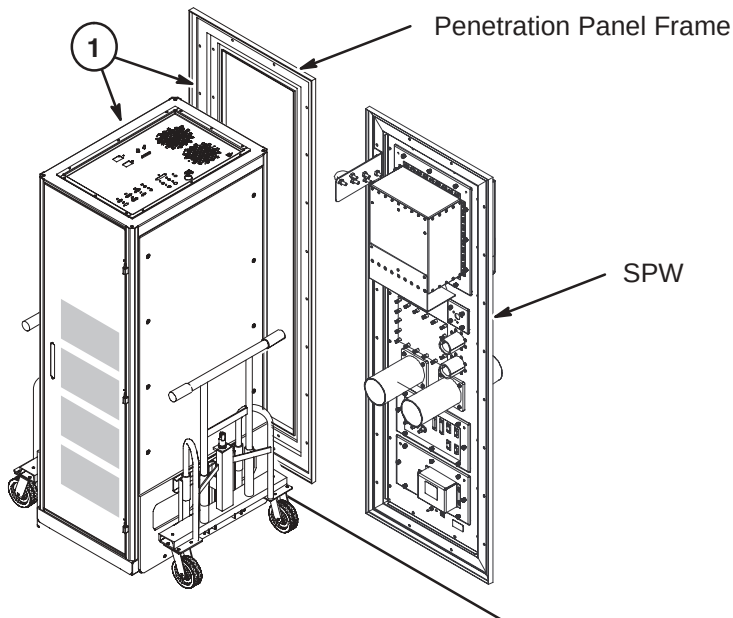
2-10 INSTALL PEN CABINET

2-10-1 Install Penetration Panel in RF Screen Room Opening

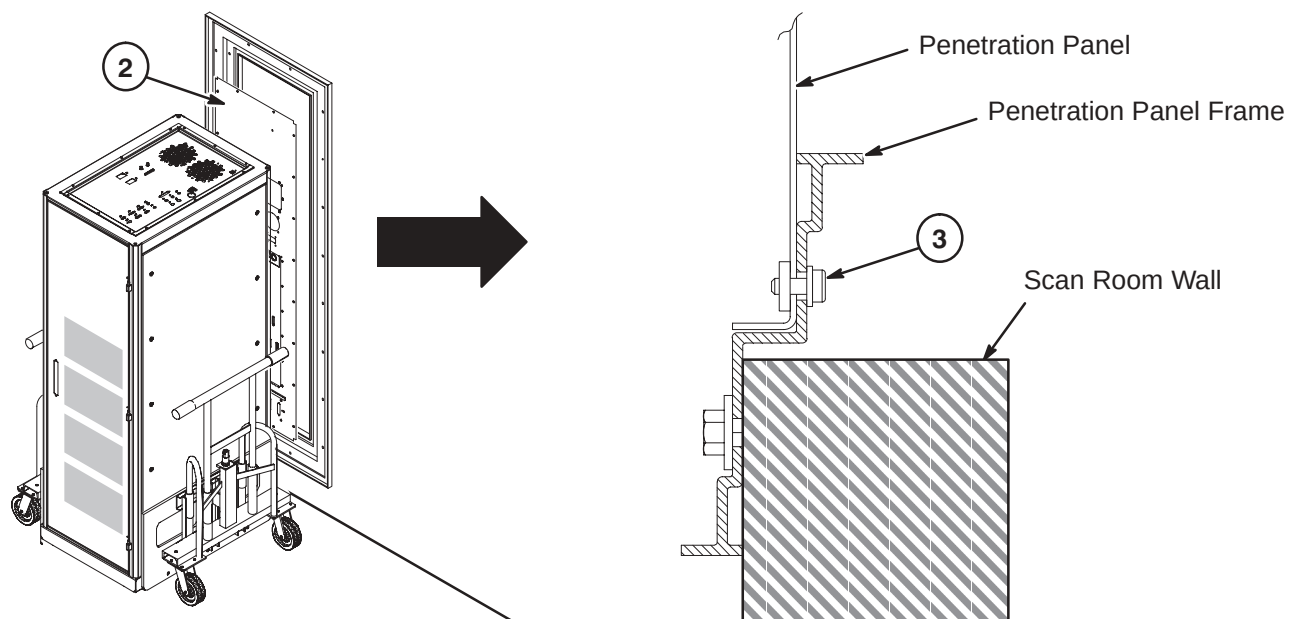
It is recommended that the following steps be completed after routing of cables in overhead cable trays.

1. Move PEN cabinet to within 2 inches (50mm, +/- 25mm) of framed opening in RF room wall.

Note: The SPW was previously installed.



2. Unfasten Penetration Panel from rear of cabinet.
3. Install and secure Penetration Panel into open frame using 20 M5 x 12mm SS Hex screws and 20 M5 SS Flat Washers. **Note:** The fasteners were delivered to site earlier with the SPW installation.

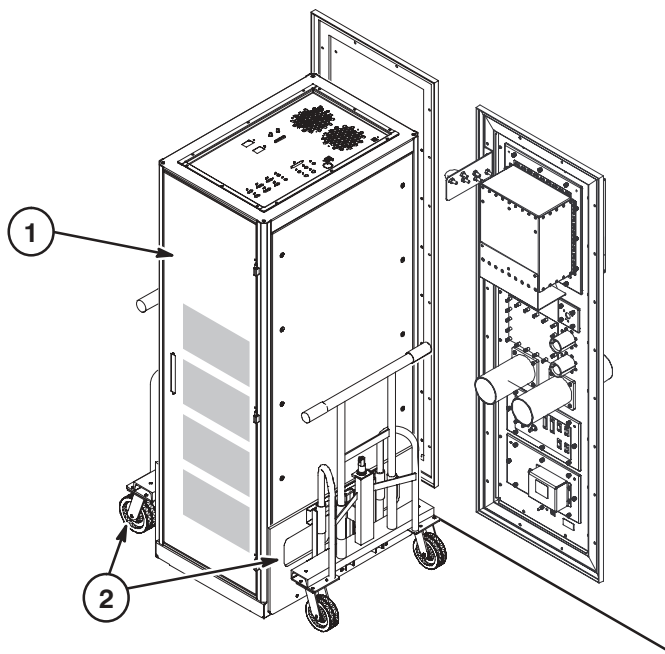


2-10-2 Position PEN Cabinet

1. Move PEN Cabinet into position and lower cabinet into place.

Note: Rear of cabinet must be positioned as close as possible to frame of Pen Panel.

2. Remove cabinet moving dollies.

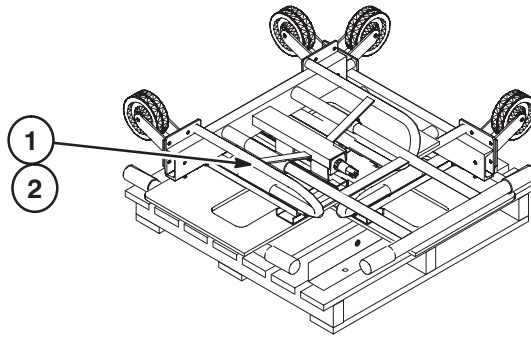


2-11 RETURN MOVING DOLLIES

Moving dollies must be returned for use on future shipments.

1. Secure dollies to one of the pallets used for shipping cabinets to site.
2. Call Tech-Trans at 1-800-851-9929 to make arrangements for dollies to be returned for next shipment.

Note: For sites outside of the United States or Canada, re-use or dispose of dollies per the local requirements.



SECTION 3 – PGR CABINET PDU SET-UP

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3-2 CIRCUIT BREAKER DIP SWITCH SETTINGS (If Required)	3-3
3-3 SYSTEM POWER CONNECTIONS	3-4

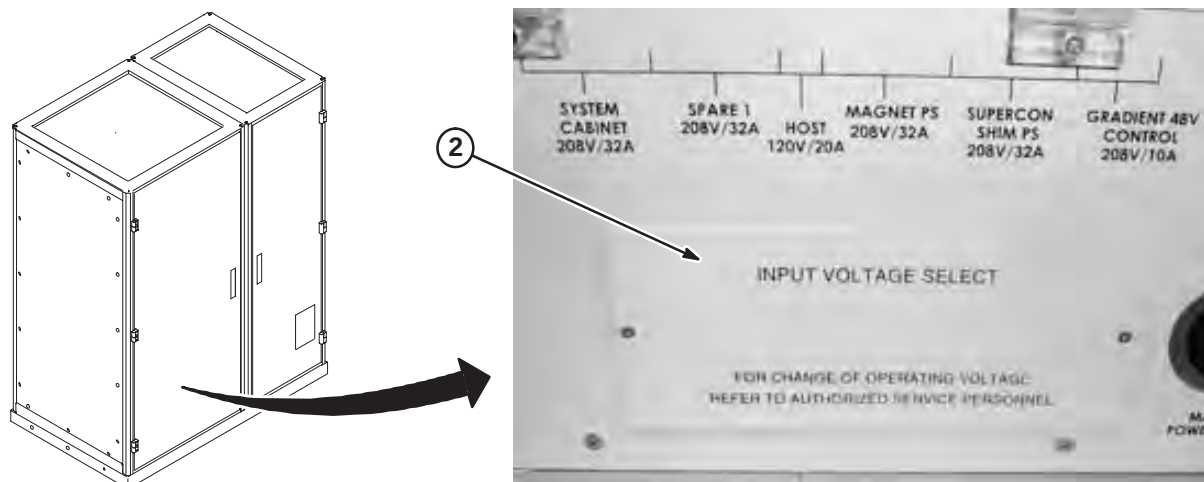


LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

3-1 INPUT VOLTAGE SELECTION

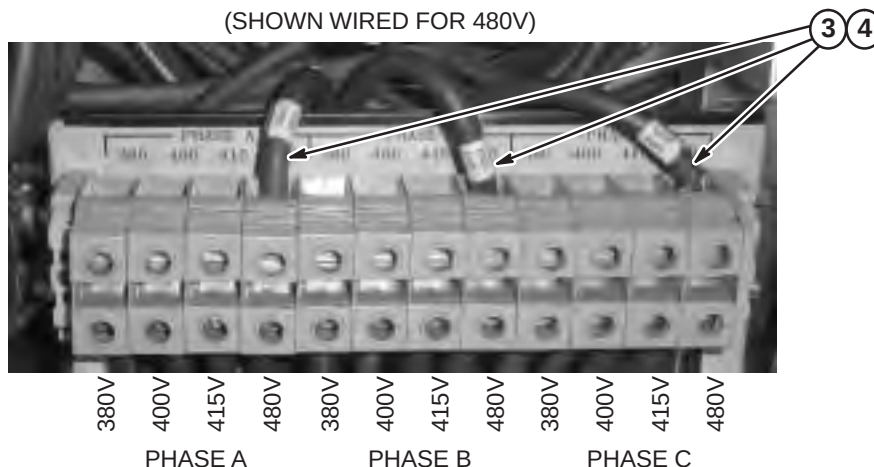
The PDU is shipped with the default voltage setting of 480 VAC.

1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT VOLTAGE SELECT.



3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.
4. If the power connections are not configured properly, ensure that the proper personnel correct the wiring connection. See *PDU Operation and Maintenance Manual* for additional details.

(SHOWN WIRED FOR 480V)

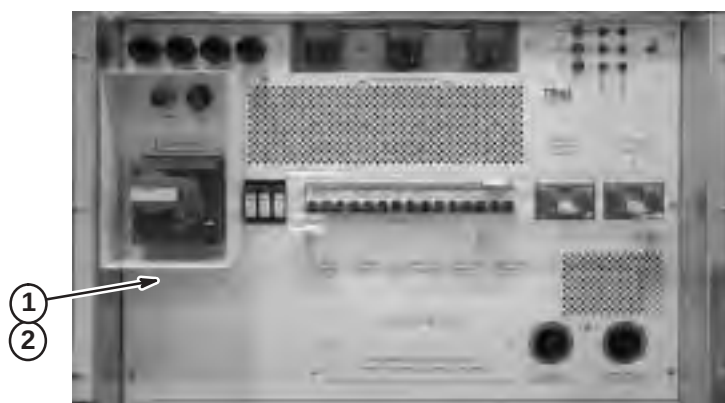


3-2 CIRCUIT BREAKER DIP SWITCH SETTINGS (If Required)

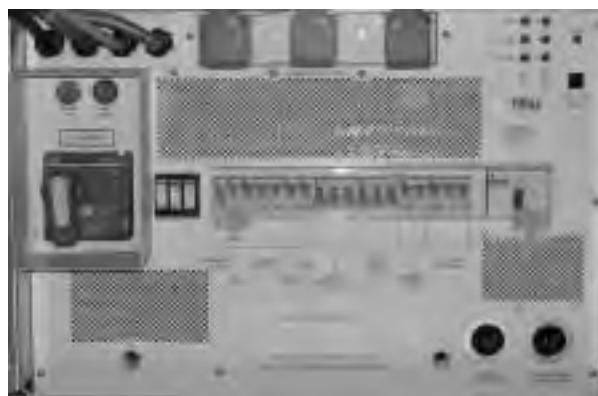
Note: This DIP Switch setting procedure is only used for PDU's installed at sites during or before Q3 of 2010. PDU's delivered after Q3 of 2010 do not have DIP switches.

The overload and short circuit trip settings for the Input Circuit Breaker must be set to the correct values corresponding to the input voltage.

1. Remove the small PDU cover plate marked CURRENT ADJUST under the main circuit breaker which is held in place by a single screw on each side.
2. Lift open the transparent plastic cover on the lower portion of the circuit breaker to give access to the DIP switches controlling the circuit breaker operation.



Teal PDU delivered before Q3 of 2010

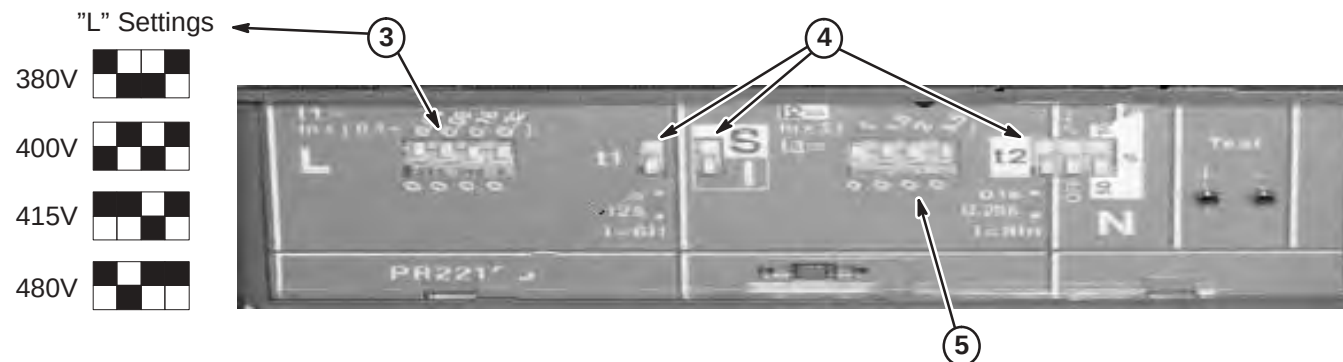


Teal PDU delivered after Q3 of 2010

The illustration below shows the position of the DIP switches for each input voltage selection. If the dip switches are not configured correctly, make sure that proper personnel correct the dip switches settings after following instructions in the *PDU Operation and Maintenance Manual*.

Note: The white portion of the DIP switch indicates the position of the switch.

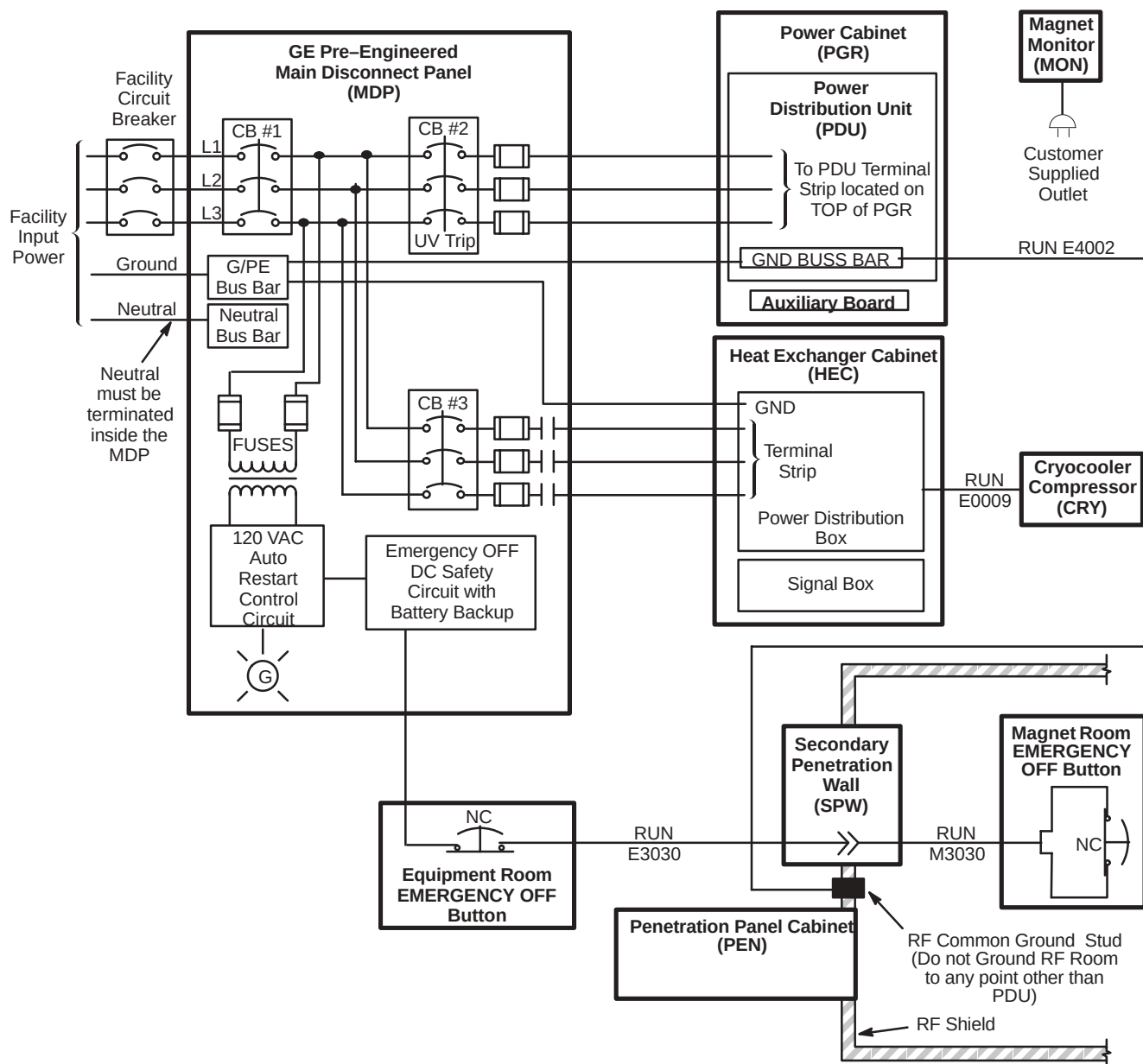
3. Set the "L" switches according to the input voltage to the left.
4. Leave the "t1", "S1" and "t2" switches in the DOWN position.
5. Leave the "I3" switches in the UP position.



3-3 SYSTEM POWER CONNECTIONS

All power connections to the Main Disconnect Panel (MDP) and other components should be completed by customer supplied licensed electrician.

- Note:**
- Runs E3030, E0009, M3030 and E4002 are GE supplied cables.
 - **All other wiring is customer supplied.**
 - All circuits require Ground wires.



SECTION 4 – CABLE & SYSTEM INSTALLATION

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4-1 CABLE INTERCONNECT DOCUMENTATION

The cable maps illustrate the Discovery MR450 or MR750 system interconnect for all supplied cables and interface with customer supplied wiring.

- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
 - Instructions for connections for optional laser cameras are supplied with option.
 - Connection procedures for other options are covered in the installation manual shipped with the option.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables when their connections are complete.

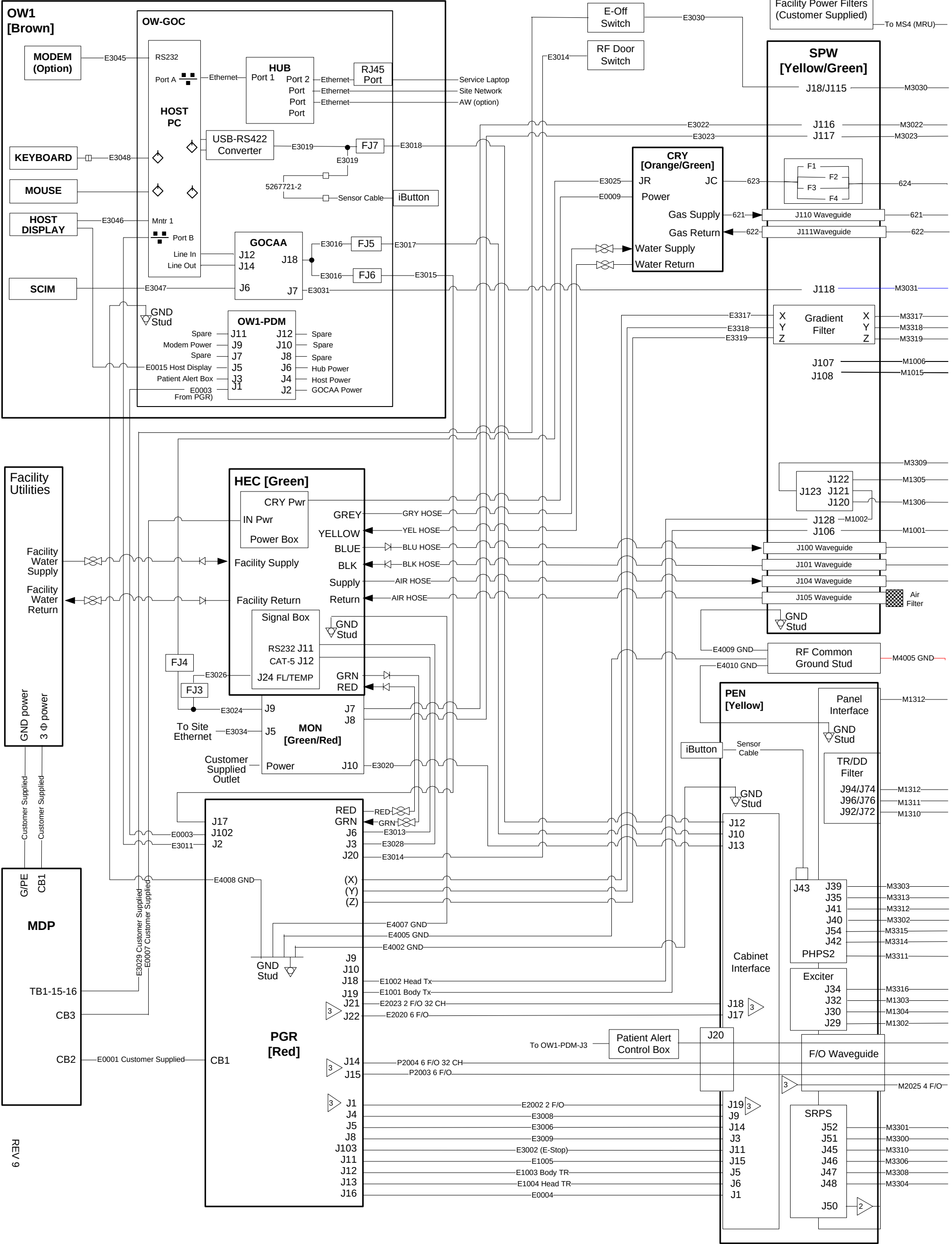
Fold-out the applicable System Interconnect Cable Maps are provided on the following pages:

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– Discovery MR450 1.5T 32 Channel without IRD System Interconnect (Equipment Room)	4-6
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– Discovery MR450 1.5T 32 Channel without IRD System Interconnect (Magnet Room)	4-9
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Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off routed cables. Additional information on each cable is located in the Appendix. Upon completion of connecting cables, return fold-out maps to the binder for future reference.

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32 Channel MR450 1.5T Interconnect Map with IRD & Electric Dock, w/o HOS – Equipment & Control Rooms



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

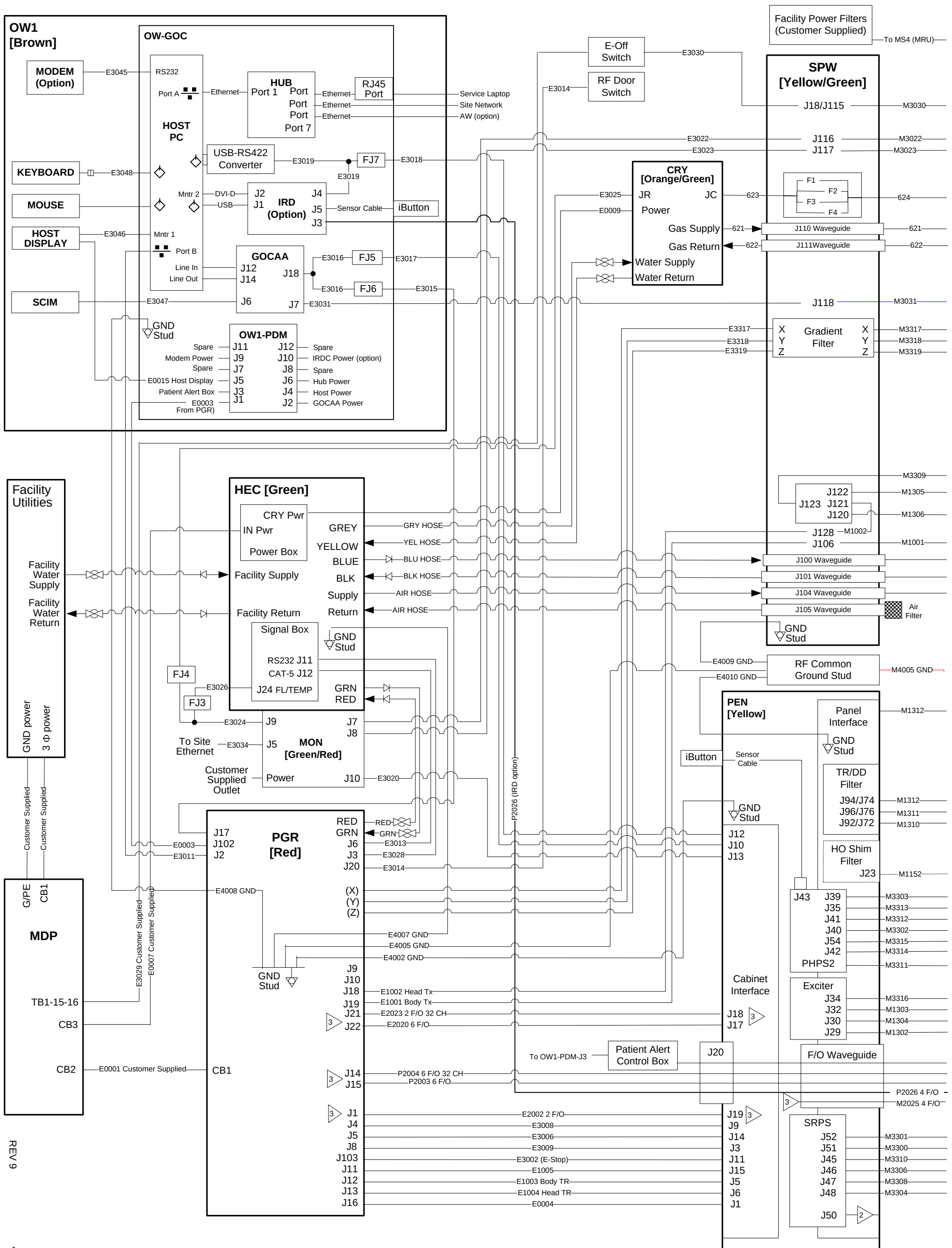
--Dotted lines for module outlines or cables indicate an option

Symbol Legend

- Ball Valve
- Check Valve

- Terminator applied at J50 of Penetration Panel.
- Refer to Install Fiber Optic Cables section for detail instructions on routing and connecting fiber optics in cabinets.

32 Channel MR450 1.5T Interconnect Map with IRD, Electric Dock & HOS – Equipment & Control Rooms



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
 - Mxxxx Runs entirely in the Magnet Room
 - Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room
- Subsequent Characters:**
- 0xxxx AC Power & Ground
 - 1xxxx Coax/RF
 - 2xxxx Fiber Optics
 - 3xxxx Signal
 - 4xxxx Other

--Dotted lines for module outlines or cables indicate an option

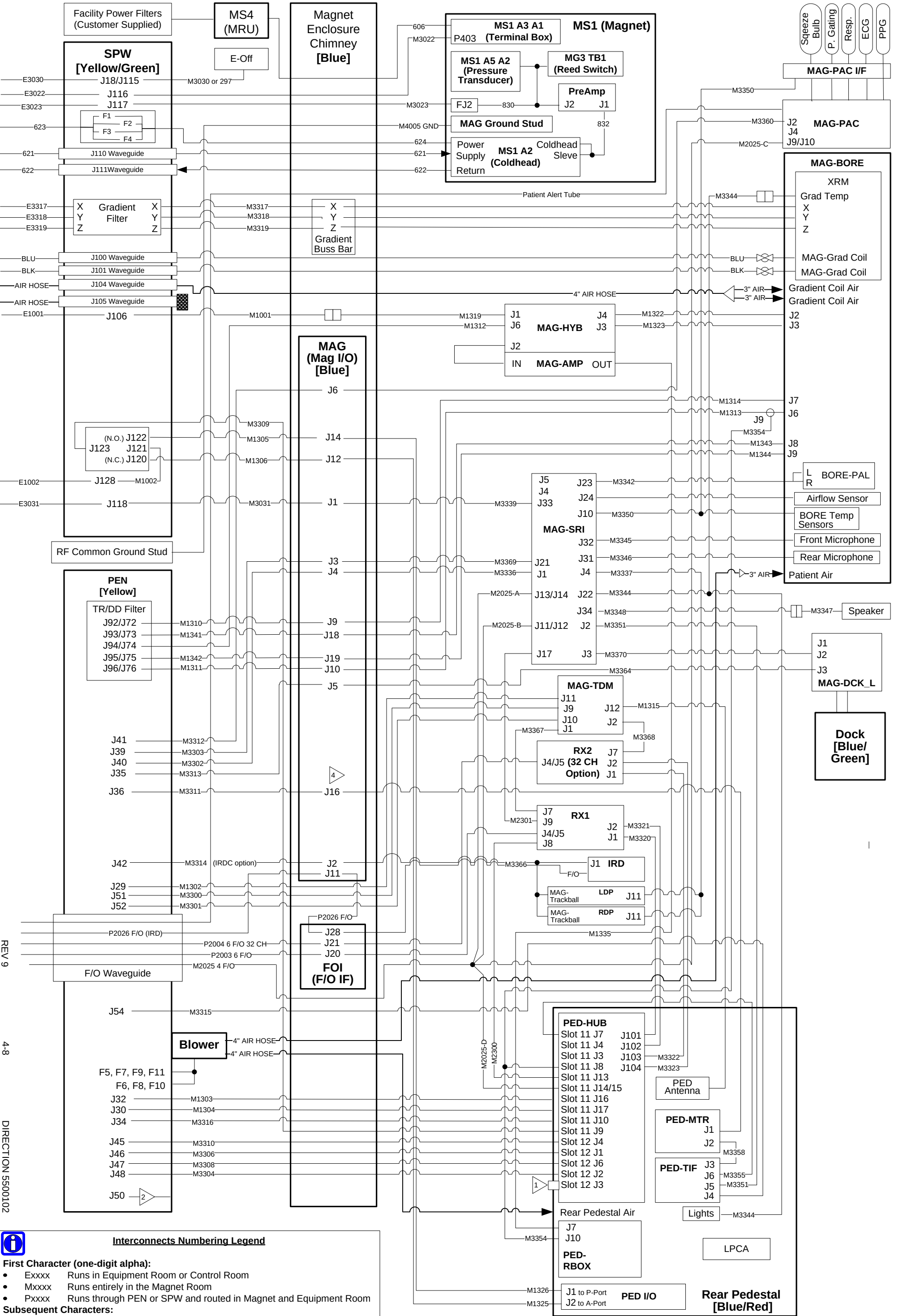
 Symbol Legend

-  Ball Valve
 Check Valve

2 Terminator applied at J50 of Penetration Panel.

3 Refer to Install Fiber Optic Cables section for detail instructions on routing and connecting fiber optics in cabinets.

32 Channel MR450 1.5T Installation Interconnect Map with IRD, w/o Electric Dock & HOS – Magnet Room





Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option



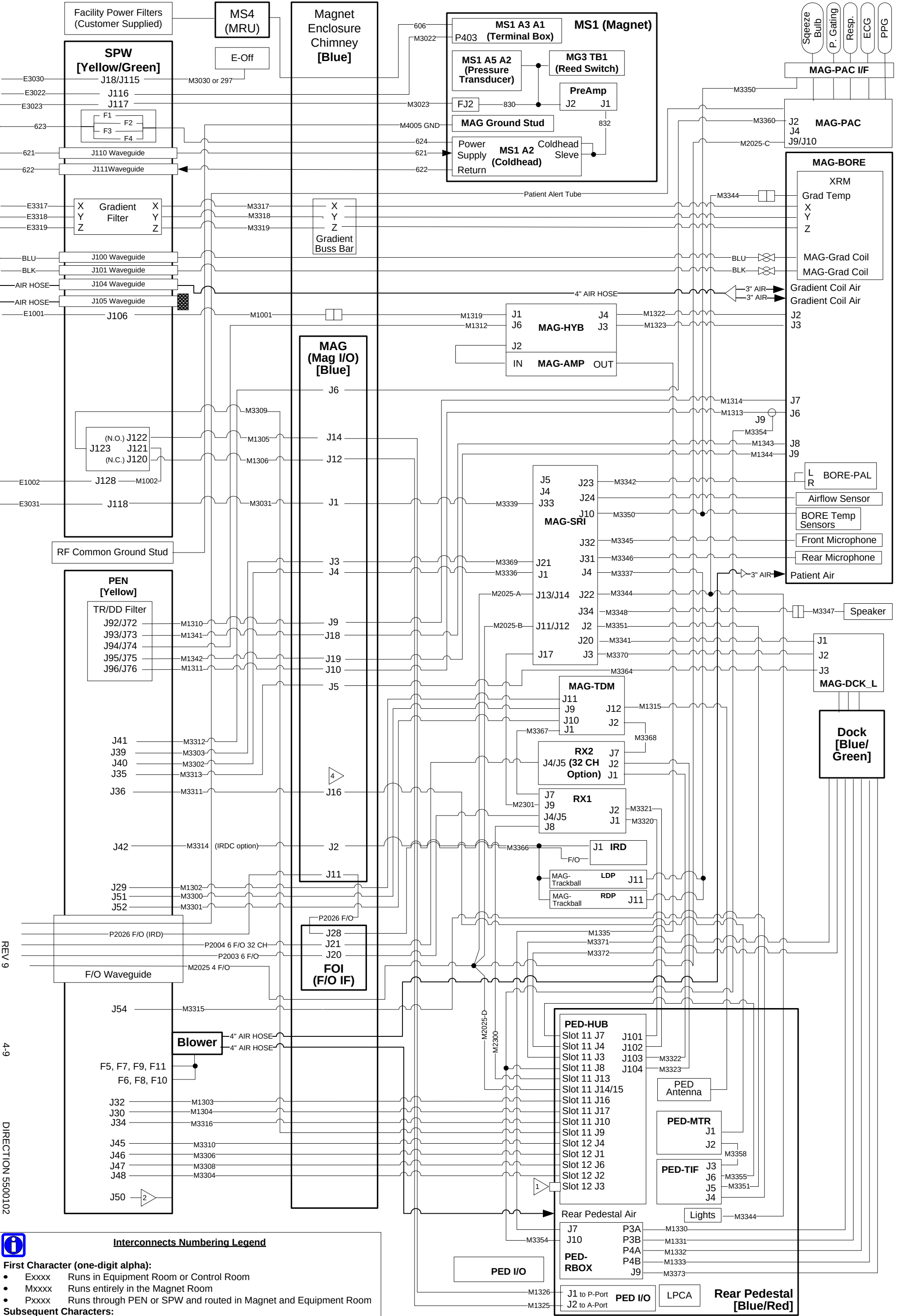
Symbol Legend

 Ball Valve

 Check Valve

-  Shunt is installed for 16 & 32 Channel systems.
-  Terminator applied at J50 of Penetration Panel.
-  For some configurations, Run M3311 will not be connected to J16 on the magnet I/O panel. It will be routed and connected directly to PED-MTR J1.

32 Channel MR450 1.5T Installation Interconnect Map with IRD, Electric Dock & HOS – Magnet Room



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option

Symbol Legend

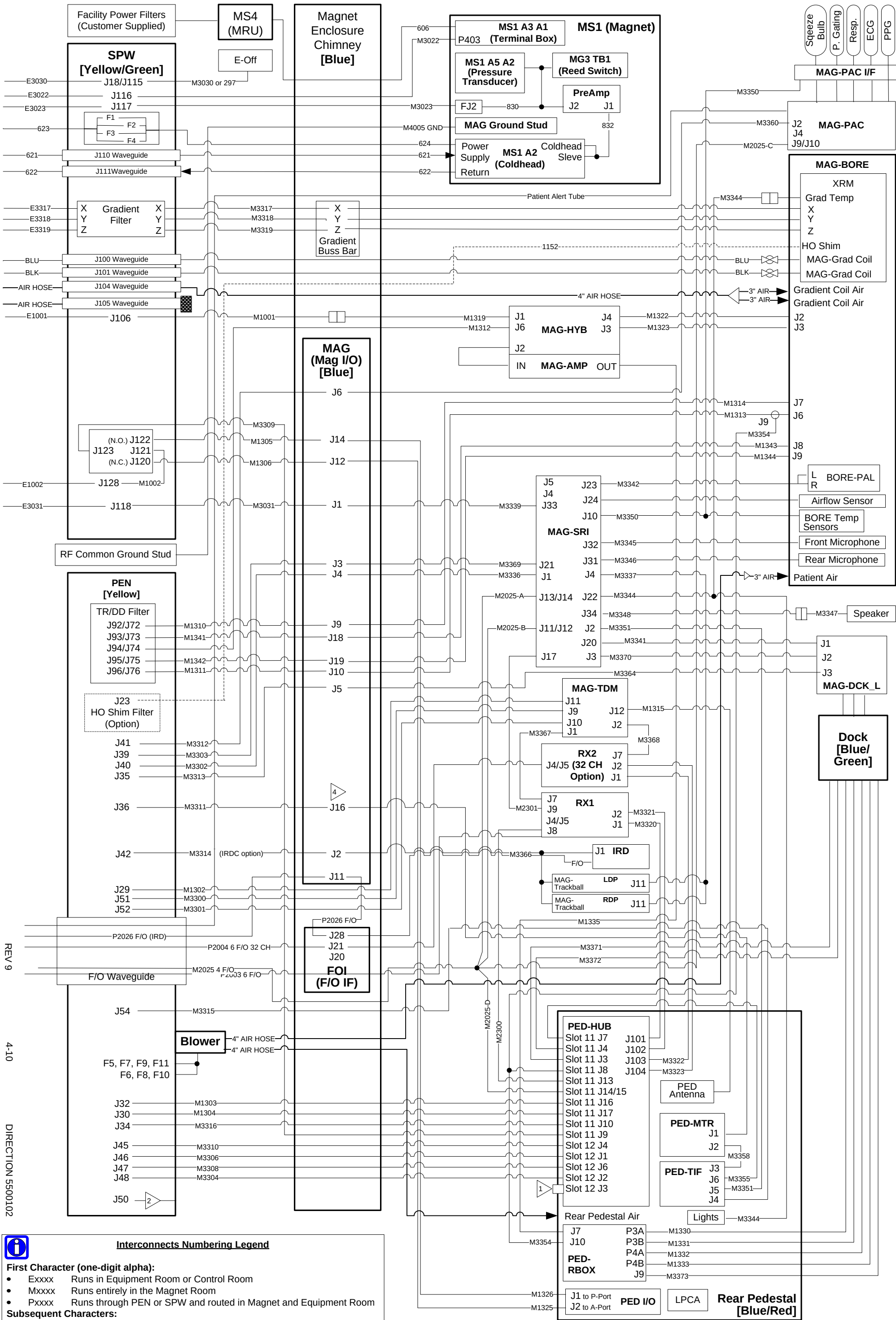
- Ball Valve
- Check Valve

1 Shunt is installed for 16 & 32 Channel systems.

2 Terminator applied at J50 of Penetration Panel.

4 For some configurations, Run M3311 will not be connected to J16 on the magnet I/O panel. It will be routed and connected directly to PED-MTR J1.

32 Channel MR450 1.5T Installation Interconnect Map with IRD, Electric Dock & HOS – Magnet Room



REV 9

 **Interconnects Numbering Legend**

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

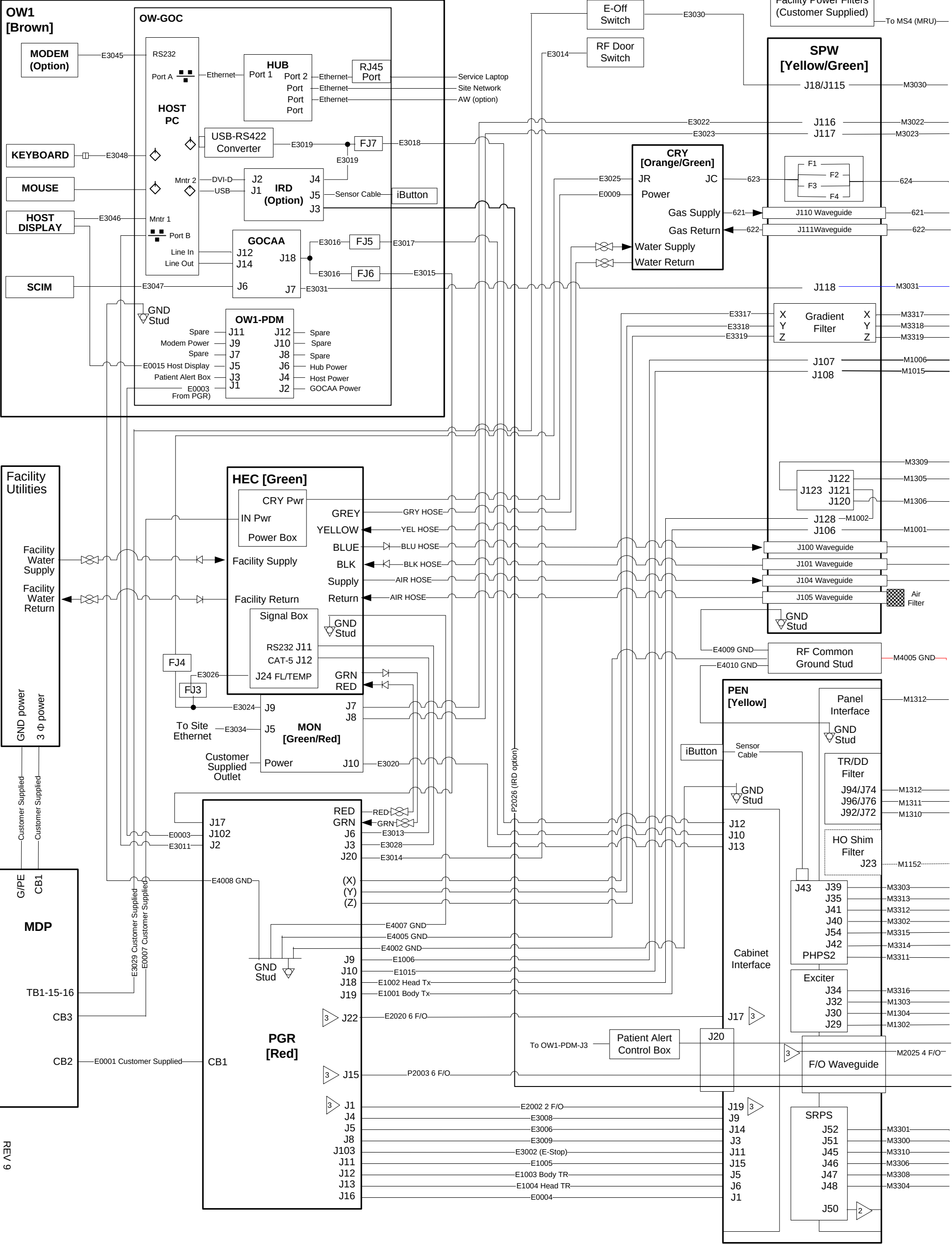
- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option

 **Symbol Legend**

 Ball Valve Check Valve

16 Channel MR750 3.0T Installation Interconnect Map with IRD – Equipment & Control Rooms



REV 9

4-11

DIRECTION 5500102



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option



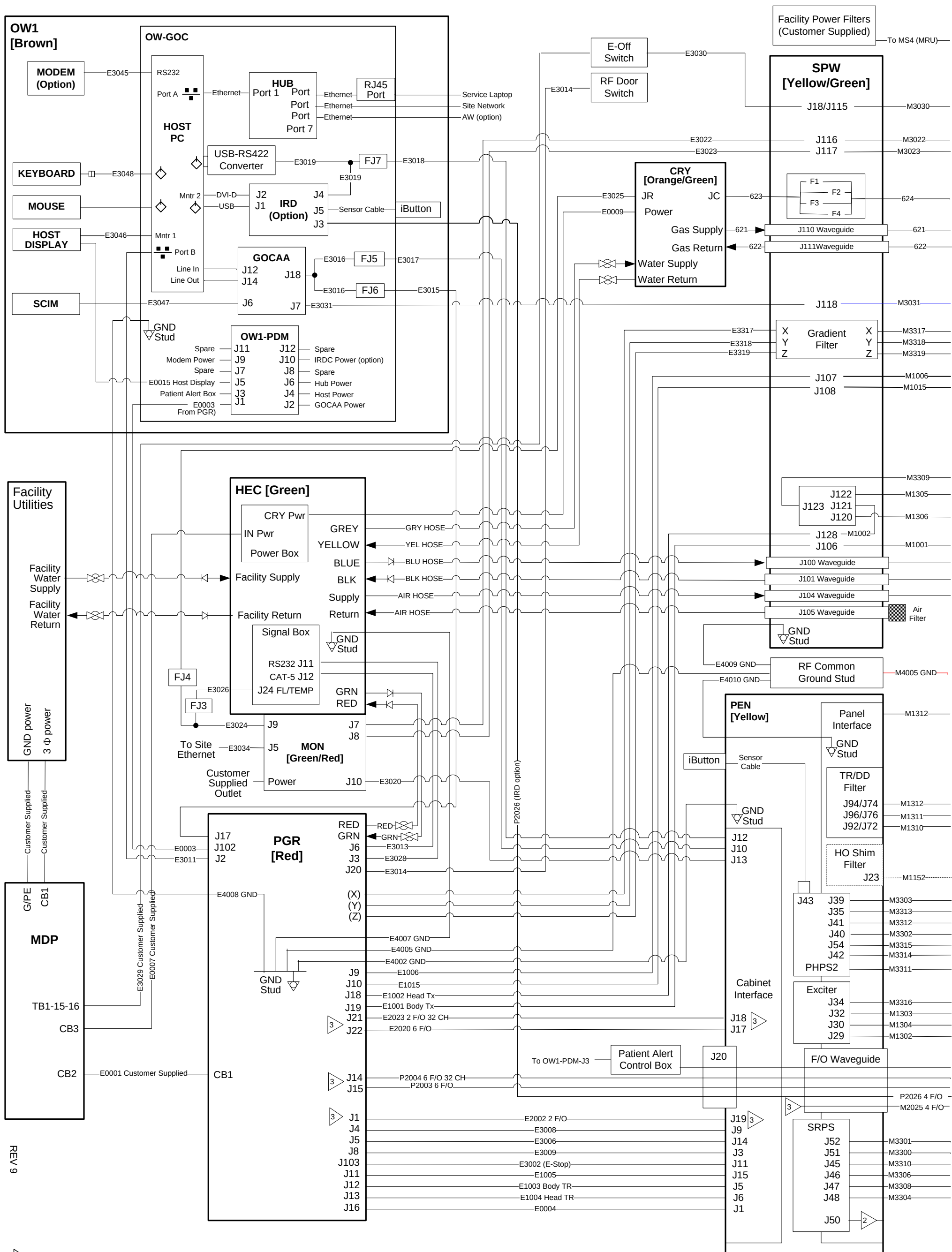
Symbol Legend

- Ball Valve
- Check Valve

Terminator applied at J50 of Penetration Panel.

Refer to Install Fiber Optic Cables section for detail instructions on routing and connecting fiber optics in cabinets.

32 Channel MR750 3.0T Installation Interconnect Map with IRD – Equipment & Control Rooms



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
 - Mxxxx Runs entirely in the Magnet Room
 - Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room
- Subsequent Characters:**
- 0xxxx AC Power & Ground
 - 1xxxx Coax/RF
 - 2xxxx Fiber Optics
 - 3xxxx Signal
 - 4xxxx Other

--Dotted lines for module outlines or cables indicate an option

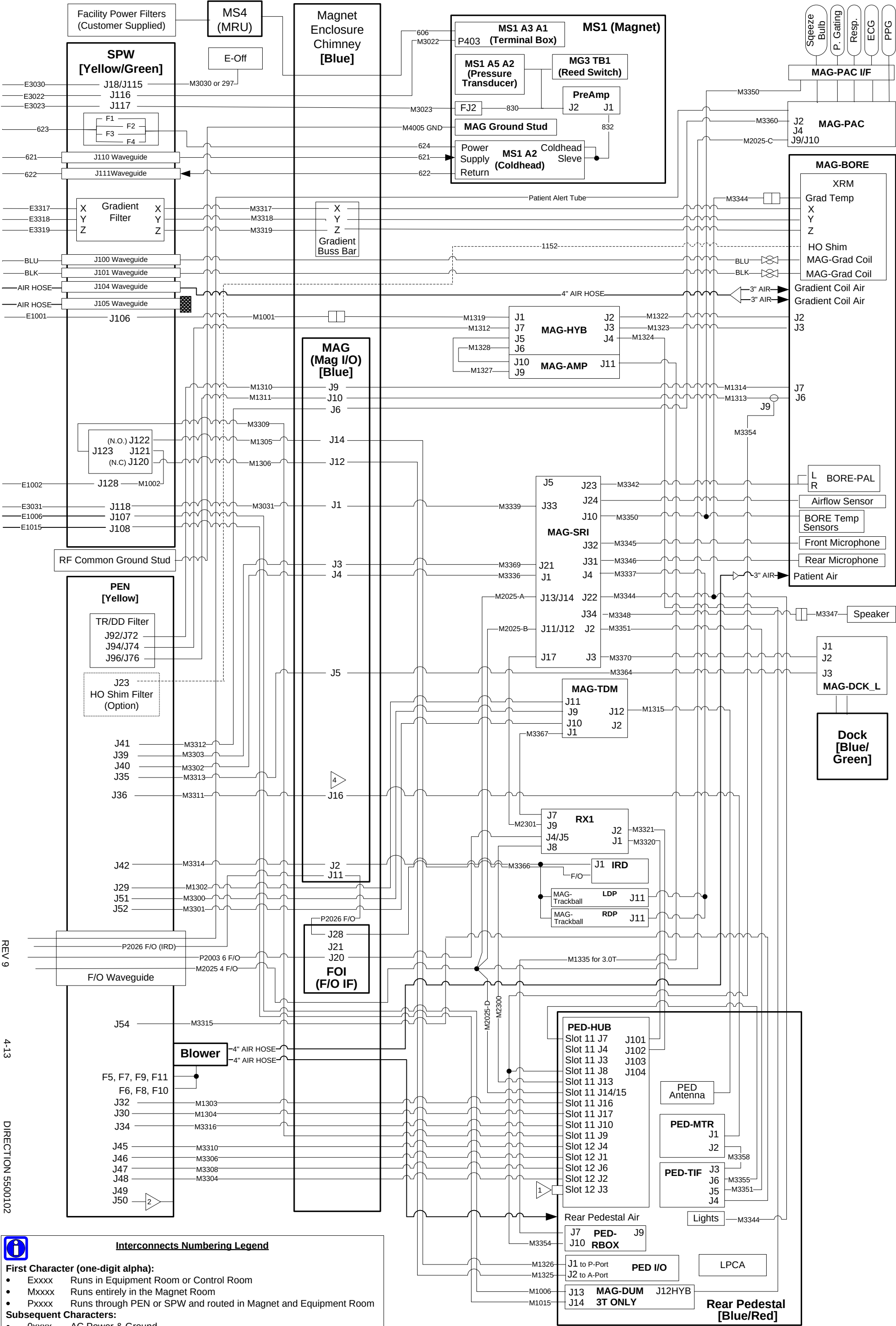
Symbol Legend

-  Ball Valve
 Check Valve

2 > Terminator applied at J50 of Penetration Panel.

3 Refer to Install Fiber Optic Cables section for detail instructions on routing and connecting fiber optics in cabinets.

16 Channel MR750 3.0T Installation Interconnect Map with IRD – Magnet Room



REV 9

4-13

DIRECTION 5500102



Interconnects Numbering Legend

First Character (one-digit alpha):

- Exxxx Runs in Equipment Room or Control Room
- Mxxxx Runs entirely in the Magnet Room
- Pxxxx Runs through PEN or SPW and routed in Magnet and Equipment Room

Subsequent Characters:

- 0xxxx AC Power & Ground
- 1xxxx Coax/RF
- 2xxxx Fiber Optics
- 3xxxx Signal
- 4xxxx Other

--Dotted lines for module outlines or cables indicate an option



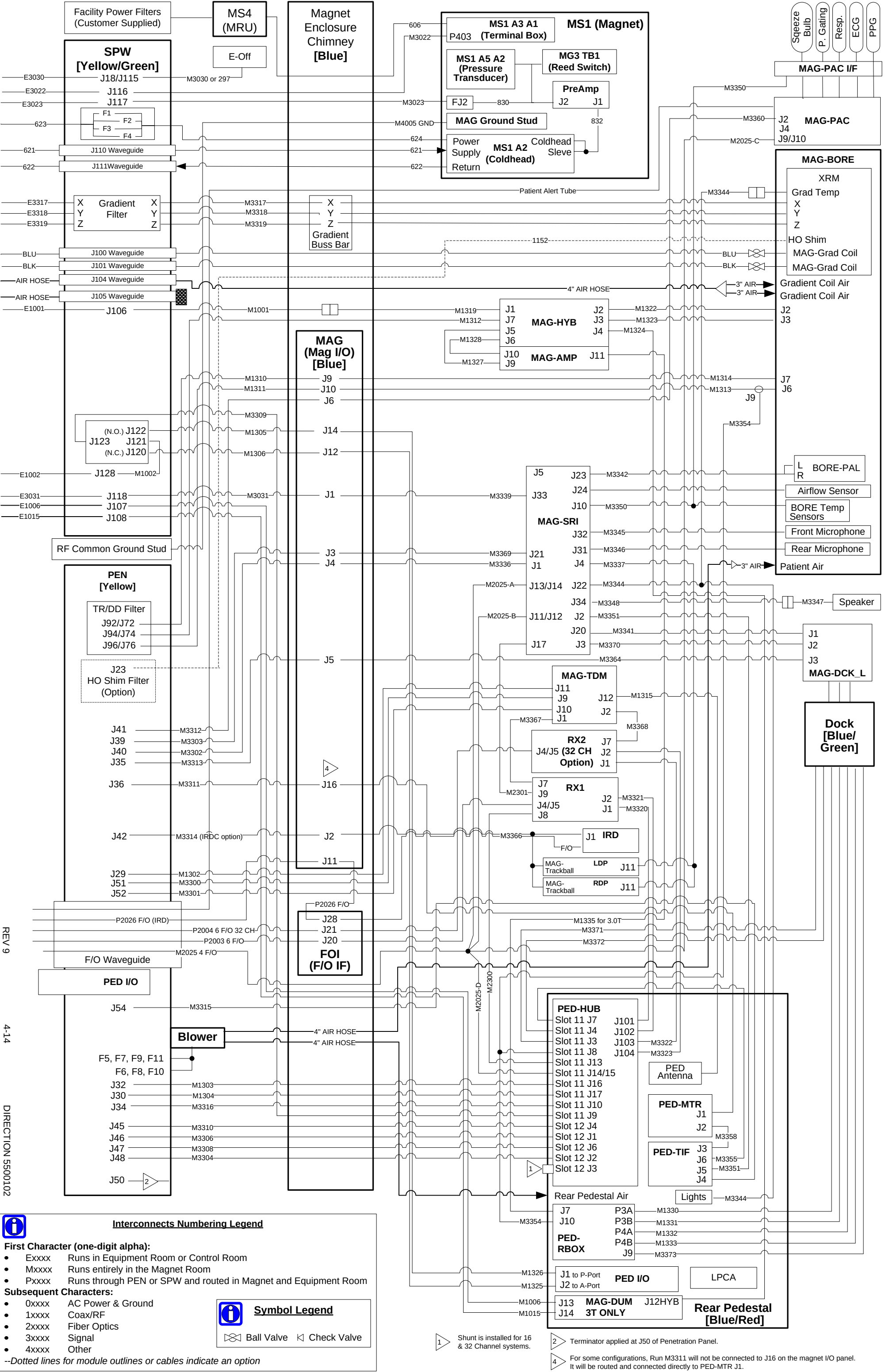
Symbol Legend

 Ball Valve

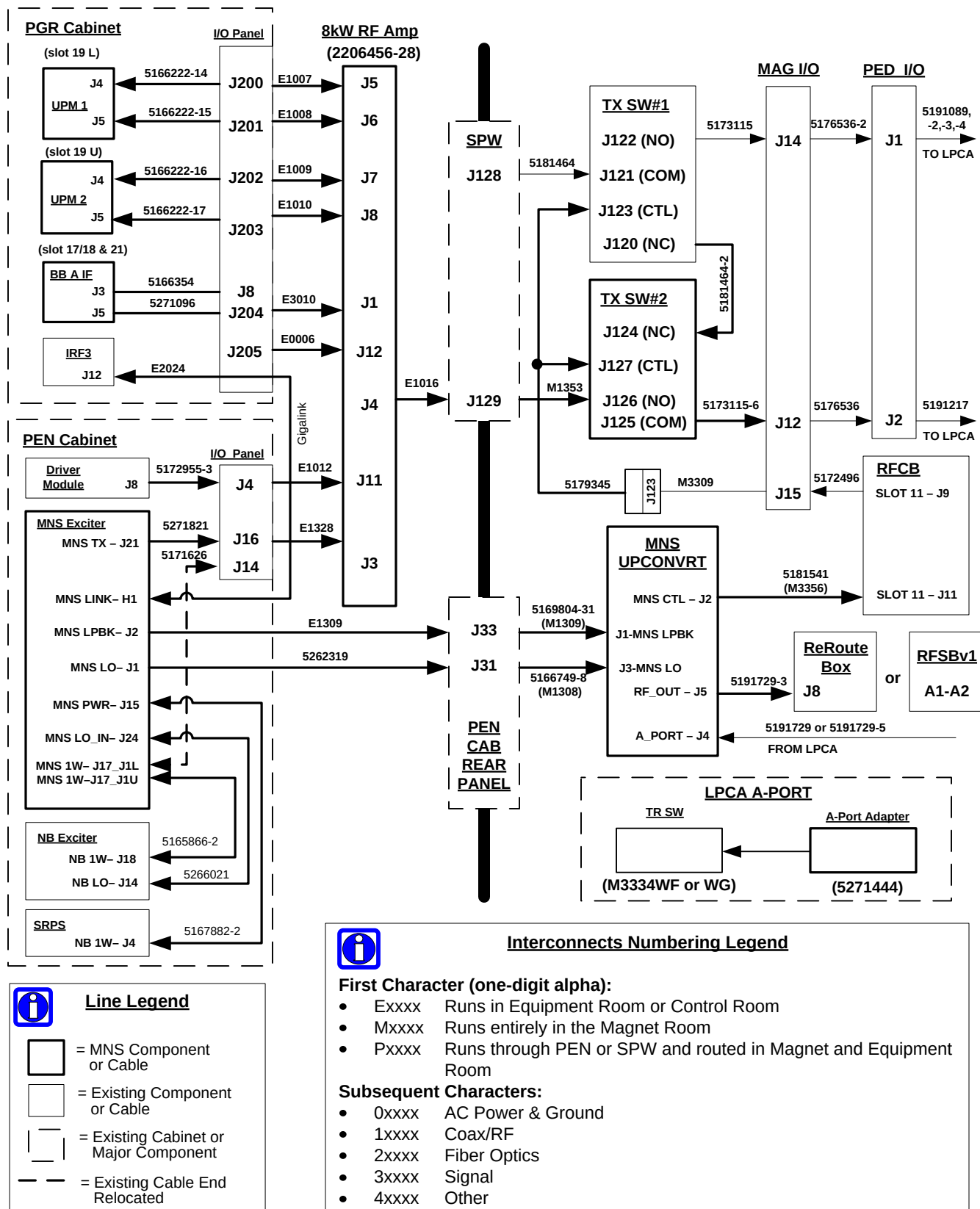
 Check Valve

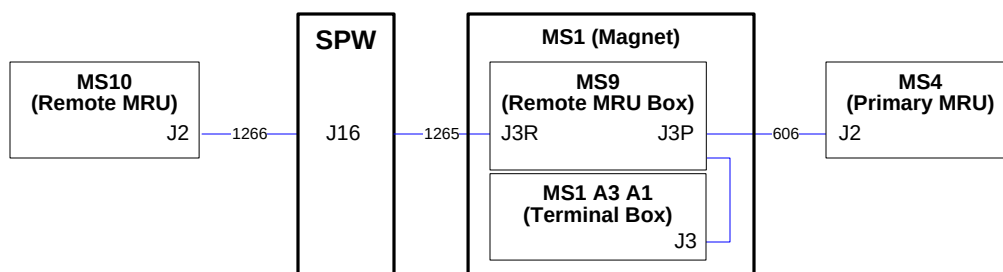
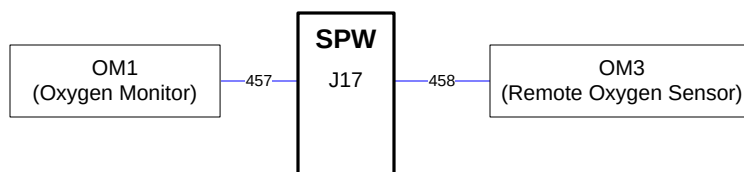
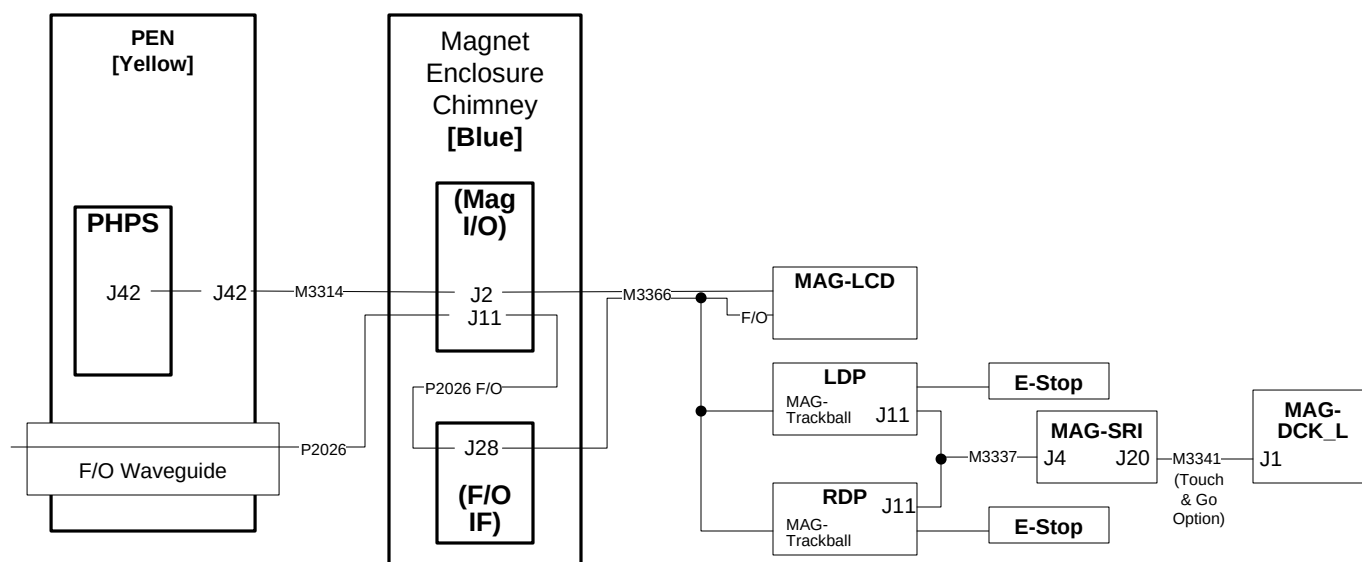
-  Shunt is installed for 16 & 32 Channel systems.
-  Terminator applied at J50 of Penetration Panel.
-  For some configurations, Run M3311 will not be connected to J16 on the magnet I/O panel. It will be routed and connected directly to PED-MTR J1.

32 Channel MR750 3.0T Installation Interconnect Map with IRD – Magnet Room



MR750 MNS CABLE INTERCONNECT MAP



REMOTE MRU OPTION CABLE MAP**OXYGEN MONITOR OPTION CABLE MAP****IN-ROOM DISPLAY CABLE MAP**

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4-2 SORTING AND ROUTING CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

Move lean cable carts into assigned room. System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the overhead cable trays.

Note: Carefully review architectural drawings to make sure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables to insure that they will route freely without twists or kinks. Route cables in accordance with the applicable Installation Interconnect Diagram.
3. The following cables can be grouped together and pulled at the same time thru the designated troughs in the overhead cable trays, as indicated in Section 4-6 and Section 4-7:

EQUIPMENT ROOM Cable Runs

- PGR to PEN
 - E3002, E3006, E3008, E3009
 - E1005, E0004, E4002, E4005
 - E1003, E1004
- PGR to SPW
 - E1006, E1015 (3T Only)
- PGR to OW
 - E0003, E3011, E3015, E4008
- PGR to HEC
 - E3013, E3028, E4007
- OW to PEN
 - E3017, E3018
- MON to SPW & PEN
 - E3020, E3022, E3023

MAGNET ROOM Cable Runs

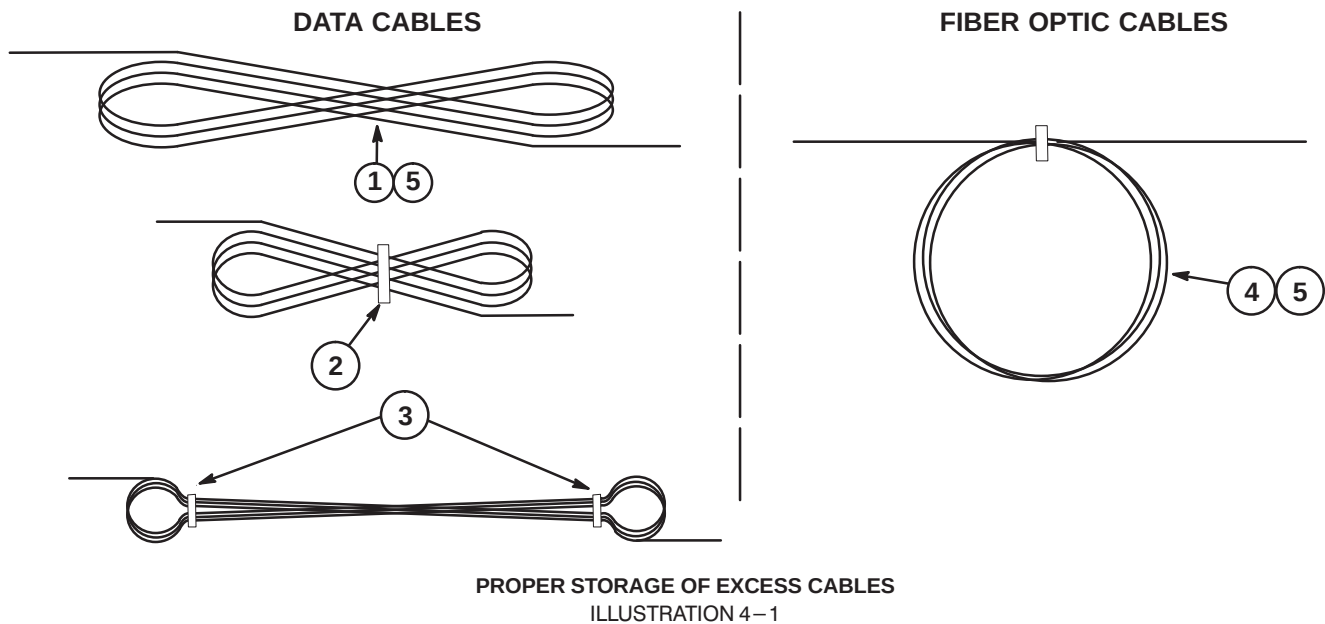
- PEN to MAG I/O
 - M1310, M1311, (M1341, M1342 1.5T Only)
 - M3302, M3303, M3312, M3313, M3314 (IRD option)
- SPW to MAG I/O
 - M1305, M1306, M3309, M3031
- SPW & PEN to MAG-TDM
 - M3300, M3301, M1302
- SPW & PEN to REAR PED
 - M1006, M1015 (3T Only)
 - M3304, M3306, M3308, M3310
 - M1303, M1304,
 - M3311, M3315, M3316,

Any cables not listed above should be routed through the overhead cable trays separately.

4-3 STORAGE OF EXCESS CABLE LENGTH

The type of coiling used to store excess length of cables depends on the type of cable. Data cables are coiled differently than fiber optic cables. Only the gradient cable will be cut to length.

- DATA CABLES – Excess length of cables must be coiled in a "Figure Eight" pattern. DO NOT COIL IN CIRCLES!
- FIBER OPTIC CABLES – Excess length of cables must be coiled in a circle pattern. DO NOT COIL IN A "FIGURE EIGHT" PATTERN!



1. Excess data cable to be stored in overhead cable trays or Penetration Panel closet must be routed in a "Figure Eight" pattern as shown.

NOTE: The minimum bend radius of any cable must not be violated no matter how the excess cable is coiled.

2. For smaller figure eight patterns, dress and tie-wrap middle of looped cables.
3. For longer figure eight patterns, dress and tie-wrap ends of looped cables.
4. Excess data cable to be stored in overhead cable trays, Penetration Panel closet, or in magnet enclosure area must be routed in a circular pattern as shown.
5. Refer to *MR450 1.5T and MR750 3.0T CABLES INSTALLED ON SITE* table in Section 8 for information on minimum bend radius for each cable.

4-3-1 Equipment Room Storage

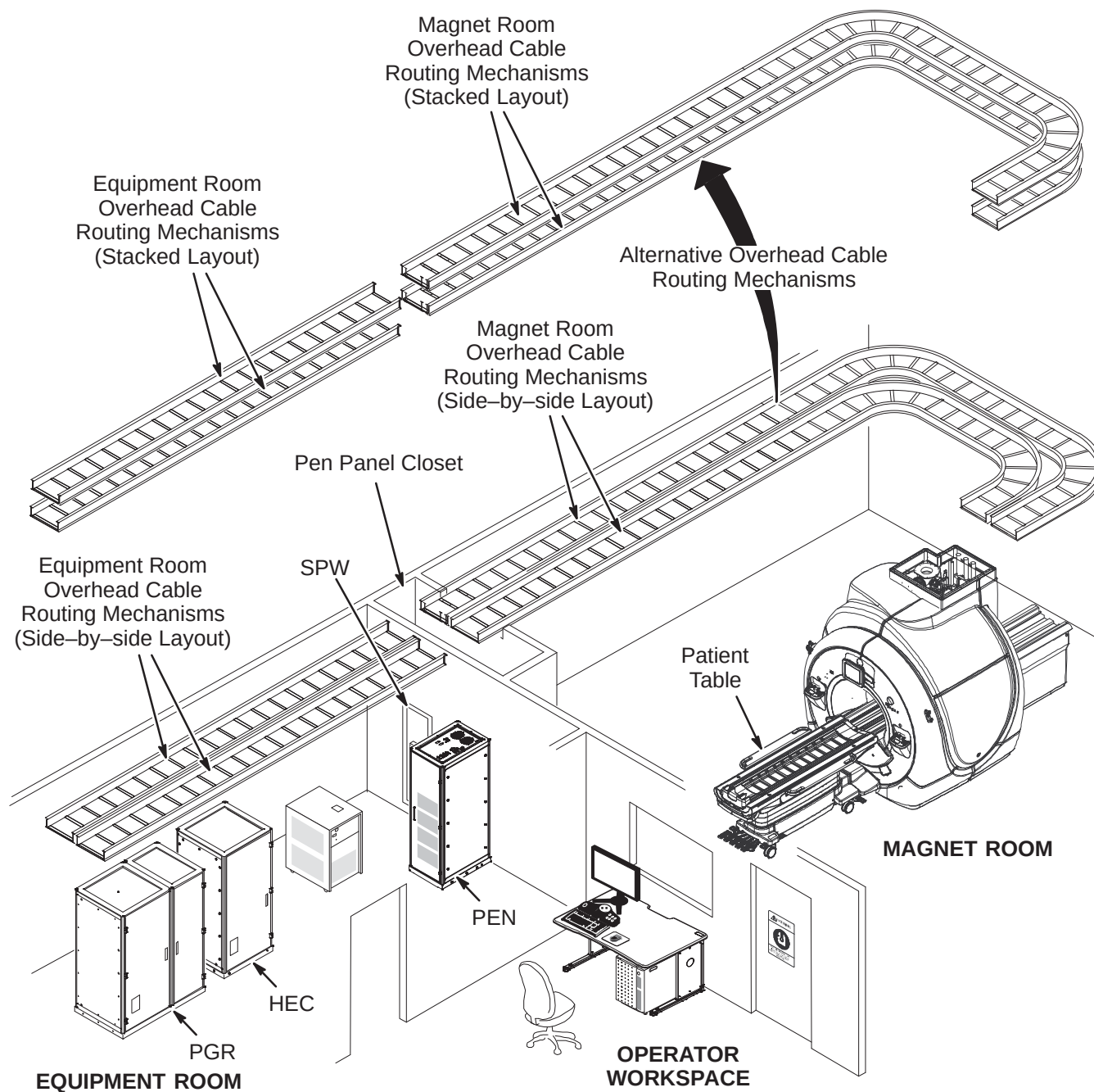
1. Route cables in provided overhead cable trays, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans.
2. Excess cable length will be stored in overhead cable trays.

4-3-2 Magnet Room Storage

1. Route cables inside magnet room thru overhead cable trays, conduit, or ducts to rear of magnet area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.
2. Excess cable length will be stored in Penetration Panel Closet.

4-4 OVERVIEW OF SYSTEM COMPONENTS

The illustration below represents a typical layout for a MR450 or MR750 system. The procedures on the following pages of this section will provide information on installing cables and completing the attachment of components to the magnet.



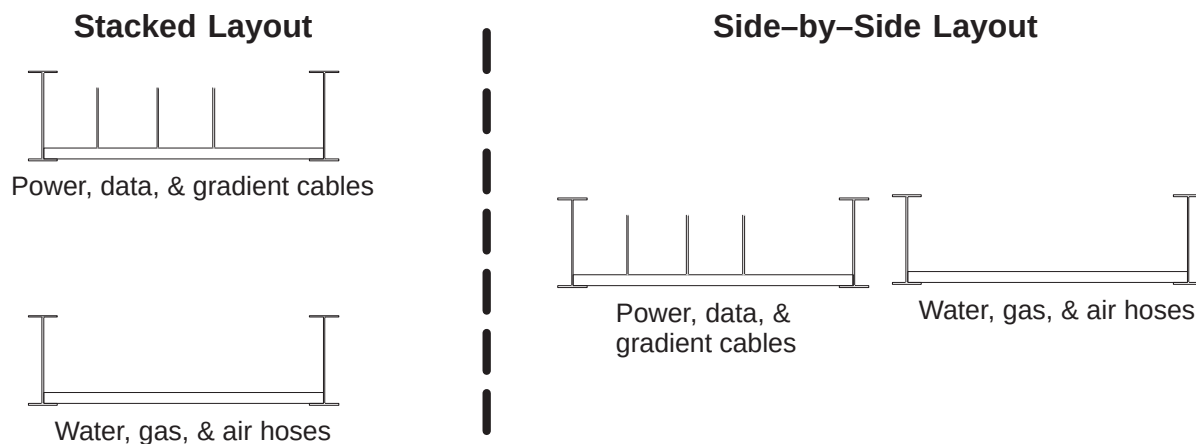
4-5 OVERHEAD CABLE ROUTING MECHANISM LAYOUTS

Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

All cables routed between cabinets and the penetration panels must use the overhead cable routing mechanisms or cable trays. For simplicity, in this manual, overhead cable routing mechanisms are referred to as overhead cable trays.

The following illustration shows the two different types of overhead cable tray layouts that can be installed at sites. The views below show the ends of the trays as viewed at the rear of the magnet.

- If the stacked layout is installed at site, the power, data, and gradient cables **MUST ALWAYS** be routed in the top cable tray. The water, gas, and air hoses must always be routed in the lower tray.
- In the side-by-side layout, the power, data, and gradient cables should be routed in the left tray.



Either layouts can be installed in the equipment room or the magnet room.

4-6 CABLE LOCATIONS IN EQUIPMENT ROOM OVERHEAD CABLE TRAYS

WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

DANGER!!

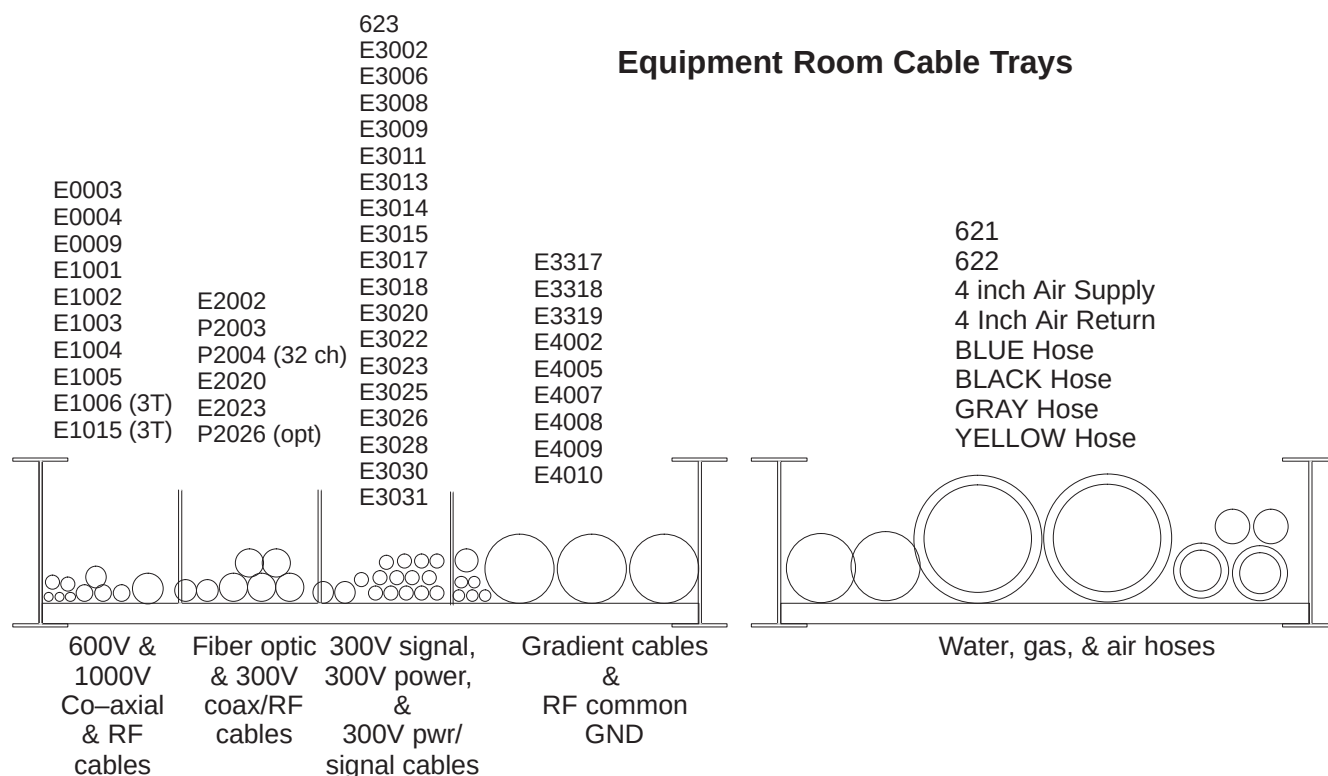
MAKE SURE ALL POWER TO CABINETS IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

All cables routed between cabinets and the penetration panels must use the overhead cable routing mechanisms or cable trays.

NOTE: The cables must be routed in their designated areas. There are two options for these areas:

- The designated areas are partitioned within one large cable tray (As shown below in the left tray)
- The designated areas are made up of four individual cable trays or other routing mechanisms

The illustration below indicates where the cables, per their Run numbers, are to be routed in a partitioned cable tray. Refer to site architectural drawings and local codes for additional specific guidelines.



4-7 CABLE LOCATIONS IN MAGNET ROOM OVERHEAD CABLE TRAYS

WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AT LEAST 10 FEET AWAY FROM THE MAGNET.

DANGER!!

MAKE SURE ALL POWER IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

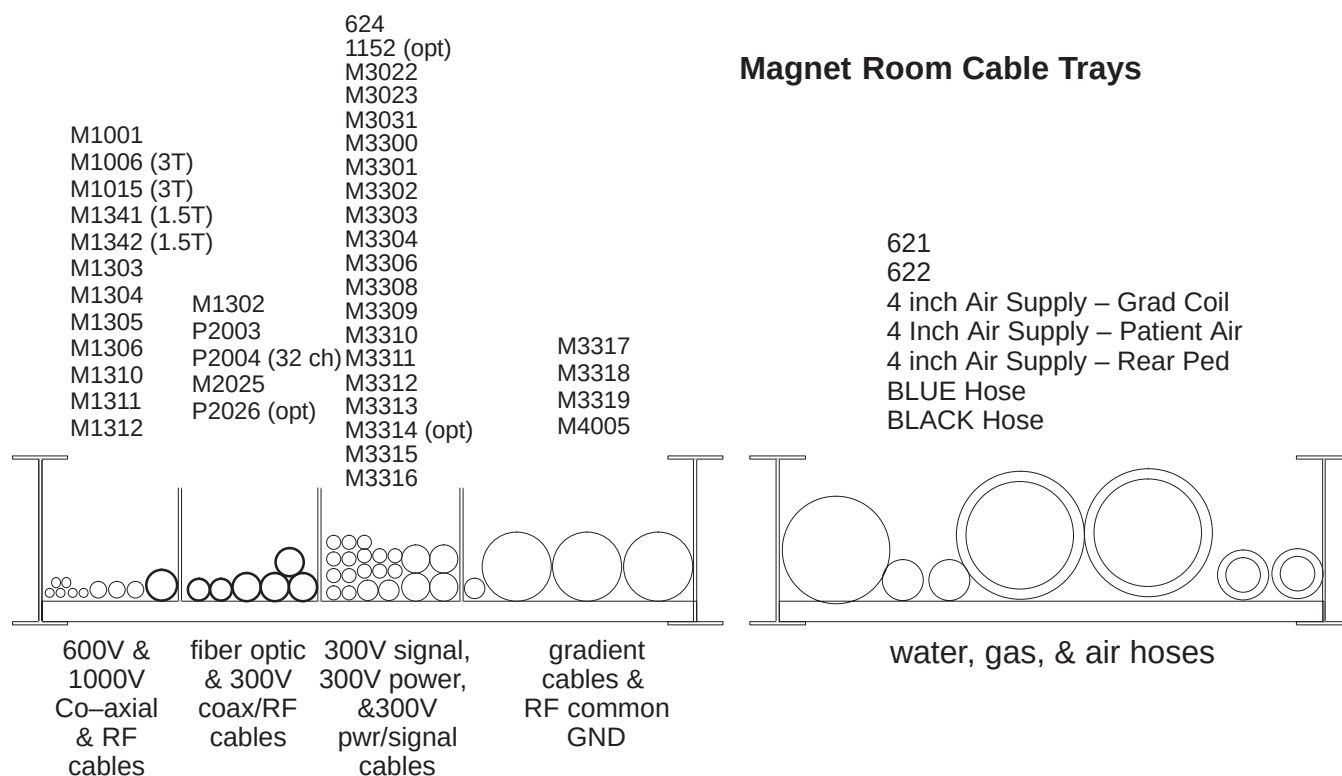
The Magnet Room may have a ceiling constructed of tiled panels. These panels will have to be carefully removed for access to overhead cable trays.

- Carefully remove and store ceiling tiles for re-installation after cables are installed.
- All excess cable will be coiled and stored in the Pen Panel Closet.
- All cables routed between magnet and the penetration panels must use the overhead cable trays.

NOTE: The cables in the cable trays must be routed in their designated areas. There are two options for these areas:

- The designated areas are partitioned within one large cable tray (As shown below in the left tray)
- The designated areas are made up of four individual cable trays or other routing mechanisms

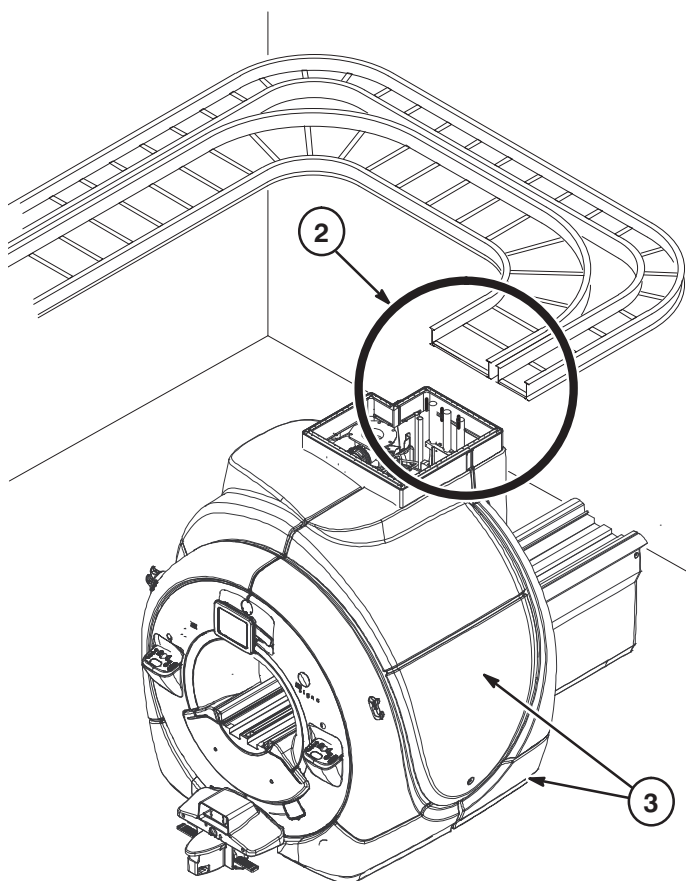
The illustration below (typical view) indicates where the cables, per their Run numbers, are to be routed in the partitioned cable trays, as seen from the magnet end of trays. Refer to site architectural drawings and local codes for additional specific guidelines.



4-8 REAR MAGNET ENCLOSURE CABLE COVER TEST-FIT

Before routing any cables in the magnet room, perform a preliminary installation of the Rear Cable Cover (5182660) at the back of the magnet.

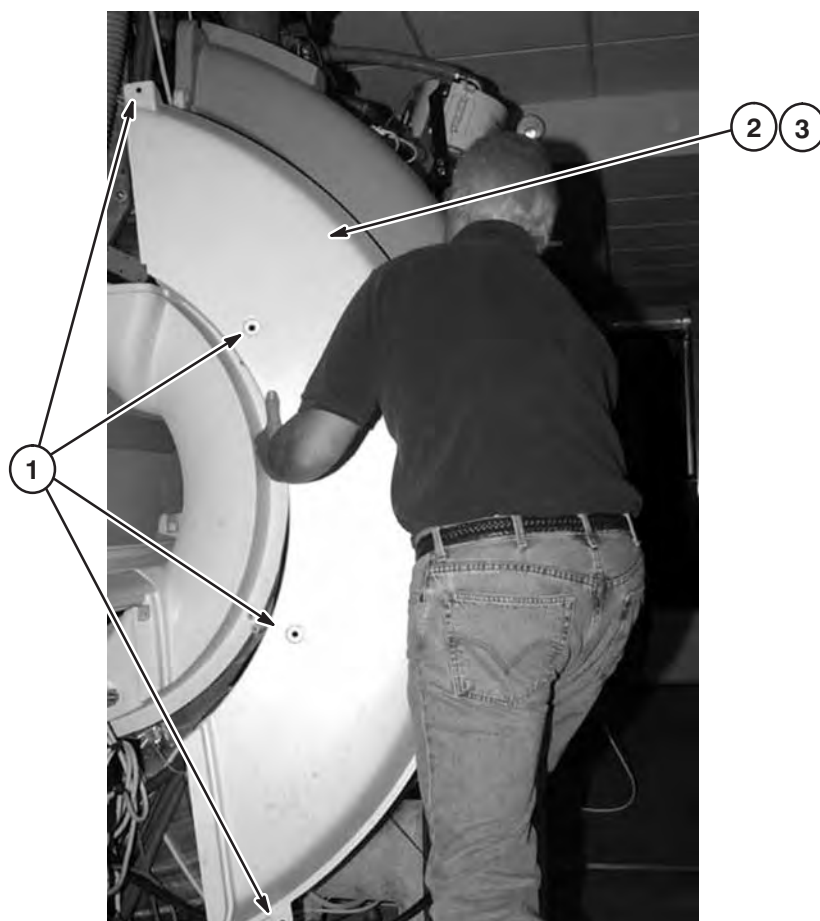
1. Install Rear Cable Cover with provided fasteners.
2. Observe and make note of relationship between end of overhead cable trays and the top edge of Rear Cable Cover. All cables and hoses routed to magnet will have to clear cover and not interfere with attaching the cover to the enclosure.
3. Remove all magnet enclosure side covers and lower trim covers if installed.



4-9 REMOVE REAR COVER

1. Unfasten and save the four (4) screws that attach the rear cosmetic cover to the magnet.
2. Pull cover away from magnet to release the cover from the poppers.
3. Store cover in safe place for reattachment later in installation process.

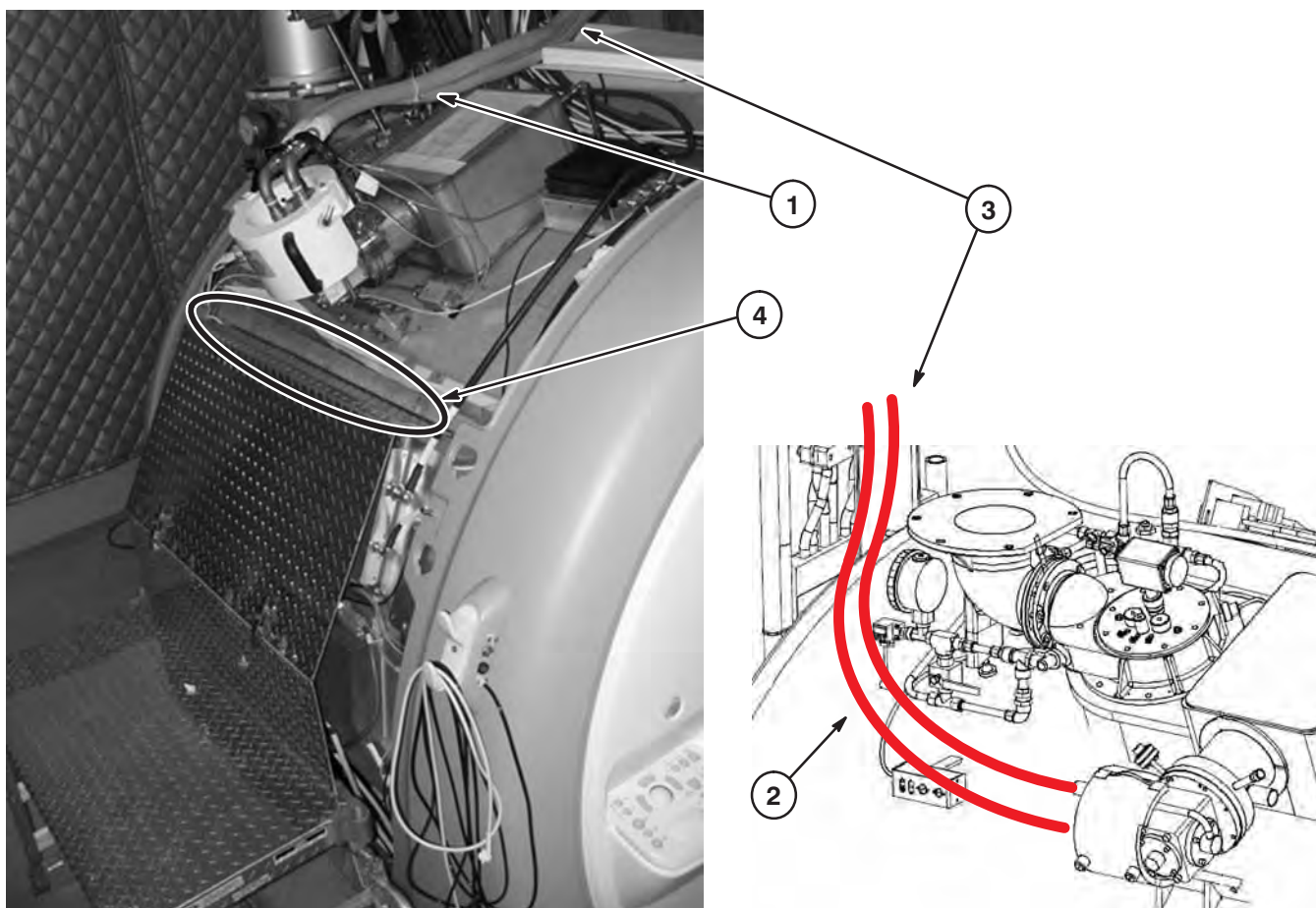
NOTE: Rear cover on other side of rear of magnet does not have to be removed.



4-10 CRYOGEN LINE POSITIONING ON MAGNET

Cryogen lines should have been routed in the correct overhead tray and connected to the shield cooler during magnet installation.

1. **For 3.0T:** Make sure that cryogen lines have been routed on near side of vent stack as shown in illustration below.
2. **For 1.5T:** Make sure that cryogen lines have been routed around vent stack as shown in illustration below.
3. Make sure that there is enough slack in the lines to avoid interference with final installation of all magnet enclosure covers.
4. **NOTE:** When routing remainder of cable, make sure to not route any cables in indicated area for both MR450 and MR750. This area on the magnet needs to be clear so that service ladder cover can be placed over modules attached to side of magnet.



4-11 GRADIENT CABLE INSTALLATION

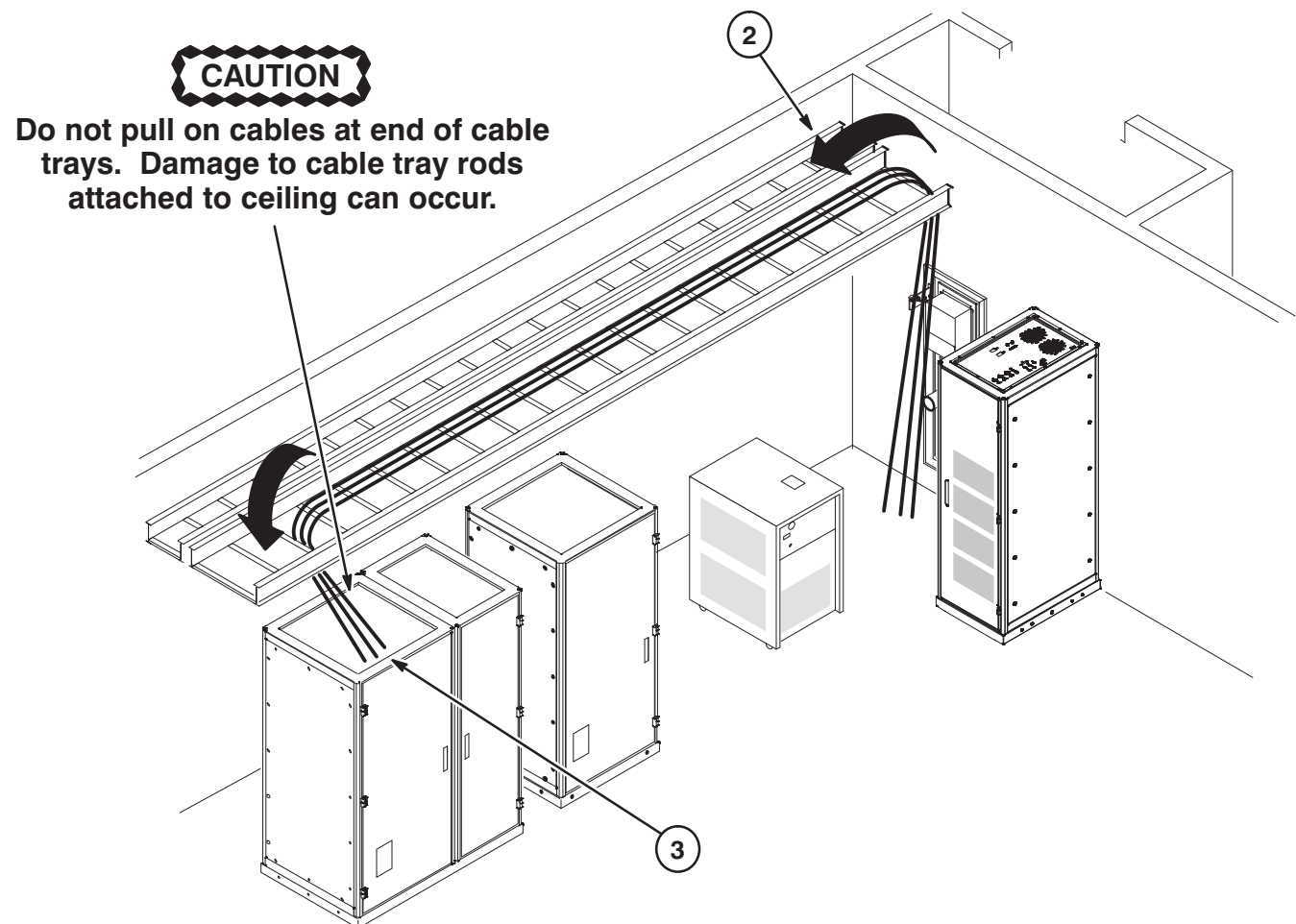
4-11-1 EQUIPMENT ROOM GRADIENT CABLES

The Equipment Room gradient cables, Runs E3317, E3318, and E3319 are delivered to site pre-terminated at one end for connection to the PGR cabinet. The other ends of the gradient cables are routed to the SPW (Secondary Pen Wall), cut to length, and terminated.

4-11-1-A Route Gradient Cables in Overhead Cable Tray

It is recommended that the three gradient cables be routed thru the overhead cable trough before any of the other cables are routed.

1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray, route the pre-terminated end of each cable through the designated overhead cable tray towards the PGR Cabinet.
 - Position in the cable tray as shown in Section 4-6.
 - Gradient cables **MUST** lay in a single layer next to each other
3. Route the pre-terminated ends of each gradient cable to the top/front of the PGR cabinet for connection to terminals.

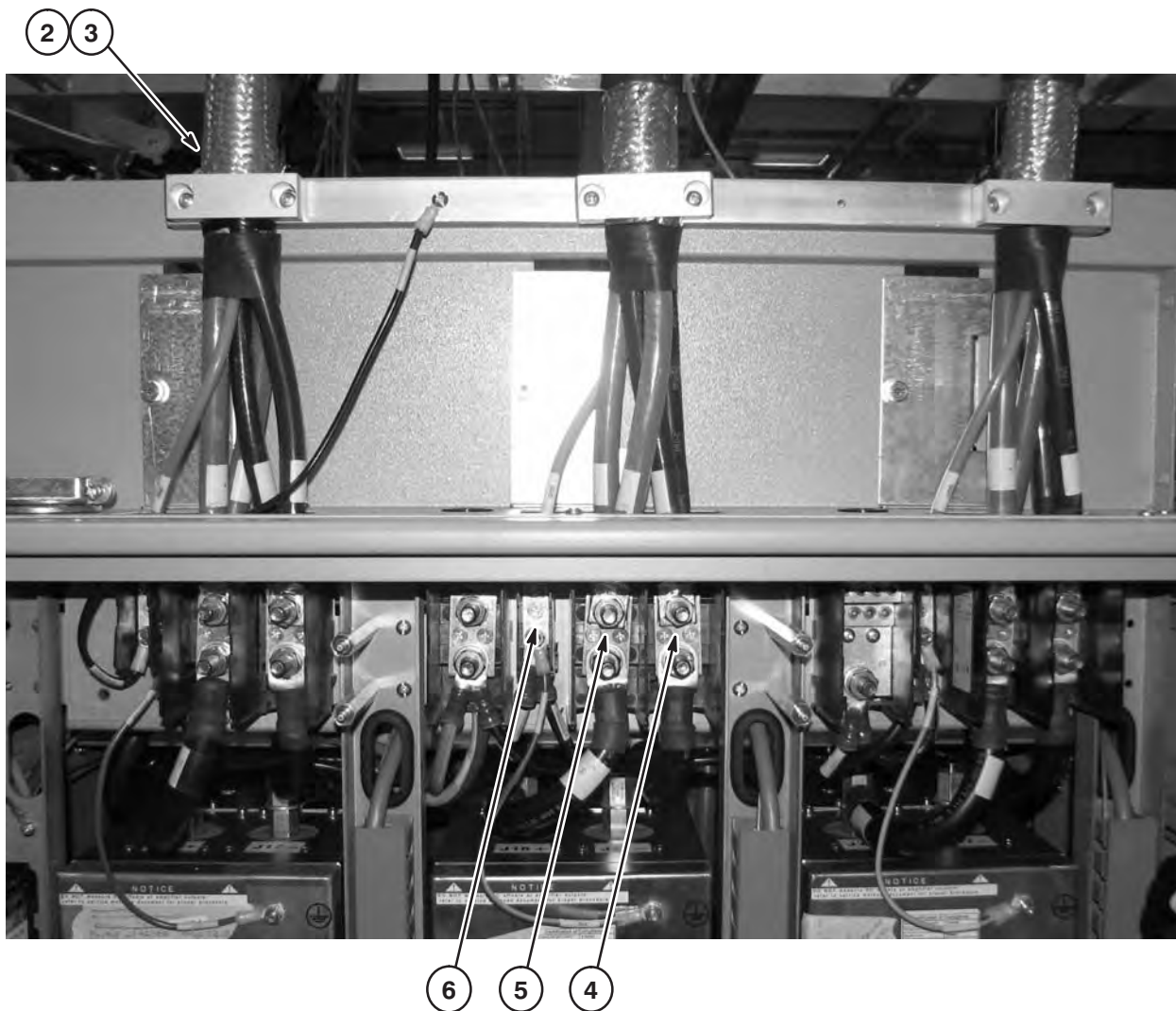


4-11-1-B Connect Gradient Cable to PGR Cabinet

The steps for this procedure install just one of the gradient cables. Repeat the steps below for installation and connection of the remaining gradient cables and the ground wires.

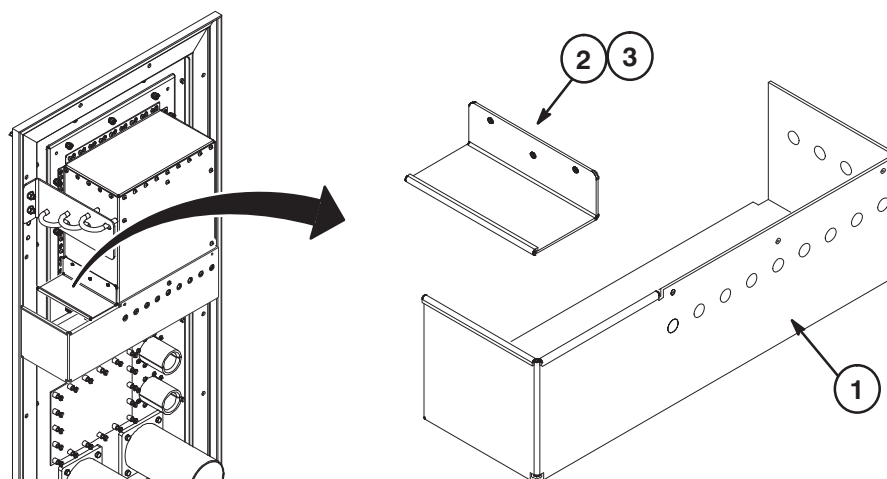
NOTE: If the gradient cable clamp bracket was removed in the earlier cabinet positioning procedures, reattach using the removed screws.

1. Remove cover at top of cabinet to access gradient cable clamps.
2. Position the braided area of the cable in the clamp at the top of the cabinet.
3. Close clamp around cable, but do not tighten completely. Adjustment may be necessary as the terminated ends of wires are attached.
4. Position the two black wires on the stud of the terminal block and fasten with nuts.
5. Position the two red wires on right stud of the terminal block and fasten with nuts.
6. Attach the ground wire on the ground stud and fasten with nut.

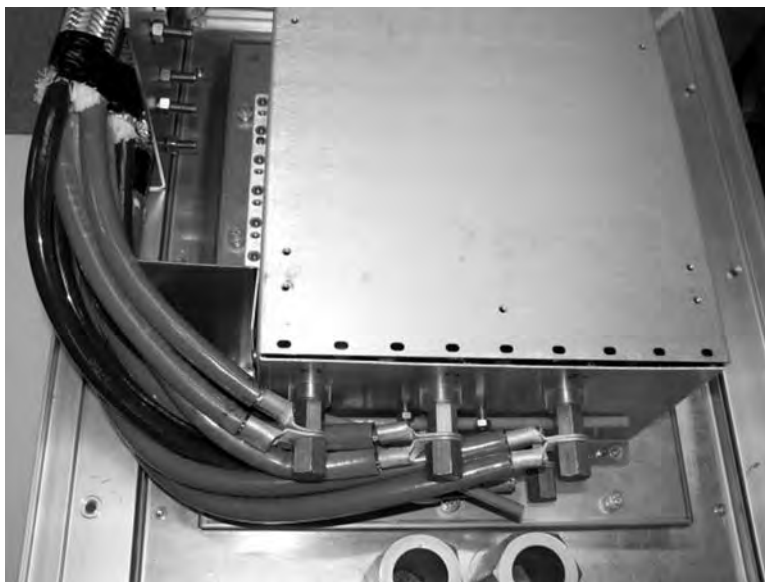


4-11-1-C Remove Gradient Filter Safety Shields

1. Unfasten three screws and remove the front safety shield. Save for re-installation after cables are attached.
2. **NOTE:** The side safety shield may or may not be removed. It is required to be in place after completion of gradient cable installation. Keeping this shield attached to gradient filter during gradient cable installation will make sure that the cables will have enough length to attach to the studs at the bottom of the gradient filter.
3. If it is necessary to remove the safety shield, unfasten three screws and remove the side safety shield. Save for re-installation after cables are attached.
4. This photograph offers a view of what the gradient cable installation will look like after connection of the cables and prior to re-installation of the front safety cover.



4



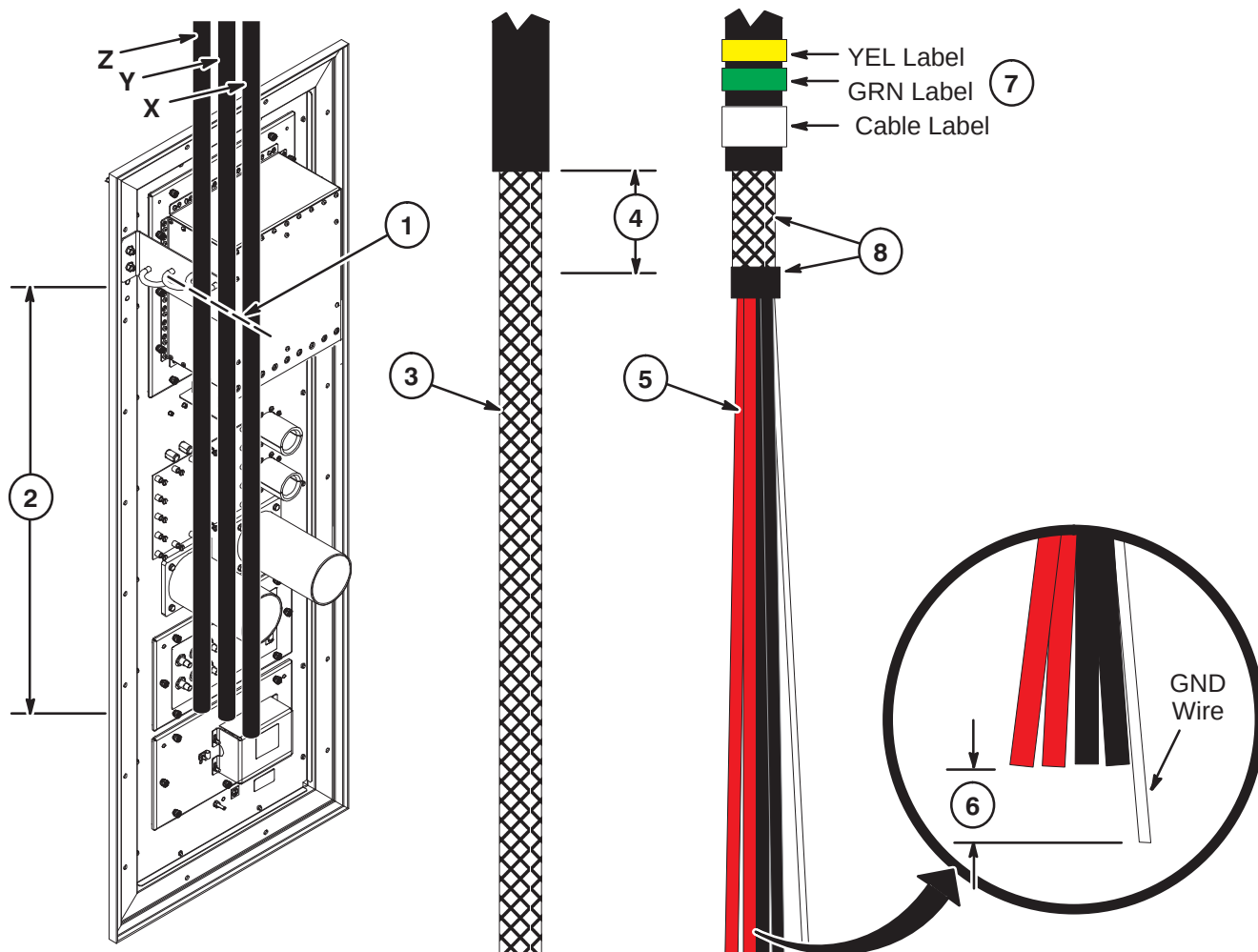
4-11-1-D Cut Gradient Cable to Length at SPW

The following steps will provide instructions for exposing the cable braid, cutting to length and terminating the red and black wires for connection to the studs on the bottom of the Gradient Filter.

1. Position cables at U-bolt clamp at side of SPW (Secondary Pen Wall) and mark each cable.
2. From the mark on each cable, measure and cut each cable according to chart below:

	X	Y	Z
Cut to length from Mark	20 Inches (508mm)	22 Inches (559mm)	25 Inches (635mm)

3. To expose braid, remove outer cable jacket to mark created in Step 2.
4. Measure and mark 3.5 inches (90mm) from edge of outer cable jacket.
5. Remove remaining braid from the mark to end of each cable exposing the two red wires, the two black wires, and the ground wire.
6. The red and black X wires can be cut back 2 inches (51mm). DO NOT cut back the ground wire.
7. On the end of each cable jacket, apply the provided yellow and green tape as shown below. Also apply the applicable cable label for each Run: Run E3317 = FILT X, RUN E3318 = FILT Y, RUN E3319 = FILT Z
8. Remove plastic covering on braid and apply electrical tape around edge of braid to prevent unraveling of braid.
9. Discard remaining cable.



4-11-1-E Connect Gradient Cable Wires to Gradient Filter

- 1. Position the X, Y, and Z gradient cables on bracket and tighten U-bolts around braided section of each cable.

CAUTION: Do not over tighten U-bolts. Damage may result to cables.

- 2. Terminal attachment locations: Black wires = X-, Y-, Z-. Red wires = X+, Y+, Z+.
- 3. Note the back-to-back position of the terminals attached to studs. Before crimping terminals onto wires, position and mark cables as to alignment of terminals for ease of fastening crimped terminals to studs.

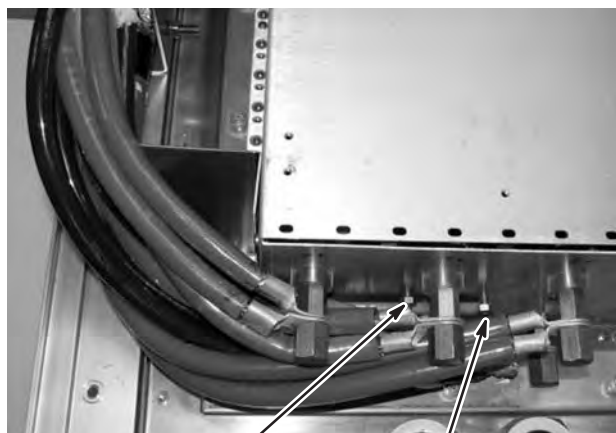
NOTE: Refer to step 7 on next page as to the order of installation of cables on studs. Adjustment to lengths of any of the red or black wires may be necessary to accomodate connections to studs.

- 4. Strip 0.75 in. (19mm) of insulation from end of gradient black and red wires.
- 5. Label each of the five wires in each gradient cable per table below. Red wires are "+", Black wires are "-".

Label 1 GRADIENT CABLE	Label 2 BLK	Label 3 BLK	Label 4 RED	Label 5 RED	Label 6 GND
RUN E3317 SPW GRAD FILT-X	X-	X-	X+	X+	GND
RUN E3318 SPW GRAD FILT-Y	Y-	Y-	Y+	Y+	GND
RUN E3319 SPW GRAD FILT-Z	Z-	Z-	Z+	Z+	GND

4-11-1-E Connect Gradient Cable Wires to Gradient Filter (continued)

6. Ground Wires – Route, mark and cut to length, strip 0.75 in. (19mm) of insulation from end of GND wires, and crimp a furnished T & B 54105 terminal on each ground wire. Connect wires as indicated. **Do not over tighten GND studs.**
7. Gradient cable wires – On end of each end, place (Do not crimp at this time) a furnished T & B 54165 terminal. Route and connect to studs using recommended order below. If too long, cut and re-strip end of individual wires or adjust braided area of cable position in bracket U-bolt.
8. After attaining correct length for each wire, crimp terminals on wires. **IMPORTANT: Make sure terminals are positioned correctly as shown in step 3.**



X & Y GND Stud

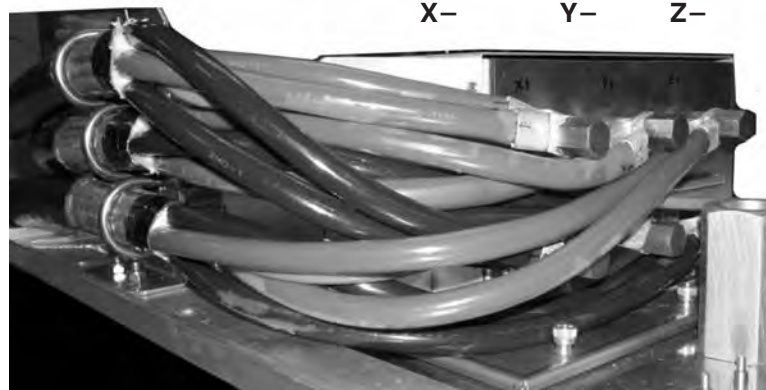
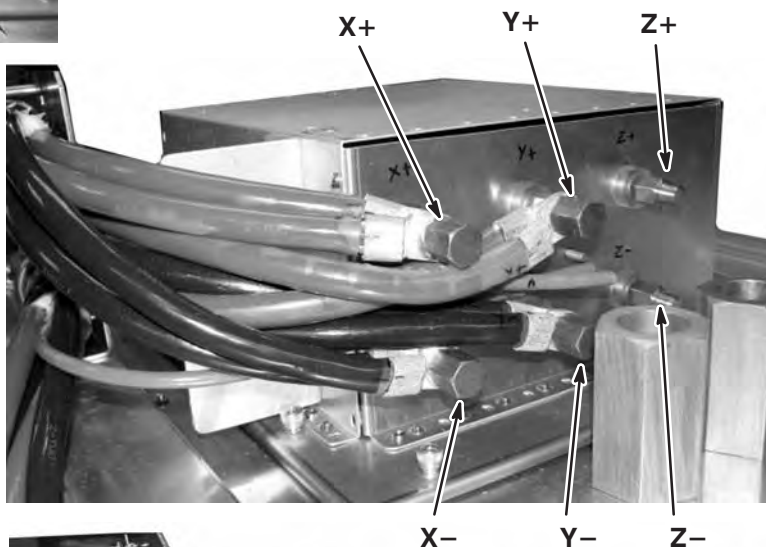
Z GND Stud

7 Recommended Order of Connecting Wires

8

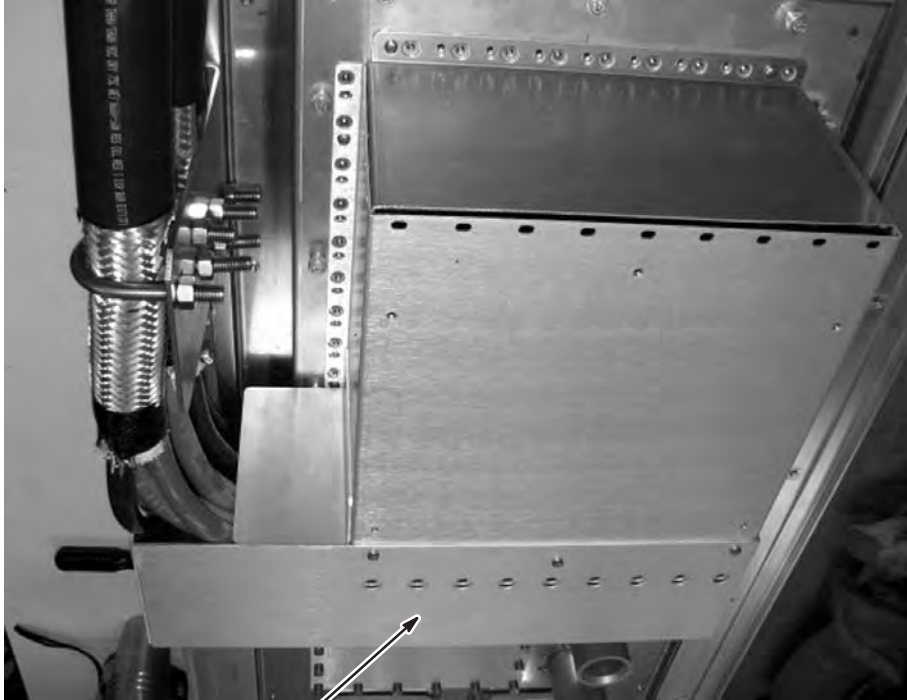
- First: Y–
- Second: Y+
- Third: X–
- Fourth: X+
- Fifth: Z–
- Sixth: Z+

Final Configuration of Wires



4-11-1-F Install Safety Cover(s)

1. If removed, re-attach the Side Safety Shield using the three previously removed screws.
2. Re-attach the Front Safety Shield with the three previously removed screws.



②



①

4-11-2 MAGNET ROOM GRADIENT CABLES

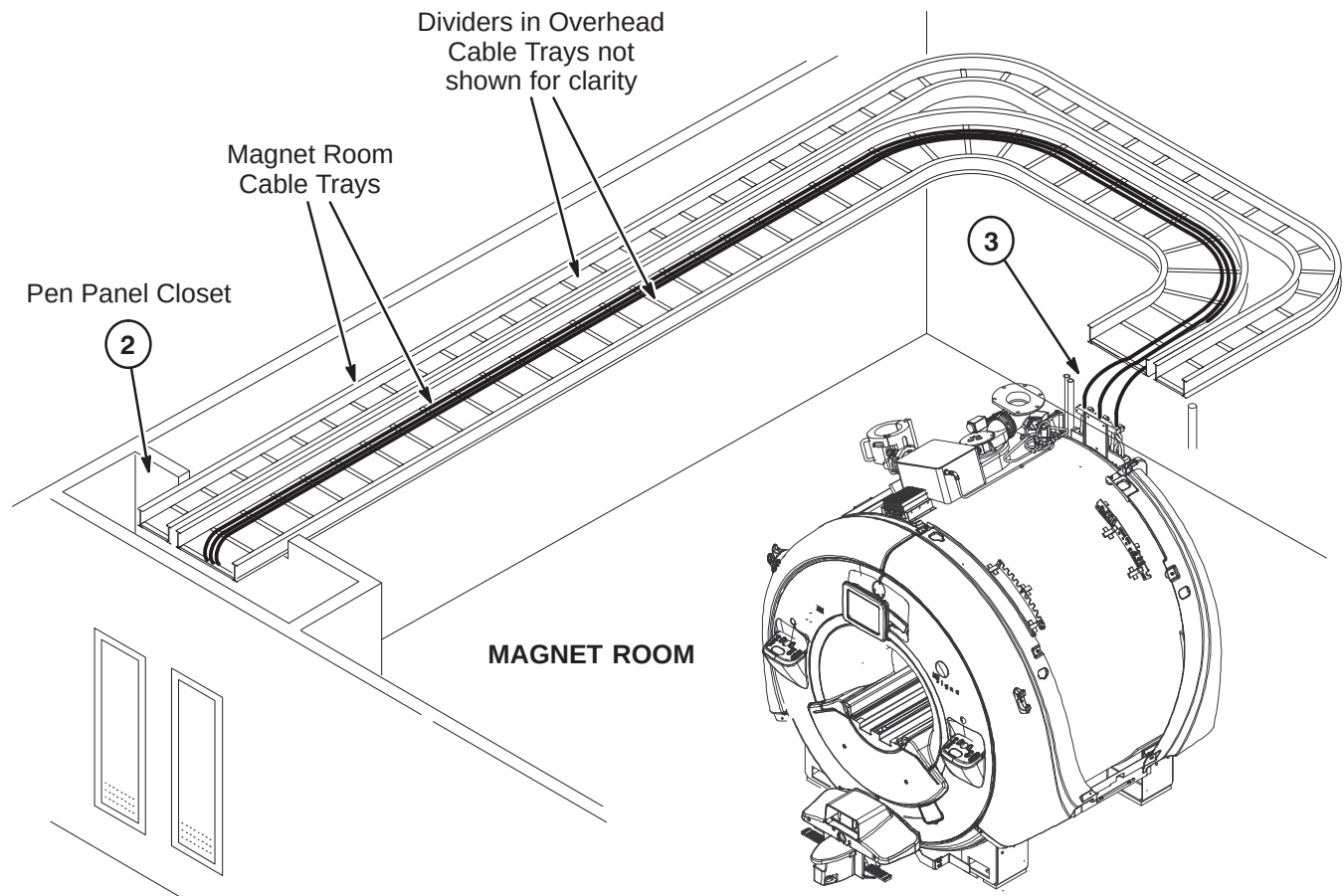
The Magnet Room gradient cables, Runs M3317, M3318, and M3319 are delivered to site pre-terminated at one end for connection to the rear of the Magnet. The other ends of the gradient cables are routed to the SPW (Secondary Pen Wall), cut to length, and terminated.

4-11-2-A Remove Magnet Enclosure Covers

Before routing any cables, remove and store in a safe place the magnet enclosure side covers, arc covers, and rear covers at top of magnet.

4-11-2-B Route Gradient Cables in Overhead Cable Tray

1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray in the Pen Panel Closet, route the pre-terminated end of each cable through the designated overhead cable tray towards the rear of the Magnet.
 - Make sure they are positioned in the cable tray as shown in Section 4-7.
 - Gradient cables MUST lay in a single layer next to each other.
3. Route the pre-terminated ends of each gradient cable to the rear of the magnet for connection to terminals as shown in Section 4-11-2-C.



4-11-2-C Connect Gradient Cables at Back of Magnet

The steps for this procedure install just one of the gradient cables. Repeat the steps below for installation and connection of the remaining gradient cables and the ground wires.

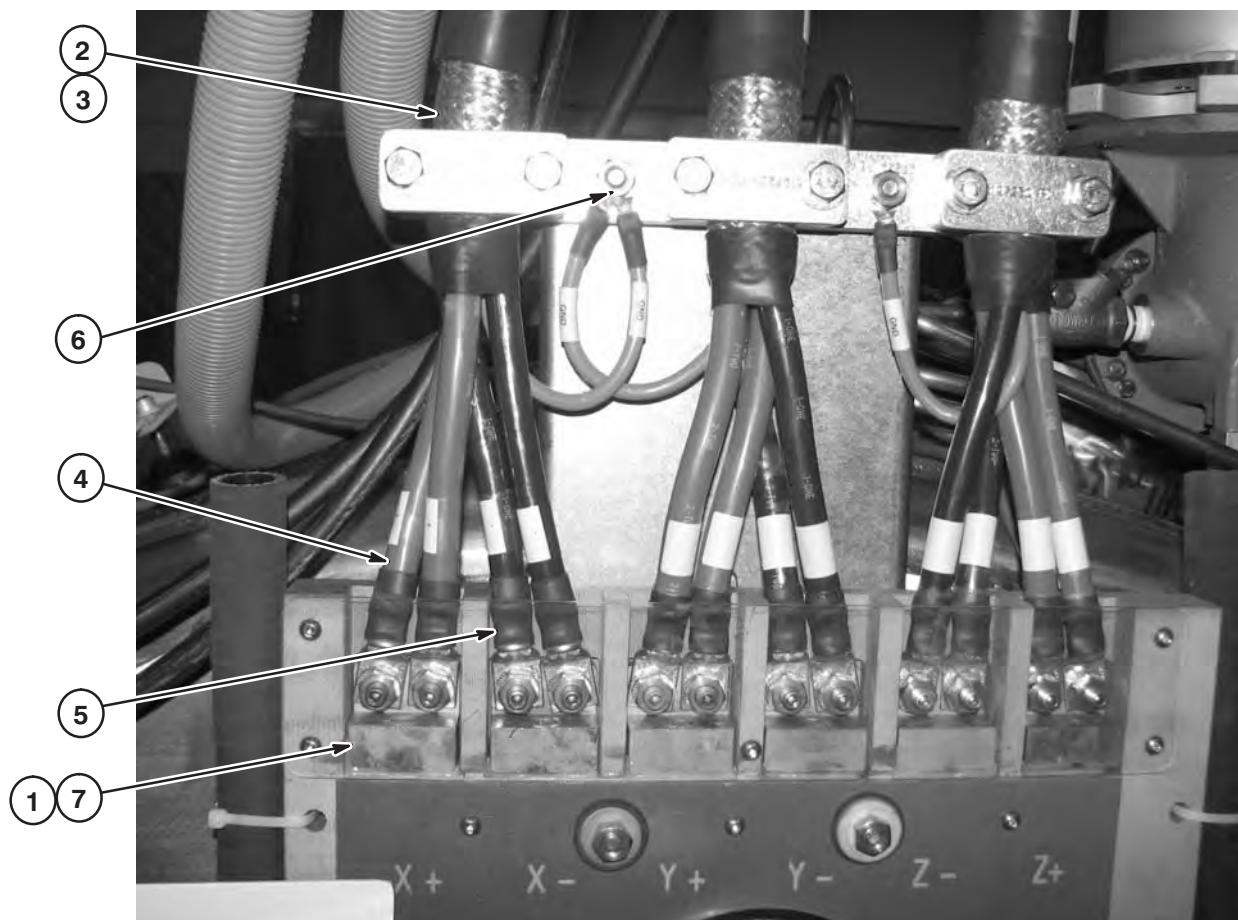
CAUTION

The RED and BLACK wires for the “Z” connection are opposite from the “X” and “Y” connections. The RED Z+ wires are on the right and the BLACK Z- wires are on the left for Z.

1. Remove safety shield.
2. Position the braided area of the cable in the clamp.
3. Close clamp around cable, but do not tighten completely. Adjustment may be necessary as the terminated ends of wires are attached.

NOTE: For the next two steps, make sure the lockwasher is installed between the terminal and nut.

4. Position the two red wires on the two “+” studs and fasten with provided lockwasher and nuts.
5. Position the two black wires on the two “-” studs of the terminal block and fasten with provided lockwashers and nuts.
6. Attach the ground wire on the ground stud and fasten with provided nut. (X and Y are connected together on one GND stud and the Z ground wire is connected to the other.
7. Re-attach safety shield.



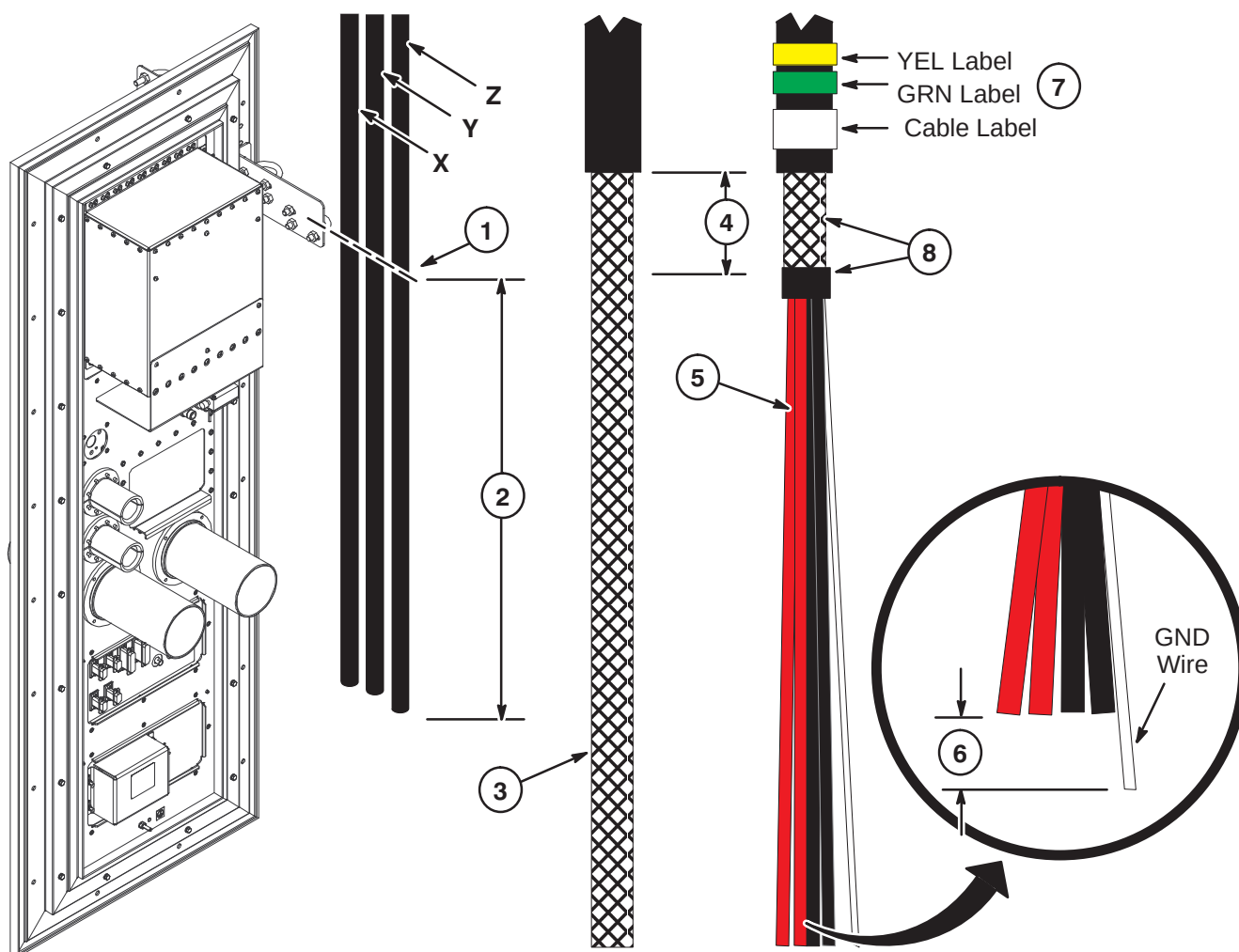
4-11-2-D Cut Gradient Cable to Length at SPW

The following steps will provide instructions for exposing the cable braid, cutting to length and terminating the red and black wires for connection to the studs on the bottom of the Gradient Filter.

1. Position cables at U-bolt clamp at side of SPW (Secondary Pen Wall) and mark each cable.
2. From the mark on each cable, measure and cut each cable according to chart below:

	X	Y	Z
Cut to length from Mark	20 Inches (508mm)	22 Inches (559mm)	25 Inches (635mm)

3. To expose braid, remove outer cable jacket to mark created in Step 2.
4. Measure and mark 3.5 inches (90mm) from edge of outer cable jacket.
5. Remove remaining braid from the mark to end of each cable exposing the two red wires, the two black wires, and the ground wire.
6. The red and black X wires can be cut back 2 inches (51mm). DO NOT cut back the ground wire.
7. On the end of each cable jacket, apply the provided yellow and green tape as shown below. Also apply the applicable cable label for each Run: Run M3317 = FILT X, Run M3318 = FILT Y, Run M3319 = FILT Z
8. Remove plastic covering on braid and apply electrical tape around edge of braid to prevent unraveling of braid.
9. Discard remaining cable.



4-11-2-E Connect Gradient Cable Wires to Gradient Filter

- 1. Remove plastic shield.
- 2. Position the X, Y, and Z gradient cables on bracket and tighten U-bolts around braided section of each cable.
Note: U-bolts are located on side of bracket away from the gradient filter.

CAUTION: Do not over tighten U-bolts. Damage may result to cables.

- 3. Terminal attachment locations: Black wires = X-, Y-, Z-. Red wires = X+, Y+, Z+.
- 4. Note the back-to-back position of the terminals attached to studs. Before crimping terminals onto wires, position and mark cables as to alignment of terminals for ease of fastening crimped terminals to studs.

NOTE: Refer to step 8 on next page as to the order of installation of cables on studs. Adjustment to lengths of any of the red or black wires may be necessary to accomodate connections to studs.

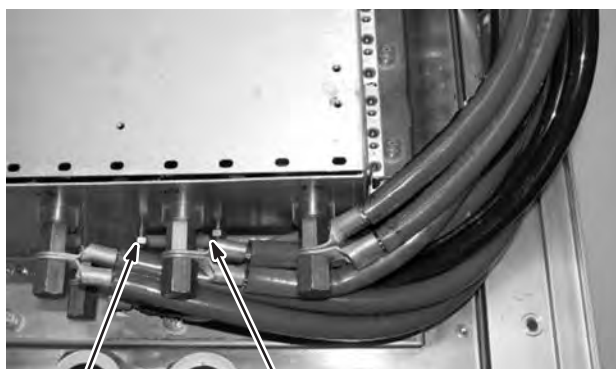
- 5. Strip 0.75 in. (19mm) of insulation from end of gradient black and red wires.
- 6. Label each of the five wires in each gradient cable wire per table below. Red wires are "+", Black wires are "-".

The diagram illustrates the connection of gradient cables to a terminal block. It shows the X, Y, and Z cables being positioned on a bracket (2) and secured with U-bolts. The terminal block has studs for X+, Y+, Z+, X-, Y-, Z+, and GND. A callout shows a wire being stripped (5) and crimped (4). A photograph shows the physical installation of the cables on the terminal block.

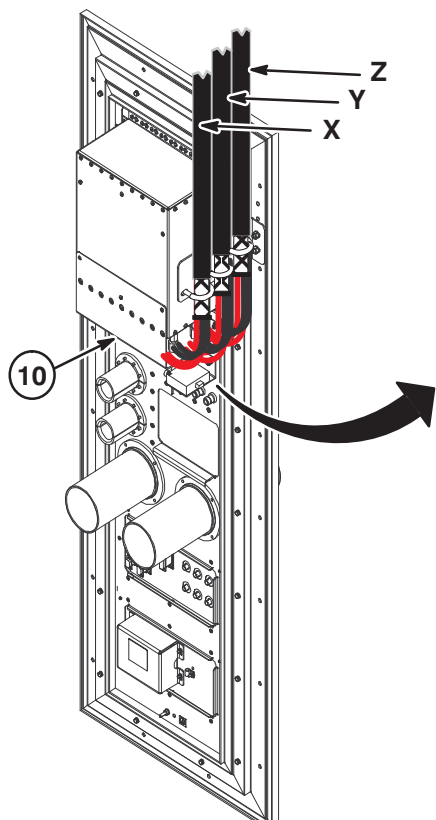
Label 1 GRADIENT CABLE	Label 2 BLK	Label 3 BLK	Label 4 RED	Label 5 RED	Label 6 GND
RUN M3317 SPW GRAD FILT-X	X-	X-	X+	X+	GND
RUN M3318 SPW GRAD FILT-Y	Y-	Y-	Y+	Y+	GND
RUN M3319 SPW GRAD FILT-Z	Z-	Z-	Z+	Z+	GND

4-11-2-E Connect Gradient Cable Wires to Gradient Filter (continued)

7. Ground Wires – Route, cut to length, strip 0.75 in. (19mm) of insulation from end of GND wires, and crimp a furnished T & B 54105 terminal on each ground wire. Connect wires as indicated.
8. On end of each gradient cable wire, place (Do not crimp at this time) a furnished T & B 54165 terminal. Route to studs using recommended order below. If too long, cut and re-strip end of individual wires or adjust braided area of cable position in bracket U-bolt.
9. After attaining correct length for each wire, crimp terminals on wires. **IMPORTANT: Make sure terminals are positioned correctly as shown in step 3.**
10. Install plastic safety shield.

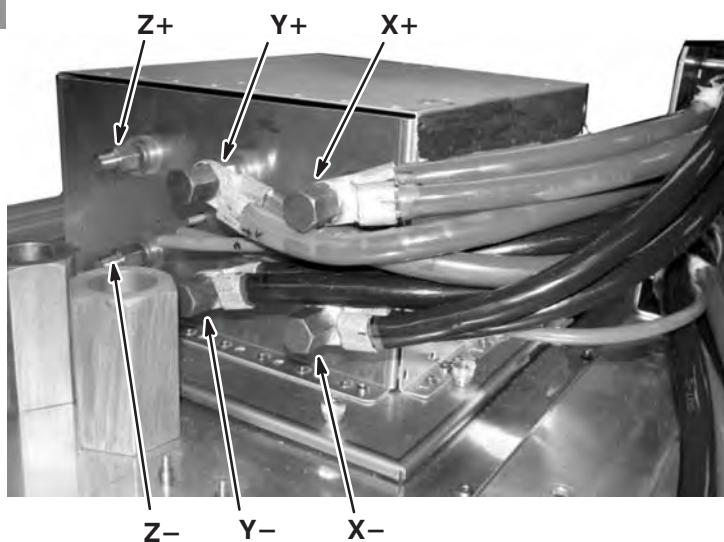


Z GND Stud X & Y GND Stud



**8 Recommended Order
of Connecting Wires**

- 9**
- First: Y–
 - Second: Y+
 - Third: X–
 - Fourth: X+
 - Fifth: Z–
 - Sixth: Z+



Final Configuration of Wires



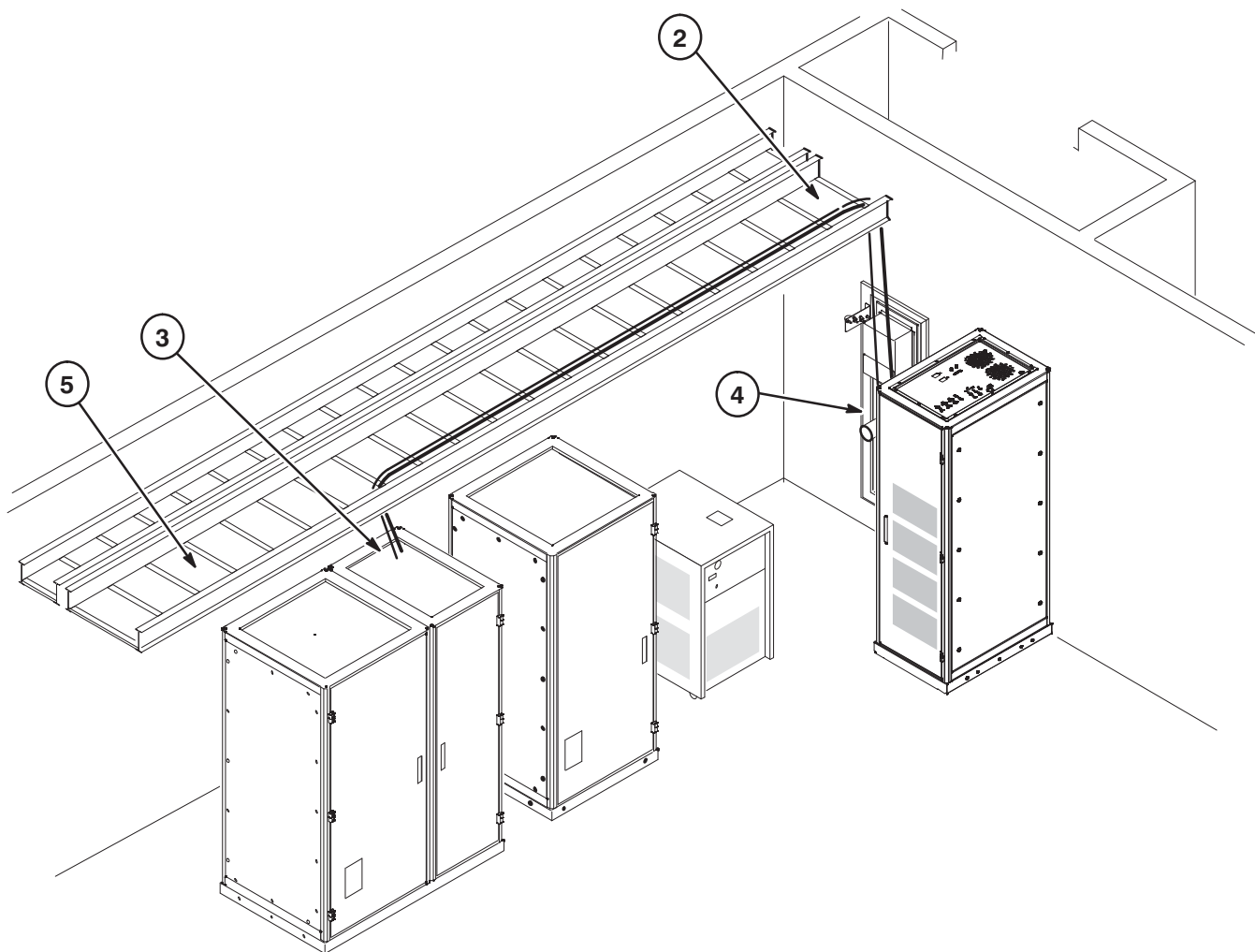
4-12 INSTALL RF TRANSMIT CABLES

4-12-1 ROUTE AND CONNECT IN EQUIPMENT ROOM

Note: The Equipment Room RF Transmit cables, Runs E1001 and E1002 are delivered to site pre-terminated and must not be cut to length.

It is recommended that the RF Transmit cables be routed thru the overhead cable tray after the gradient cables have been routed and before any of the other cables are installed.

1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray, route the Head and Body cables marked for connection to the PGR cabinet through the 24 inch overhead cable tray towards the PGR Cabinet. Position in the cable tray as shown in Section 4-6.
3. Route the ends of the cable to the top of the PGR cabinet and connect as marked to J18 and J19.
4. Connect other ends of RF Transmit cables as marked to J106 and J128 on SPW.
5. Coil and store excess cable length in overhead cable tray.

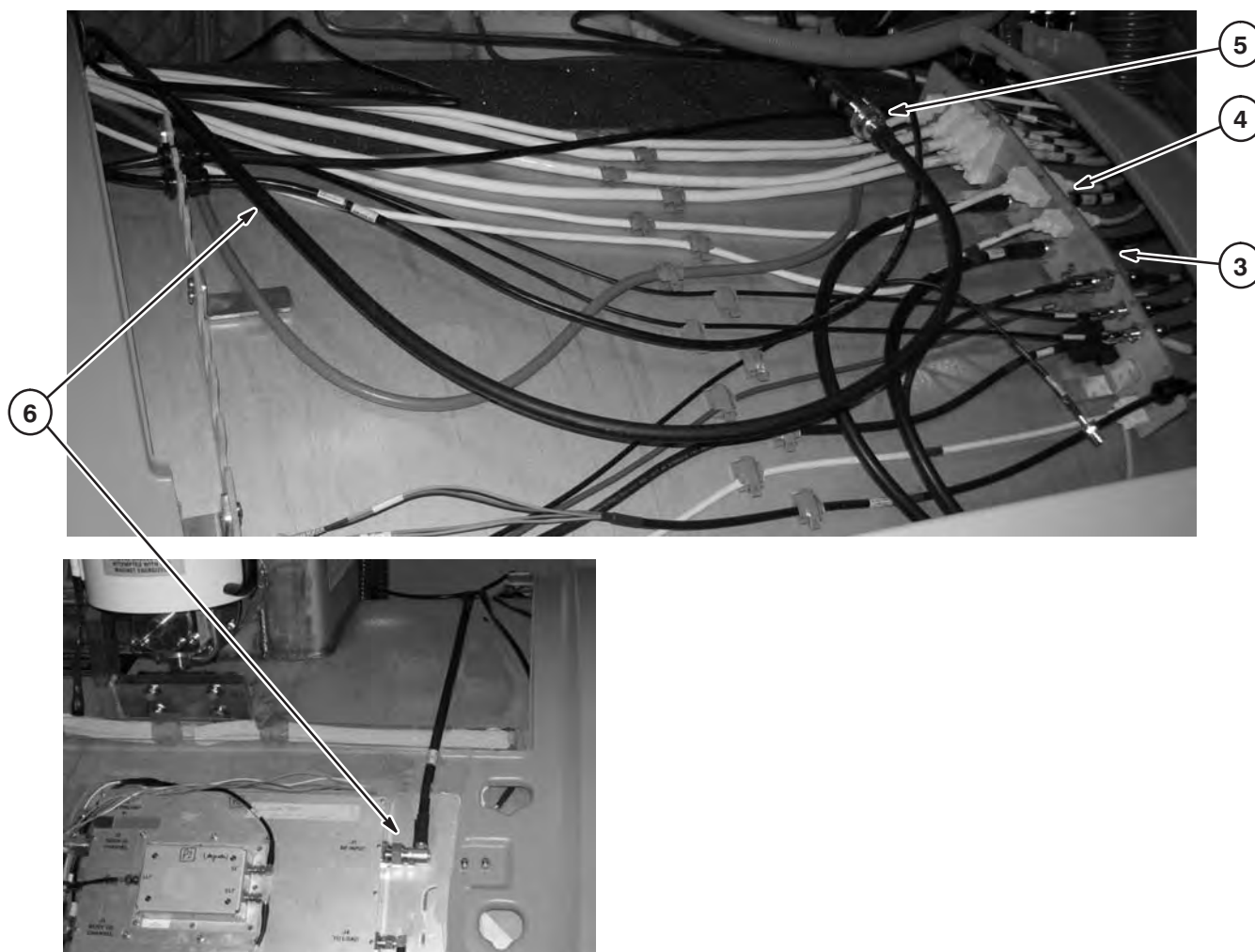


4-12-2 ROUTE AND CONNECT IN MAGNET ROOM

Note: The Magnet Room RF Transmit cables, Runs M1001, M1305, and M1306 are delivered to site pre-terminated and **MUST NOT BE CUT TO LENGTH**.

It is recommended that the RF Transmit cables be routed thru the overhead cable tray after the gradient cables have been routed and before any of the other cables are installed.

1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Starting at the SPW end of the cable tray, route Body cable M1001 and Head cables M1305 and M1306 through the designated overhead cable tray towards the rear of the magnet. Position in the cable tray as shown in Section 4-7.
3. Connect as marked M1305 to J14 on magnet I/O panel.
4. Connect as marked M1306 to J12 on magnet I/O panel.
5. Connect as marked M1001 to extension cable M1319 on top of magnet.
6. Route M1319 over top of magnet and connect to J1 on MAG-HYB.



4-12-2 ROUTE AND CONNECT IN MAGNET ROOM (continued)

7. Connect Run M1001 to J106 on SPW. Two people, one on each side of SPW, are needed to connect M1001 and E1001 to SPW.

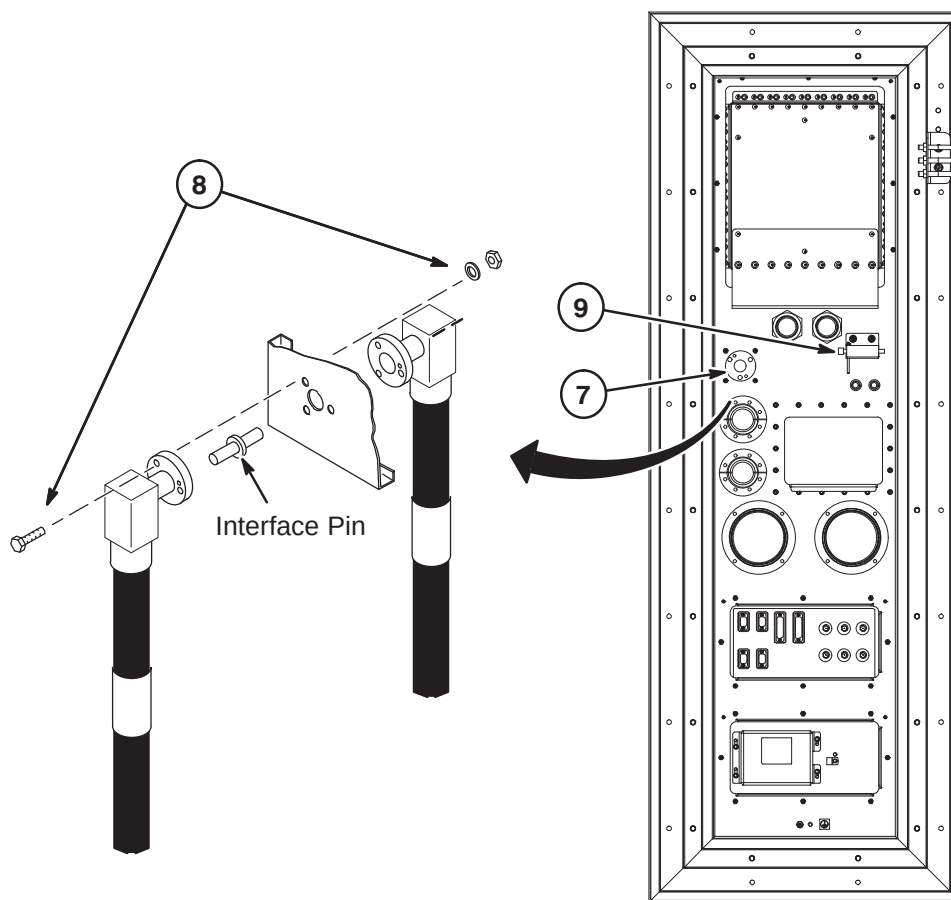
CAUTION

Do NOT install O-Ring or Grease that may be provided with Cable Termination Kit. Installation of these parts will prevent the ground from making contact resulting in RF leaks.

8. Insert Interface Pin into one of the connectors. At J106, align connectors on each side of Penetration Panel and push together. Fasten together using three of the following:

- 1/4-20 x 1.25 Hex Head Bolt (brass)
- 1/4 Screw Size Lock Washer (phos bronze)
- 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)

Note: The flat washer provided with the termination kit may be discarded to facilitate installation of connectors to Penetration Panel.

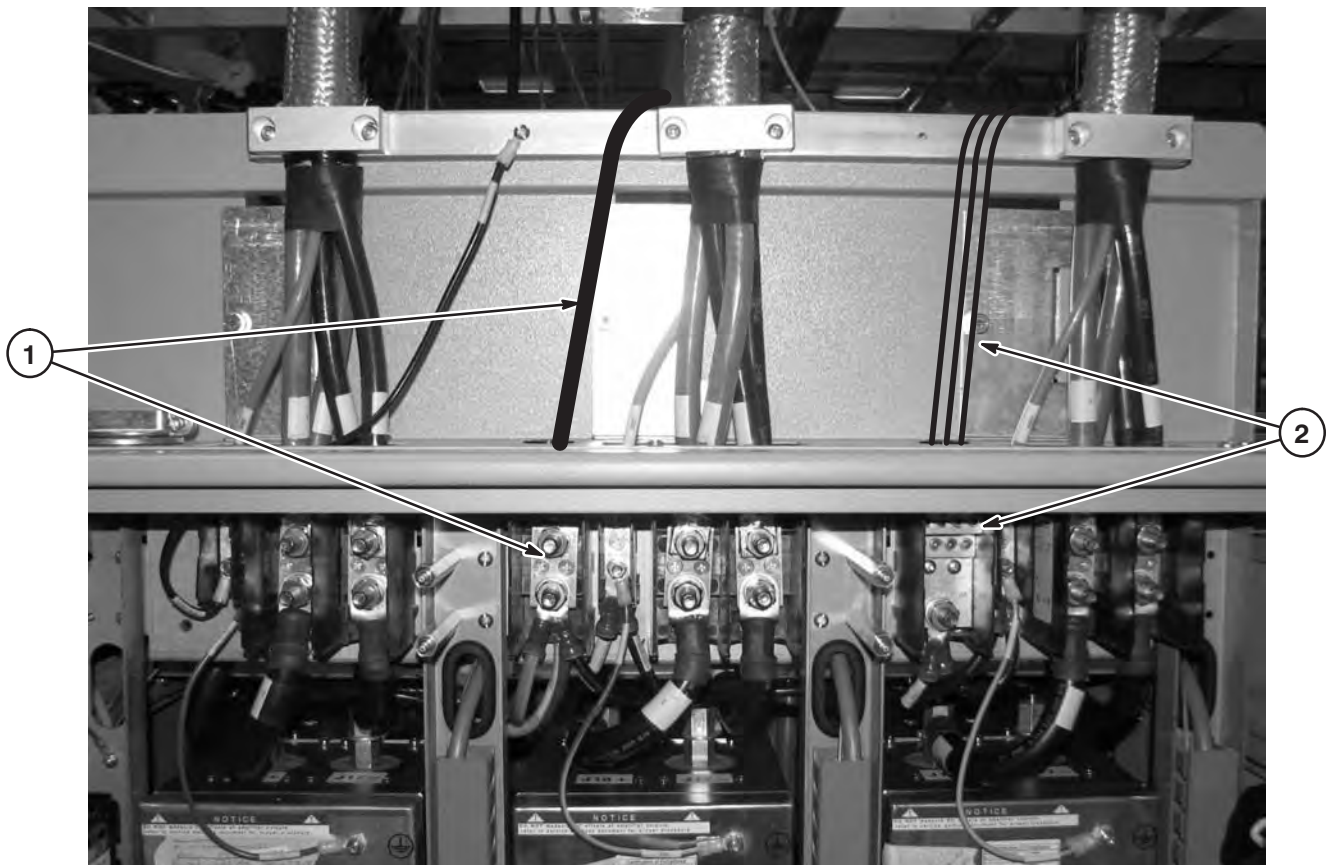


9. Connect Run M1305 to J122 (N.O.) and Run M1306 to J120 (N.C.) on the Transmit Switch.
10. Coil and store excess cable length of M1001, M1305, and M1306 on cable hooks in Penetration Panel closet.
- Note:** The coil radius must be greater than the minimum bend radius for these cables.

4-13 ROUTE AND CONNECT GROUND CABLES

4-13-1 PGR CABINET GROUND CABLES

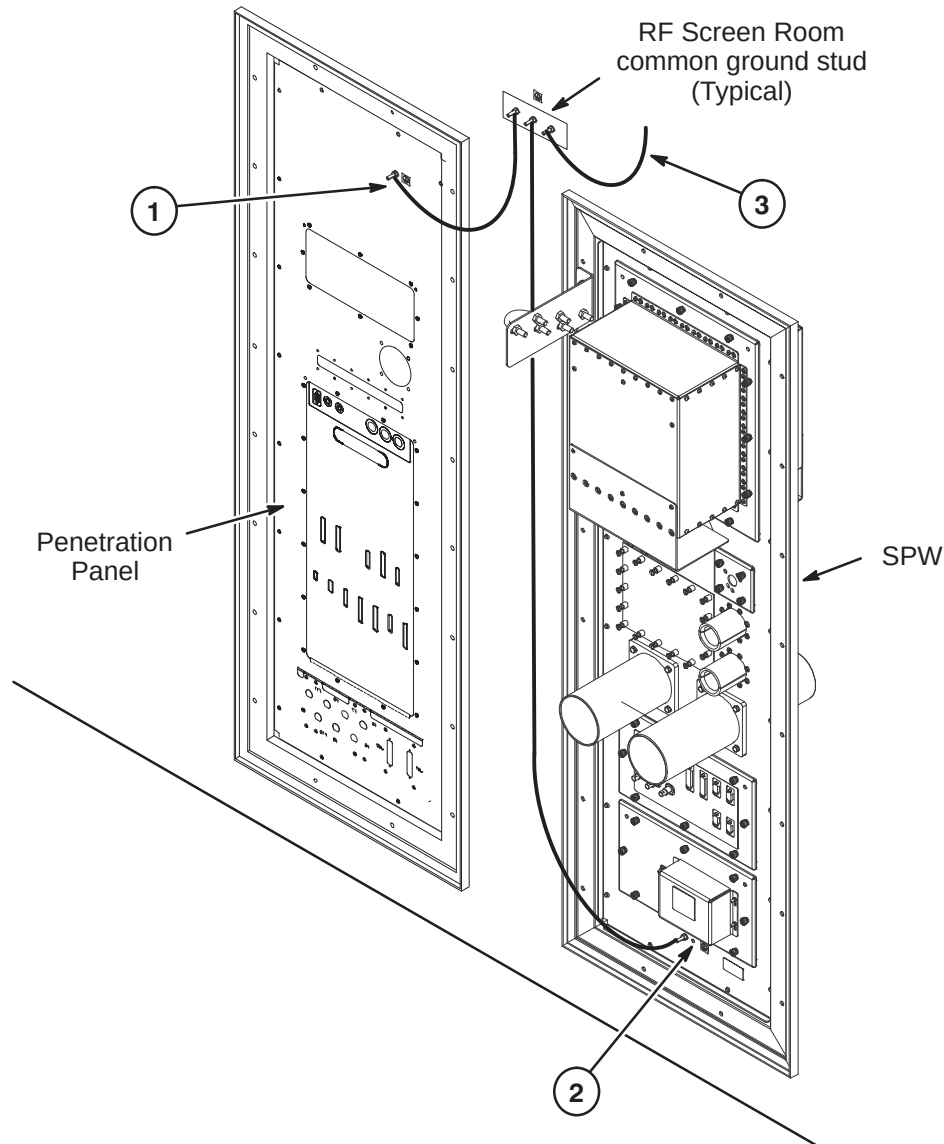
1. Route E4005 from RF common ground stud on screen room and connect to ground stud in PGR cabinet.
2. Route and connect the following to the PGR ground terminal as shown below:
 - E4002 from PEN cabinet.
 - E4007 from HEC cabinet.
 - E4008 from OW.



4-13-2 PENETRATION PANEL AND SPW GROUND CABLES

The ground cables in the following two steps are installed on the Equipment Room side of the panels.

1. Route and connect E4010 from RF common ground stud on screen room to ground stud in Penetration Panel.
2. Route and connect E4009 from RF common ground stud on screen room to ground stud at bottom of SPW.
3. Route and connect E4005 from RF common ground stud on screen room to ground stud in PGR cabinet.



4-14 ROUTE AND CONNECT EQUIPMENT ROOM POWER CABLES

To perform the Grounding and Leakage Current Checks procedure, only the power and ground cables can be connected. The test must be completed before any of the data cables can be connected.



Do not connect data cables until the grounding and leakage test has been completed.

NOTE: For efficiency purposes, all the cables can be routed thru the overhead cable trays but only the power and ground can be connected for this test.

Ground cables were routed and connected in Section 4-13. Connect the following power cables.

- E0003, PGR J102 to OW-PDM J1
- E0004, PGR J16 to PEN J1
- E0009, HEC CRY power to CRY power
- MagMon3 power cable to HEC
- Customer supplied power from MDP CB2 to PGR CB1
- Customer supplied power from MDP CB3 to HEC power

Section 4-15-1 thru Section 4-15-4 provide instructions on routing and connecting all cables in the Equipment Room.

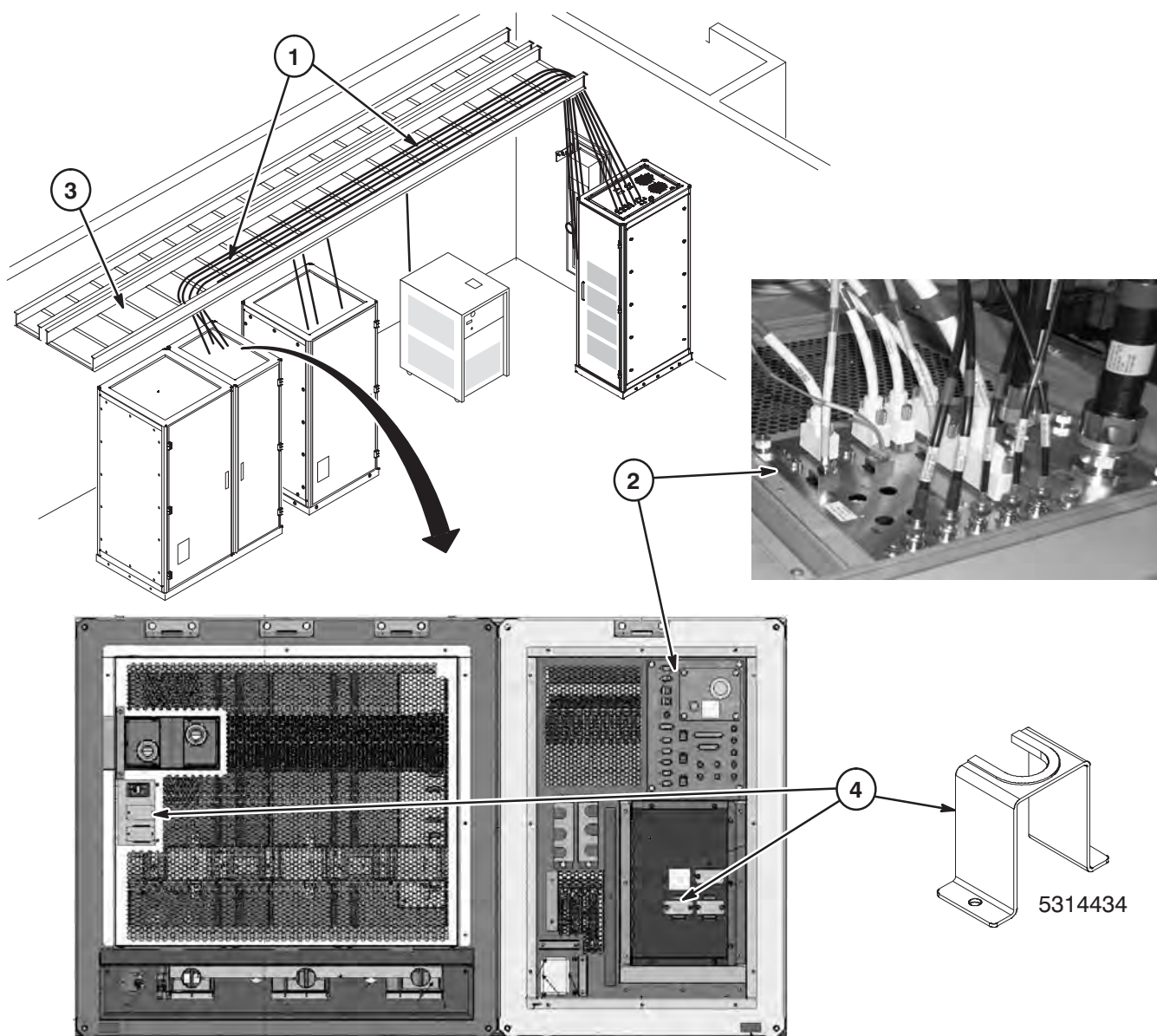
Section 4-18 provides instructions on routing and connecting all cables in the Operator Workspace.

4-15 ROUTE AND CONNECT CABLES IN EQUIPMENT ROOM

Refer to Section 4-6 for location of cables and hoses routed in Equipment Room in the overhead cable trays. Make sure cables are in their designated areas of the trays.

4-15-1 PGR CABINET CABLES

1. Route the cables thru the assigned channel in the 24 inch overhead cable routing mechanism (cable tray) to the cables final connection point.
2. Refer to cable map and connect remainder of data cables to top of PGR Cabinet after Grounding and Leakage test has been completed.
3. Refer to Section 4-2 for coiling of any excess cable in the overhead cable trays.
4. At J16 and J102, install Connector Bracket (5314434) on mounting standoffs to hold power cable Runs E0004 and E0003 in place. Bracket is part of Desktop Collector, 2321156-26.



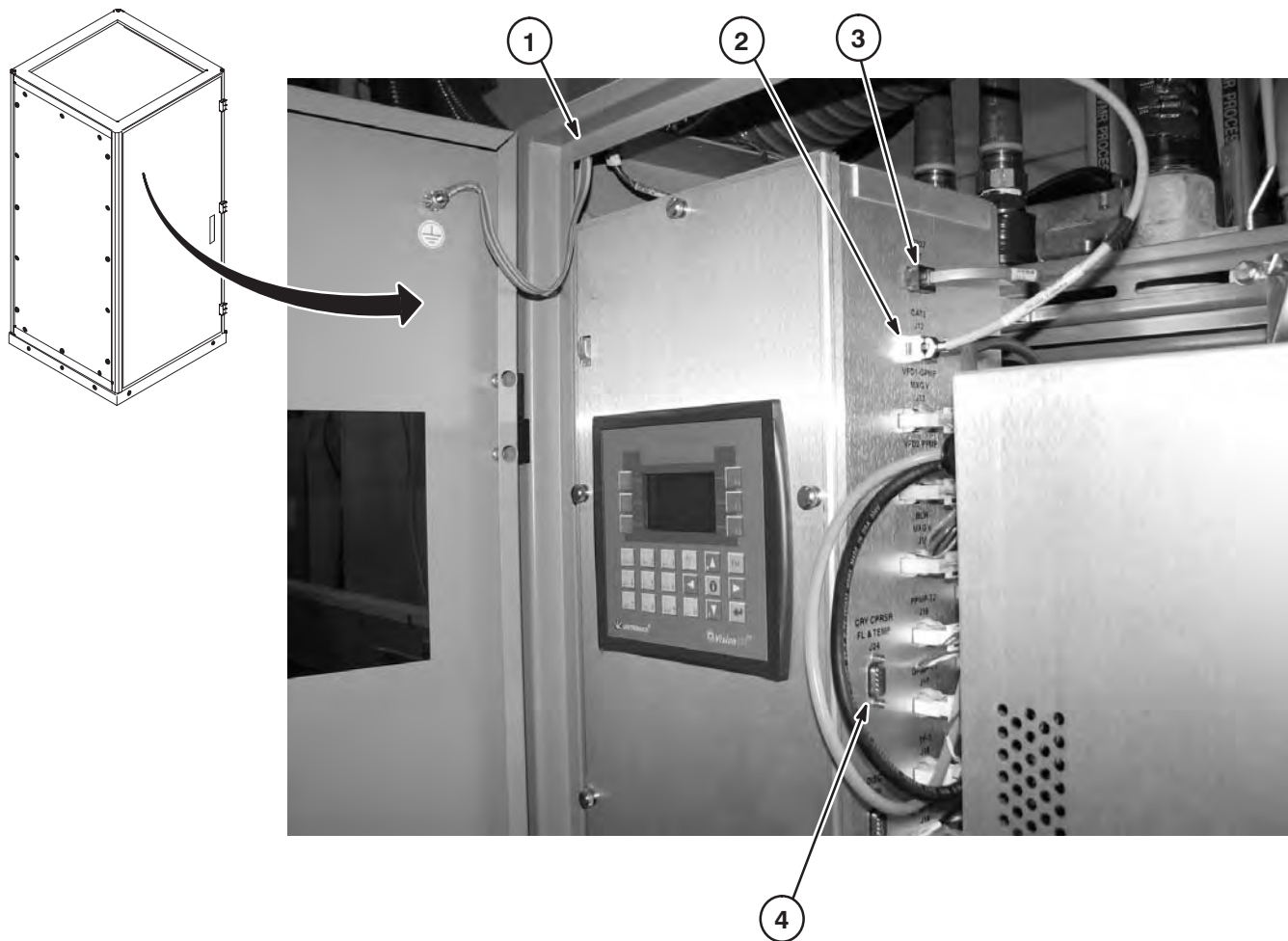
4-15-2 HEC CABINET CABLES

Refer to cable map and route remainder of cables to HEC Cabinet. Connect data cables after Grounding and Leakage test has been completed..

1. Connect the ground wire E4007 from the PGR cabinet to ground stud at top of cabinet.

Note: If ground wire is difficult to attach at location indicated in illustration, then connect E4007 to ground stud on opposite wall of cabinet.

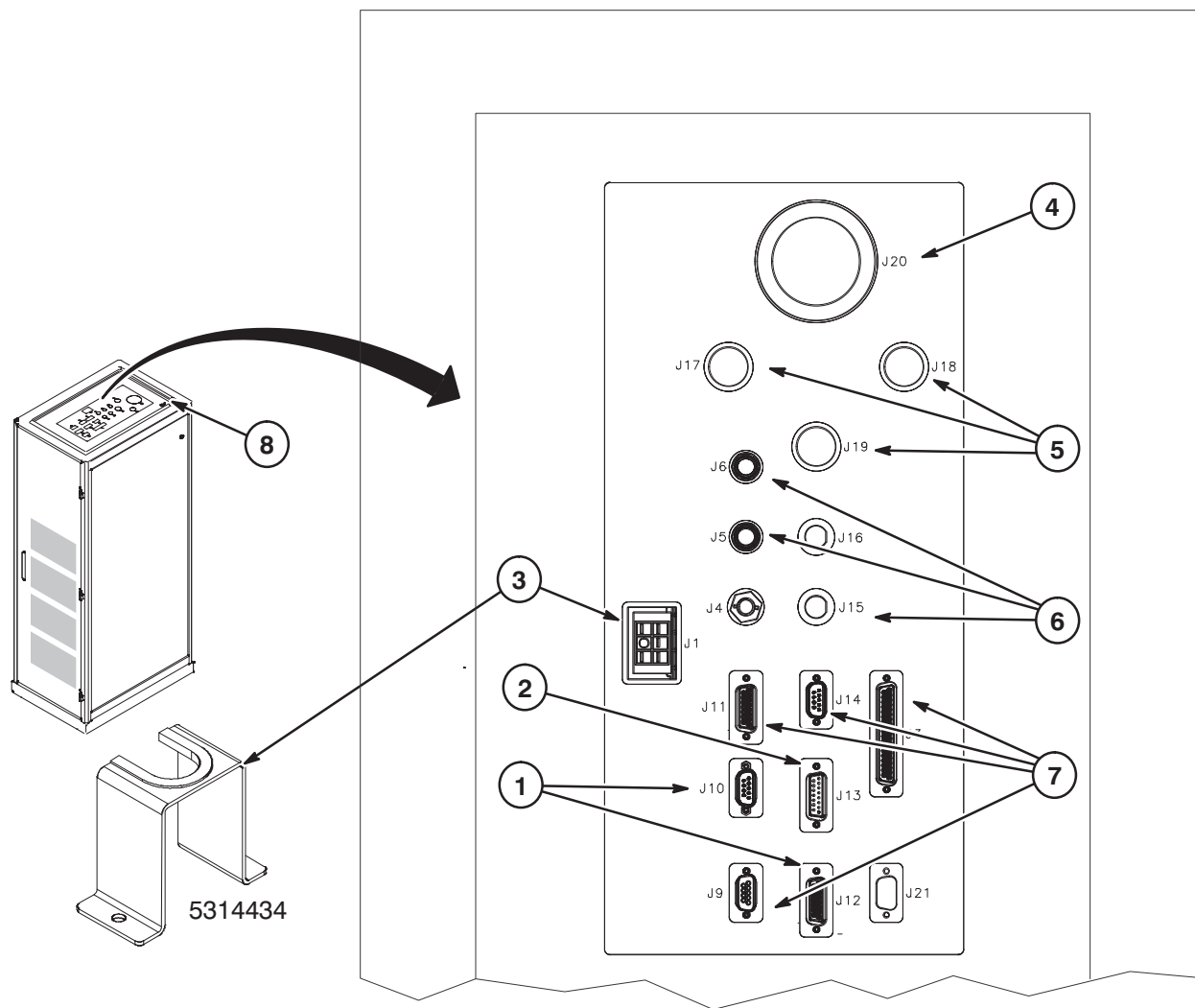
2. Connect Run E3013 to J12 CAT5.
3. **For 3.0T ONLY:** Connect Run E3028 to J11 RS232.
4. Run E3026 should have already been connected to J24 CRY CPRSR FL & TEMP.
5. Magnet Monitor power cable, Run E3006, should have already been connected to the customer supplied power outlet.



4-15-3 PEN CABINET CABLES

Refer to cable map and route cables to PEN Cabinet. Connect after Grounding and Leakage test has been completed.

1. Route Runs E3017 and E3018 from OW and connect to J10 and J12 as marked.
2. Route Runs E3020 from MagMon and connect to J13 as marked.
3. Route Power cable Run E0004 from PGR Cabinet and connect to J1. Install Connector Bracket (5314434) on mounting standoffs to hold E0004 in place. Bracket is part of Desktop Collector, 2321156-26.
4. Route fiber optic cable Runs P2003 and P2004 thru J20 and thru wave guide in Pen Panel to magnet.
5. Route Fiber Optic cable Runs E2002, E2020, and E2023 thru J19, J17, and J18 as marked. Refer to Section for connection of fiber optic wires inside of PEN Cabinet.
6. Route Coax/RF Runs E1003, E1004, and E1005 from PGR Cabinet and connect to J5, J6, and J15.
7. Route Signal cable Runs E3002, E3006, E3008 and E3009 from PGR Cabinet and connect to J11, J14, J9, & J3.
8. Route and connect GND cable Run E4002 to ground stud.



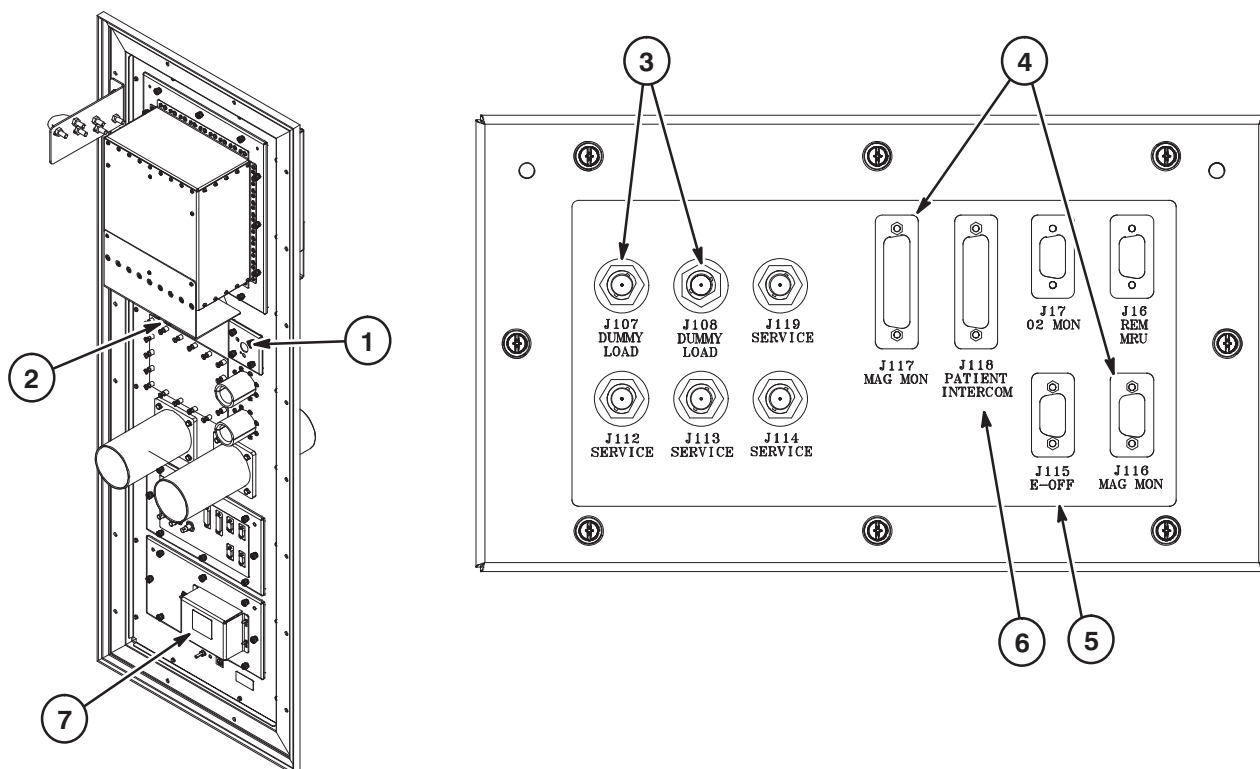
4-15-4 SPW CABLES

The SPW (Secondary Pen Wall) was delivered to site and installed prior to magnet delivery.

- The gas lines, Runs 621 and 622, were routed thru the J110 and J111 waveguides when the magnet was installed.
- The gradient cables were routed and connected to the gradient filter in a previous procedure in the section.
- The coolant lines have been routed thru J100 and J101 waveguides.
- The 4 inch air hose from the PGR cabinet has been routed to J104 waveguide.

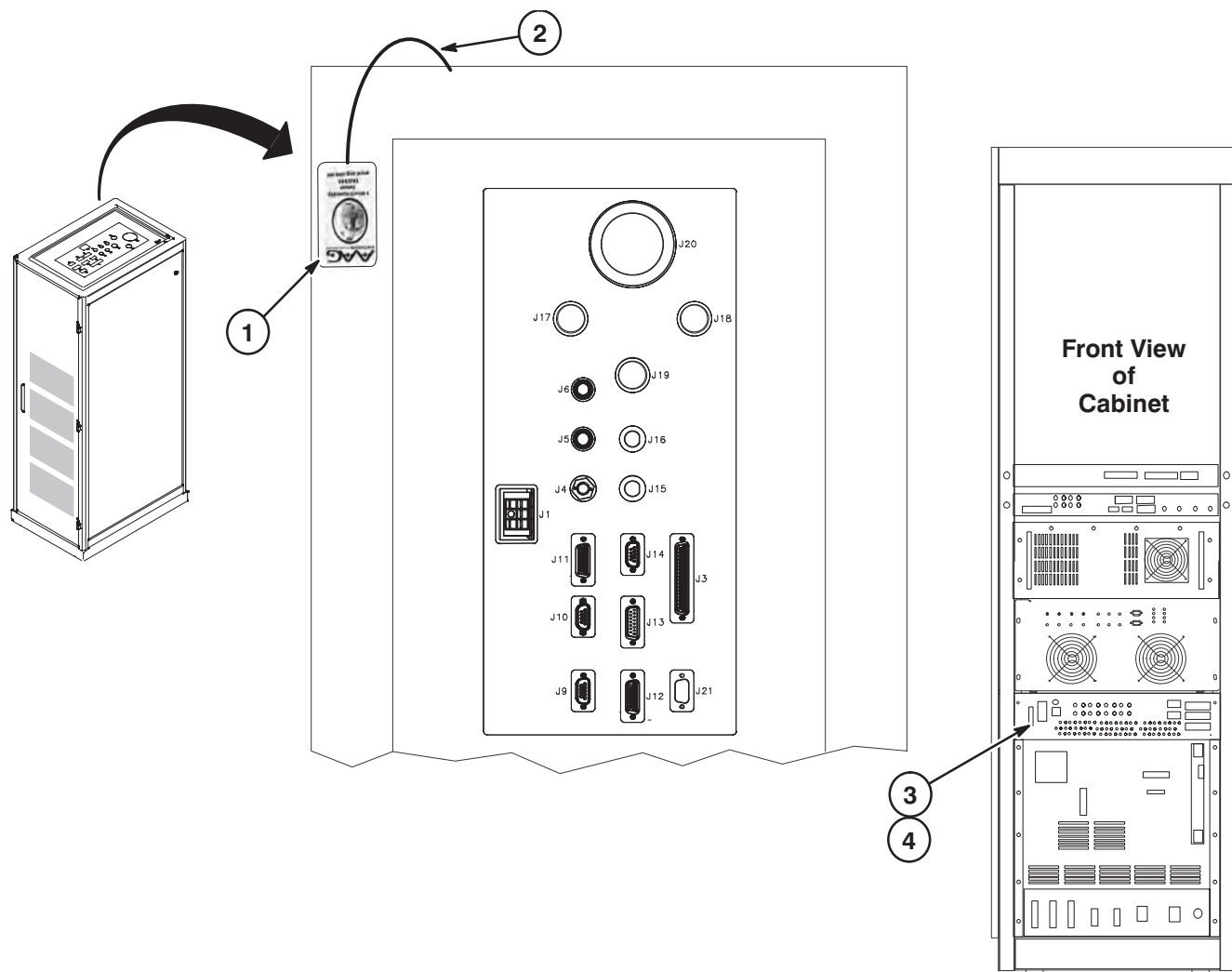
Refer to cable map and connect remaining cables to SPW after Grounding and Leakage test has been completed.

1. Route and connect Run E1001, Body TX, to J106.
2. Route and connect Run E1002, Head TX, to J128
3. Route and connect Coax/RF cables Runs E1006 and E1015 to J107 and J108 (3.0T only).
4. Route and connect Runs E3022 and E3023 from MagMon to J116 and J117.
5. Route and connect Run E3030 from E-Off Switch to J115.
6. Route and connect Run E3031 from GOC to J118.
7. Remove safety cover. Route and connect Run 623 to F1 and F3. Replace safety cover.



4-16 INSTALL TEMPERATURE/HUMIDITY SENSOR COMPONENTS ON PEN CABINET

1. Retrieve the iButton sensor (5317298) from the Desktop Collector kit (2321156-26) and attach to top of PEN cabinet using dual locking tape.
2. Connect RJ11 sensor cable (5317299) to iButton and route to rear of PHPS2 module in PEN Cabinet.
3. At J43 on rear of PHPS2 module, install DB9 to RJ11 Adapter (5318097).
4. Connect end of RJ11 sensor cable to RJ11 Adapter.



4-17 INSTALL RUN E3014 – RF DOOR SWITCH

Run E3014 is a 100 ft cable with a 9-pin subminiature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminiature D plug.

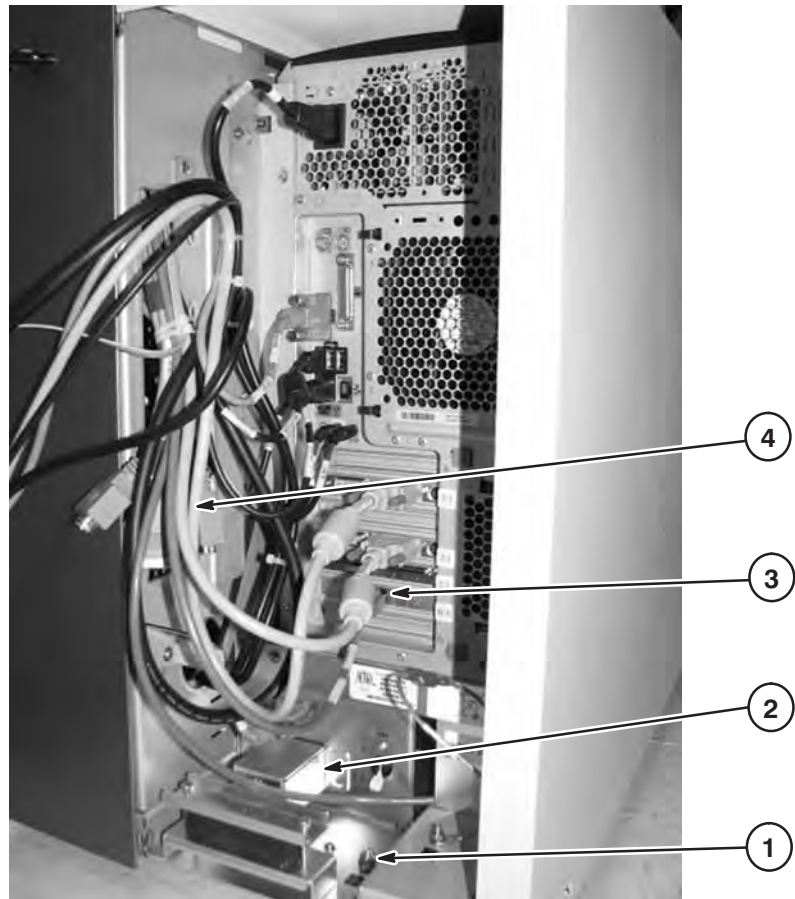
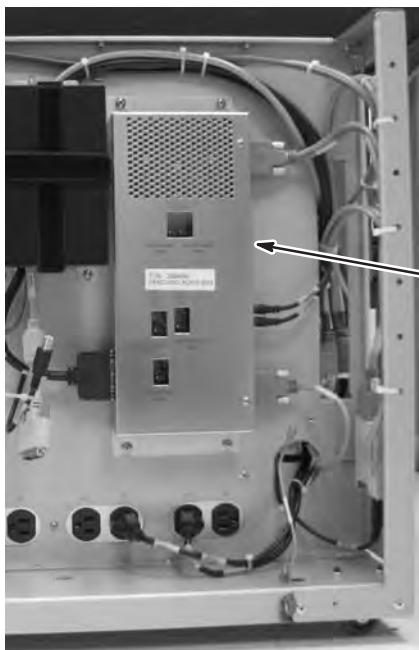
Note: If system being installed is in an upgraded site, existing Run 701 can be re-used and re-connected if the information on the cable label is changed to RUN E3014, PGR-J20. If this is done, then steps 2 through 5 below do not have to be completed.

1. Connect Run E3014 to J20 on PGR Cabinet and route to RF Door Switch.
2. Connect black lead on Run E3014 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run E3014 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run E3014 to RF Door Switch N.C. (normally closed) contact.

4-18 ROUTE AND CONNECT CABLES TO OPERATOR WORKSPACE

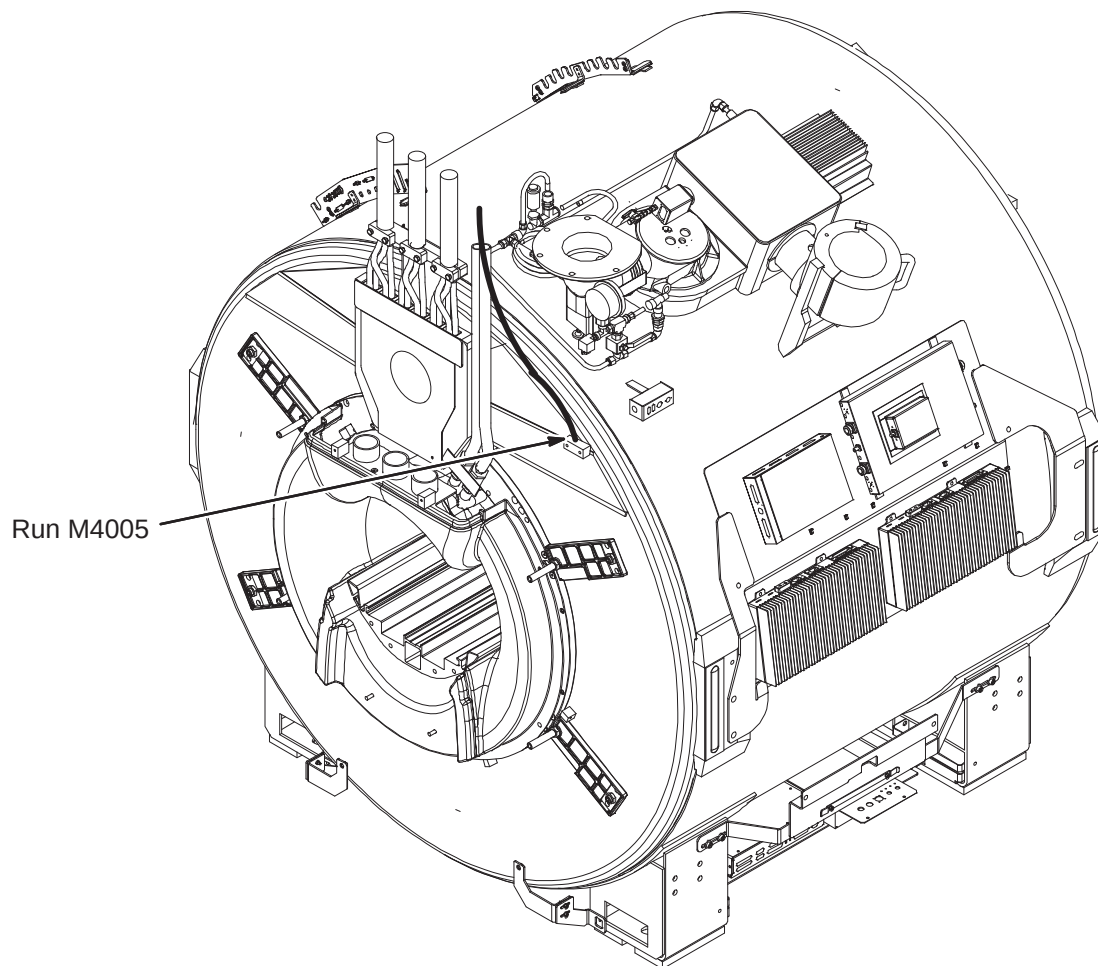
Refer to Section 6 for instructions on setting up the Operator Workspace area.

1. Connect Ground wire E4008 to GND stud.
2. Unfasten cover and connect power cord Run E0003 from PGR J102 to OW1-PDM J1.
3. Connect Run E3011 to ethernet connection "B".
4. Connect Runs E3015, E3017, and E3018 to FJ6, FJ5, and FJ7 as marked.
5. Route and connect Run E3031 to GOCAA J7.



4-19 CONNECT MAGNET GROUND, RUN M4005

Make sure magnet ground cable, Run M4005 is connected to GND stud at back of magnet. Other end of ground cable is routed thru overhead cable trays and connected to RF Screen Room common ground.

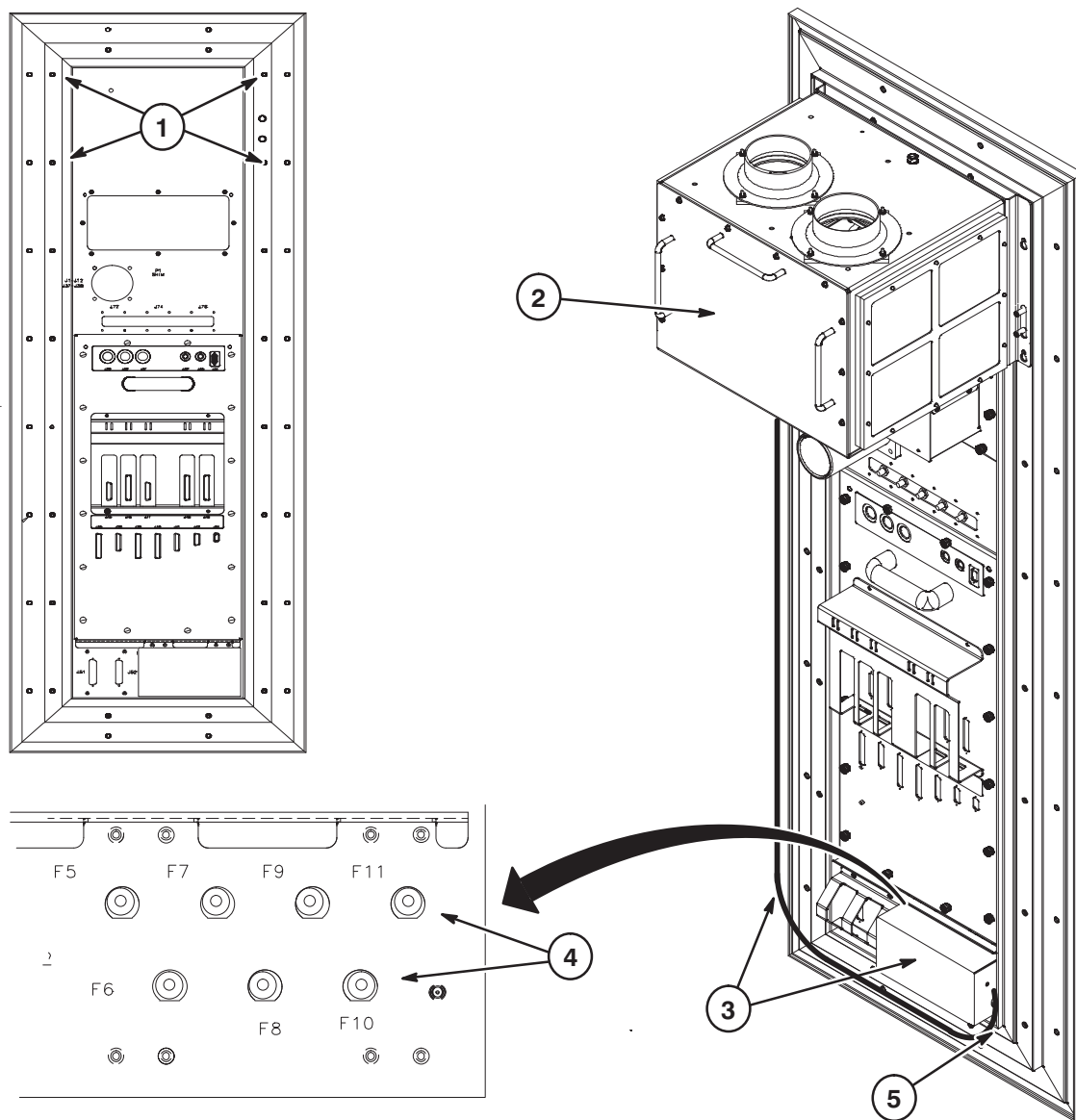


4-20 INSTALL BLOWER BOX ON PENETRATION PANEL

1. Loosen, but do not remove, four screws securing top of Pen Panel.

Note: If slot on mounting flange does not provide enough clearance for placing over existing flange studs, remove above mounting screws, position blower box on SPW, and attach with removed screws. At least two people are required for this procedure.

2. Lift and slide Blower Box (5213397) mounting holes over loosened screws. Tighten the four screws to secure Blower Box to Pen Panel frame.
3. Route power cord to bottom of Pen Panel and remove safety power cover.
4. Connect power cord wires as marked to the seven connection points.
5. Re-attach safety power cover making sure that power cord is routed thru access hole.

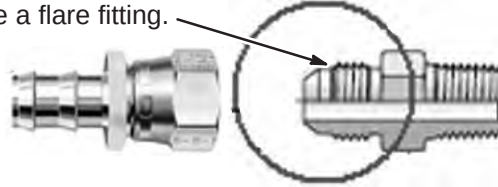


4-21 INSTALL AIR AND WATER HOSES

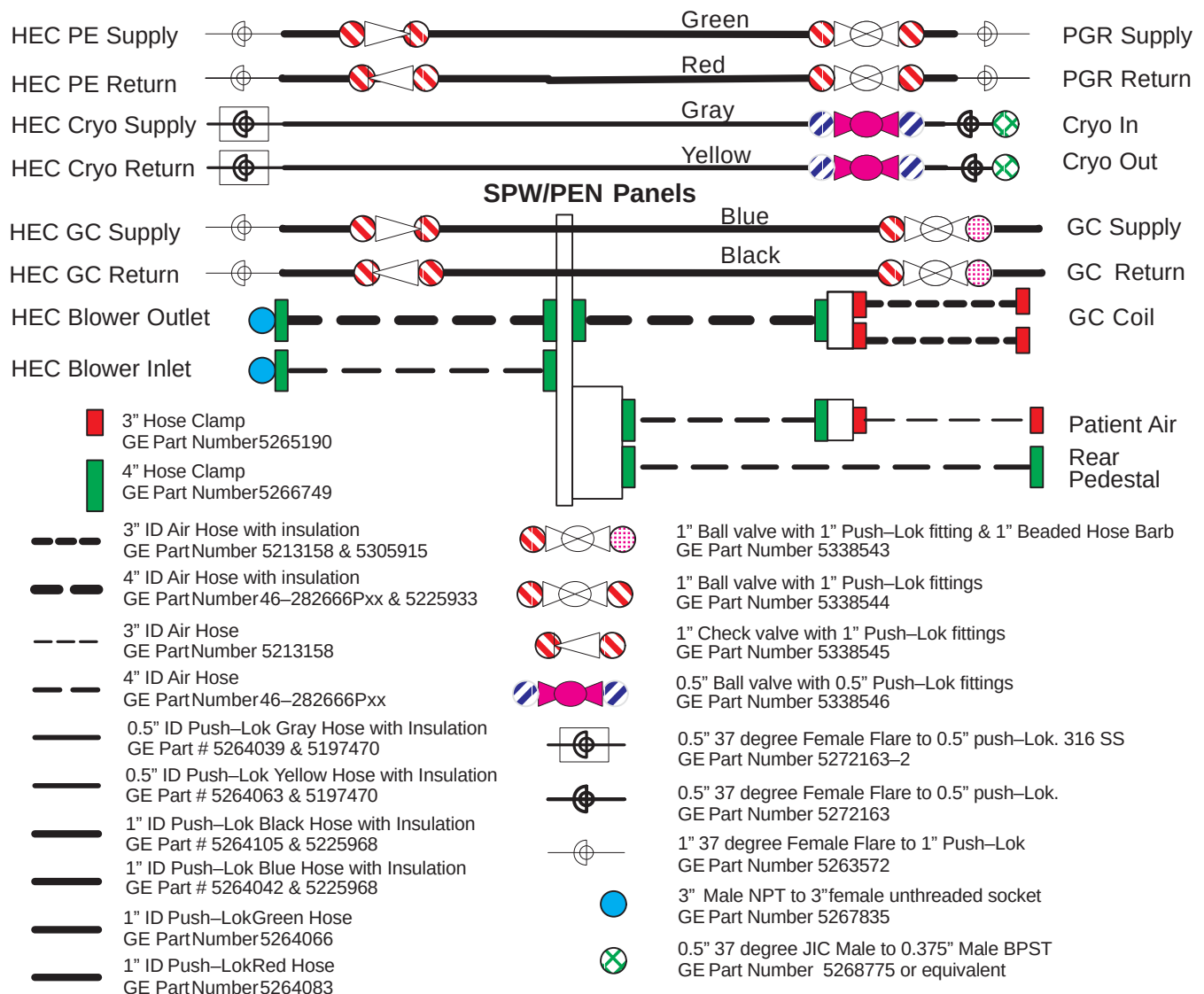
Refer to Section 4-6 for location of cables and hoses routed in Equipment Room in the overhead cable trays. Make sure cables are in their designated areas of the trays.

Hose and Fitting Installation Notes:

- To prevent cracking of fittings, hold fittings in the HEC stationary with wrench while flare connections are being made.
- Trim extra length off of hose clamps after applying clamp to the 3 and 4 inch hoses. Smooth rough edges.
- A small amount of water may be applied to push-lok barbs to facilitate insertion into hoses.
- DO NOT use teflon tape on threads that have a flare fitting.



Air and Water Hose Installation Schematic:



4-21-1 ROUTE AND CONNECT COOLANT HOSES BETWEEN PGR and HEC

Refer to Air and Water Hose schematic in Section 4-21 for part numbers and description of hose hardware.

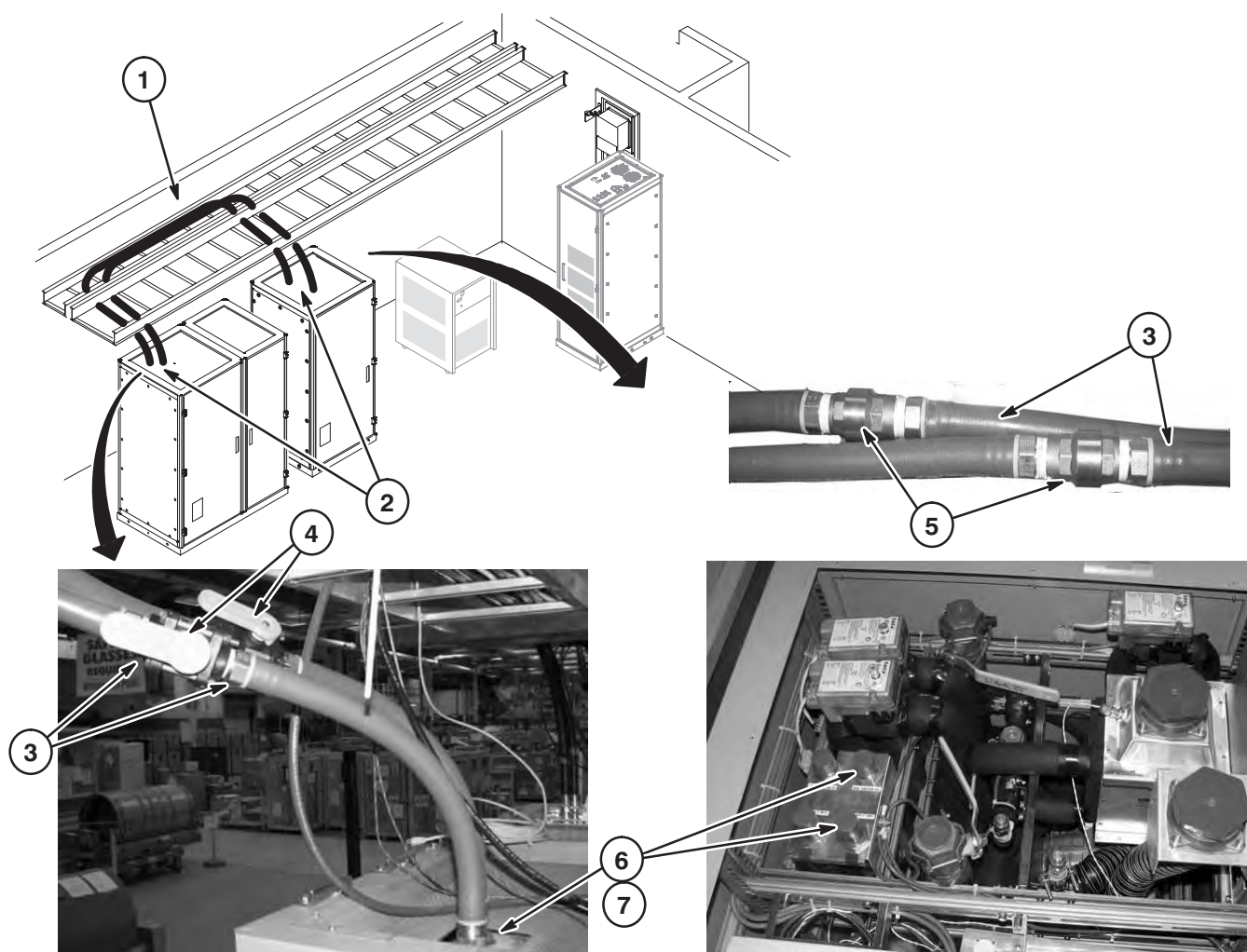
1. Route RED (5264033) and GREEN (5264066) coolant hoses from HEC Cabinet thru the designated overhead cable tray to PGR Cabinet.
2. Cut to length.

NOTE: At each cabinet, ball and check valves are to be installed in each hose at a location of 1 to 3 feet (300 to 900mm) outside the cabinet. **Make sure the distance measured is outside the perimeter of the top of the cabinet.**

3. Cut each hose at the appropriate 1 to 3 foot mark outside the cabinet, as mentioned above.
4. At PGR, install 1" ball valve assembly (5338544).
5. At HEC cabinet, install 1" check valve assembly (5338545).
6. Insert 1" Flare to Push-Lok connector (5263572) at ends of each hose. **DO NOT** use teflon tape on threads.

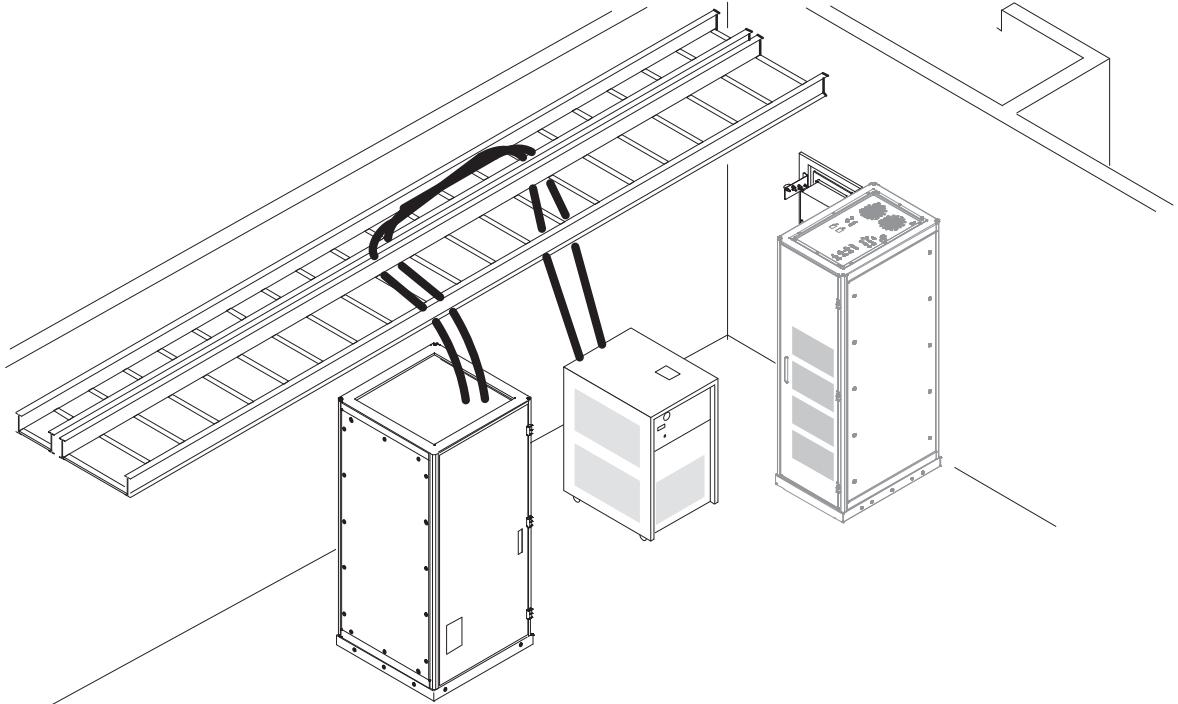
NOTE: To prevent cracking of fittings when attaching flare connectors to cabinet connection points, hold fittings in the HEC stationary with wrench while flare connections are being made.

7. Connect RED and GREEN hoses to top of PGR and HEC cabinets as marked.



4-21-2 ROUTE AND CONNECT COOLANT HOSES BETWEEN CRYOGEN COMPRESSOR and HEC

The installation procedures for installing the hoses, barbs, and the insulaton sleeves between the Cryogen Compressor and the HEC was part of Direction 5500097, *Pre-Magnet Delivery System Installation* manual. Refer to that manual for instructions, if needed.

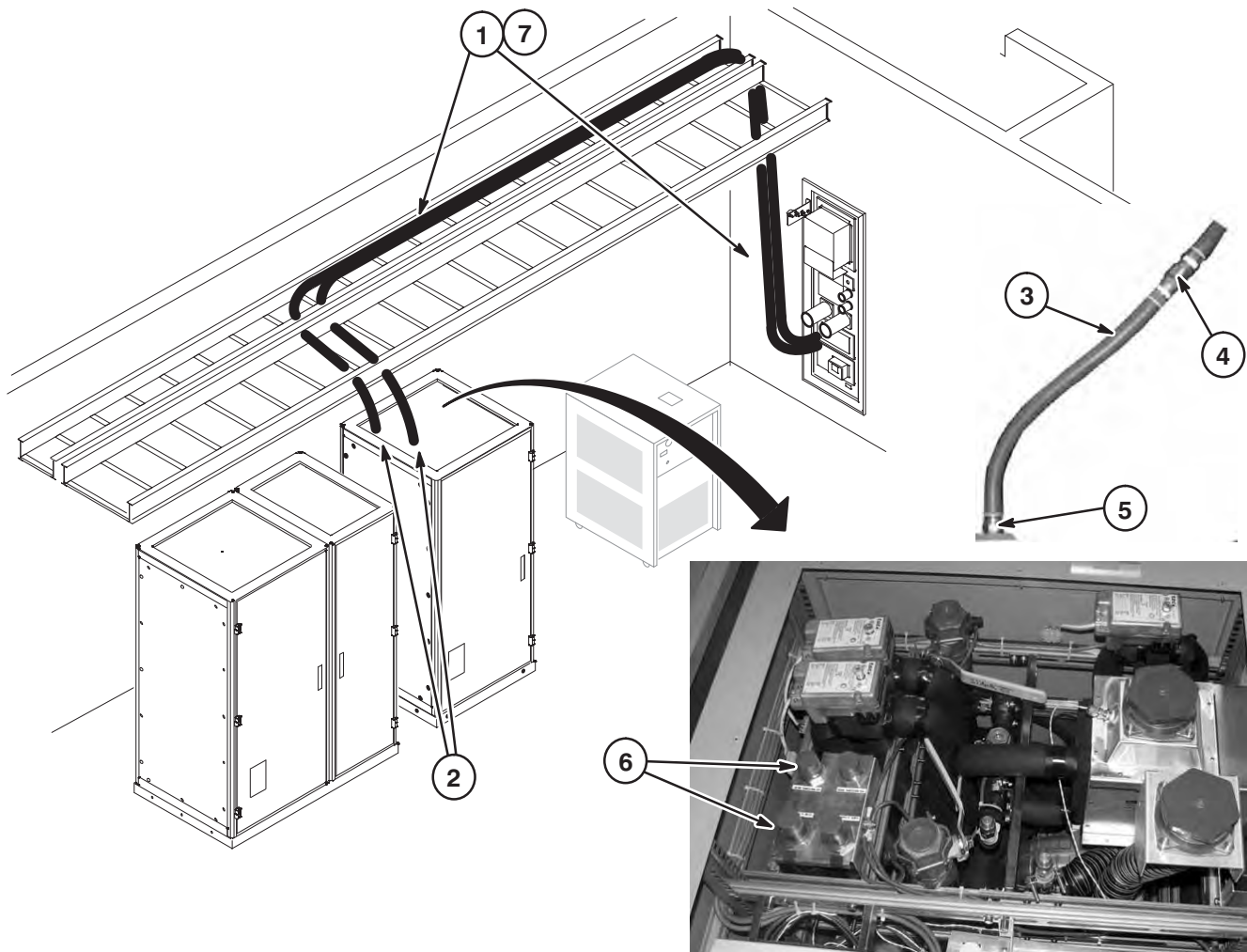


4-21-3 ROUTE AND CONNECT GRADIENT COIL WATER COOLANT HOSES IN EQUIPMENT ROOM

1. Route BLUE and BLACK coolant hoses from HEC Cabinet the designated overhead cable tray to wave guides J100 and J101 in SPW. Route thru wave guides to rear of magnet. Refer to Section 4-21-4 for routing and connection of Gradient Coil cooling water hoses in magnet room.
2. Cut coolant hoses to length at HEC Cabinet.
3. At the HEC cabinet, for both the BLUE and BLACK hoses (only blue hose shown below), check valves are to be installed in each hose at a location of 1 to 3 feet (300 to 900mm) outside the cabinet. **Make sure the distance measured is outside the perimeter of the top of the cabinet.** Cut each hose.

NOTE: To prevent cracking of fittings when attaching flare connectors to cabinet connection points, hold fittings in the HEC stationary with wrench while flare connections are being made.

4. Install Check Valve assembly (5338545) for each hose between cut ends. Make sure valve is installed in the correct direction.
5. At the end of each hose, install a 1 inch 37 degree Female Push-Lok Swivel fitting (5263572). **DO NOT** use teflon tape on threads.
6. Connect each hose to designated RETURN inlet (BLACK) and SUPPLY outlet (BLUE) at top of HEC Cabinet.
7. Install insulation sleeves (5225968) around 1 inch water coolant hoses in both the Magnet and Equipment rooms. Using black duct tape, tape the seams between sections of insulation to prevent gaps.

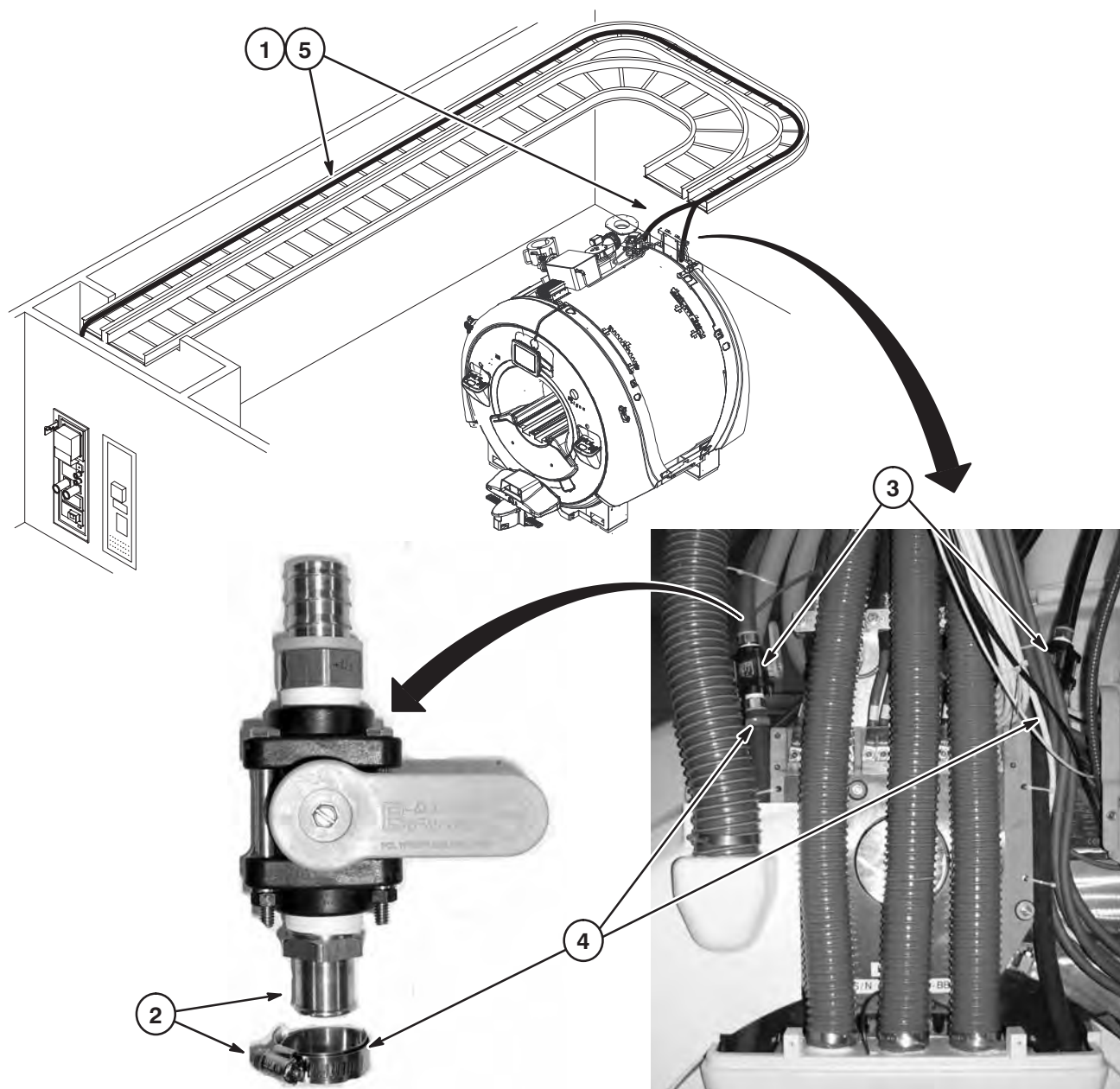


4-21-4 ROUTE AND CONNECT GRADIENT COIL WATER COOLANT HOSES IN MAGNET ROOM

One Inch Ball Valve assemblies (5338543) will be installed in each hose.

1. Hoses were routed in Section 4-21-3 from HEC thru SPW to rear of magnet and cut to length.
2. **NOTE:** It is important that the ball valve assembly with beaded hose barb and clamp (5338543) be inserted and clamped to hose routed from the gradient coil. Orientation shown below.
3. Attach each ball valve and fitting assembly to BLACK and BLUE hoses as shown.
4. Install hose clamp on gradient coil hose below the ball valve on the BLACK and BLUE hoses.
5. Install insulation sleeves (5225968) around 1 inch water coolant hoses in both the Magnet and Equipment rooms. Using black duct tape, tape the seams between sections of insulation to prevent gaps.

NOTE: When handle of valve is perpendicular to hose, ball valve is closed.

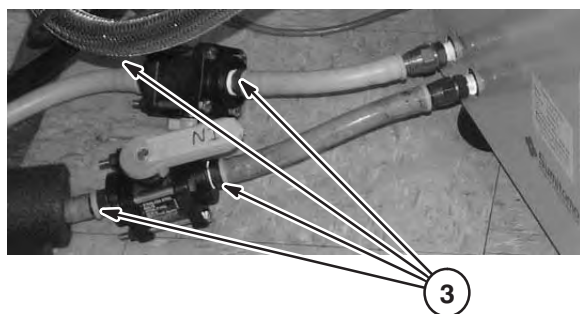
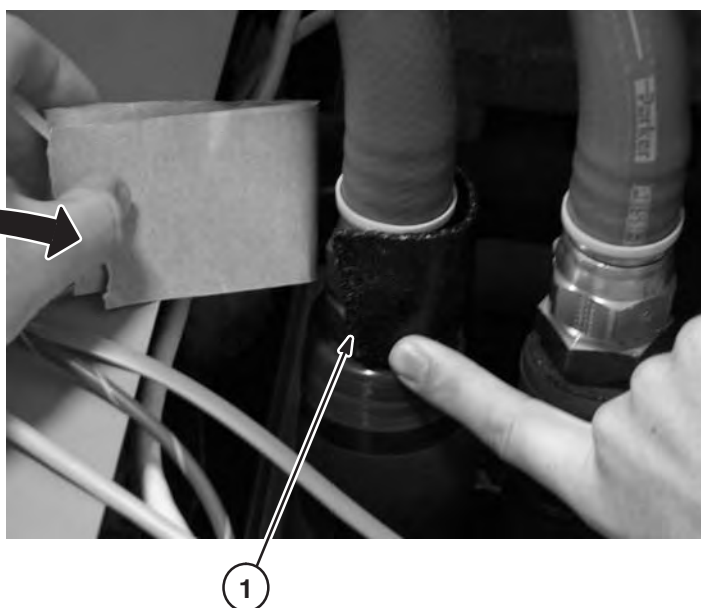
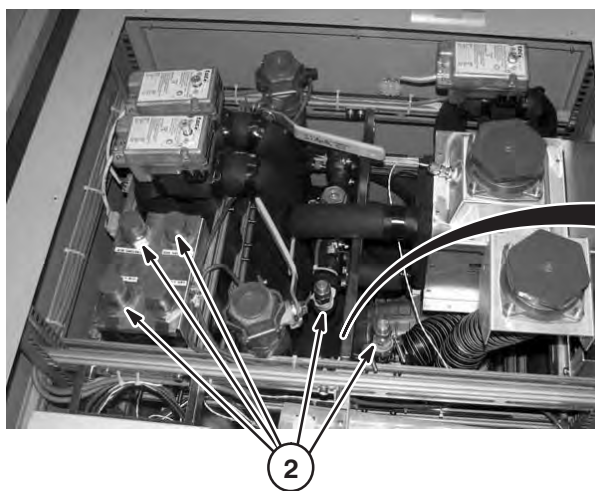


4-21-5 APPLY INSULATION ON FITTINGS

IMPORTANT! Do not apply the Cork/Rubber Insulation Wrap until after all leak checks have been completed by FE.

The Cork/Rubber Insulation Wrap (5268212) must be applied to all gradient coil and cryogen compressor hose connection points.

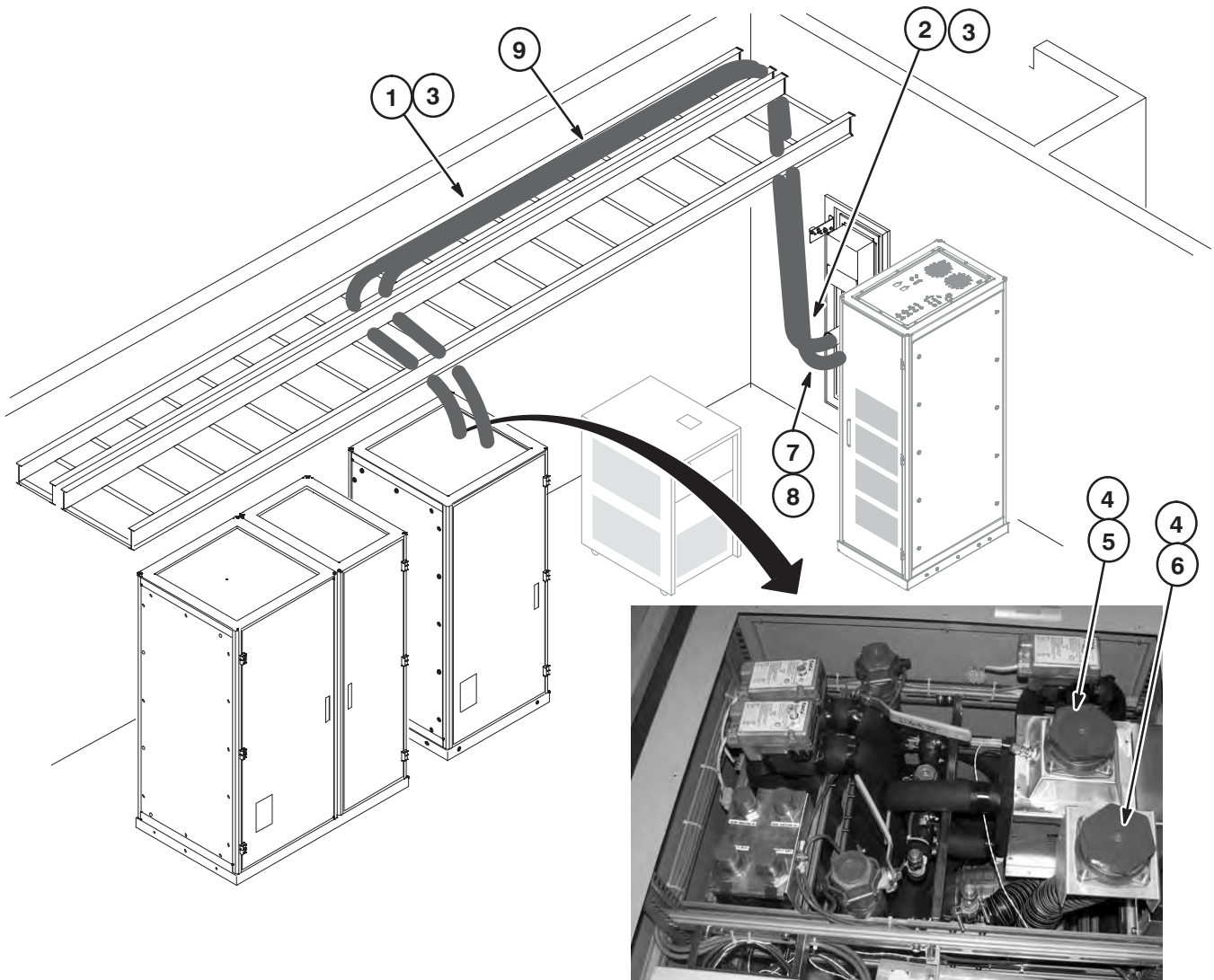
1. Apply and secure a length of Insulation Wrap to cover the BLUE hose connection.
2. Repeat process in Step 1 for the GREEN, BLACK, RED, GREY, and YELLOW hoses at the top of the HEC cabinet.
3. Repeat process in Step 1 for the GREY and YELLOW hoses at the Cryogen Compressor.
4. Repeat process in Step 1 for the BLUE and BLACK hoses at the rear of the magnet connecting to the Gradient Coil.



4-21-6 ROUTE AND CONNECT AIR COOLANT HOSES BETWEEN HEC and SPW

Note: Trim extra length off of hose clamps after applying clamp to the 3 and 4 inch hoses. Smooth rough edges.

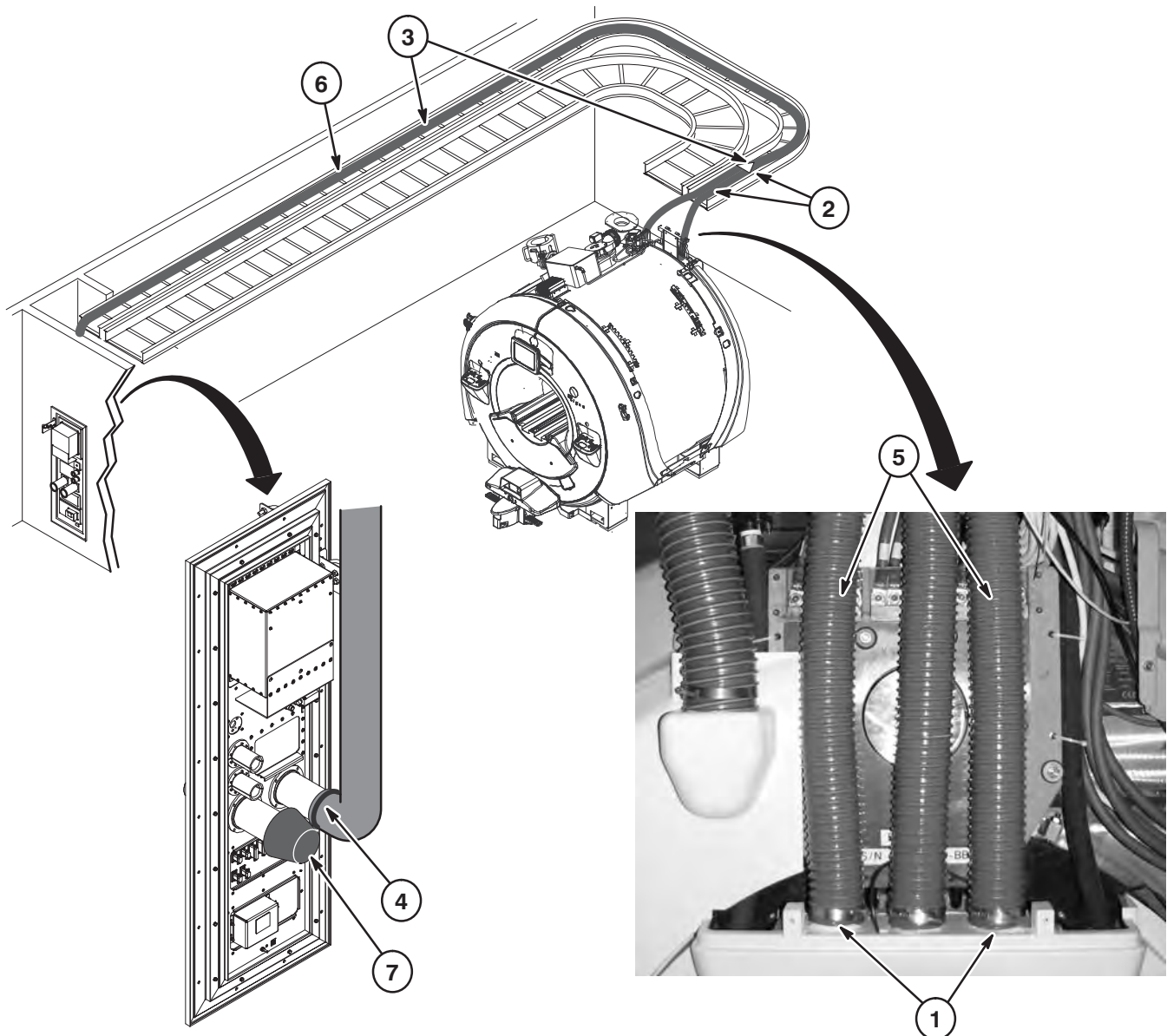
1. Route 4 inch SUPPLY air hose (46-282666Pxx) in the designated overhead cable tray from SPW to the HEC.
2. Make sure there is enough length for insulation that will be applied around the hose in Step 9. Cut to length.
3. Repeat steps 1 and 2 for routing and cutting 4 inch RETURN air hose.
4. Remove and discard cap and install 3inch PVC Socket Connection (5267835).
5. Connect 4 inch SUPPLY air hose to top of HEC cabinet marked SUPPLY with 4 inch hose clamp (5266749).
6. Connect 4 inch RETURN air hose to top of HEC cabinet marked RETURN with 4 inch hose clamp (5266749).
7. Connect other end of 4 inch SUPPLY air hose to J104 on SPW and secure with 4 inch hose clamp (5266749).
8. Connect other end of 4 inch RETURN air hose to J105 on SPW and secure with 4 inch hose clamp (5266749).
9. Attach insulation sleeves (5225963) to SUPPLY 4 inch hose. Using black duct tape, tape the seams between sections of insulation to prevent gaps. Insulation sleeve on RETURN hose is optional.



4-21-7 ROUTE AND CONNECT AIR COOLANT HOSES BETWEEN SPW and BODY COIL

Note: Trim extra length off of hose clamps after applying clamp to the 3 and 4 inch hoses. Smooth rough edges.

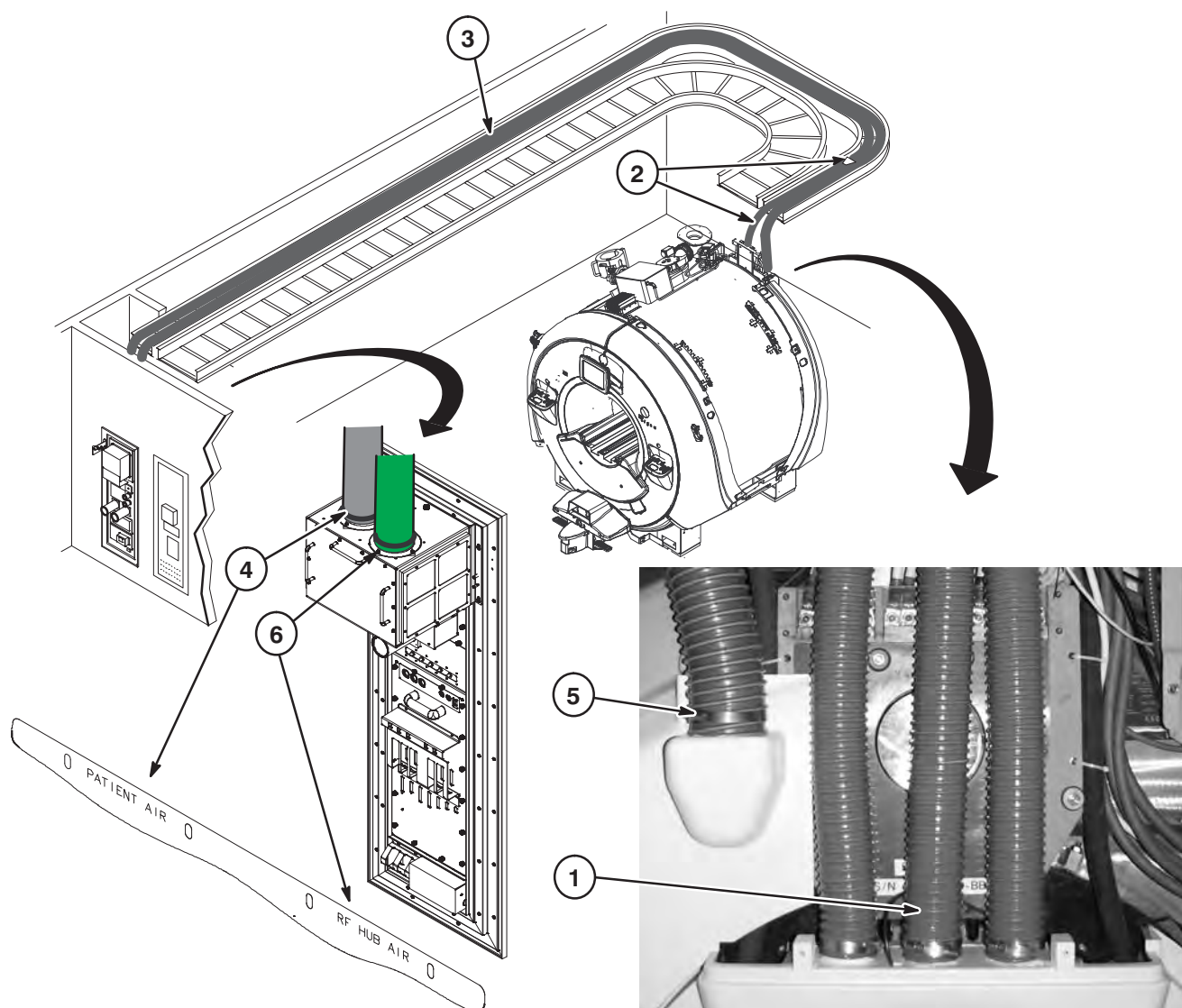
1. At rear of magnet, install two 3 inch, 3 meter long, Air Hoses (5213158) on the two outer sockets. Secure with 3 inch hose clamp (5265190) on each hose.
2. Route entire length of each air hose into overhead cable trays and attach to Air Y-Fitting (5196137) with 3 inch hose clamp (5265190) on each hose.
3. Route 4 inch Air Hose (46-282666Pxx) in overhead cable tray and attach to Air Y-Fitting with 4 inch hose clamp (5266749).
4. At J104 waveguide on SPW, cut air hose to length and attach to waveguide with 4 inch hose clamp (5266749).
5. Install insulation sleeves (5305915) around 3 inch air hoses.
6. Install insulation sleeves (5225963) around 4 inch air hose.
7. Air Filter (5290917) should have been installed with SPW and attached with Hose Clamp (5263772-3).



4-21-8 ROUTE AND CONNECT AIR HOSES BETWEEN BLOWER BOX and MAGNET

Note: Trim extra length off of hose clamps after applying clamp to the 3 and 4 inch hoses. Smooth rough edges.

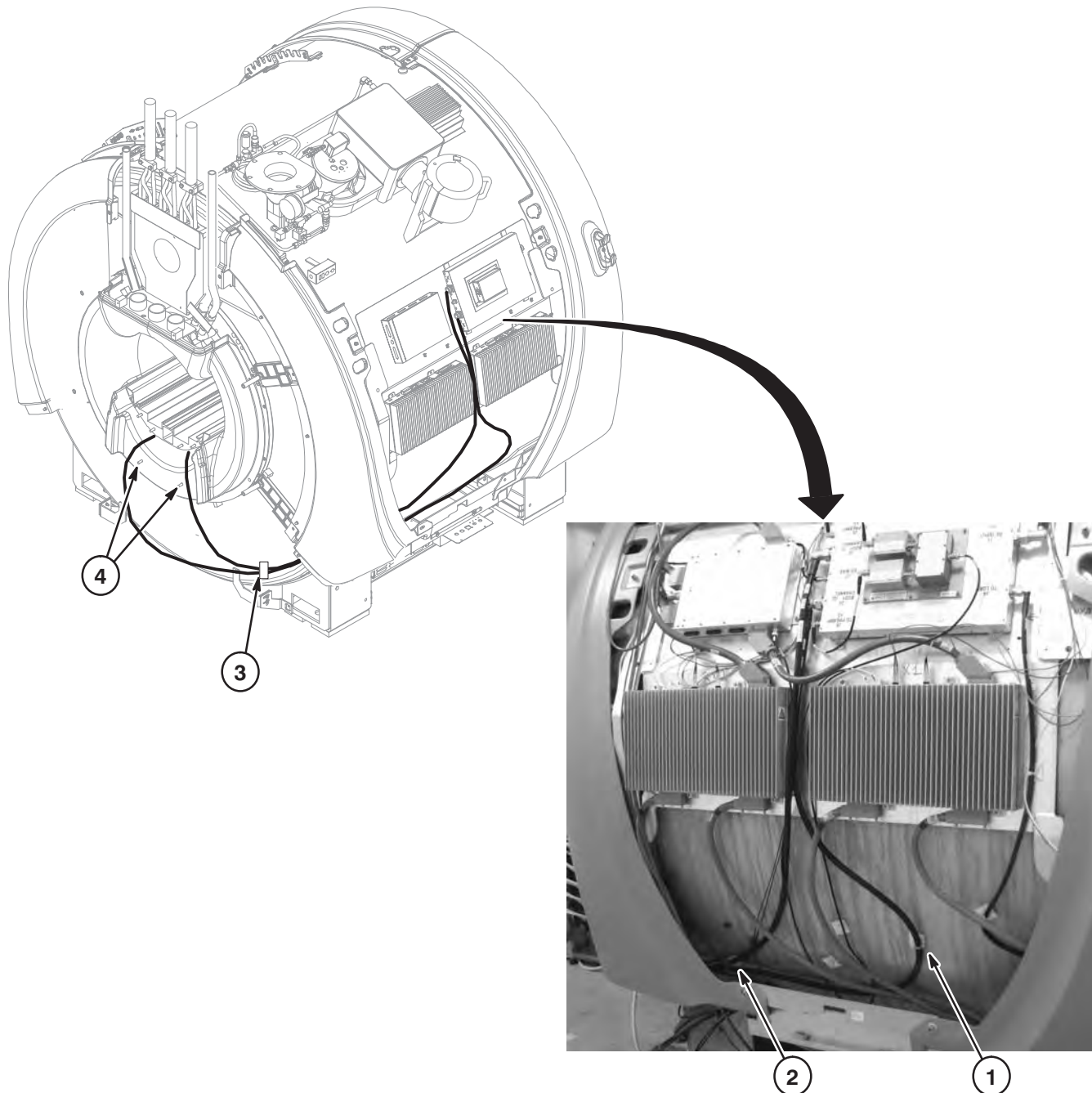
1. At rear of magnet, install 3 inch, 3 meter long, PATIENT AIR Hose (5213158) on the center socket. Secure with 3 inch hose clamp (5265190).
2. Route entire length of 3 inch air hose into overhead cable trays and attach to PVC Reducer (5198521) with 3 inch hose clamp (5265190).
3. Route 4 inch, 11 meter long, Air Hose (5199506-2) in overhead cable tray and attach to PVC Reducer with 4 inch hose clamp (5266749).
4. At the Blower Box, cut patient comfort air hose to length and attach to the left side flange labeled PATIENT AIR with 4 inch hose clamp (5266749).
5. Route 4 inch, 14 meter long, RF HUB AIR Hose (5199506-3) in overhead cable tray and attach to socket on rear cover with 4 inch hose clamp (5266749).
6. At the Blower Box, cut rear pedestal air hose to length and attach to the right side flange labeled RF HUB AIR with 4 inch hose clamp (5266749).



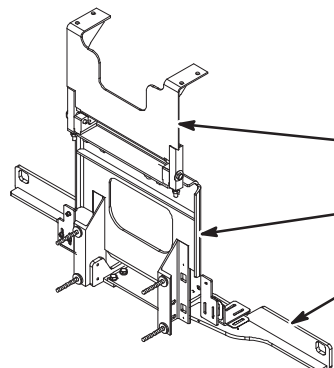
4-22 SECURE ROUTING OF I AND Q CABLES

Proper routing of the “I” and “Q” cables on side of magnet will aid in spacing and routing all cables when Rear pedestal is connected to magnet enclosure.

1. Route “I” cable, Run M1322, and secure to magnet with cable-tie at location indicated.
2. Route “Q” cable, Run M1323, and secure to magnet with cable-tie at location indicated.
3. Tie-wrap cables together at this location.
4. Make sure cables are routed to the outside of the Rear Pedestal mounting studs on the magnet enclosure.



4-23 BRIDGE & REAR PEDESTAL INSTALLATION



The Bridge Front Support Bracket installed in this section is an assembly of three brackets:

- Bridge Support Upper Asm (5335143)
- Bridge Support Lower Asm (5334546)
- Bridge Support Magnet Mount Asm (5337932)

The installation of this Bridge Support Assembly eliminates the requirement of the bracket that connects the Rear Pedestal to the Rear End Bell.

The Rear Pedestal has been delivered to site with instructions on how to unload it from the shipping pallet. After unloading the Rear Pedestal, move it to the rear of the magnet.

WARNING!

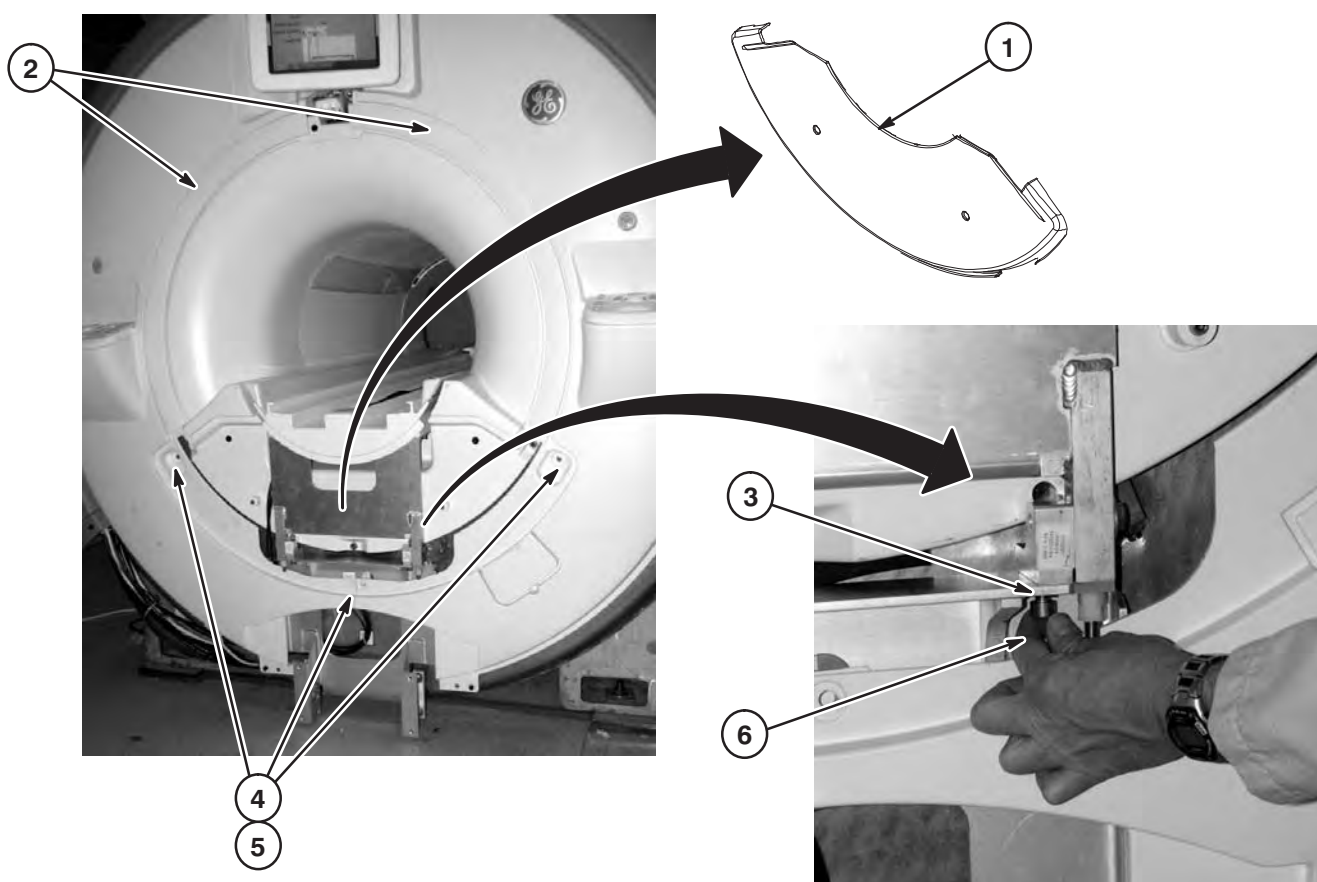
FERROUS MATERIAL HAZARD! If magnet is ramped, the wheel assemblies attached to the Rear Pedestal for shipping and delivery MUST be removed before moving Rear Pedestal into Magnet Room.

CAUTION

The weight of the Rear Pedestal requires at least **THREE** people to move Rear Pedestal into position at rear of magnet.

4-23-1 PREPARATION FOR ATTACHING LOWER BRIDGE SUPPORT (If not already installed)

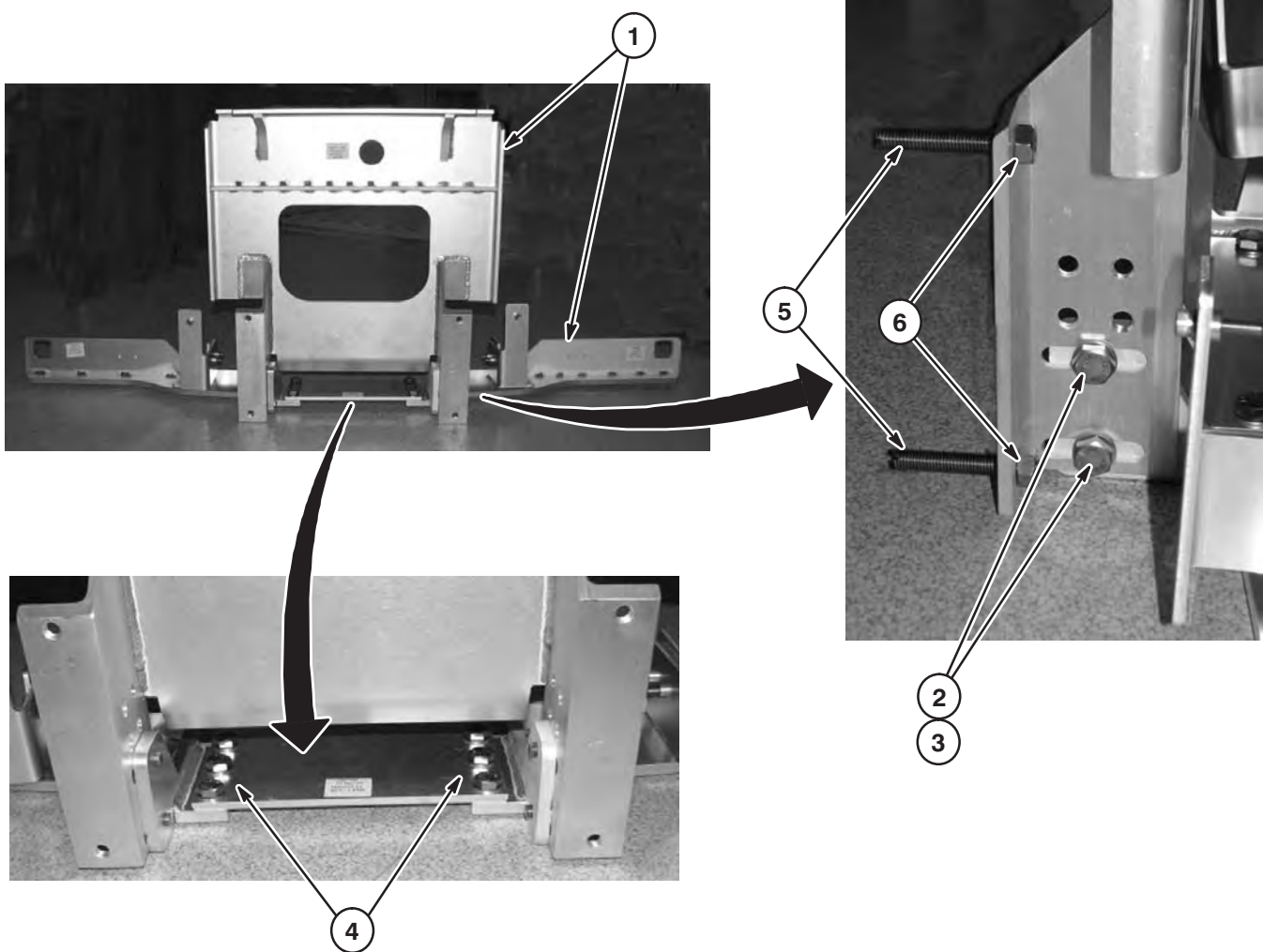
1. Remove Front Bridge Cover.
2. Remove front Enclosure Rings.
3. Remove (4) M10 x 25mm Socket Head Cap Screws and Washers that connect the Bridge Support Upper Asm to the Bridge Support Lower Asm. Two on each side.
4. System with In-Room Display – Remove three M8 screws that connect the bottom portion of the left and right Front Enclosure covers to the magnet.
5. System without In-Room Display (not shown below) – Remove the bottom two M8 screws on front cover. Next, remove the left and right control panels. Remove the M8 screws under each control panel.
6. Move Bridge Support Lower Asm away from magnet. This is required to complete next procedure of attaching Bridge Support Magnet Mount.



4-23-2 ATTACH BRIDGE SUPPORT MAGNET MOUNT (If not already installed)

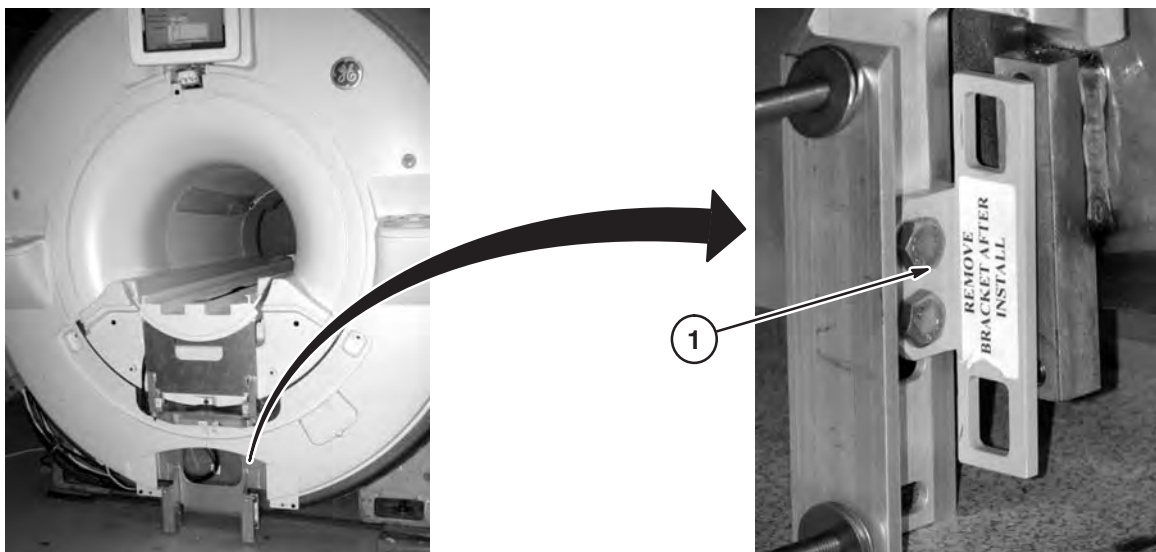
NOTE: Some international shipments may not ship with the Bridge Support Magnet Mount bracket pre-installed. If installed, proceed to Section 4-23-3.

1. Position the Bridge Support Lower in between the C-channels of the Bridge Support Magnet Mount Asm (5337932).
2. Attach Bridge Support Lower to the Bridge Support Magnet Mount Asm with four (4) M10 x 20mm Hex Head Screws (2109866-19), four (4) M10 Washers (2109878-4), and Loctite 242 (46-170686P1).
3. Tighten the four screws after centering the screws in the C-channel slotted holes.
4. **NOTE:** If the two C-channels of the Support Lower do not fit properly between the Bridge Support Magnet Mount Asm, loosen the six (6) M10 screws and adjust the distance between the interface surfaces of the Bridge Support Magnet Mount Asm until the C-channels firmly touch.
5. Apply Loctite 271 (46-170684P1) to the four (4) M10 x 65mm Studs (2116325-2) and thread into the two C-channels on the Support Lower.
6. Lock each of the four studs in place with an M10 Nut (2109875-4).

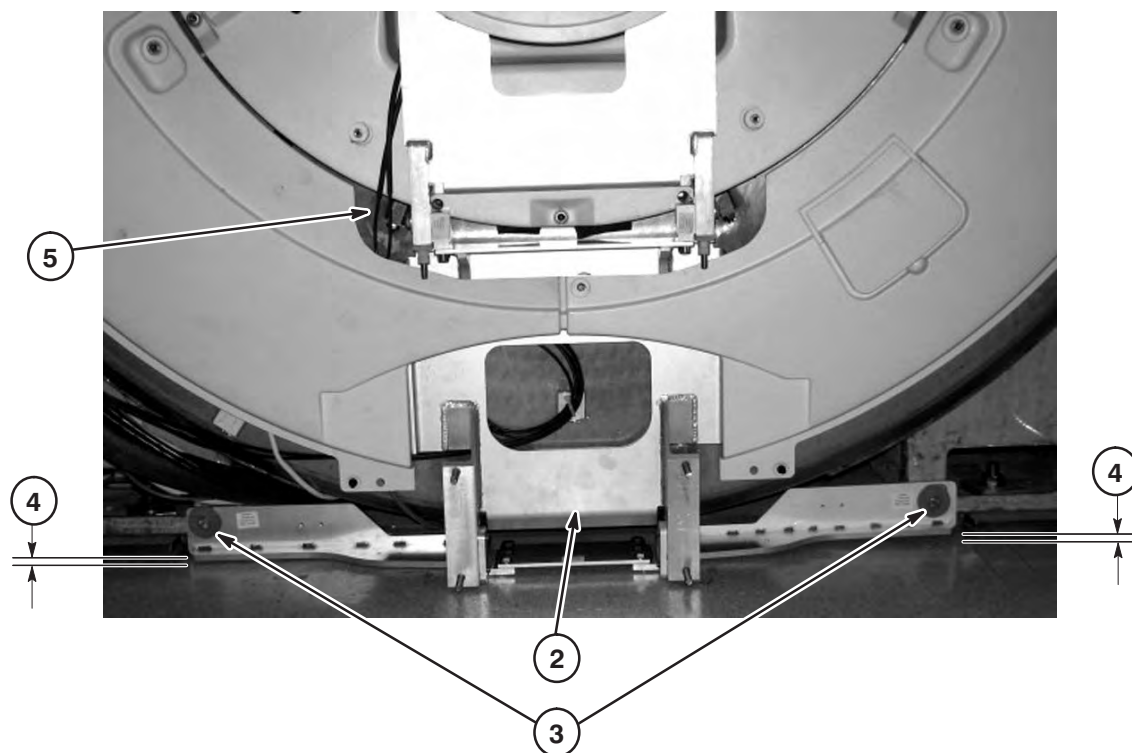


4-23-2-A ATTACH FRONT BRIDGE SUPPORT BRACKET TO MAGNET

1. If still in place, remove yellow shipping brackets from front of magnet.

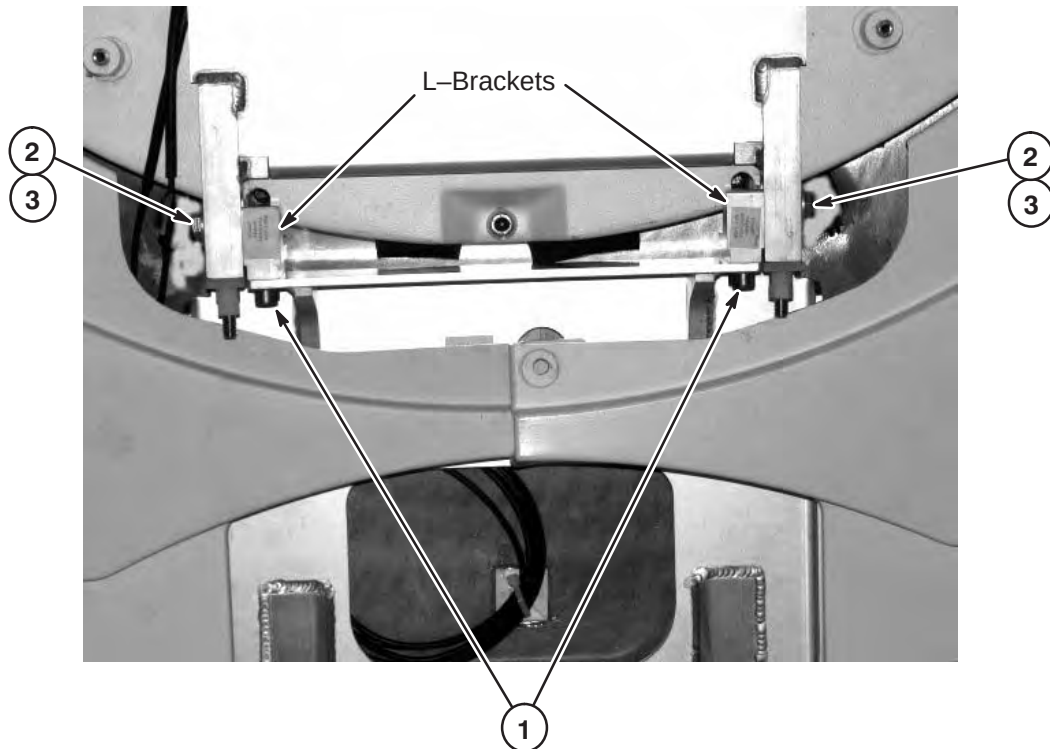


2. Slide the Bridge Support Bracket Asm from the side between the magnet and the front covers.
3. Mount bottom of bracket to left and right magnet feet using two (2) Phenolic Washers (5146165) and two (2) M10 x 25mm Screws (46-318508P20), one of each on each side. DO NOT TIGHTEN HARDWARE.
4. **IMPORTANT:** Make sure that the lower Bridge Support Magnet Mount Assembly bottom surface, only by the magnet feet, does not come in contact with the magnet room floor (recommend a 2mm minimum clearance).
5. **NOTE:** Front RF body coil connections should be routed to the left side of new bridge support bracket.



4-23-2-B ATTACH BRIDGE UPPER AND LOWER SECTIONS

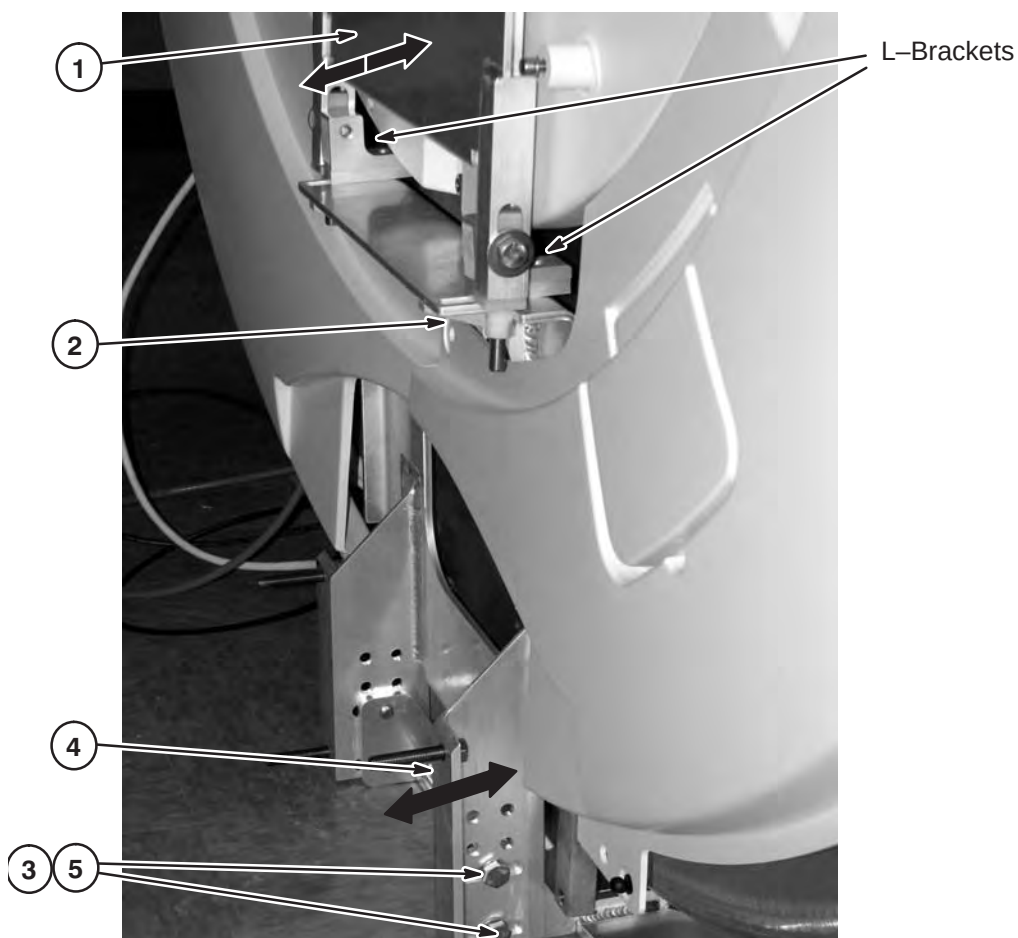
1. Re-attach Bridge Support Upper Asm to Lower Asm with the previously removed four (4) M10 x 25mm Socket Head Cap Screws (2109867-23), four (4) M10 Washers (2109878-4), and Loctite 242 (46-170686P1). Leave fastening hardware loose.
2. Install the two (2) M10 x 50mm Hex Head Cap Screws (46-318508P25) that tie the L-Bracket to the square hollow tube supports by approximately one inch (25mm).
3. Add Loctite 242 to the M10 threaded hole in the L-Brackets and screw back in the two M10 x 50mm Hex Head Cap Screws. Leave hardware loose.



4-23-3 Z-DIRECTION BRIDGE ALIGNMENT

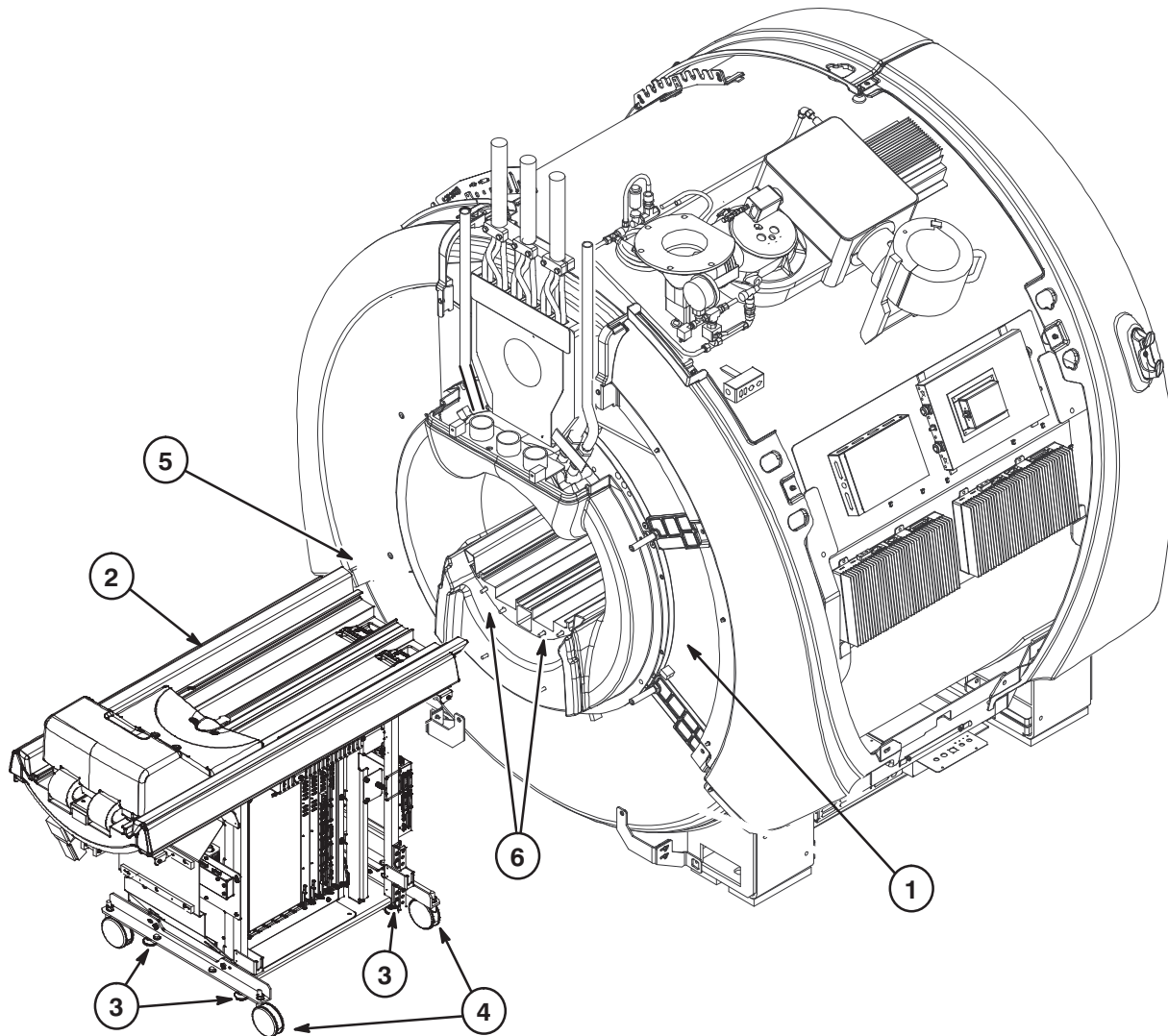
Align the Bridge in the Z-direction (in/out of the magnet bore) until front surface of Bridge aligns with front surface of end bell to within ± 3 mm.

1. Alignment to front end bell is done by pushing or pulling Bridge with Support Upper attached which allows the L-Brackets to slide toward or away from the magnet.
2. When aligned, tighten the four M10 screws attaching L-Bracket to the Support Lower.
3. **NOTE:** If more adjustment is needed for the Z-axis adjustment, loosen the four bolts (two on each side) holding the Support Lower to the Bridge Support Magnet Mount Asm.
4. Move the Support Lower until the ± 3 mm bridge-to-end bell alignment has been achieved.
5. Re-tighten bolts.



4-23-4 POSITION REAR PEDESTAL AT REAR OF MAGNET

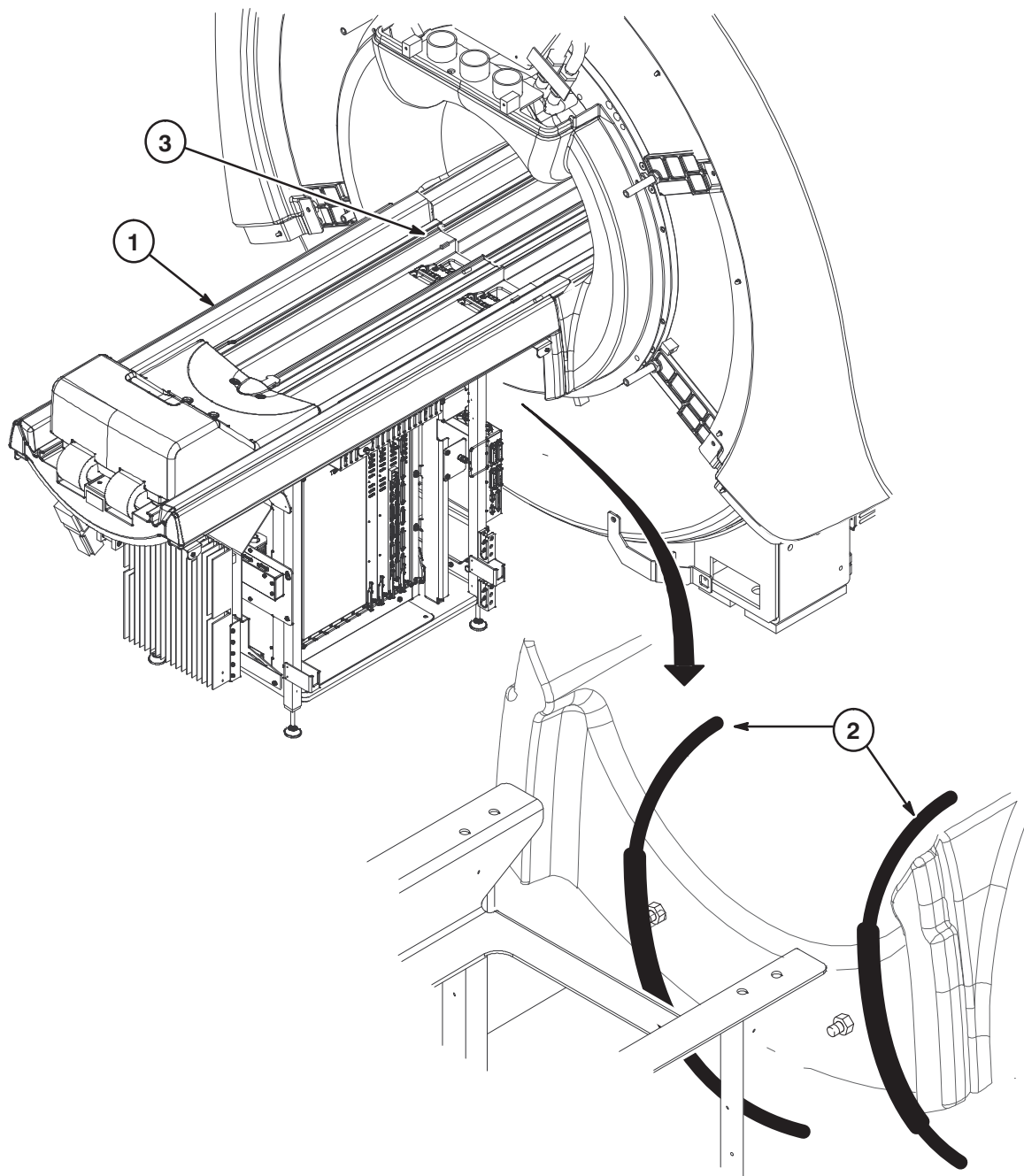
1. Remove Left Rear Cover Assembly for access to cable routing. (cover already removed in illustration)
2. Position Rear Pedestal near rear of magnet.
3. Lower the four leveling feet until wheels of front and rear shipping brackets are not touching the floor.
4. Remove front and rear wheeled shipping brackets.
5. Move Rear Pedestal closer to rear of magnet and adjust leveling feet to align top of rear split bridge with top of front split bridge in magnet bore.
6. Remove four M10 nuts from studs on Front Split Bridge.



4-23-5 ATTACH REAR PEDESTAL TO BRIDGE

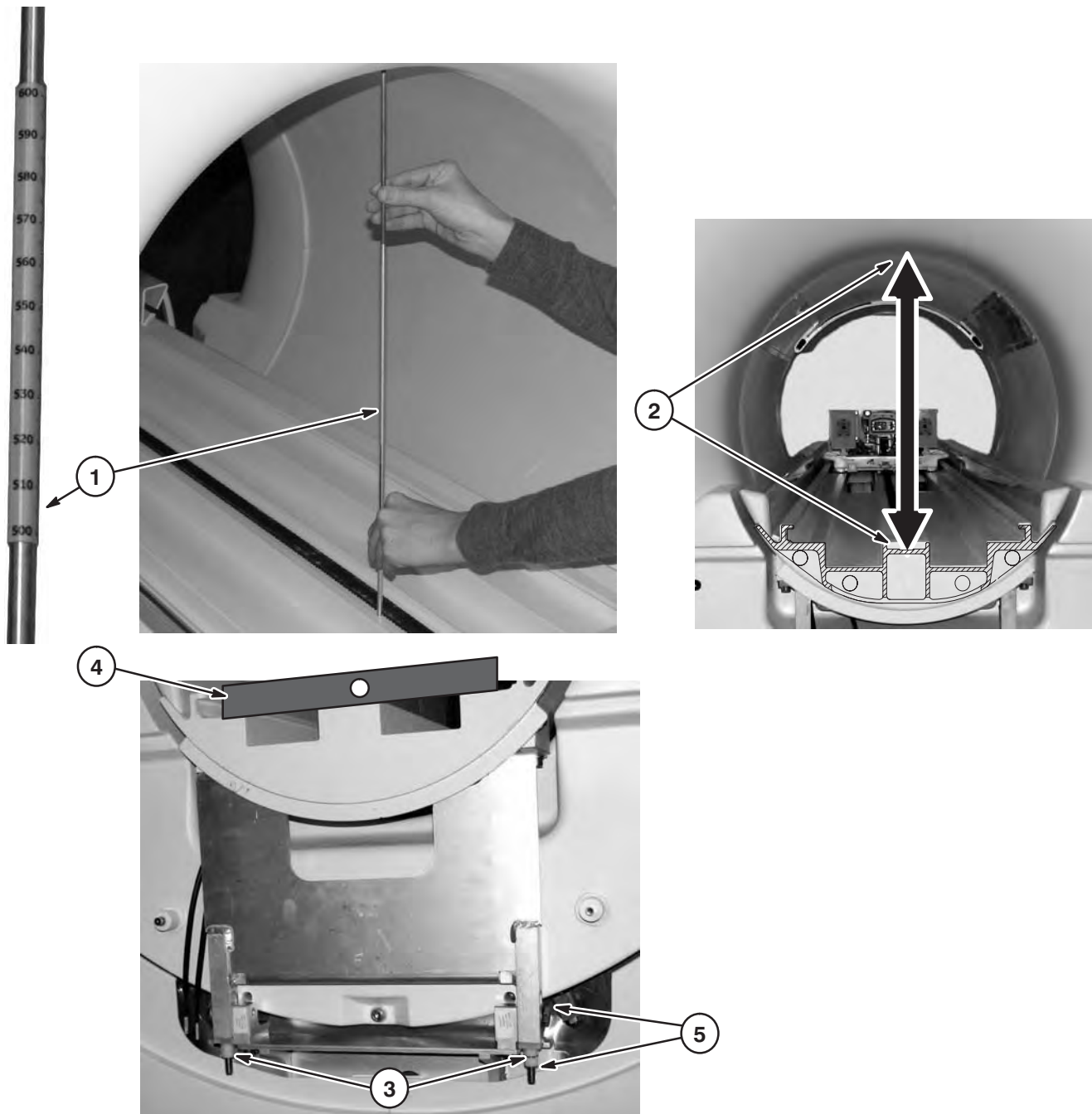
1. Move Rear Pedestal forward and adjust height to align with four studs on Front Split Bridge.
2. Make sure the I and Q cables are routed on the outside.
3. Move Rear Pedestal forward and engage four studs on Front Split Bridge. Secure with four nuts previously removed.

NOTE: The connection braces between the Rear Pedestal and the Rear End Bell are not installed. The Bridge Front Support Bracket eliminates the need for these braces to connect the Rear Pedestal to the Rear End Bell.



4-23-6 Y-DIRECTION BRIDGE ALIGNMENT (Front of Magnet)

1. Align the Bridge in the Y-direction (up/down in the magnet bore) to within 502mm – 505mm using the bridge height gauge (delivered to site packaged with the Dock assembly).
2. Make sure the distance measured is between the top of the RF body coil and the bridge surface under the belt.
3. Adjust the M10 Nylon nuts at the bottom of the Bridge Support Upper bracket to raise and lower Bridge to attain the 502mm – 505mm distance in between the top of the patient bore and the surface under the belt on the bridge.
4. A level can be used to make sure bridge is level in the X-direction. Adjust M10 Nylon nuts as necessary.
5. Tighten the two (2) M10 bolts and the M10 Nylon Hex Nuts on each side.

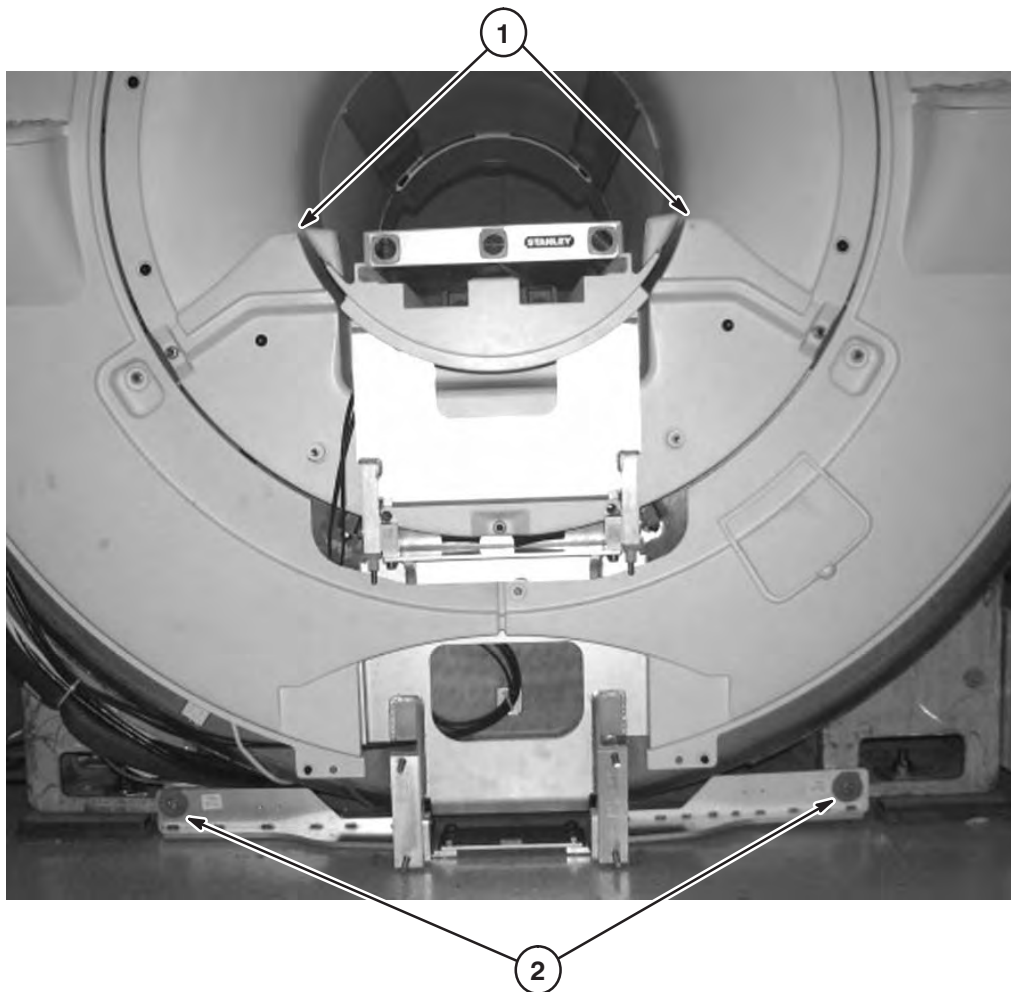


4-23-7 X-DIRECTION BRIDGE ALIGNMENT

1. Align the Bridge in the X-direction (left/right in the magnet bore) by shifting entire front Bridge Support Assembly until the distance between Bridge and **RF Body Coil** is equal on both sides of the Bridge.

IMPORTANT: Bridge must not be in contact with the RF Body Coil along its entire length.

2. Tighten the two (2) M10 bolts that fasten the Bridge Support Assembly to the magnet feet.
3. Re-attach front bridge cover.



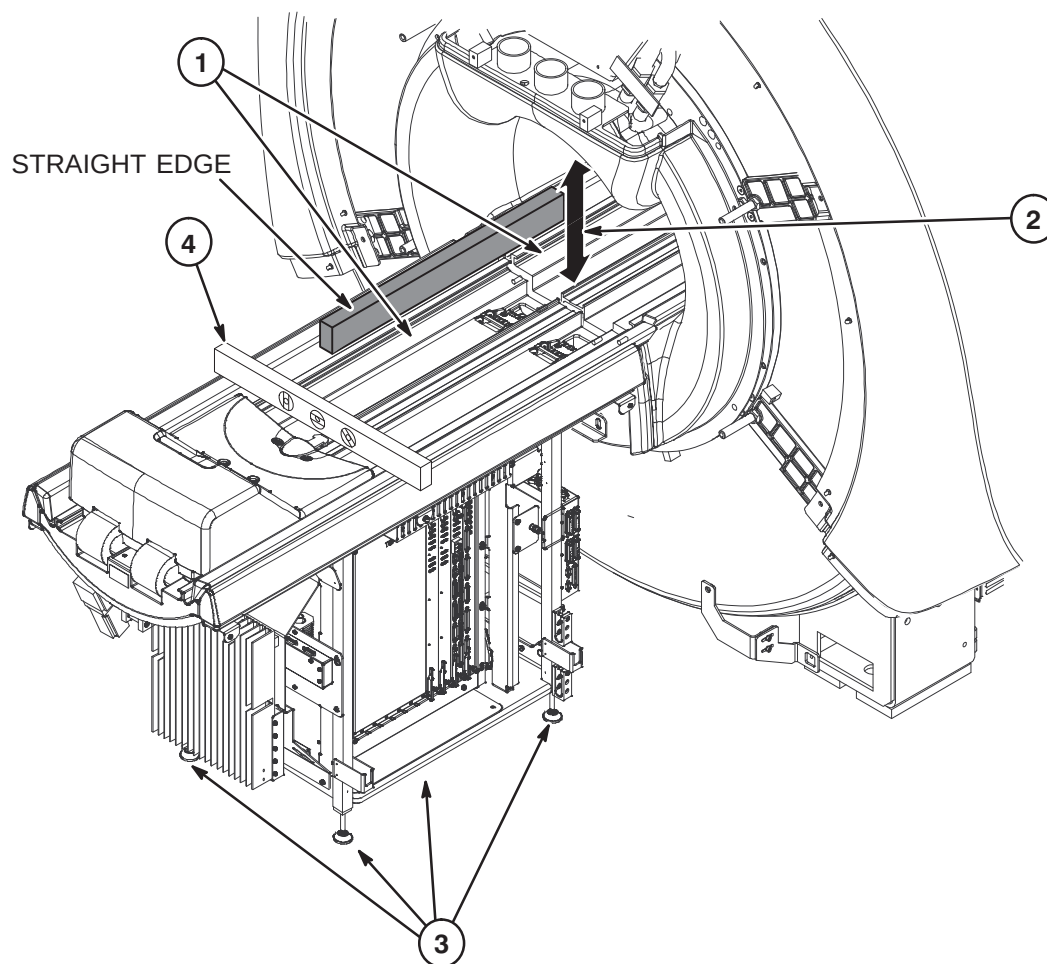
4-23-8 REAR SPLIT BRIDGE HEIGHT ADJUSTMENT (Rear of Magnet)

Refer to Section 4-23-6 for distance between bridge and top of the RF Coil.

Note: The Bridge does not need to be “level” to Earth in the Z-direction in achieving the alignment of bridge surfaces.

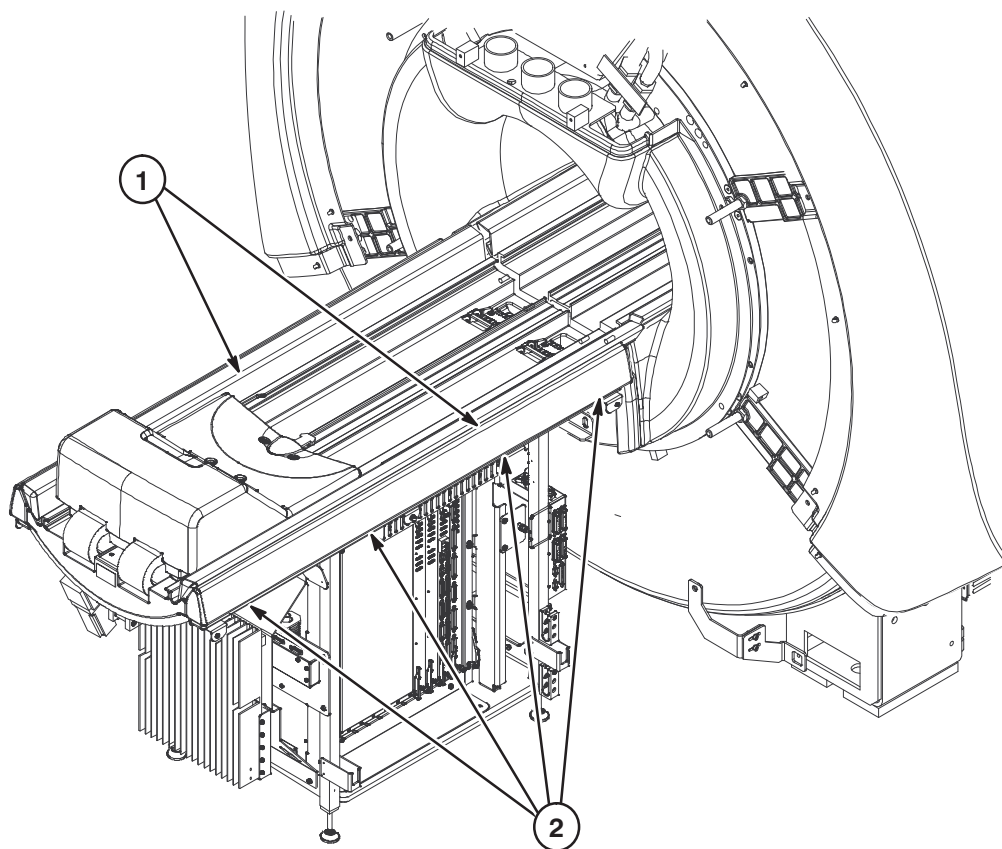
Therefore, you do **NOT** need a level at this point of the installation, you **DO NEED** a straight edge. (Though you can use a level as a straight edge)

1. Use a Straight Edge to make sure that the front and rear Bridge surfaces, on both sides of Bridge where the LPCA wheels roll, are even. This will provide smooth travel for the Carriage wheels. If uneven, refer to Step 3.
2. Align the bridge in the Y-direction (up/down in the magnet bore) to within 502mm–505mm using the bridge height gauge (delivered to site packaged with the Dock assembly) at the rear of the magnet. Make sure the distance measured is between the top of the RF body coil and the bridge surface under the belt. If adjustment is required, refer to Step 3.
3. To adjust for evenness and/or height, turn the feet of Rear Pedestal to raise and lower Bridge.
4. At this point in adjusting the Rear Pedestal, use a level to make sure the Bridge on the rear pedestal is level in the X-direction. Adjust Rear Pedestal feet as necessary.



4-23-9 FRONT AND REAR SPLIT BRIDGE ALIGNMENT ADJUSTMENT

1. Gap between Rear Split Bridge and left and right Rear Pedestal Bumpers should be uniform along entire length.
2. Adjustment for left and right Rear Pedestal Bumpers is made by loosening four screws underneath each bumper and moving bumper until uniform gap is made.

**4-23-10 BRIDGE HEIGHT ADJUSTMENT**

1. Make sure the distance between the top of the patient bore and the surface under the belt on the bridge is between 502mm to 505mm at both ends of the magnet. Adjust bridge as necessary to meet distance.
2. Measure again after completion of Rear Pedestal installation procedures. Adjust Bridge and Rear Pedestal height adjustments as necessary.

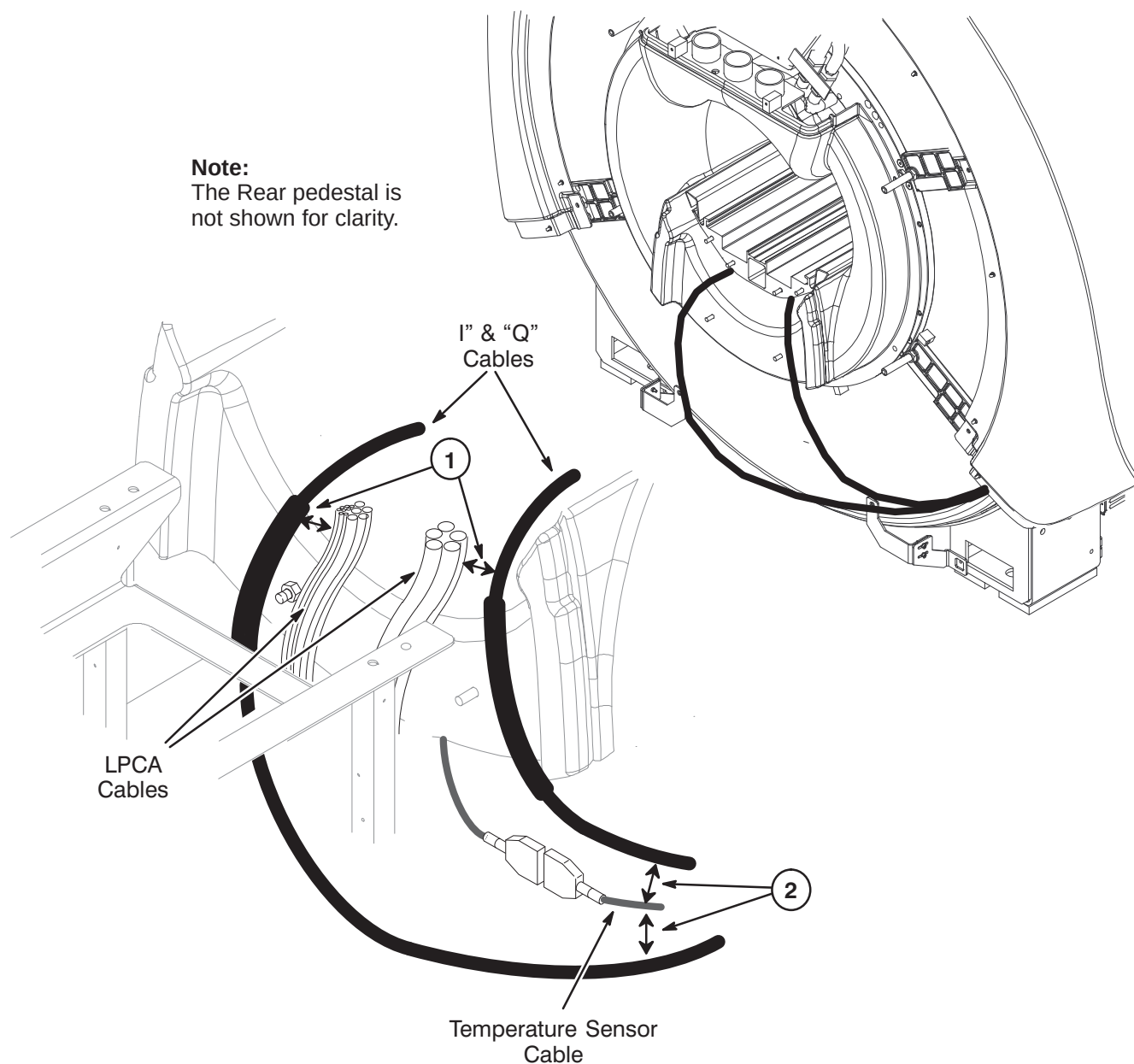
4-23-11 CABLE SEPARATION AT REAR PEDESTAL CONNECTION

For optimal performance of system, the "I" and "Q" cables must be spaced a minimum of 75mm (3 inches) from cables exiting the cable tracks from the LPCA and 30mm (1.2 inches) from the temperature Sensor cable routed from the gradient coil.

1. Make sure the Cable Track cables are separated at least 75mm from the "I" and "Q" cables.
2. Make sure the Temperature Sensor cable is separated at least 30mm from the "I" and "Q" cables.

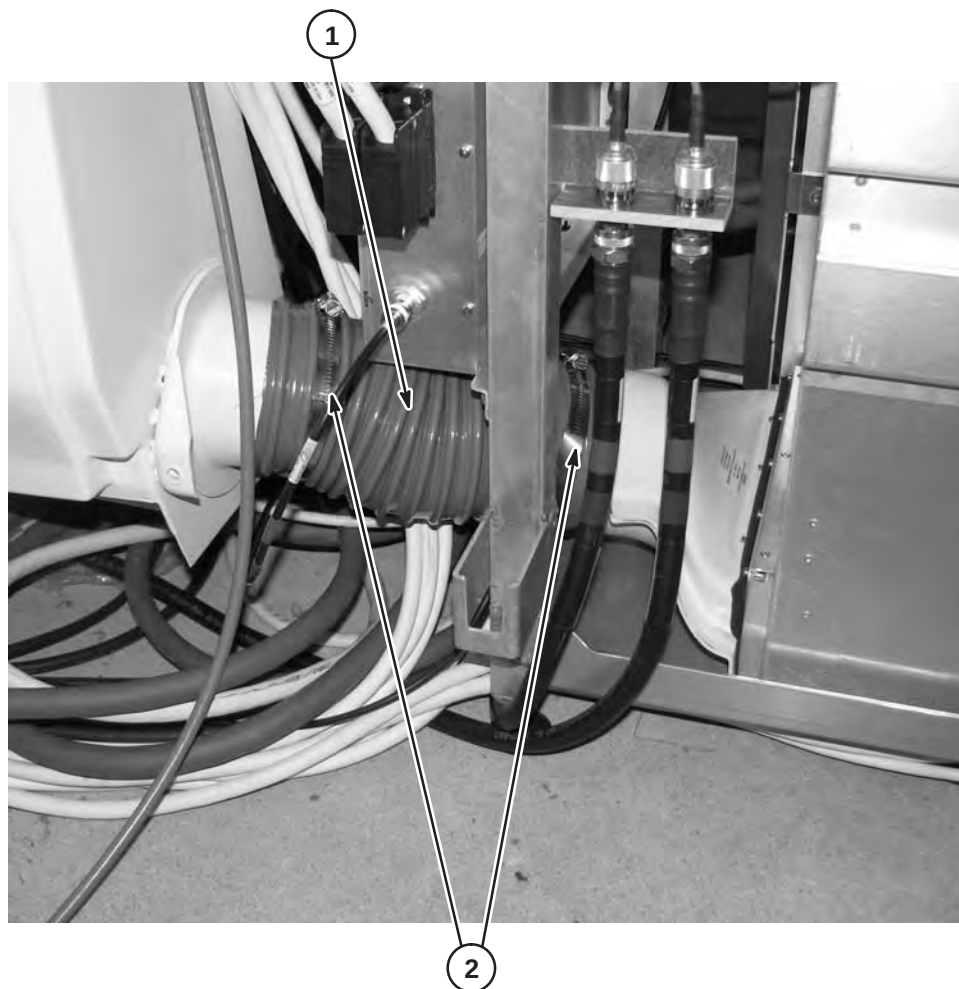
Note:

The Rear pedestal is not shown for clarity.



4-23-12 ATTACH AIR HOSE TO REAR PEDESTAL

1. Attach 4 inch x 254mm long air hose (5199506-4) between magnet enclosure cover and RF Hub duct.
2. Secure in place with 4 inch hose clamp (5266749) at each end.

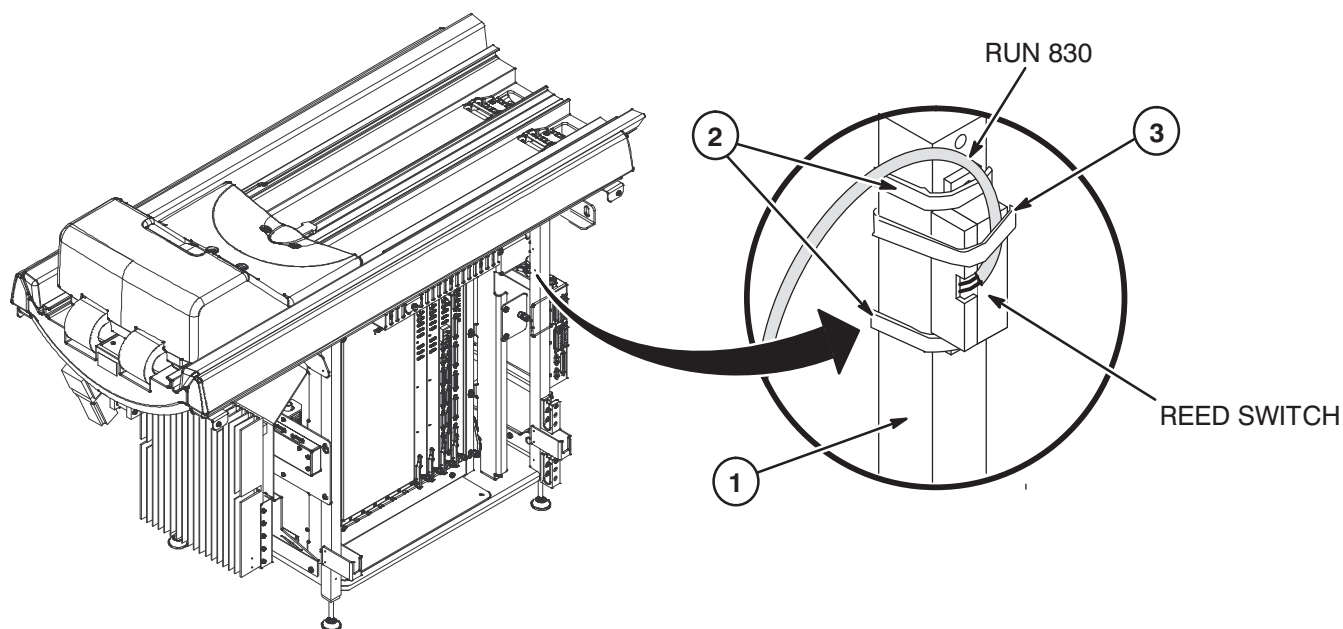


4-23-13 REED SWITCH INSTALLATION (Optional)

Note: The Magmon3 does not use this Reed switch. Installation is optional.

Note: The Reed Switch (2108790) is pre-assembled to Run 830, the Magnet Room Interface Cable (2222797). The redesigned Run 830 will not have the Reed Switch, therefore it will not be installed.

1. At the rear pedestal, test the Reed Switch (2108790) to see where it activates. Usually you can place the switch vertically on one of the Rear Pedestal legs.
2. Wrap two Cable Ties (46-208758P3) around the top and bottom of the Reed Switch and the support strut to hold the Switch in place.
3. Use an additional Cable Tie to secure the cable to the strut to prevent the wires from being pulled out of the Switch. Move the ends of the Cable Ties to the back of the strut.



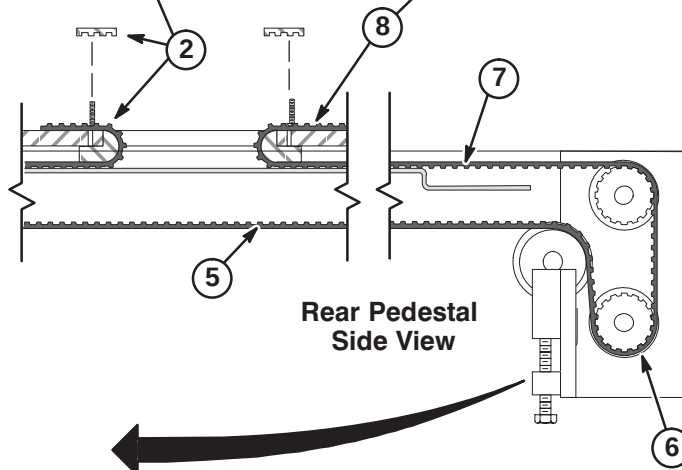
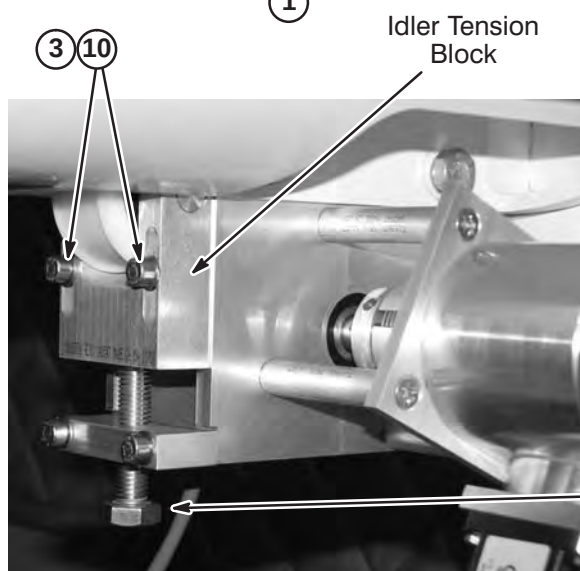
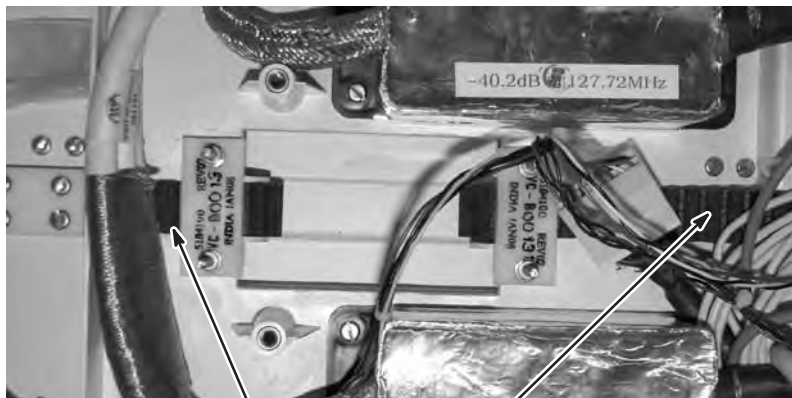
4-24 INSTALL LPCA

4-24-1 ROUTE AND ATTACH DRIVE BELT

1. Unfasten four screws from top of LPCA and remove top cover.
2. Route one end of Drive Belt, (7080mm long) delivered with Front Split Bridge, thru opening in LPCA Base Plate and attach with clamp. Make sure that 3 teeth are exposed after clamp is attached.
3. Under bridge, loosen the two socket head screws on Idler Tension Block.
4. Lower the Idler Tension Block with pulley to its lowest position by unscrewing hex head bolt.
5. Route other end of drive belt from front of Bridge towards Tension Pulley.
6. Feed back end of Belt underneath Bridge, around Tension Pulley and rear pulleys into Bridge.

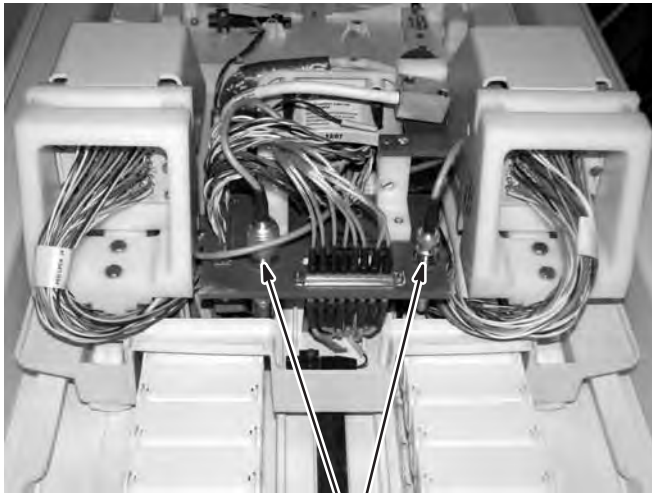
NOTE: Check to verify that belt is not caught around spacers near rear pulley.

7. Route Belt under and through opening to top of Carriage.
8. Position Belt under Clamp Plate. There should be 15 teeth exposed past the clamp. Secure in place.
9. Tension the belt by raising the Idler Tension Block. Turn the hex head bolt until the two socket head cap screws are at the bottom of the slot in the idler Tension Block.
10. Tighten the two socket head screws.

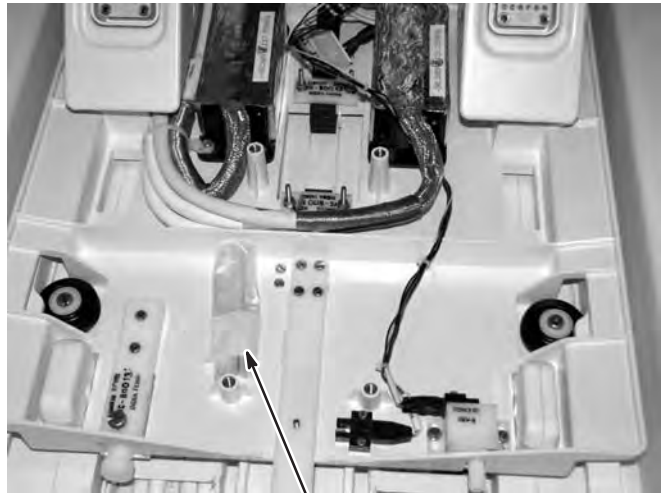


4-24-2 RE-INSTALL LPCA COVER

1. Check all connectors and make sure they are seated properly.
2. Unfasten plastic bag from surface of LPCA and remove spring inside.
3. Set spring on Cradle Latch stud.
4. Carefully replace LPCA cover and fasten with four screws previously removed.



1



2



3



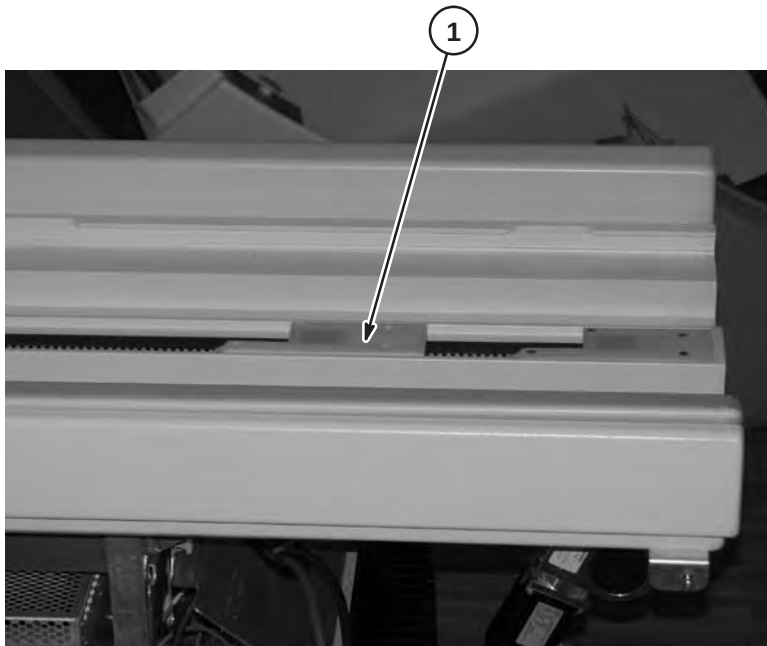
4

4-24-3 LPCA TRAVEL ADJUSTMENT

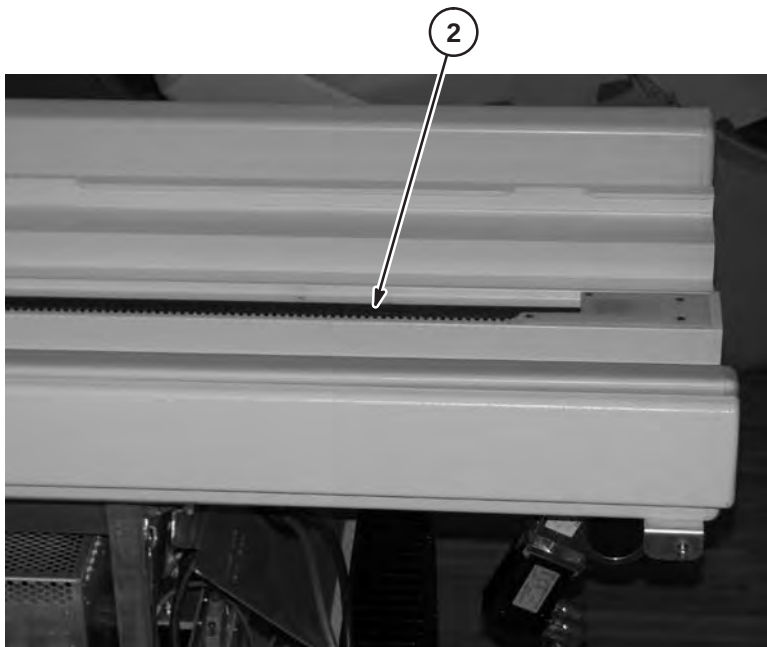
The distance the LPCA can travel at the rear of magnet depends on the distance between the end of the rear split bridge and the wall. The cable track must not touch the scan room wall.

There are two options for LPCA travel distance:

1. If travel of LPCA to rear of bridge needs to be shortened to avoid the cable tracks from touching the wall, leave the plate secured to bridge as shown.



2. If the LPCA can travel the full length of the bridge, remove the plate.



4-25 IN-ROOM DISPLAY INSTALLATION

For systems installed with In-Room Display (IRD), the IRD Monitor and the Operator Control Panels (OCP) need to be installed and connected to the pre-routed cables.

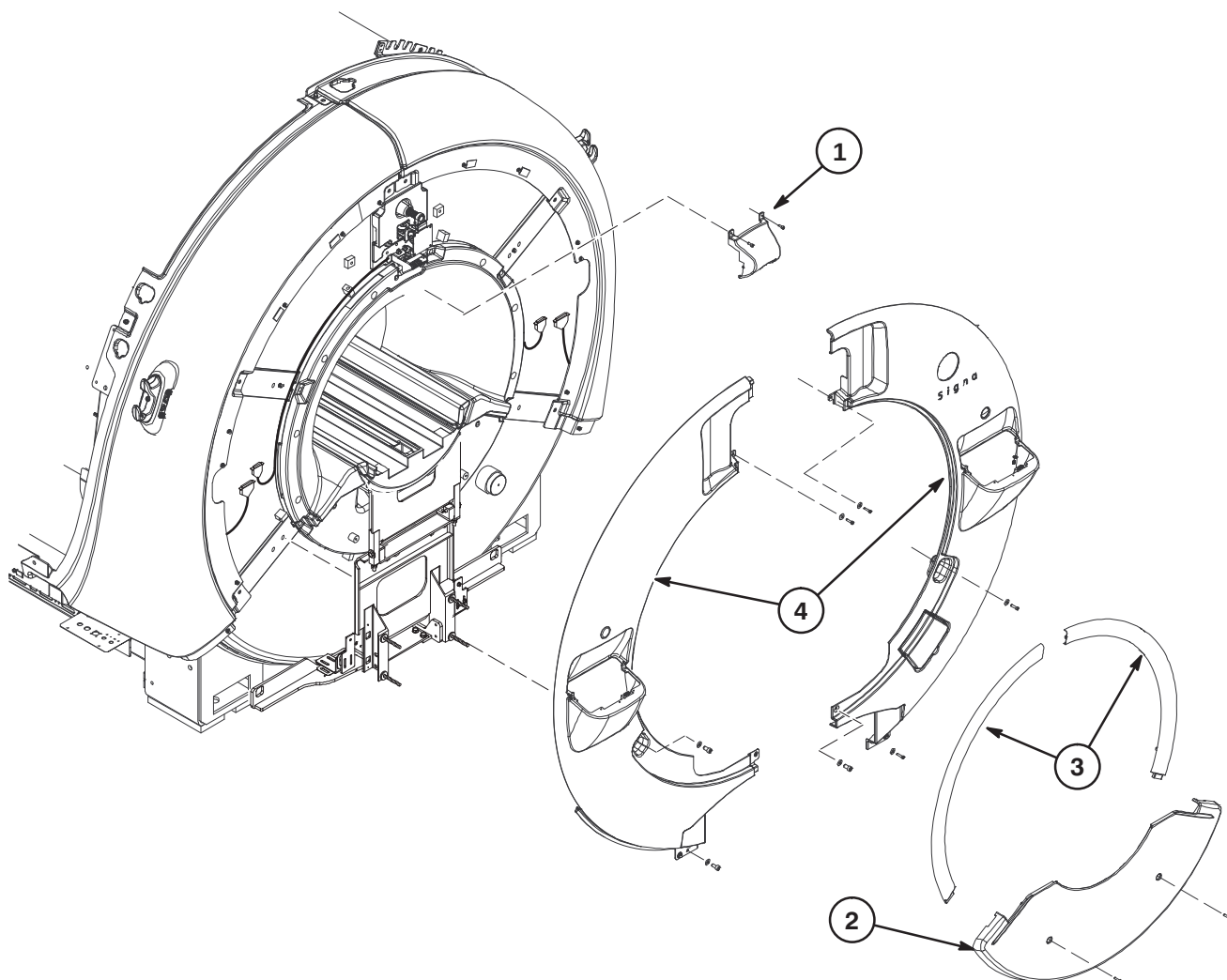
The system installed has one of two types of Front Bridge Supports installed. The type of support can affect the installation steps of the IRD.

- If the system has the type of support secured to the front face of the magnet, then all steps in this procedure, 4-25-1 to 4-25-4 can be completed at one time.
- If the system has the type of support secured to left and right legs of the magnet, then sections 4-25-3 and 4-25-4 should be completed after the Dock is installed.

NOTE: Two people are required to perform this procedure. It should be performed before dock is installed.

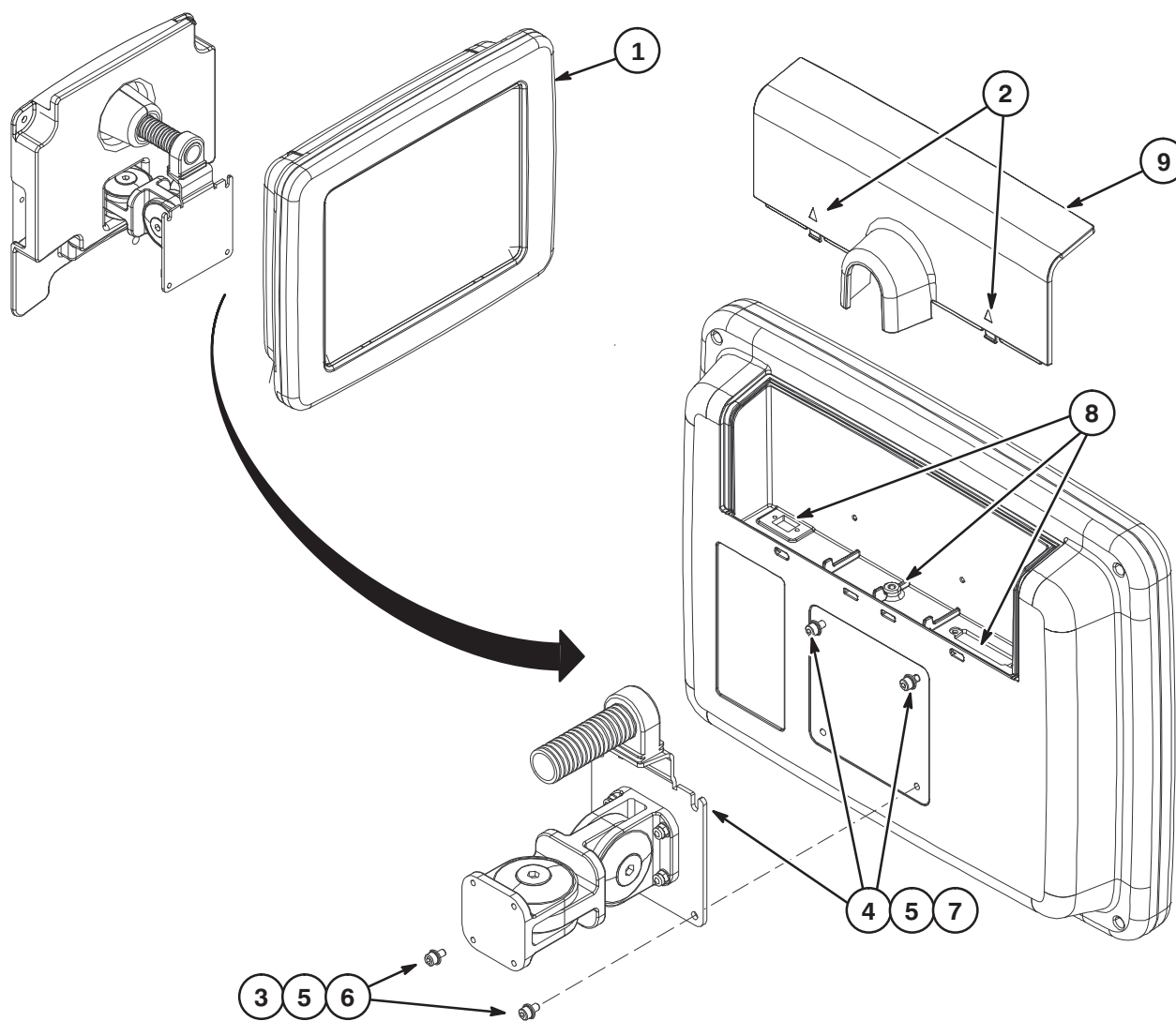
4-25-1 REMOVE FRONT COVERS

1. Unfasten two M6 x 16mm screws and remove Laser Cover.
2. Unfasten two M6 x 16mm screws and remove Front Bridge Cover.
3. Carefully remove left and right Front Trim Rings.
4. Unfasten the seven M10 x 20mm screws and nylon washers. Remove the left and right Front Cover assemblies.



4-25-2 INSTALLATION OF IN ROOM DISPLAY MONITOR

1. Locate the IRD Monitor (5133768).
2. Remove the top cover on the back of the monitor by pushing in at the arrows and sliding upwards.
3. Remove the two bottom screws on the back of the Monitor.
4. Loosen, but do not remove, the top two screws on back of Monitor and position Monitor on IRD Mount plate.
5. Apply Loctite 242 to mounting screws.
6. Fasten the two bottom screws and hand tighten the two top screws to secure Monitor to mounting plate. Tighten with screwdriver 1/4 turn.
7. Hand tighten the two top screws, then tighten with screwdriver 1/4 turn.
8. Connect the three Run M3366 cables: ground, fiber optic, and power, to the back of the Monitor.
9. Replace top cover on to monitor.

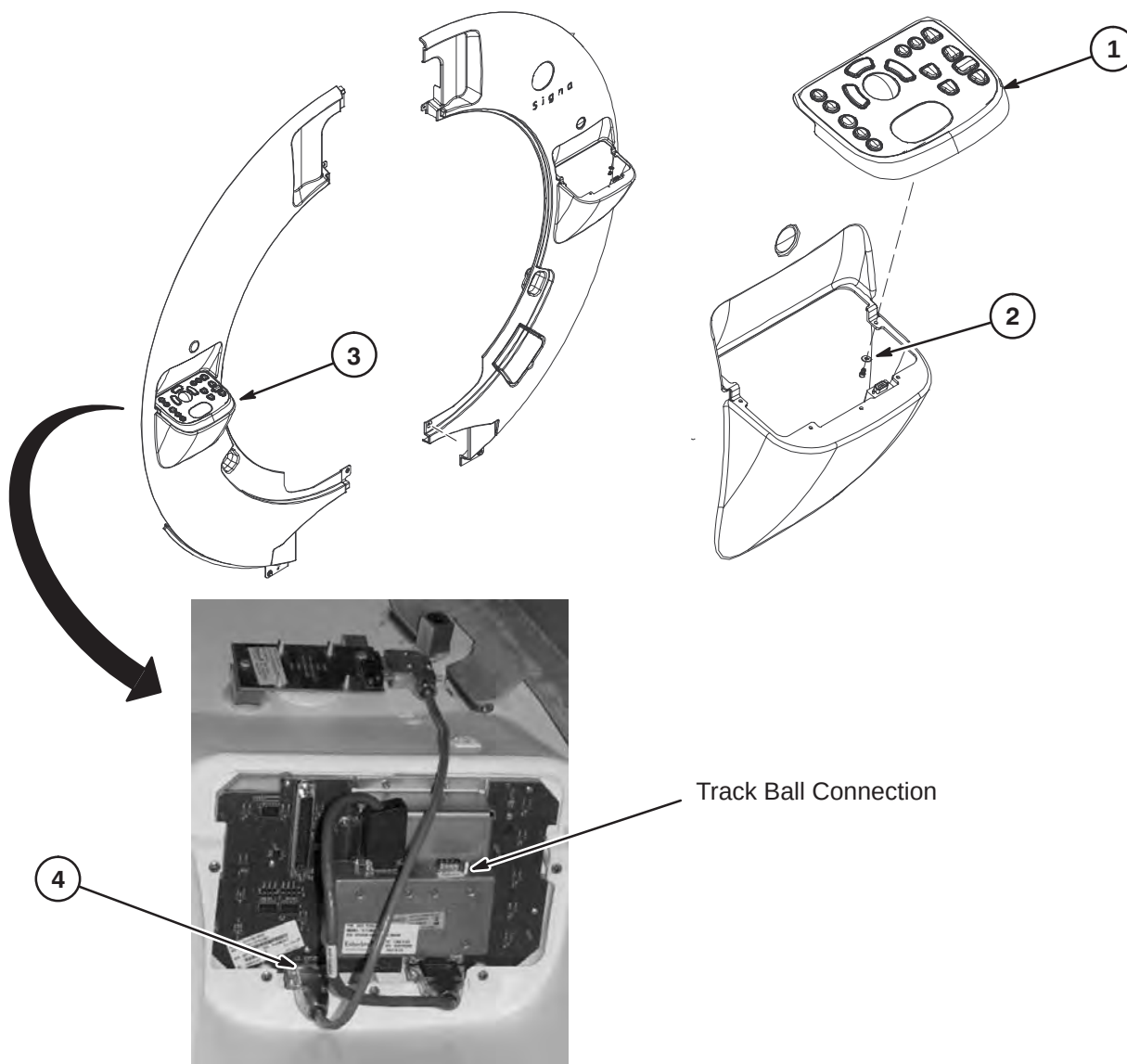


4-25-3 INSTALL LEFT AND RIGHT OPERATOR CONTROL PANELS

1. Locate the first Operator Control Panel (5177062).
2. Fasten to right Front Cover using the provided five M4 screws and M4 washers.
3. Repeat steps 1 and 2 for installing the second OCP to the left Front Cover.
4. Connect E-Stop cables to J3 on left and right Control Panels on front Covers.

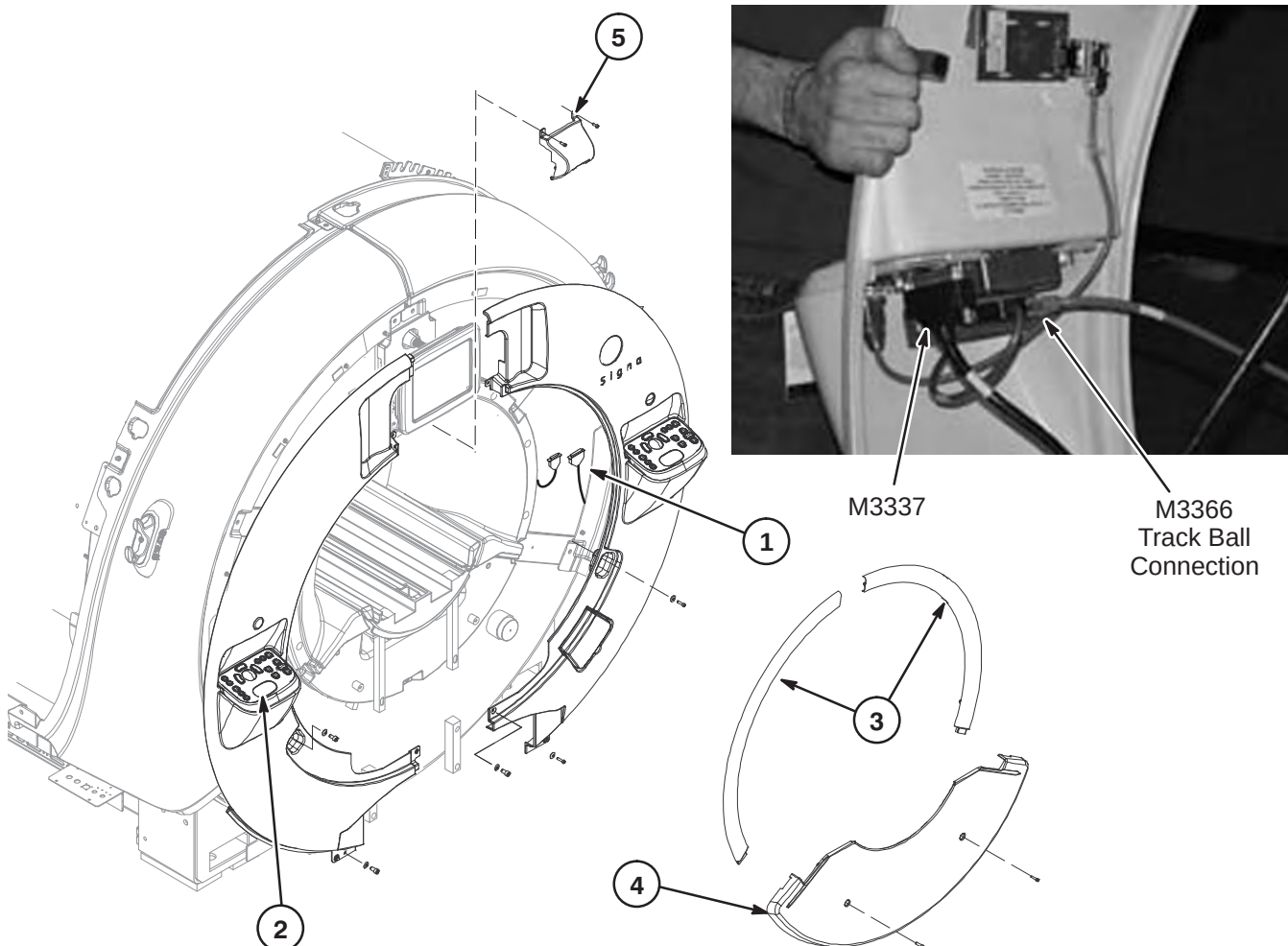
CAUTION

Take special care in connecting Runs M3366 and E-Stop cables. Any reverse connection of these 9-pin Sub-D connectors will mean at system power-on, it will not be possible to recover the system from E-Stop condition.



4-25-4 CONNECT CABLES AND ATTACH COVERS

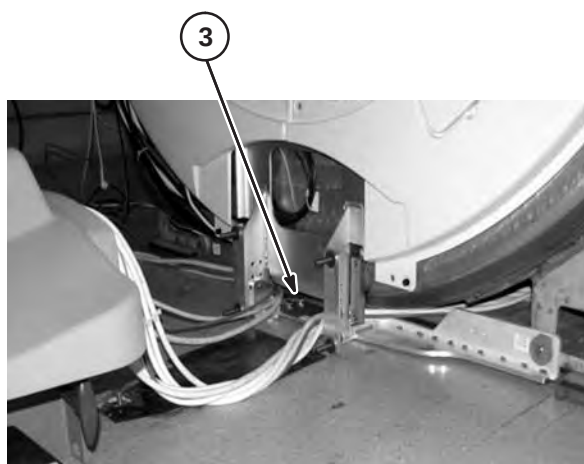
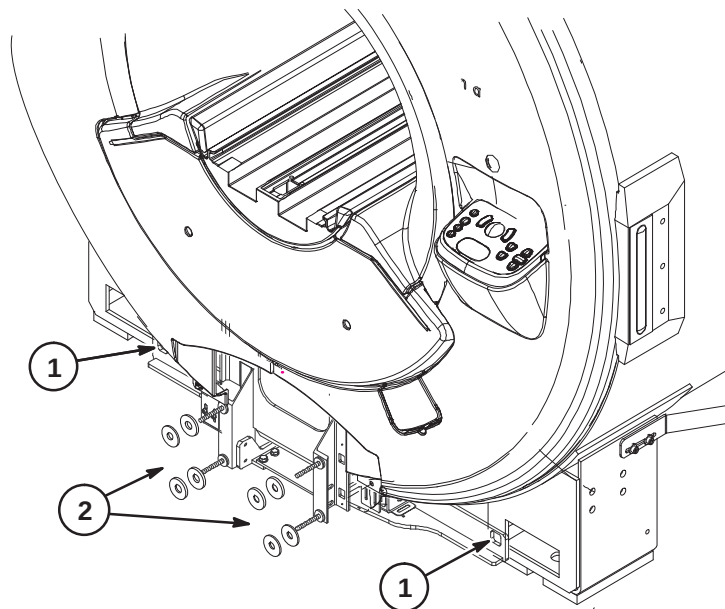
1. Move right cover to front of magnet and attach Run M3337 to J11 and Run M3366 to MAG-Trackball. Attach Front Cover to magnet using fasteners previously removed.
2. Move left cover to front of magnet and attach Run M3337 to J11 and Run M3366 to MAG-Trackball. Attach Front Cover to magnet using fasteners previously removed.
3. Attach left and right Front Trim Rings.
4. Install Front Bridge Cover using fasteners previously removed.
5. Install Laser Cover using fasteners previously removed.



4-26 DOCK INSTALLATION

4-26-1 FASTEN TO MAGNET

1. Make sure Dock Mounting Bracket is securely attached to the magnet legs.
2. Start with two (2) 11mm washers on each stud, both sides, between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is achieved.
3. Cables will be routed thru lower Dock Mounting Bracket as Dock is moved into place.

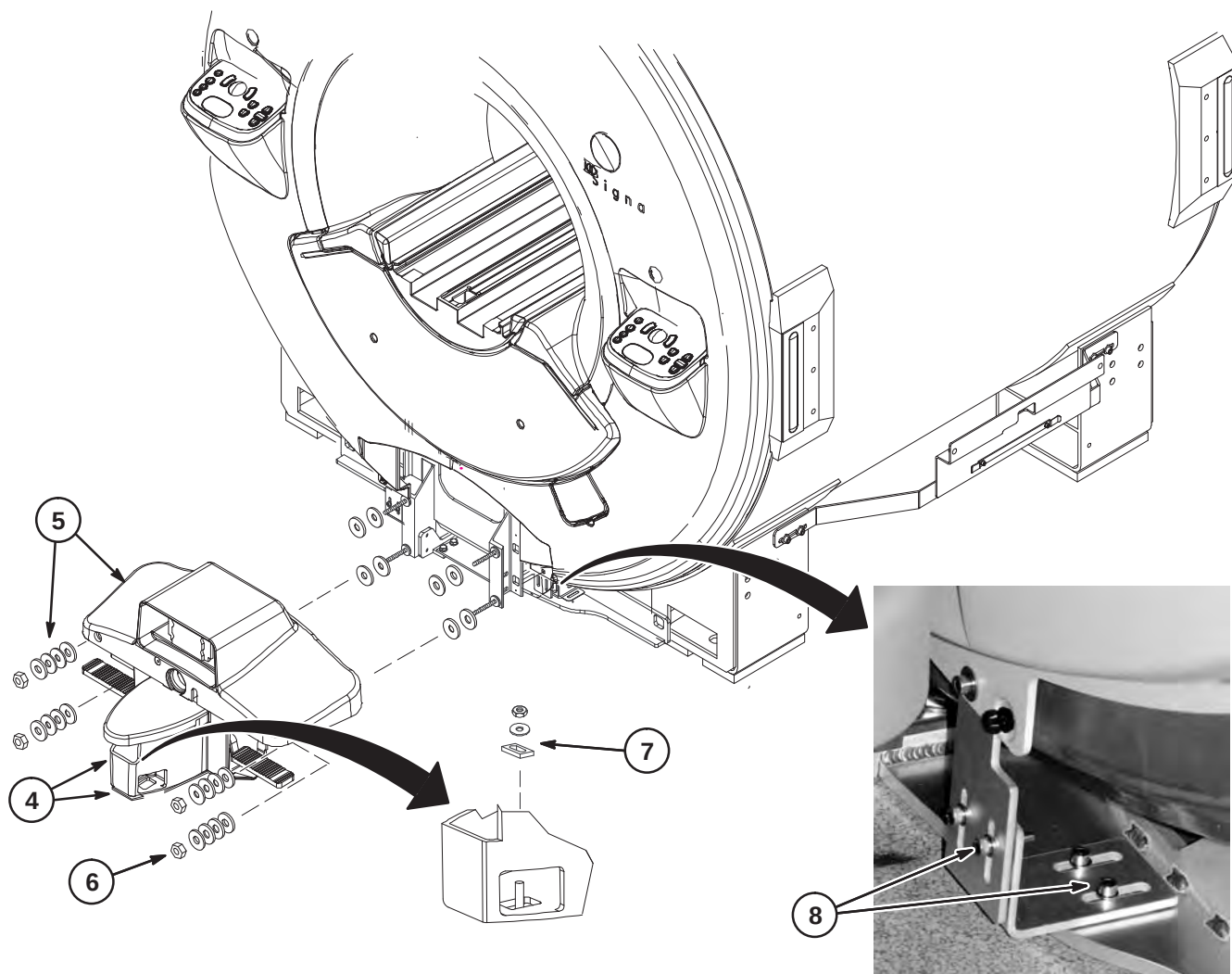


4-26-1 FASTEN TO MAGNET (continued)

4. Position Dock in front of magnet and align Rubber Pad (2230001) under dock.
5. Move Dock into position on studs and place the remaining washers equally on each of the four mounting studs.
6. Fasten with provided nuts, one on each stud.
7. Fasten Dock Asm to floor mounted stud with: (1) Dock Clamp Plate (2178874), (1) 11mm ID Washer (2109878-4), and (1) 10mm Hex Nut (2109875-4). Apply Loctite 242 as required.

Note: If adjustment is required for alignment of patient table, refer to calibration and setup procedures on Service Methods DVD until proper gap is achieved.

8. On each side, adjust brackets as needed and tighten screws.



4-26-2 DOCK CABLE CONNECTIONS – LEFT SIDE

1. Attach cable mounting bracket on magnet leg using two M10 x 25 screws (46-318508P20), M10 Washers (2109878-4) and M10 Nuts (2109875-4).
2. Route and connect cables from dock as shown.
3. Insert and loop excess cable thru hole at front of magnet leg.
4. Loosen bottom of cover and slide cables behind cover. This is required to install front Trim Covers.



4-26-3 DOCK CABLE ROUTING – RIGHT SIDE

For systems that have the electrical dock, an extra set of cables will be routed from the Dock along the right side of the magnet.

1. Route cables from dock thru bottom opening of Bridge Support Magnet Mount Asm as shown.
2. Continue routing cable under magnet towards the Rear Pedestal.
 - Refer to Section 4-27-3 for dressing cables under magnet
 - Refer to Section 4-27-7 for routing cables at rear of magnet and connections to the Rear Pedestal

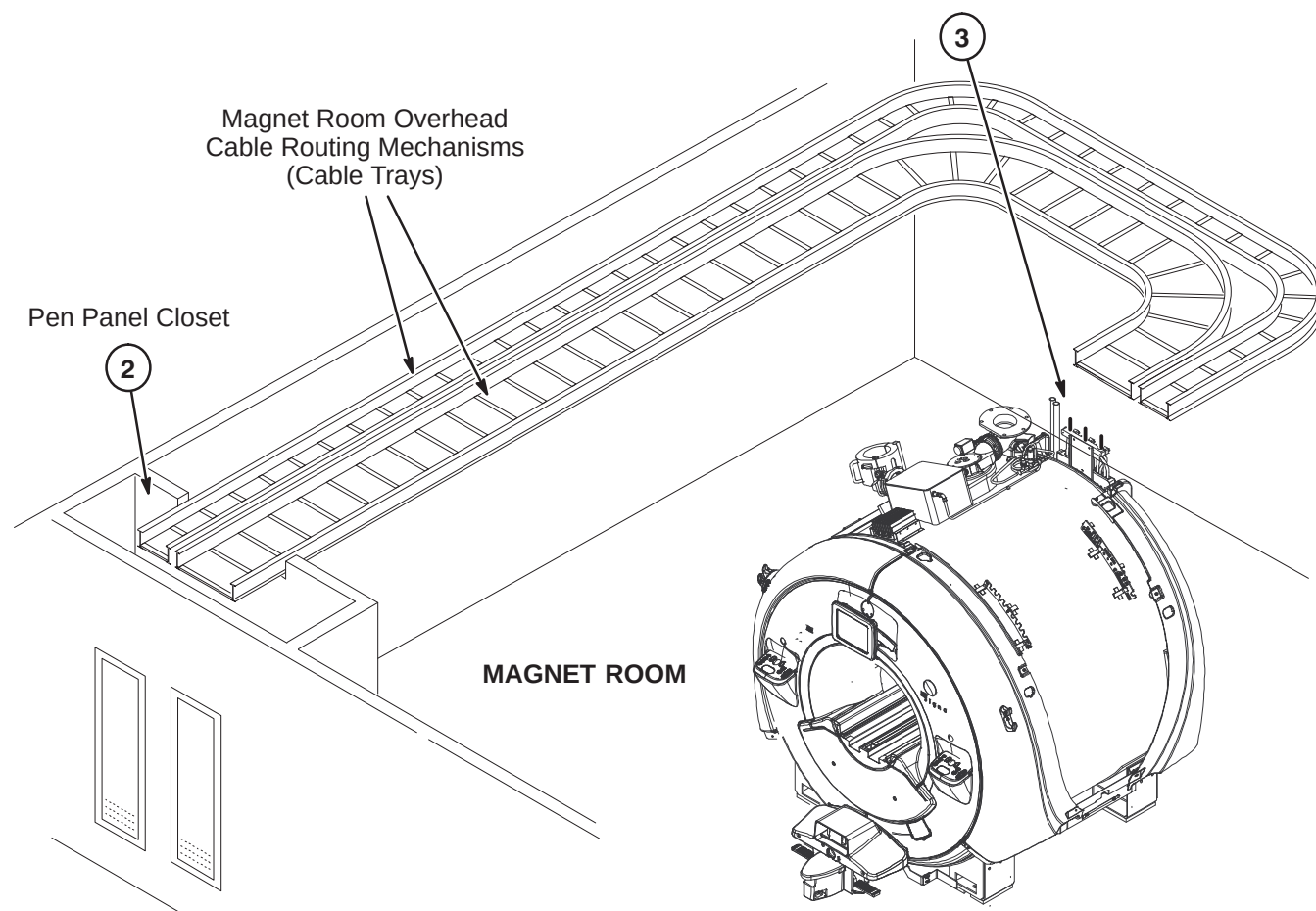


4-27 ROUTE AND CONNECT CABLES IN MAGNET ROOM

Refer to applicable Magnet Room cable map and illustration in Section 4-7 for location and routing of the remaining cables in the overhead cable trays.

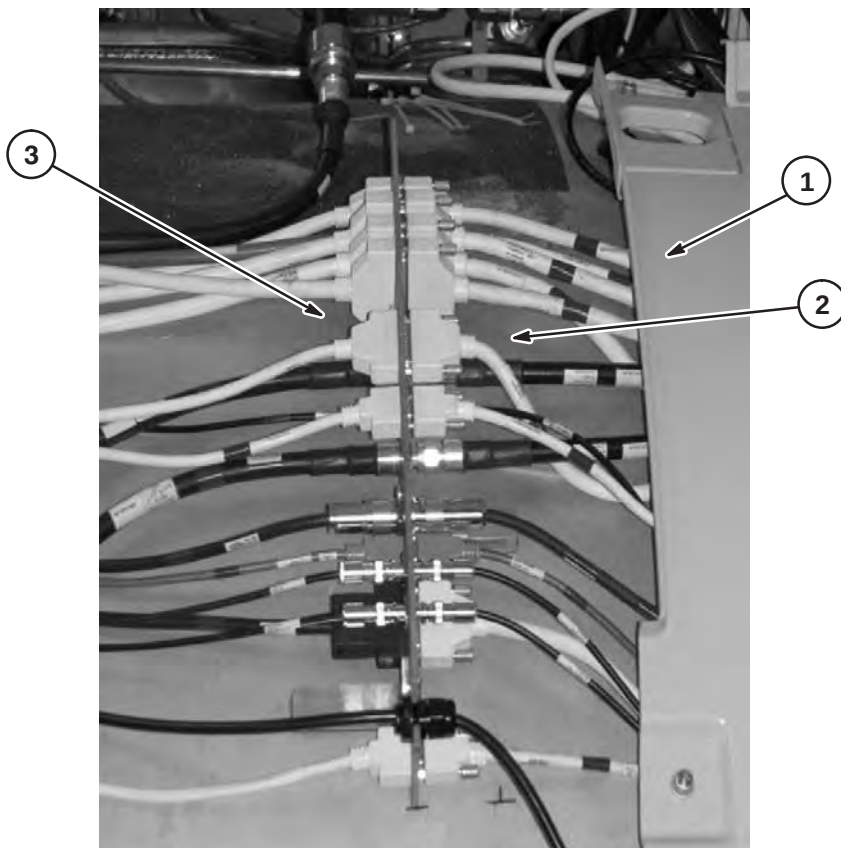
4-27-1 ROUTE CABLES IN OVERHEAD CABLE TRAY

1. Make sure all non-magnetic tools, ropes, and/or ladders that may be required for routing of cables are assembled and ready for use.
2. Route cables from Pen Panel Closet towards magnet. Excess cable length will be stored in Pen Panel Closet.
3. Refer to the following sections for connections and routing of cables around magnet.



4-27-2 CONNECT RUNS TO MAGNET I/O PANEL

1. Route cables from overhead cable tray, under rear arc cover, to magnet I/O panel.
2. Refer to applicable cable map and connect the following Runs to the I/O Panel at the top of the magnet:
 - M3031 (J1)
 - M3314 (J2) (IRD Option)
 - M3303 (J3)
 - M3302 (J4)
 - M3313 (J5)
 - M1312 (J6)
 - M1310 (J9)
 - M1311 (J10)
 - P2026 (J11) (IRD Option)
 - M1306 (J12)
 - M1305 (J14)
 - M1341 (J18) (1.5T only)
 - M1342 (J19) (1.5T only)
3. The cables on the other side of the I/O panel have been pre-connected.



4-27-3 CONNECT CABLES AT DOCK BRACKETS

1. Connect and route the cable Runs M3341 (option), M3364, and M3370 from SRI and Magnet I/O Panel to MAG_DCK_L bracket. Dress and neatly store excess cable under magnet.

Note: The following steps are performed if the system being installed is a 32 Channel.

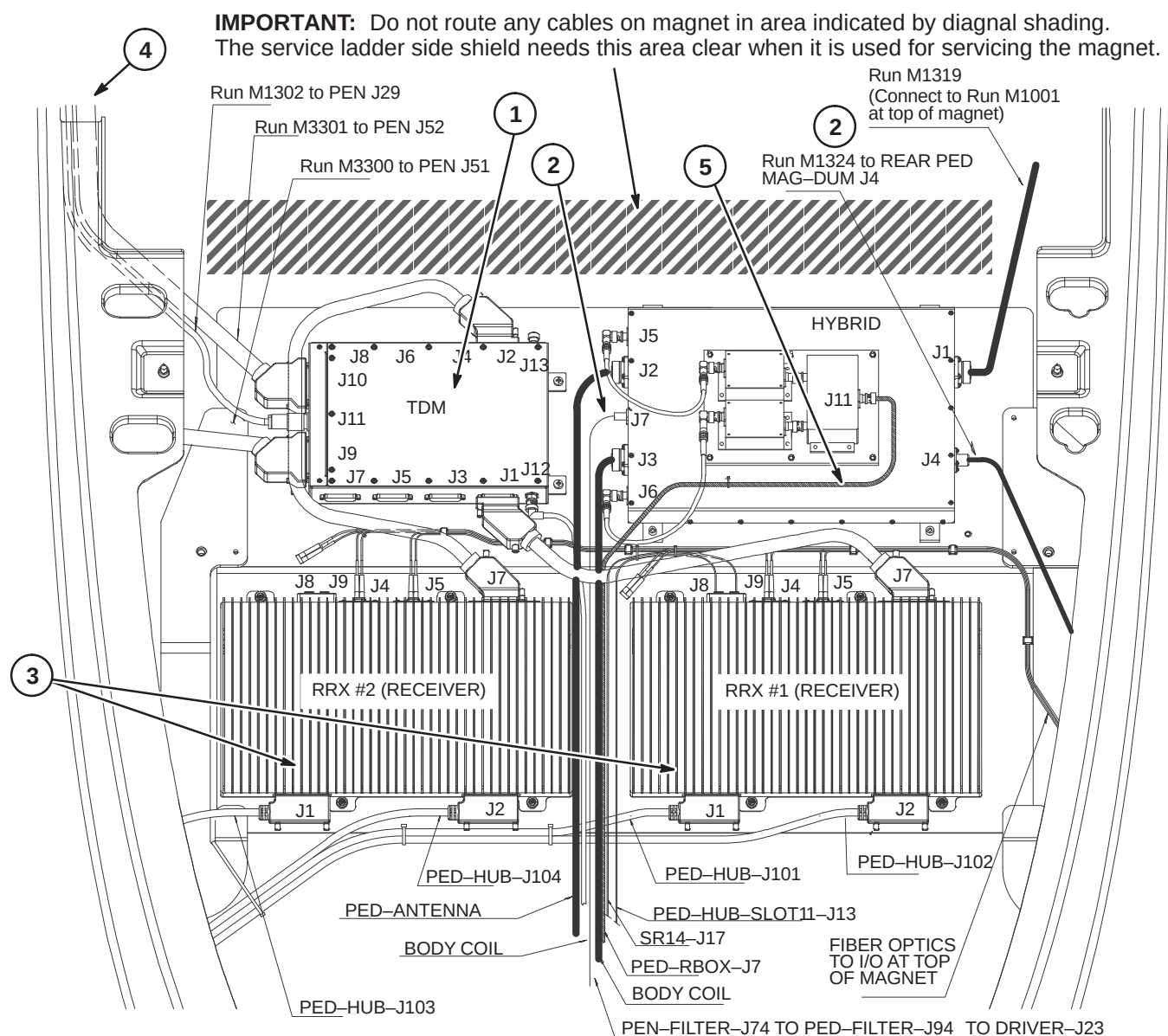
2. Dress and neatly store excess cable under right side of magnet towards the Rear Pedestal.



4-27-4 ROUTE AND CONNECT CABLES to MAG-TDM, MAG-SRI, MAG-DUM, and 3.0T MAG-HYB

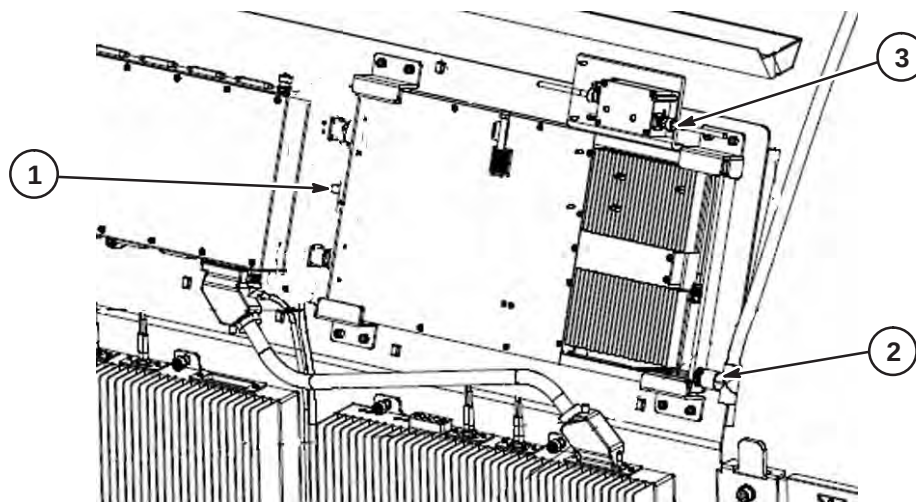
Refer to Section 4-27-5 for connection of cables to 1.5T Hybrid. Cable connections to other modules remains the same as shown below.

1. Refer to applicable cable map to route and connect the following Runs on the side of the magnet to MAG-TDM: M3300, M3301, and M1302
2. Route and connect cables to MAG-HYB: M1324 (3.0T only), M1312, and M1319.
3. Route and connect cables to RRX1 (and RRX2, 32 Channel only): M3320, M3321, M3322, and M3323
4. Route Runs M1302, M3300, and M3301 under the rear arc cover toward the top of the magnet and then to the overhead cable trays.
5. Route and connect Run M1335 from J11 on Pre-amp Combiner (3.0T), or OUT on Pre-amp (1.5T), to J7 on Rear Pedestal PED-RBOX.



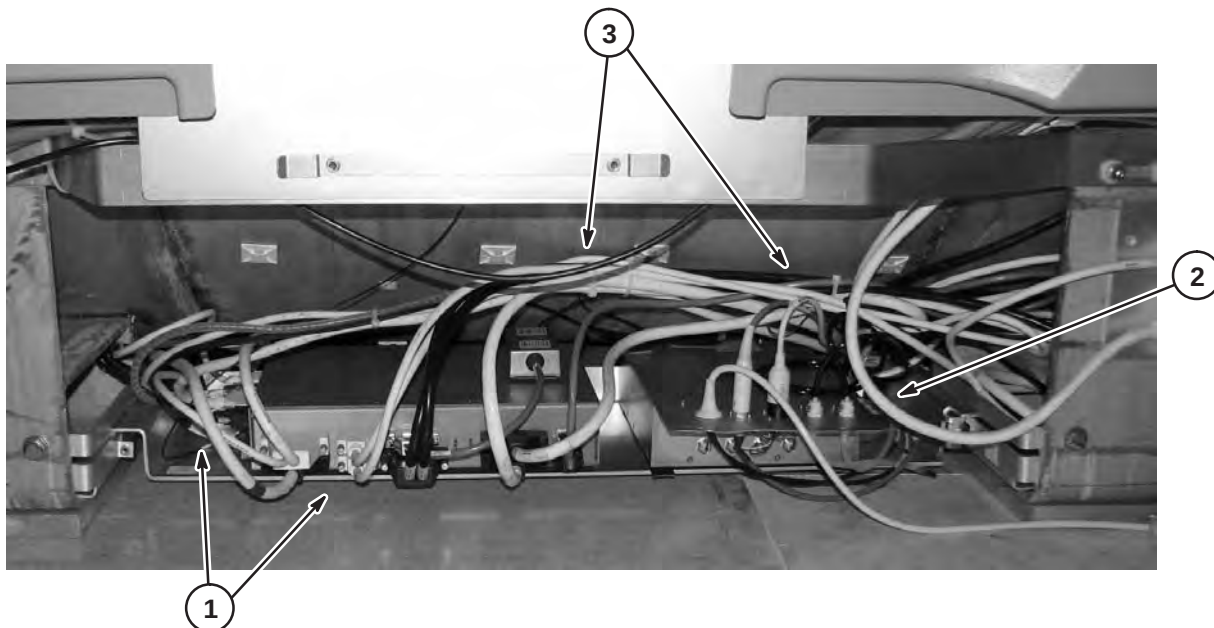
4-27-5 ROUTE AND CONNECT CABLES to 1.5T MAG-HYB

1. Route and connect Run M1312 to J6 for 1.5T system.
2. Connect Run M1319 to J1 and route to top of magnet and connect to Run M1001.
3. Route and connect Run M1335 from OUT on Pre-amp to J7 on Rear Pedestal PED-RBOX.

**4-27-6 ROUTE AND CONNECT CABLES to MAG-SRI, and MAG-PAC**

Refer to applicable cable map and connect the following runs.

1. Route and connect cables from the Magnet I/O Panel to the MAG-SRI: M3336, M3339, M3369, M2025-A, and M2025-B
2. Route and connect to the MAG_PAC: M3360, M3363, M2025-C, and the Patient Alert Air Tubing
3. A cable service loop of at least 3 feet (1000mm) is required if SRI/PAC platform is pulled forward for servicing. Dress and store neatly behind or above SRI/PAC, and under magnet any excess cable lengths.



4-27-7 ROUTE AND CONNECT CABLES TO REAR PEDESTAL

NOTE: A minimum 3 foot (1000mm) service loop must be provided between the magnet and the Rear Pedestal. This is required for service of the magnet.

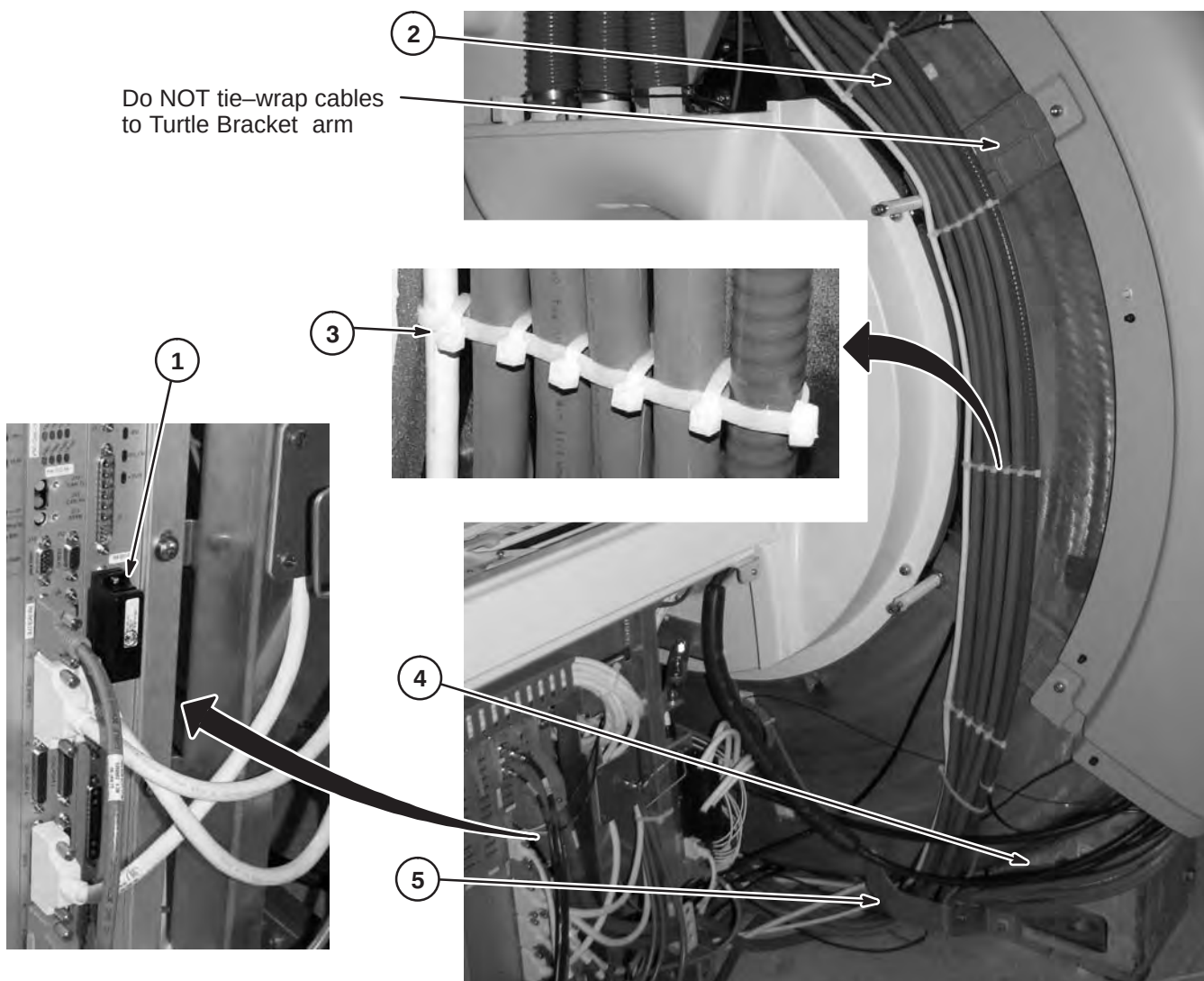
1. Install Shunt Connector (5183786) to Slot 12 J3 before connecting any cables to Rear Pedestal.
2. Refer to cable map and route and connect the following Runs down the rear of the magnet to the Rear Pedestal: M1006 (3.0T only), M1015 (3.0T only), M3310, M3306, M3308, and M3304.

NOTE: Run M3307 may not be installed, depending on product configuration. Refer to Step 13 in this section for installation procedure and details.

3. Tie-wrap cables together.

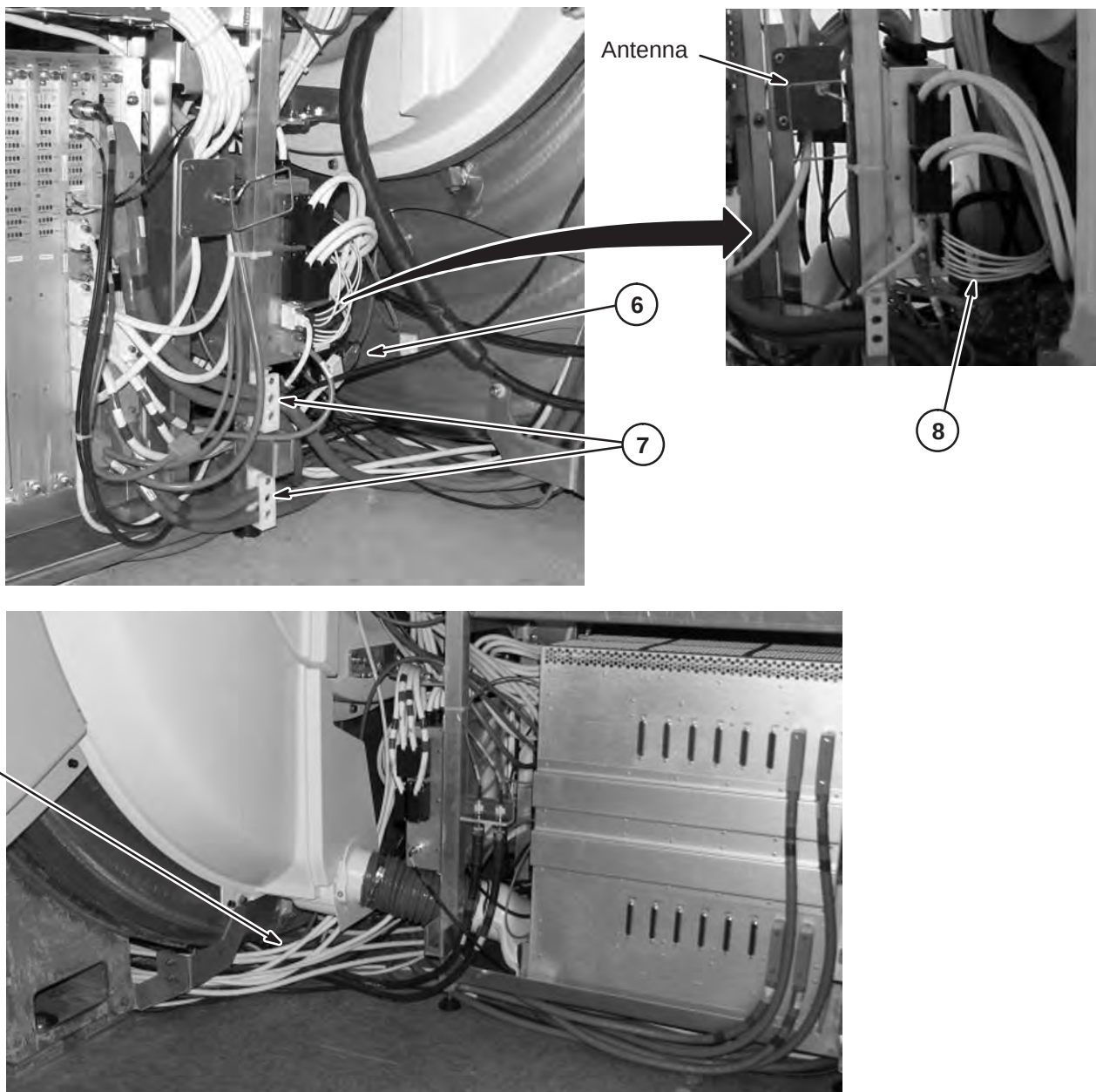
CAUTION: Do not tie-wrap to Turtle Bracket Arms. The Arms may break from attached excess weight.

4. Route and connect cables from modules on side of magnet and PAC/SRI:
M1324 (3.0T only), M2300, M3320, M3321, M3322, M3323, M3315, M1335, M3354, M3340, and M3351
5. Make sure cables are routed between Trim Skirt mounting bracket and magnet.



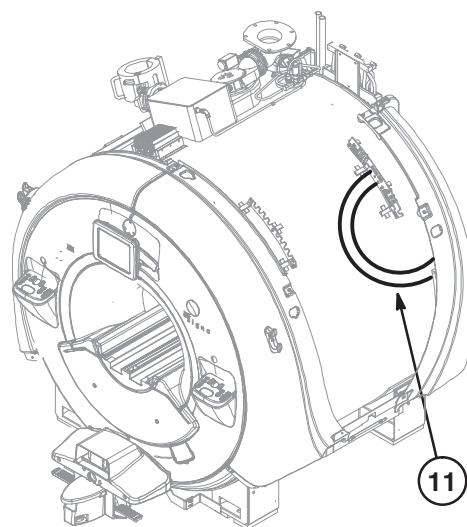
4-27-7 ROUTE AND CONNECT CABLES TO REAR PEDESTAL (continued)

6. Connect one end as marked of Run M3344 to Gradient Temperature cable.
7. Use cable clamps on side of Rear Pedestal for securing cables.
8. **IMPORTANT!** Make sure that the 8w8 Port A cable is dressed and routed between Rear Pedestal and Magnet as shown. Do not route next to antenna.
9. Route and connect cables routed down the right side of magnet from the Pen Panel to the Rear Pedestal: M1303, M1304, M3309, M3311 and M3316. (early systems used M1317, M1316, M3352, M3353 and M3357 routed from the I/O panel)
10. For 32 Channel system, route and connect cables from Dock, to the Rear Pedestal: M1330, M1331, M1332, M1333, M3371, M3372, M3373



4-27-7 ROUTE AND CONNECT CABLES TO REAR PEDESTAL (continued)

11. Route and connect Runs M1325 and M1326 to PED I/O J2 and J1 as marked.
12. If cables are not long enough to make a good connection, reposition cables on side of magnet within the existing mounted cable-ties or install new mounted cable-ties to facilitate new routing position of Runs M1325 and M1326.

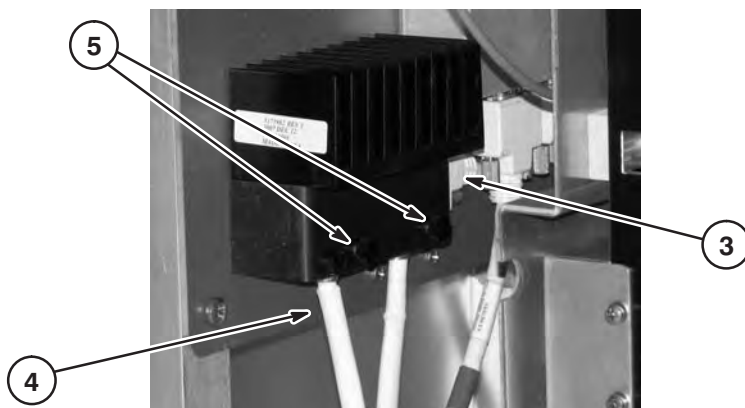
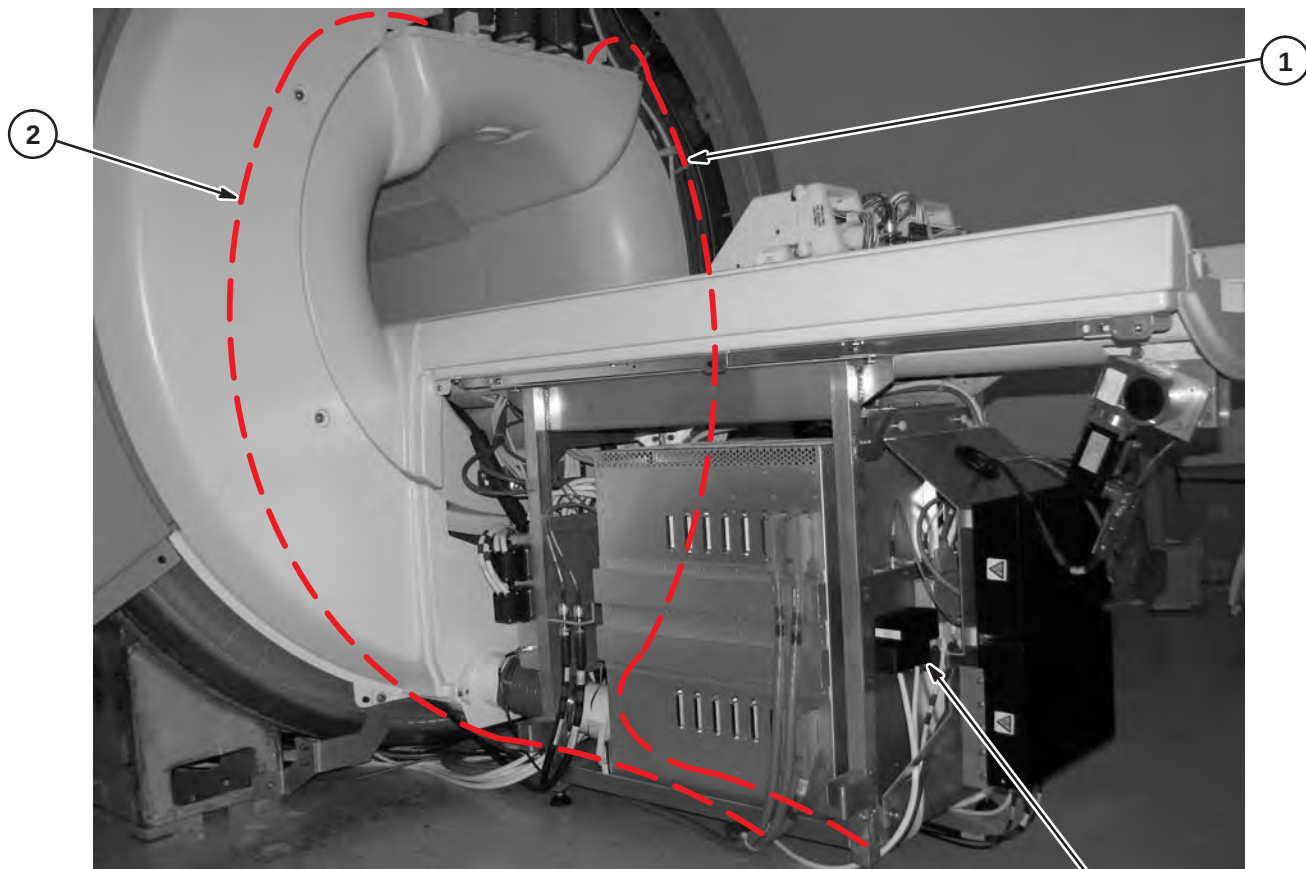


13. Run M3307 may not be installed, depending on product configuration. Refer to the following for guidance:
 - If Power Distribution Board has a connector in Slot 12–J5, then connect cable, route to PEN Panel and connect to J50. If Terminator is installed at J50, remove terminator and discard. Connect Run M3307. Done.
 - If Slot 12–J5 does not have connector (area where J5 would be found is blank), then:
 - Discard Run M3307.
 - Terminator (5172838–2) should be installed at PEN Panel.
 - Go to PEN Panel.
 - If PEN Panel has Terminator installed at J50, Done.
 - If PEN Panel does not have Terminator at J50 (blank is installed), then go to front of PEN Cabinet.
 - At front of cabinet, remove the four screws securing the SPRS module, pull module forward, and check to see if Terminator is installed at rear of SPRS module.
 - If YES, done.
 - If NO, then Terminator 5172838–2 needs to be ordered through SOI process.

4-27-8 ROUTE AND CONNECT BORE LIGHT CABLES

The Bore Light optic cables are part of the Rear End Bell and are located in the top openings on each side of End Bell.

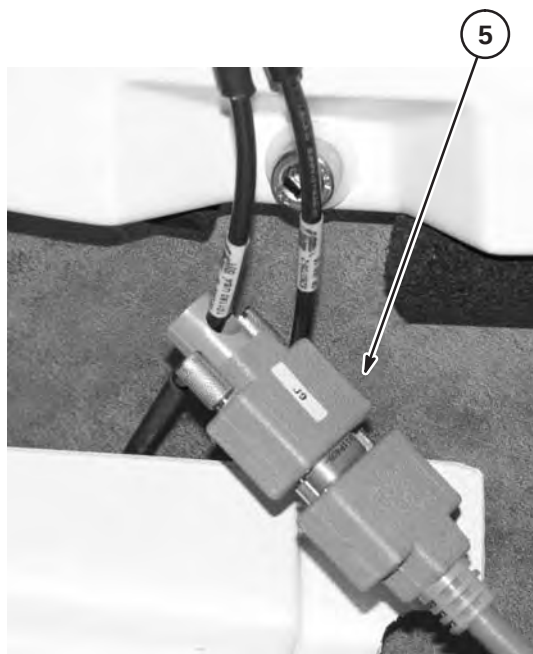
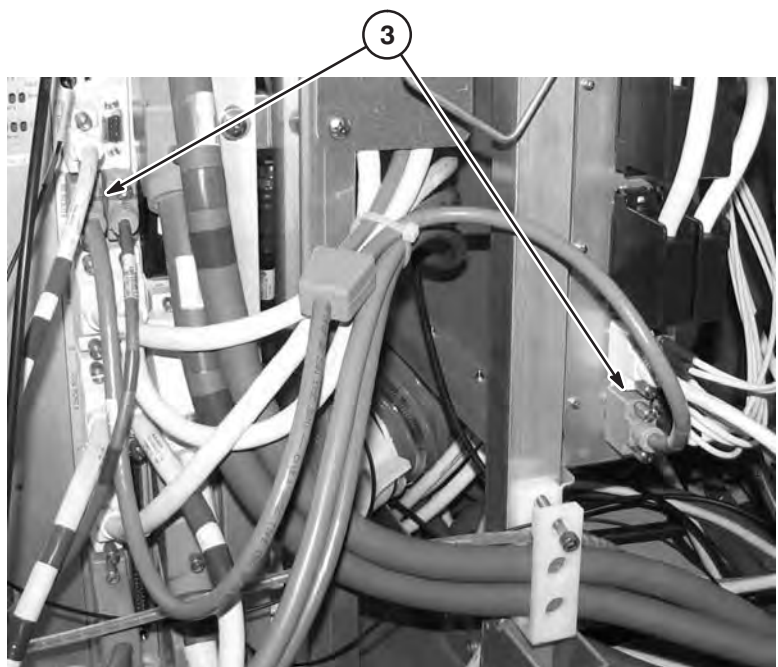
1. Route first Bore Light optic cable from top of Rear End Bell, along side previously routed cables, under Rear Pedestal to Bore Light Module.
2. Route second Bore Light optic cable from top of Rear End Bell, in back of and thru right Rear Cover, under Rear Pedestal to Bore Light Module.
3. Connect end of Run M3315 (early systems used M3340 from I/O panel) to Bore Light Module.
4. Insert ends of Bore Light optic cables into bottom of Bore light module.
5. Secure in place by tightening screws. Do NOT over-tighten. This can damage cables.



4-27-9 ROUTE CONNECT CABLES AT FRONT OF MAGNET

Refer to cable map and route and connect the following Runs at the front of the magnet:
M1313, M1314, and M3354

1. Route and connect Runs M1313 and M1314 from magnet I/O and connect to front RF Bias cables J6 and J7 at front of magnet.
2. **For 1.5T ONLY:** Route and connect Runs M1343 and M1344 from magnet I/O and connect to front RF Bias cables J8 and J9 at front of magnet.
3. Connect Rear Ped end of Run M3354 to Slott 11 J8 on PED-HUB and J10 on PED-RBOX.
4. Route other end of Run M3354 underneath magnet to Dock area.
5. Connect to J9 located on RF Bias cable.



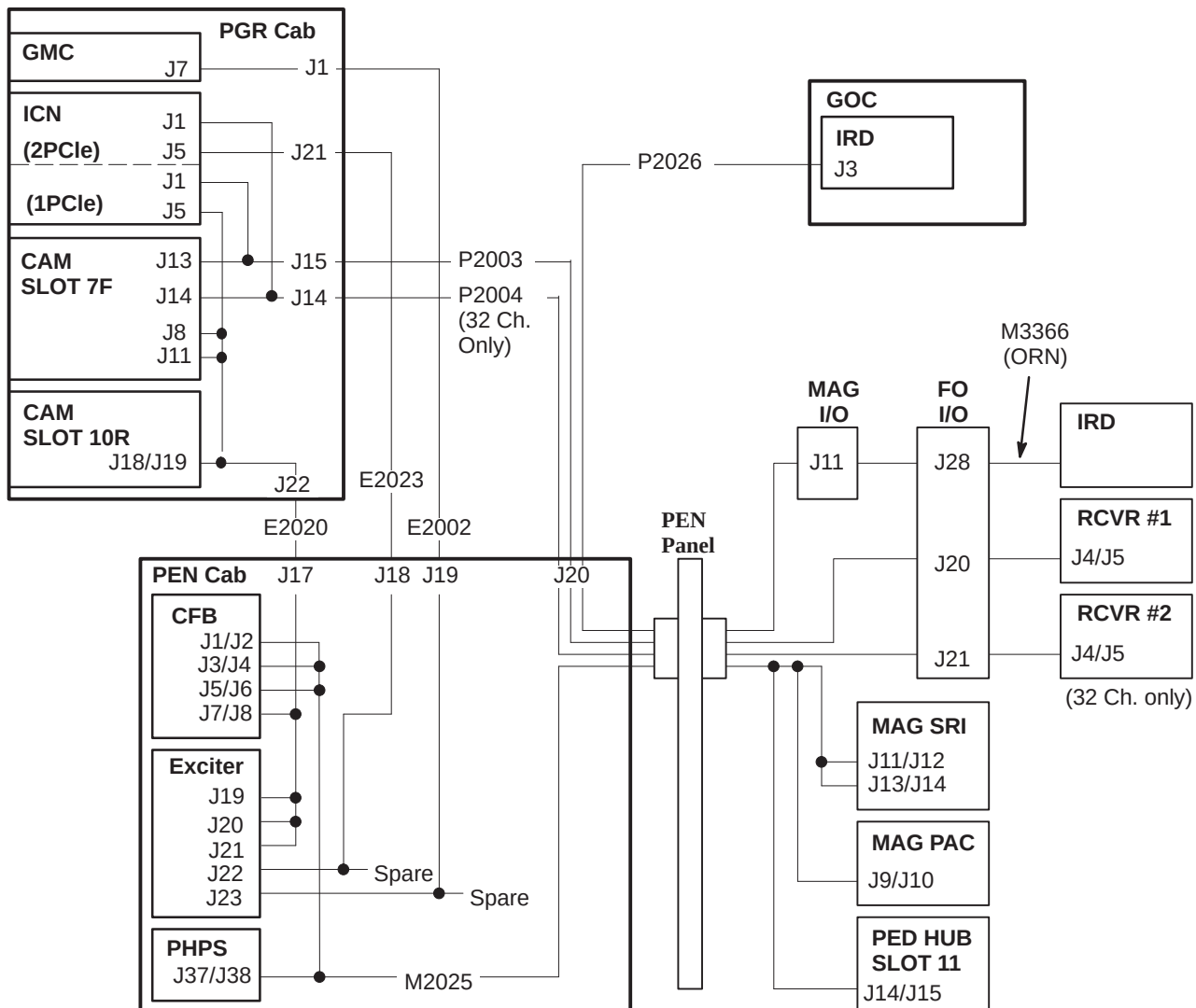
4-28 INSTALL FIBER OPTIC CABLES

Refer to Section 4-6 for location of cables routed in Equipment Room and Section 4-7 for location of cables in magnet Room in the overhead cable trays. Make sure cables are in their designated areas of the trays.

CAUTION

FIBER OPTIC CABLES ARE EASILY DAMAGED! POORLY ROUTED CABLES CAUSE INTERMITTENT SYSTEM PROBLEMS! Handle and route fiber optic cables very carefully. Do not bend cables to radius smaller than 7 Inches (178mm) or 14 inches in diameter (356mm).

1. Route M2025 in overhead tray from top of magnet, thru fiber optic waveguide in PEN Panel to PEN cabinet.
2. Route P2003 from J20 and P2004 (32 Channel only) from J21 on Fiber Optic I/O panel on top of magnet, thru fiber optic waveguide in PEN Panel, thru J20 at top of PEN cabinet to PGR cabinet.
3. Route P2026 (if IRD is installed) from J28 on Fiber Optic I/O panel on top of magnet, thru fiber optic waveguide in PEN Panel, thru J20 at top of PEN cabinet to OW.
4. Route E2002, E2020, and E2023 from top of PEN cabinet to PGR cabinet

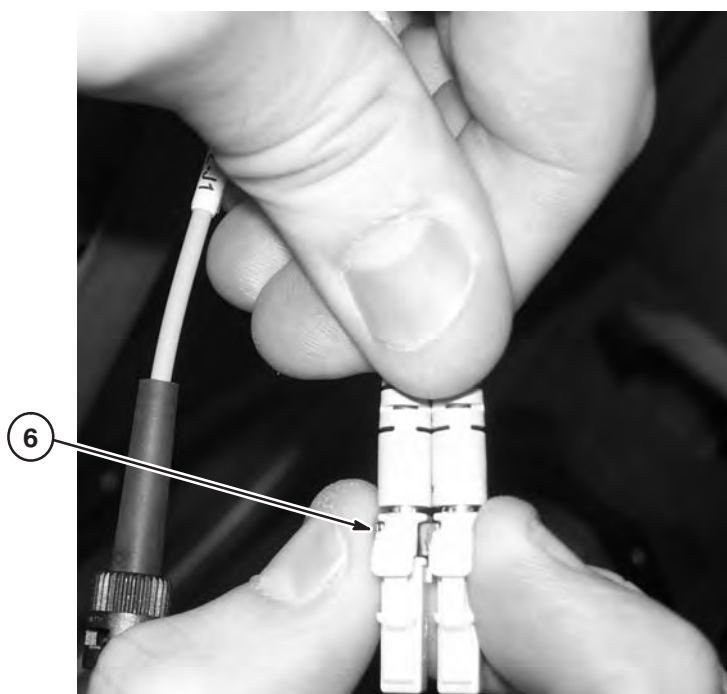
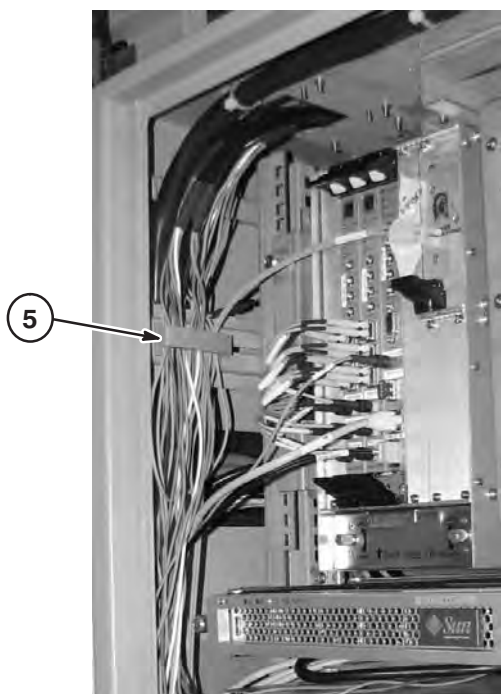


4-28-1 ROUTE AND CONNECT FIBER OPTIC CABLES TO PGR CABINET

1. Starting at the PEN Cabinet, route fiber optic Runs P2003, P2004, E2002, E2020, and E2023 thru overhead cable trays to top of PGR Cabinet.
2. Connect cables as marked to fiber optic connection panel at top of PGR Cabinet.
3. Route fiber optic wires thru opening as shown.
4. Secure in place with clamps.



5. Route fiber optic wires in PGR Cabinet behind guides.
6. These fiber optic connections are installed by pressing on each side of connector while inserting.

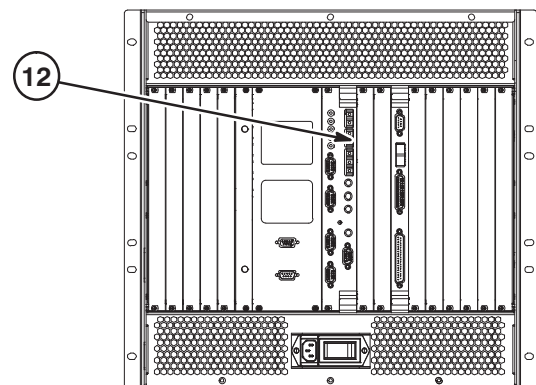
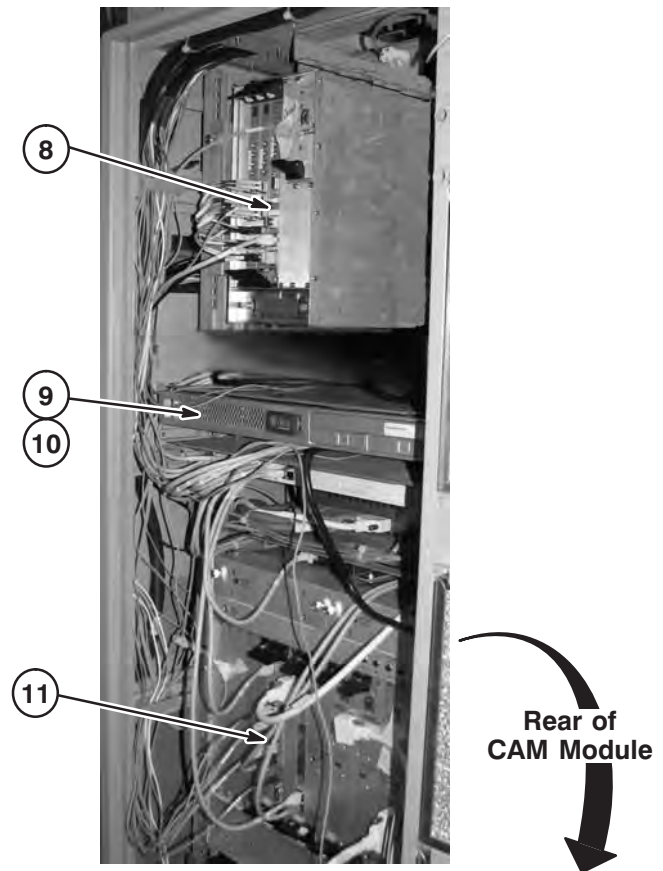
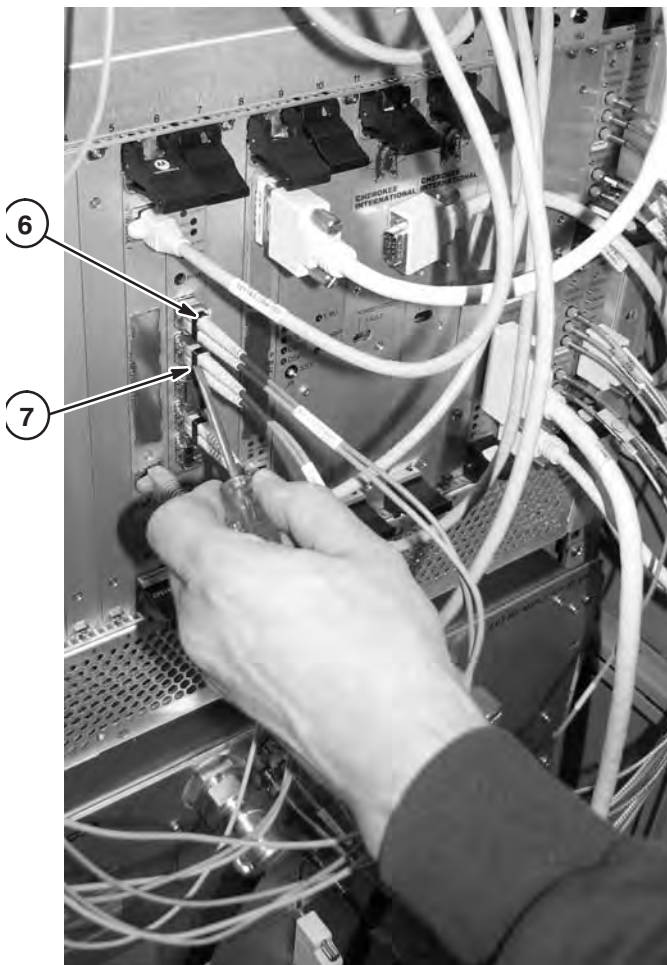


4-28-1 ROUTE AND CONNECT FIBER OPTIC CABLES TO PGR CABINET (continued)

7. For connections that do not provide space for fingers on the side, place a flat blade screwdriver between wire covers and insert using a small amount of force.

Refer to fiber optic cable map in Section 4-28 for remainder of steps.

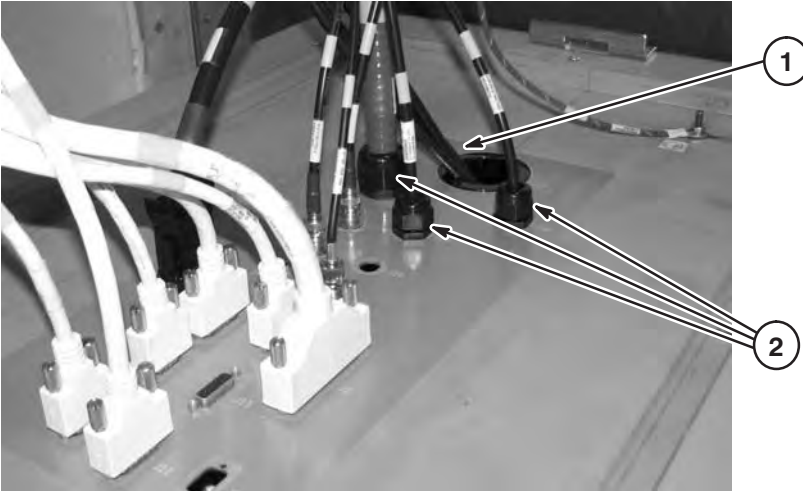
8. Connect to GMC J7 as marked.
9. Connections to ICN are at the back of the ICN module. Pull module forward and connect applicable single fiber optic to J5 on each VRF card (two cards for 32 channel).
10. Connect double fiber optic cable to J1 on the back of ICN for each VRF card.
11. Connect remaining fiber optics to J8, J11, J13, and J14 in CAM Slot 7F at front of CAM module.
12. Connection for CAM Slot 10R is at back of CAM module. Pull module forward and connect to J18/J19.



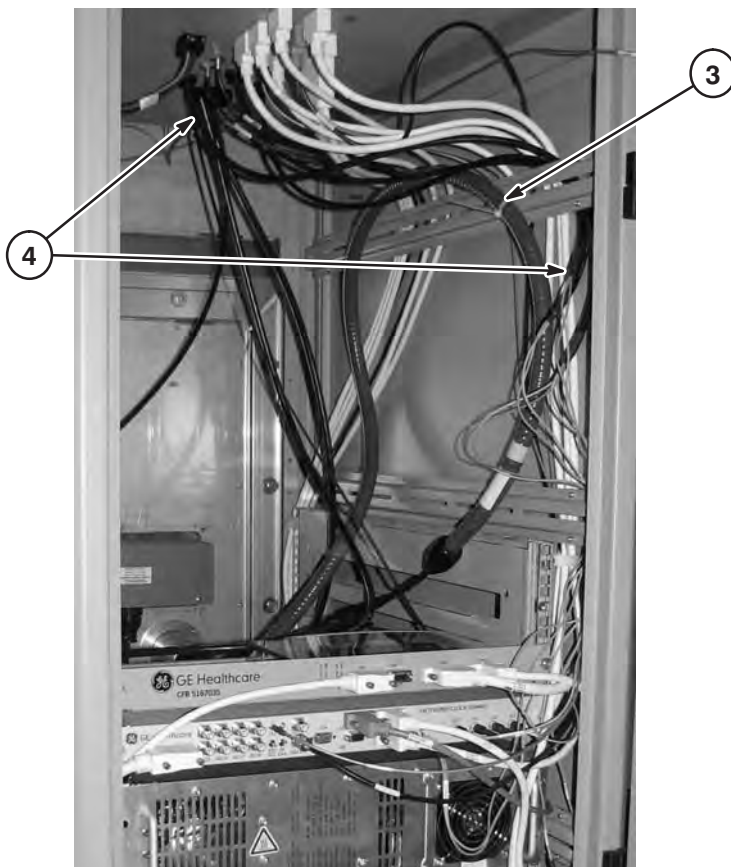
4-28-2 ROUTE AND CONNECT FIBER OPTIC CABLES IN PEN CABINET

Route and connect fiber optic cables as shown below.

1. Runs P2003, P2004, and P2026 routed thru opening at top of cabinet.
2. Connect E2002, E2020, and E2023 to top of PEN cabinet as marked.



3. Route and secure with tie-wraps Run M2025 to side of cabinet
4. Route and secure with tie-wraps Runs E2002, E2020, and 2023 to side of cabinet

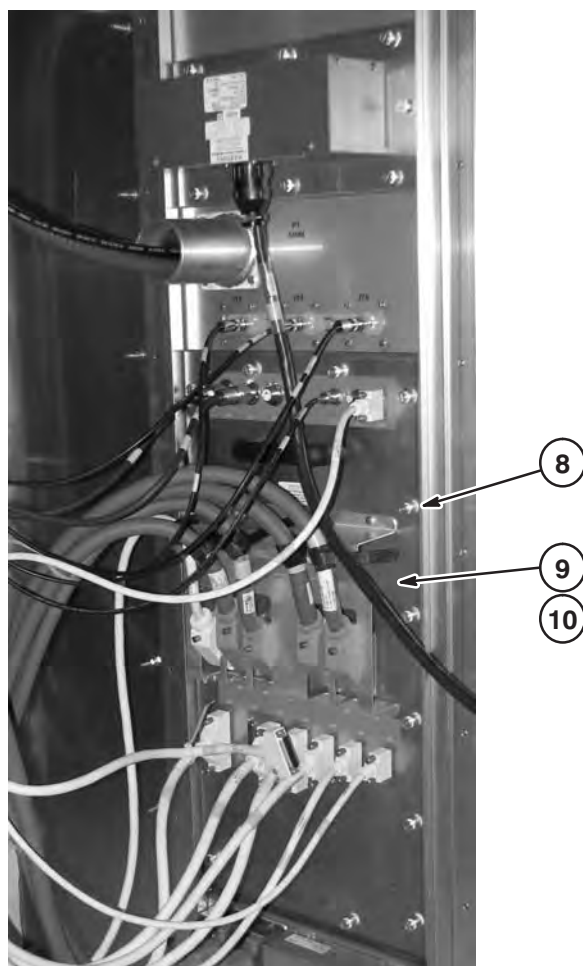
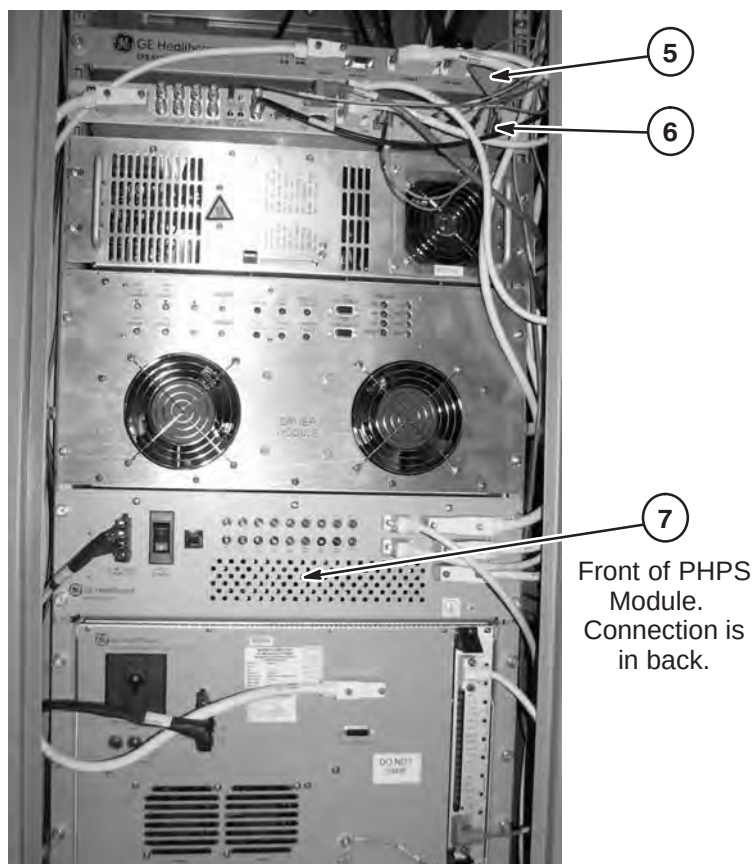


4-28-2 ROUTE AND CONNECT FIBER OPTIC CABLES IN PEN CABINET (continued)

5. Connect fiber optic wires as marked to the CFB module.
6. Connect fiber optic wires as marked to the Exciter module.
7. J37/J38 is located on the back of the PHPS module. Refer to next steps for access and connection of wire.
8. On Magnet room side of PEN panel, unfasten the 16 fasteners attaching the access plate to the PEN Panel.

Note: Cables are attached on the interior side of the access plate.

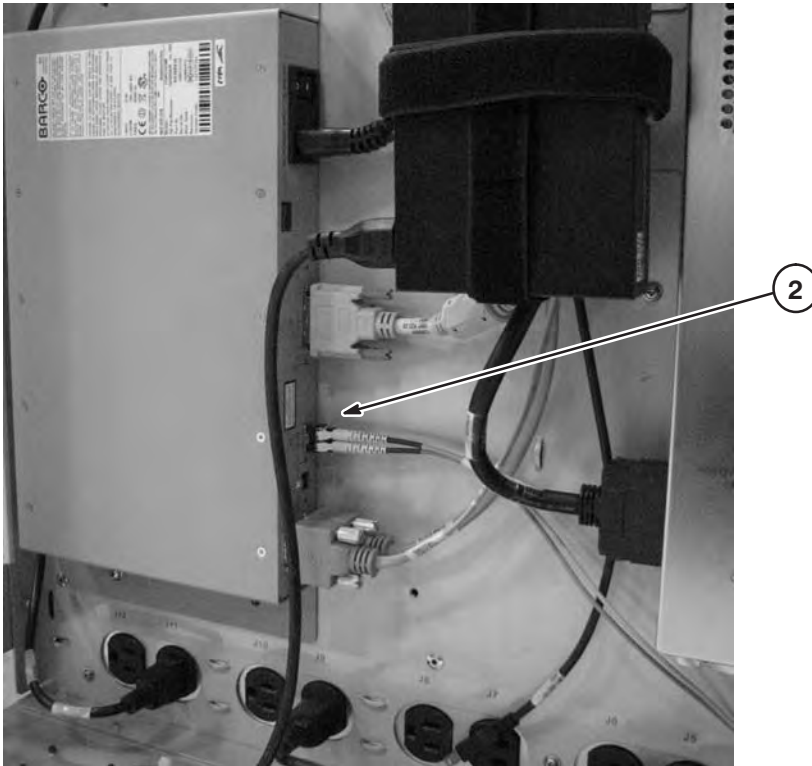
9. Remove access plate enough to reach in and connect M2025 wire to J37/J38.
10. Reattach access plate to PEN Panel.



4-28-3 ROUTE AND CONNECT FIBER OPTIC CABLE IN OW

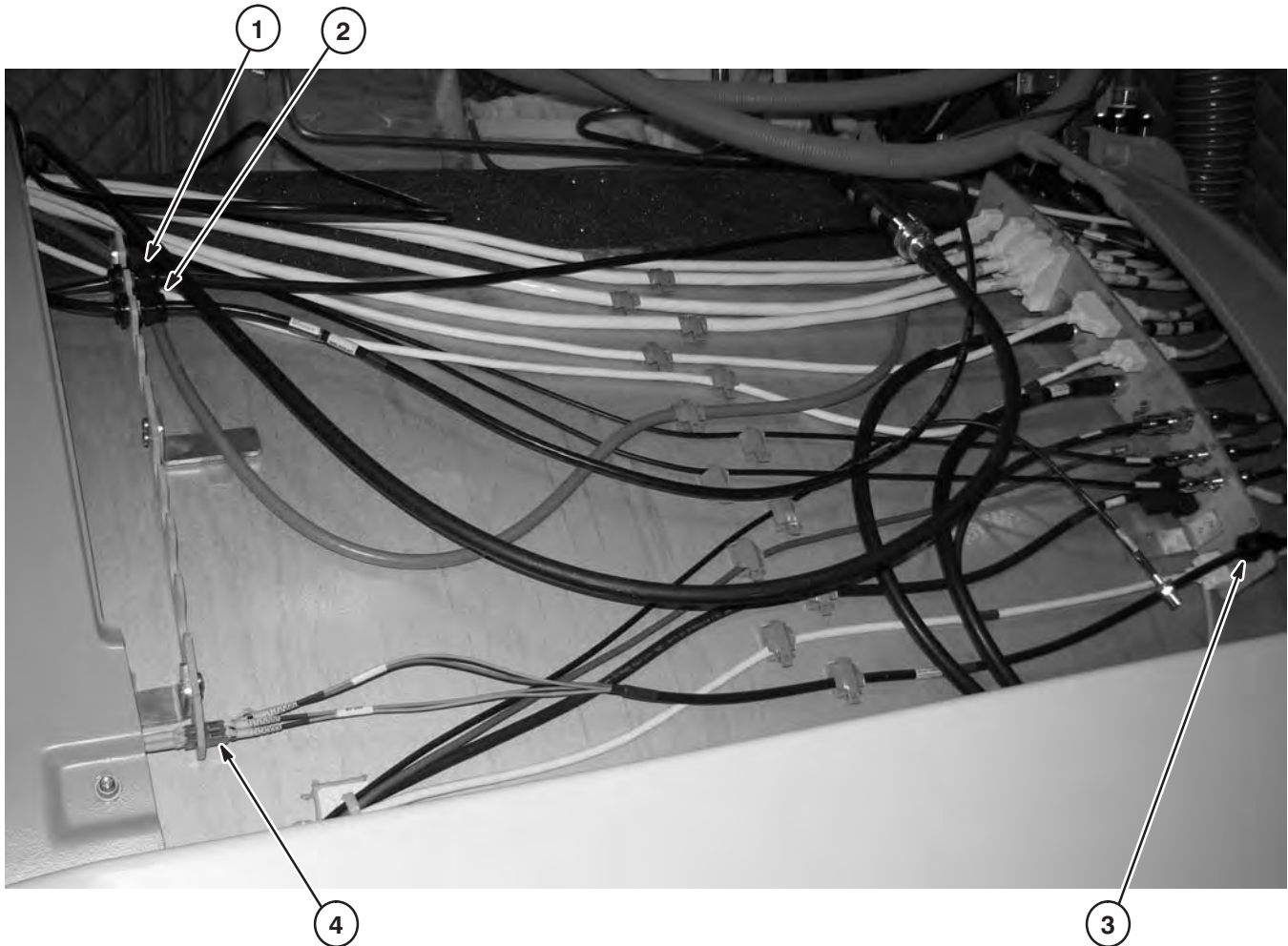
Note: This cable is installed only if system has IRD (In Room Display) as part of installation.

1. Route and attach fiber optic connector to bracket at rear of GOC cabinet.
2. Route and connect P2026 wire to J3 on IRD Module.



4-28-4 ROUTE AND CONNECT FIBER OPTIC CABLES IN MAGNET ROOM

1. Route and attach P2003 fiber optic connector to J20 on FO I/O on magnet.
2. Route and attach P2004 fiber optic connector to J21 on FO I/O on magnet.
3. If IRD is installed, route and attach P2026 fiber optic connector to J11 on Magnet I/O.
4. Connect P2026 fiber optic to J28 on FO I/O.



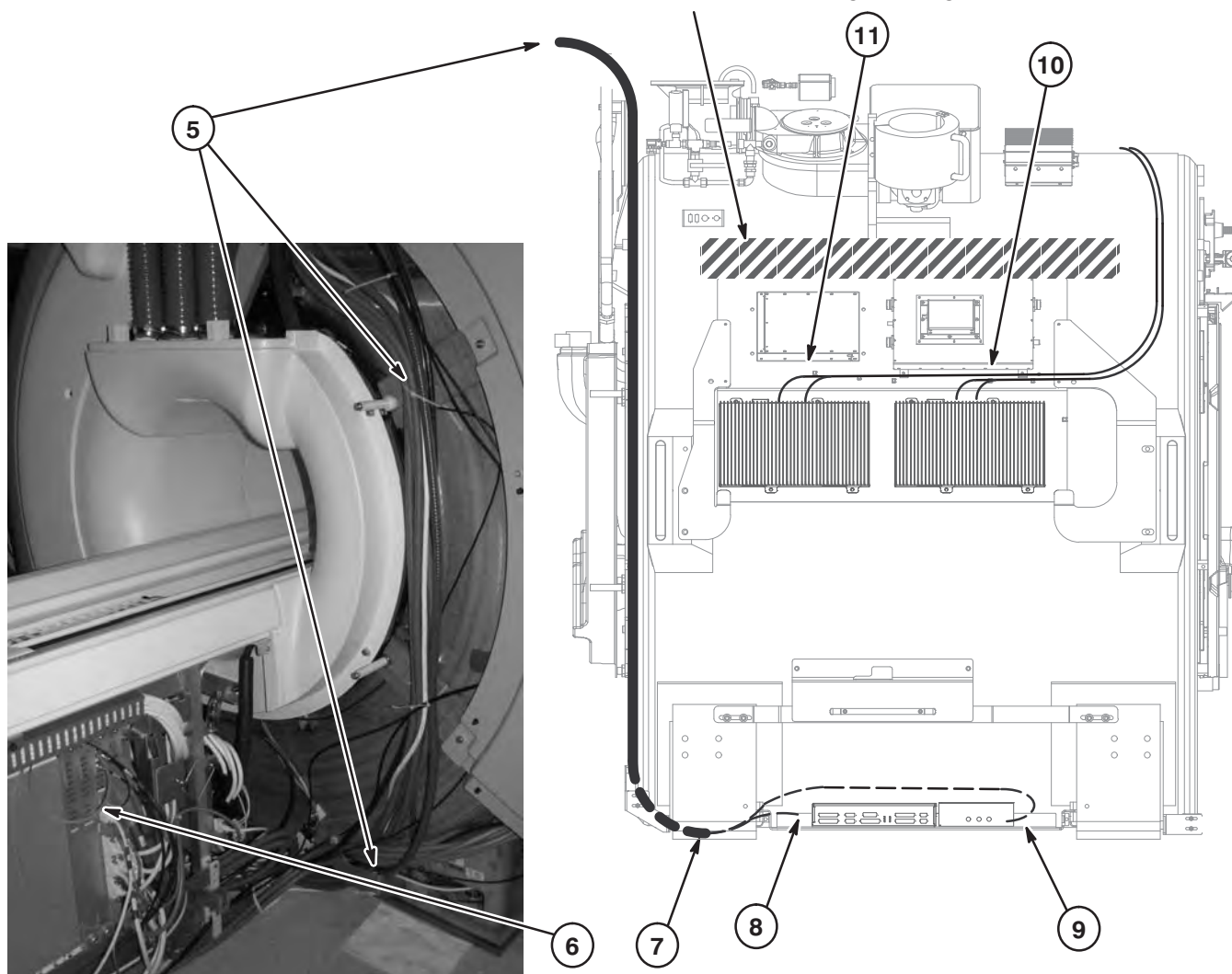
4-28-4 ROUTE AND CONNECT FIBER OPTIC CABLES IN MAGNET ROOM (continued)

5. Route M2025 at rear of magnet along side of cables routed to Rear Pedestal. Make sure conduit is routed behind trim cover bracket.

Note: Secure in place with tie-wraps after all system cables have been connected and adjusted for cable service loop to Rear Pedestal

6. Route and connect fiber optic wire to PED HUB SLOT 11 – J14/J15 as marked.
7. Route the reamaining three fibere optic wires under magnet to PAC/SRI platform.
8. Connect wires marked J11/J12 and J13/J14 to SRI.
9. Connect wires marked J9/J10 to PAC.
10. Route P2003 fiber optic wires over top of magnet and connect to J4/J5 on RCVR #1.
11. If system is 32 channel, route P2004 over top of magnet and connect to J4/J5 on RCVR #2.

IMPORTANT: Do not route any cables on magnet in area indicated by diagonal shading. The service ladder side shield needs this area clear when it is used for servicing the magnet.



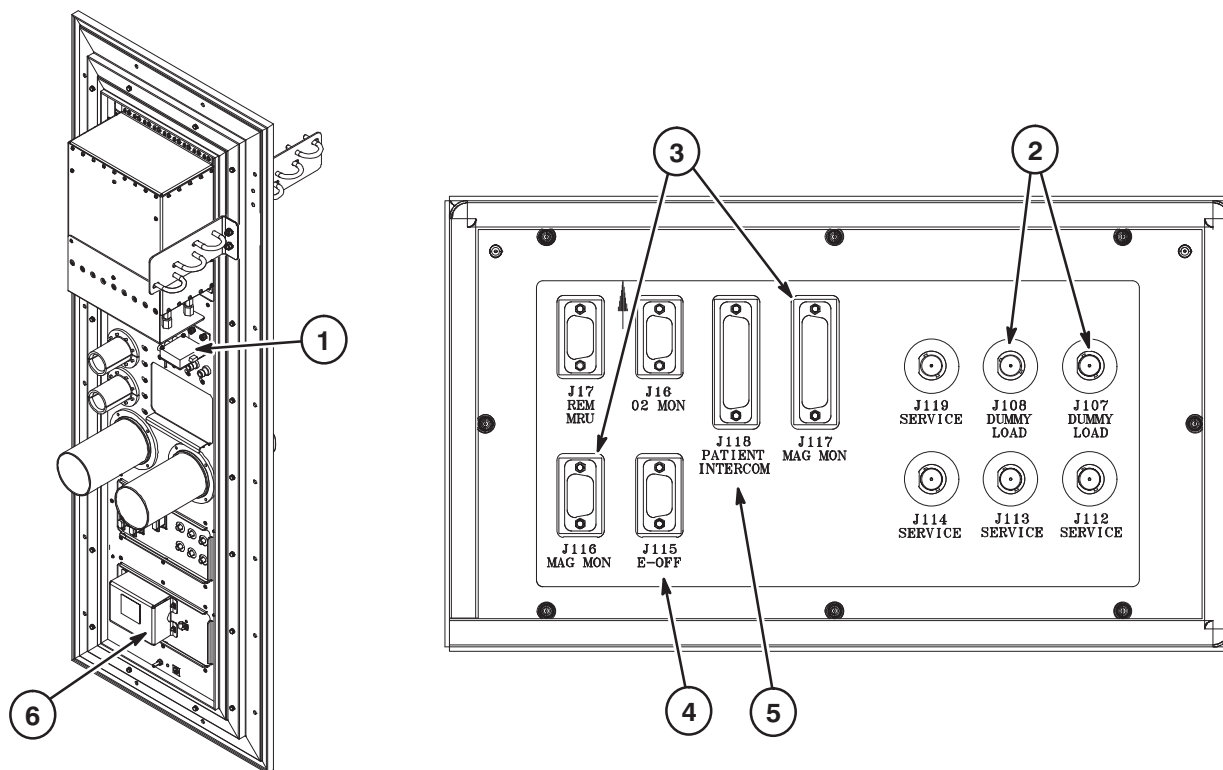
4-29 CONNECT SPW CABLES

The SPW (Secondary Pen Wall) was delivered to site and installed prior to magnet delivery.

- The gas lines, Runs 621 and 622, were routed thru the J110 and J111 waveguides when the magnet was installed.
- The gradient cables were routed and connected to the gradient filter in a previous procedure in the section.
- The coolant lines have been routed thru J100 and J101 waveguides.
- The 4 inch air hose from the PGR cabinet has been routed to J104 waveguide.

Refer to cable map and connect remaining cables to SPW after Grounding and Leakage test has been completed.

1. Route and connect Run M3309 to J123 on the Transmit Switch.
2. For 3.0T Only: Route and connect Coax/RF cables Runs M1006 and M1015 to J107 and J108.
3. Route and connect Runs M3022 and M3023 from MagMon to J116 and J117.
4. Route and connect Run M3030 from E-Off Switch to J115.
Note: If system being installed is in an upgraded site, re-use and re-connect magnet room Run 297 to J115/J118.
5. Route and connect Run E3031 from GOC to J118.
6. Remove safety cover. Route and connect Run 623 to F1 and F3. Replace safety cover.



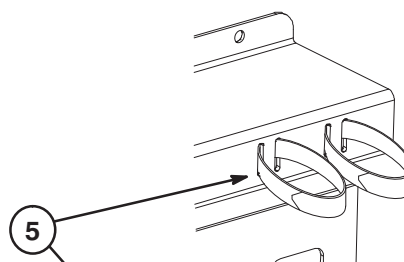
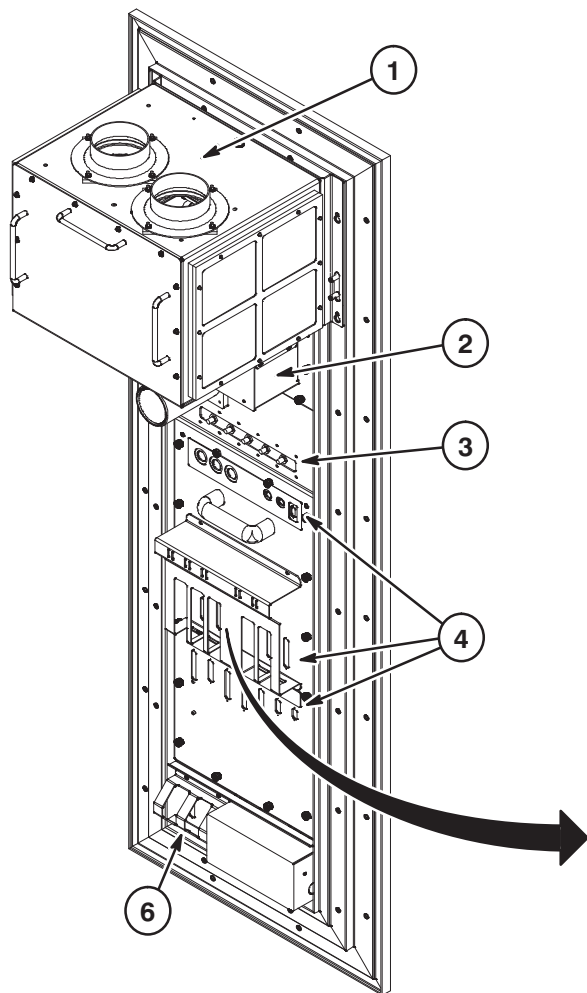
4-30 CONNECT PENETRATION PANEL CABLES

1. The Blower Box hoses and cables should have been installed and connected in previous procedures.
2. Connect Run M1152 to HOS module if HOS option is installed at site.
3. Connect cables to TR/DD Filter:

M3310 (J72)	M1341 (J73) (1.5T only)
M3311 (J76)	M1342 (J75) (1.5T only)
M3312 (J74)	
4. Connect cables to access panel:

Top Row –	M1320 (J29)	M1304 (J30)	M1303 (J32)	M3316 (J34)
Middle Row –	M3310 (J45)	M3306 (J46)	M3308 (J47)	M3304 (J48)
Bottom Row –	M3313 (J35)	M3311 (J36)	M3303 (J39)	M3302 (J40)
	M3312 (J41)	M3315 (J54)	M3314 (J42) (option)	
5. Make sure the middle row cables are routed in the correct orientation, connected, and secured in place with straps as shown.
6. Connect TDM power cables:

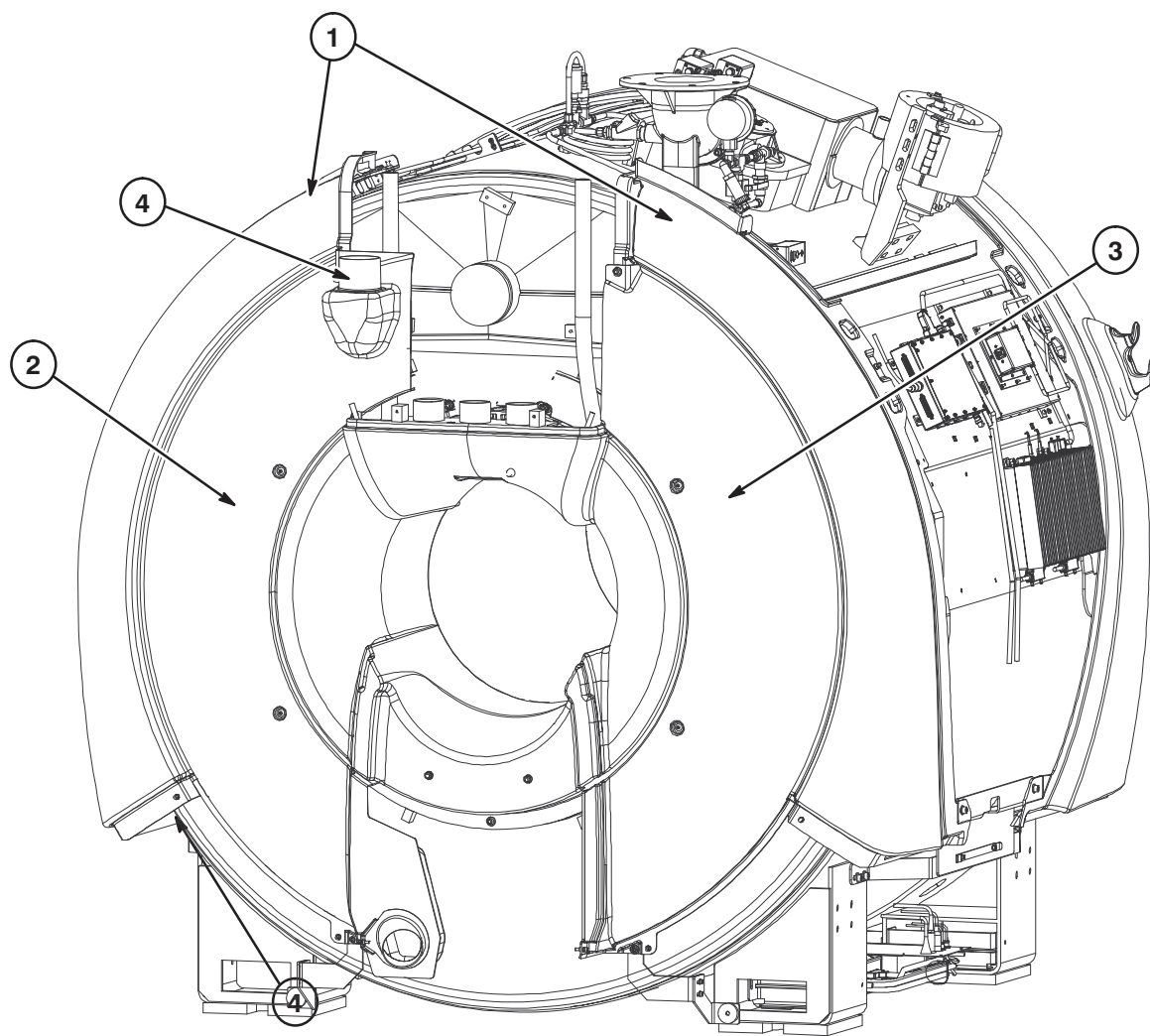
M3300 (J51)	M3301 (J52)
-------------	-------------



4-31 INSTALLATION OF MAGNET ENCLOSURE REAR COVERS

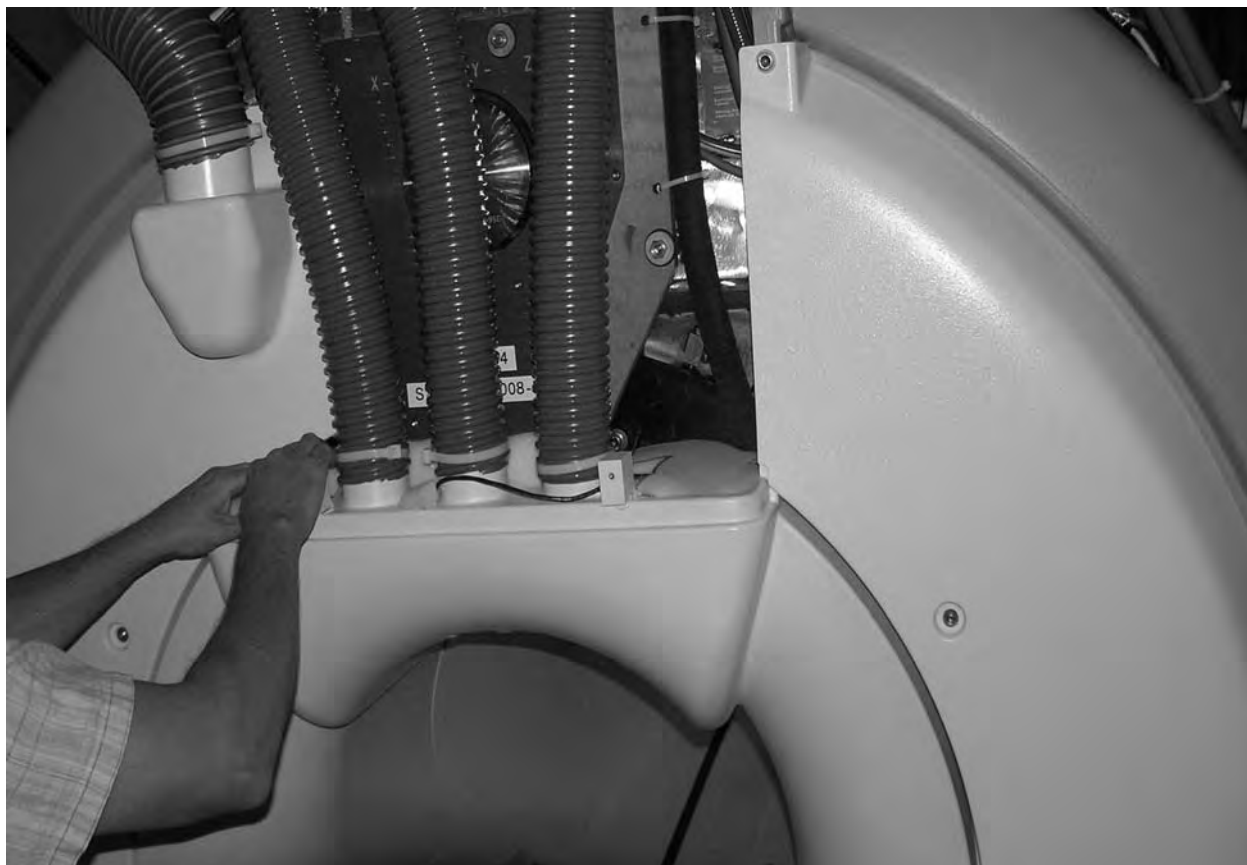
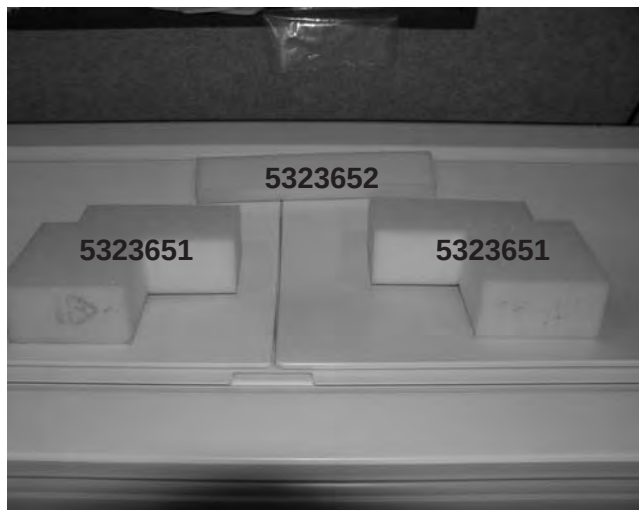
1. If previously removed, install left and right Rear Arc covers.
2. If previously removed, install right Rear Cover using the previously removed Button Head screws.
3. Install left Rear Covers using the previously removed Button Head screws.
4. If not already connected, route 4 inch, 14 meter long, RF HUB AIR Hose (5199506-3) in overhead cable tray and attach to socket on rear cover with 4 inch hose clamp (5266749).

NOTE: For clarity, rear pedestal, cables and hoses are not shown.



4-32 INSTALLATION OF REAR END BELL FOAM INSULATION

If previously removed, install the left, center, and right foam insulation (5323651 & 5323652) around the hoses at the top of the rear end bell



4-33 COIL AND DRESS CABLES IN PEN PANEL CLOSET

Inside the Pen Panel Closet, there are cable hooks for the purpose of storing excess cable lengths.

1. Separate, coil, and neatly dress any excess cable lengths and place them on available hooks.
2. Refer to Section 4-2 for information on coiling cables.



4-34 PNEUMATIC PATIENT ALERT INSTALLATION

The customer and users should be involved in the following decisions:

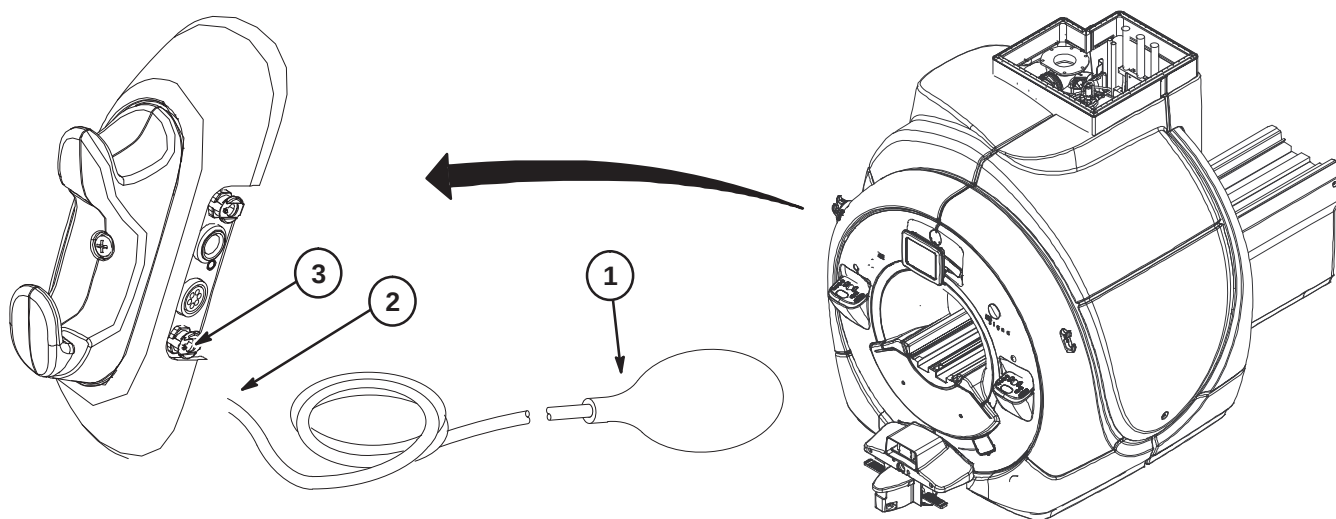
- Routing of Pneumatic Tubing
- Location of Control Box

CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

4-34-1 SQUEEZE BULB INSTALLATION

1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the "CALL" connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.
3. IConnect end of tubing to "CALL" connection on Remote PAC Interface Assembly.



4-34-2 MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

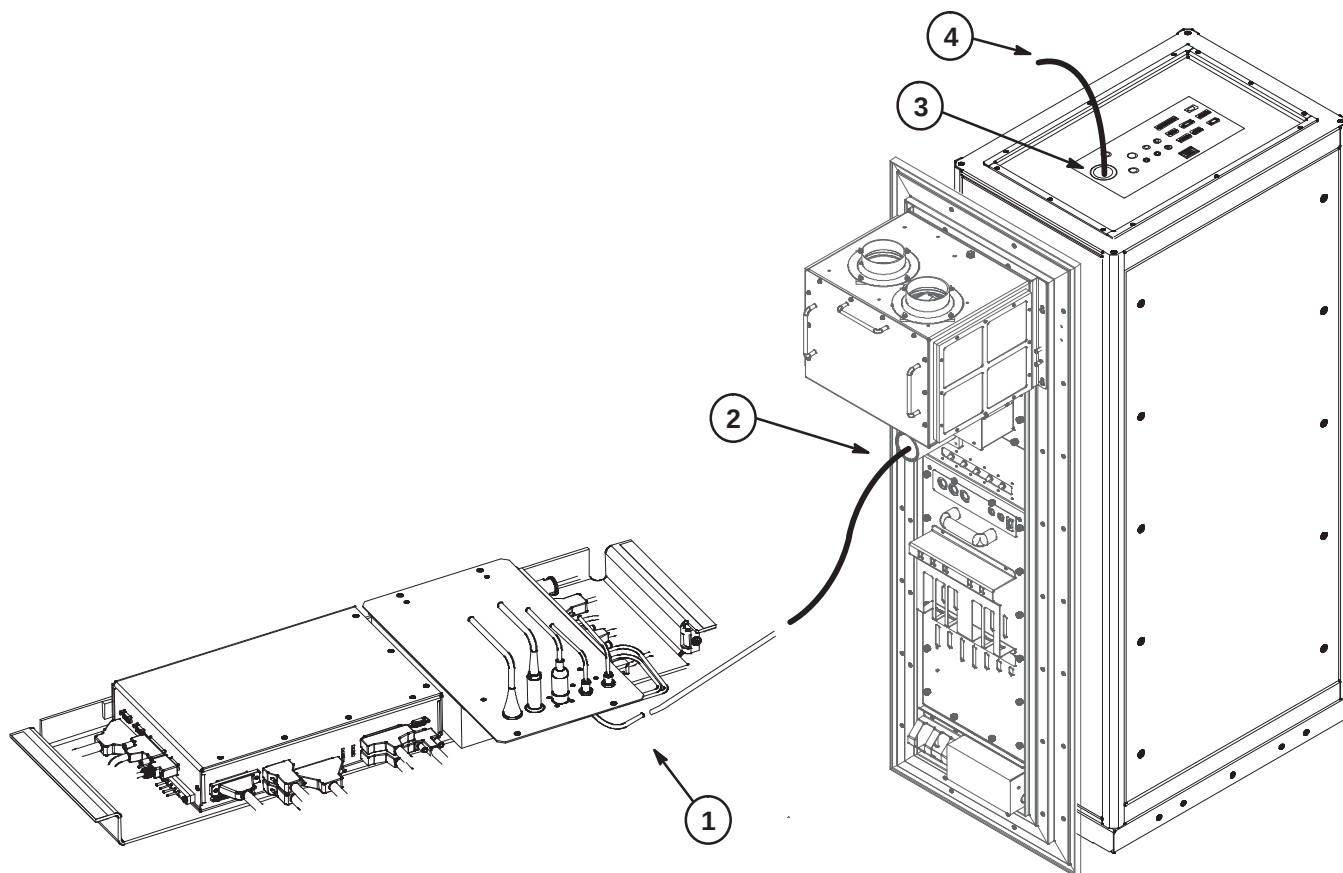
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

Also, make sure that the pneumatic tubing can be routed from the PAC connector through the overhead cable tray, and to the Penetration Panel without being pinched along the route by other cables.

1. Push end of pneumatic tubing onto PAC tubing connector located under right side of magnet.

Note: If the pneumatic tubing is to be pulled thru fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.

2. Route Pneumatic Tubing from PAC, thru the overhead cable tray to waveguide on Penetration Panel.
3. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide, thru the fiber optic opening at the top of the PEN Cabinet to where the Control Box will be installed.
4. Refer to Section 4-34-3 to determine if Extender Box is required. If Extender box is not required, route pneumatic tubing thru overhead cable tray in Equipment room to Control Box location.



4-34-3 DETERMINING NEED OF EXTENDER BOX

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46-317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:**
The extender is not needed. Proceed to Procedure 4-34-6, CONTROL BOX INSTALLATION.
- **If the pneumatic tubing is not long enough to reach control box:**
The extender is needed. Proceed to Procedure 4-34-4, INSTALLATION OF EXTENDER KIT.

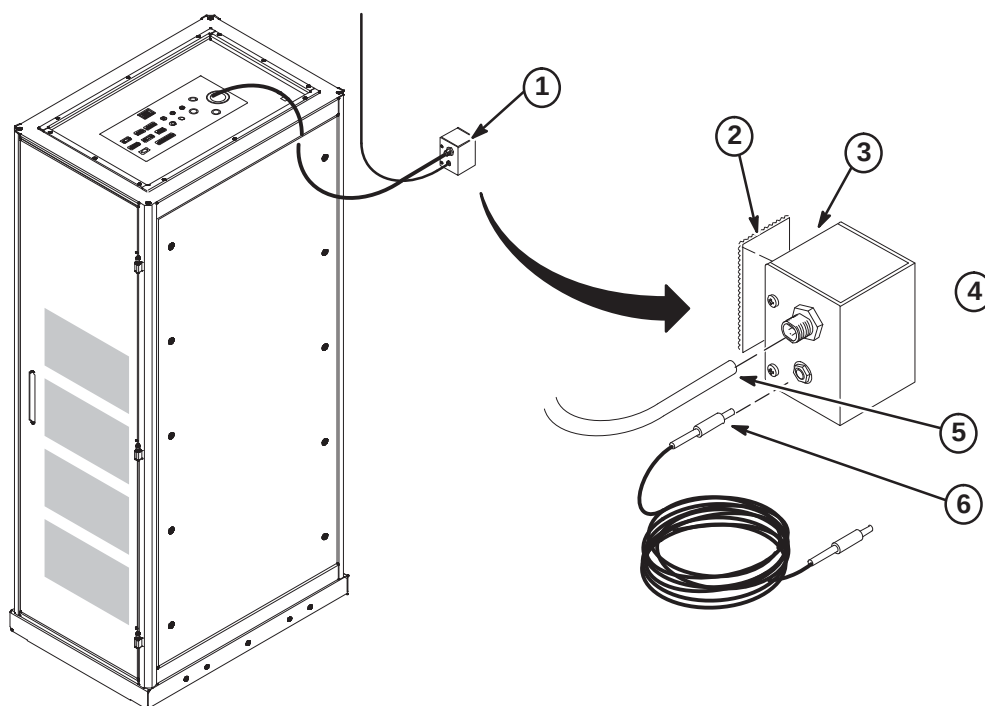
4-34-4 INSTALLATION OF EXTENDER KIT (46-317758P2)

CAUTION

Do not mount Extender Box in Magnet Room because the box contains ferrous parts.

Install Extender Box

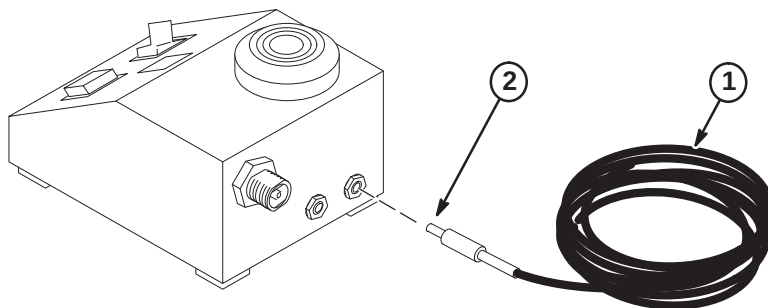
1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed-thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.



4-34-4 INSTALLATION OF EXTENDER KIT (46-317758P2) (Continued)

Connection of Extender Wire to Control Box

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.



4-34-5 CONTROL BOX INSTALLATION OPTIONS

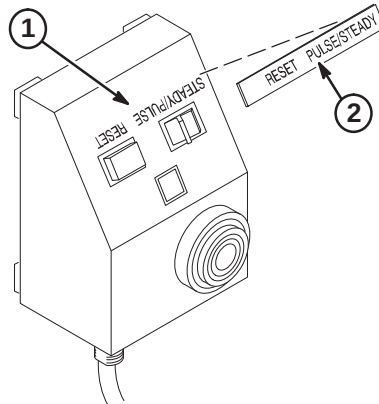
Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

- Mount control box on a wall or other vertical surface per Procedure **4-34-6**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Mount control box under shelf per Procedure **4-34-6**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

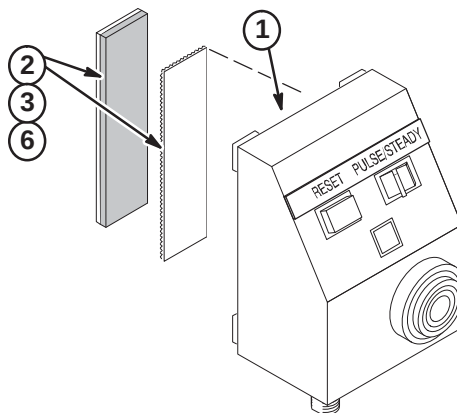
Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

4-34-6 CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate and apply over silk screened lettering as shown.
3. Select one of the following for installation of Control Box:
 - If control box will be mounted to WALL or SHELF with Velcro, go to Procedure **4-34-7**
 - If control box will be mounted to WALL with screws, go to Procedure **4-34-8**
 - If control box will be mounted to SHELF with screws, go to Procedure **4-34-9**

**4-34-7 Control Box Mounted With Velcro**

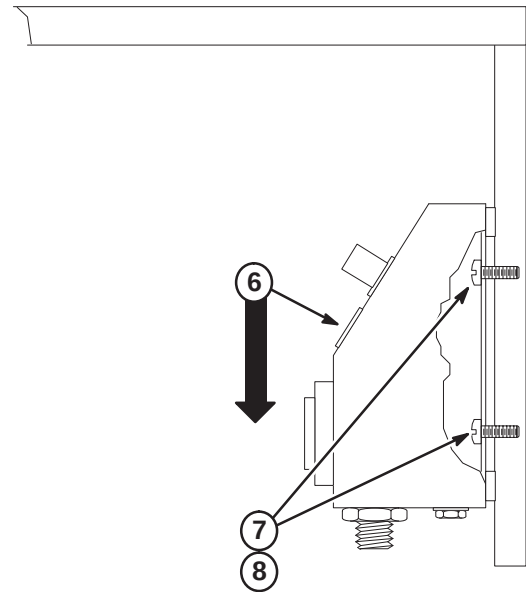
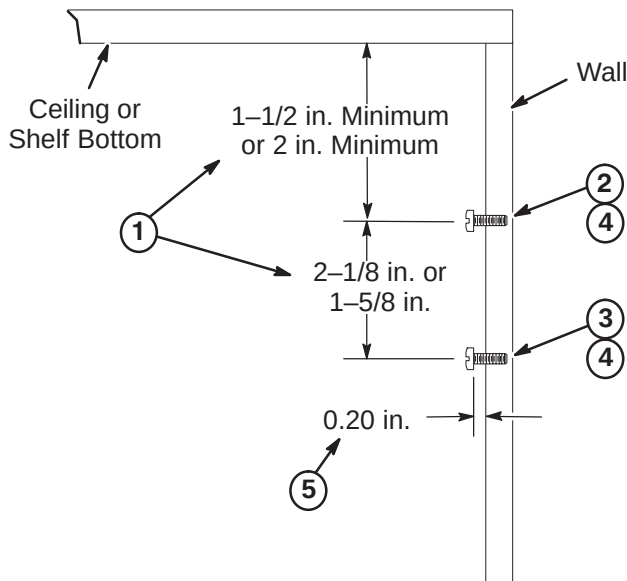
1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. Mount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to Procedure **4-34-10, FINAL CONNECTIONS TO CONTROL BOX**.



4-34-8 Installing Control Box with Mounting Screws on Vertical Surface

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

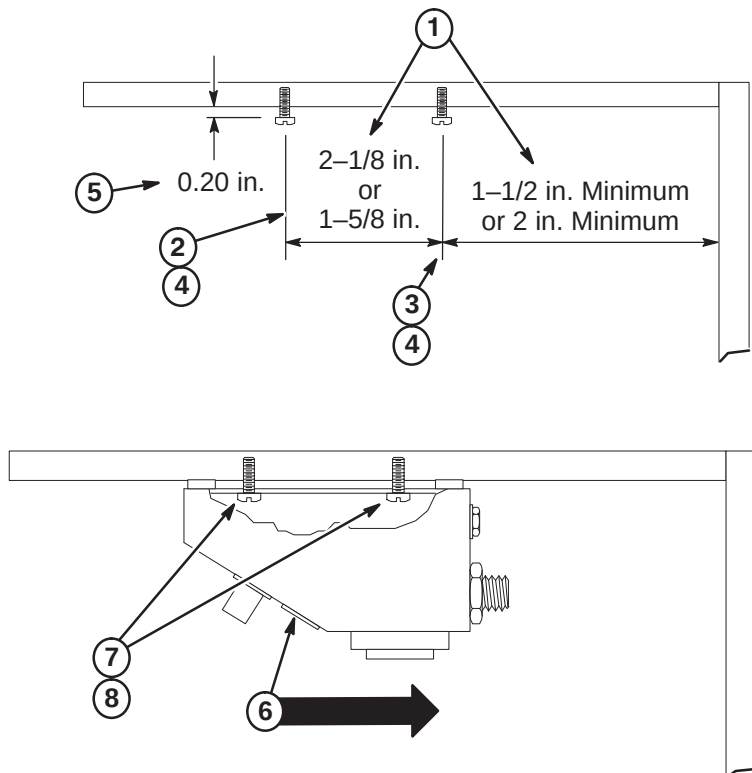
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4-34-10**, FINAL CONNECTIONS TO CONTROL BOX.



4-34-9 Installing Control Box with Mounting Screws Under Shelf

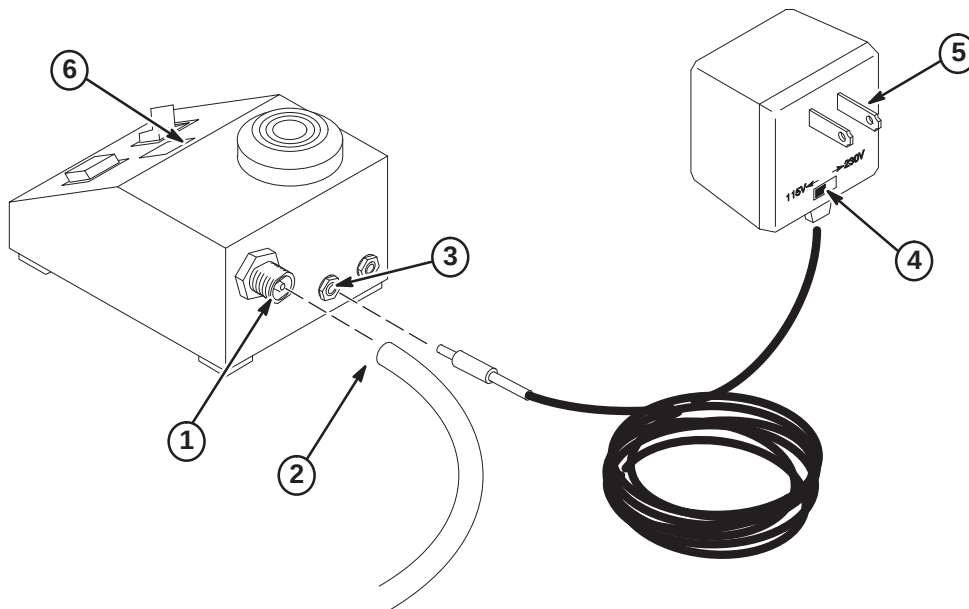
Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4-34-10**, FINAL CONNECTIONS TO CONTROL BOX.



4-34-10 FINAL CONNECTIONS TO CONTROL BOX

1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed-thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is lighted.



4-35 EMERGENCY OFF CONNECTIONS

Two Emergency Off (E-off) pushbuttons are supplied with the system. One pushbutton is for installing in the magnet room. The second pushbutton is optional and can be installed in the equipment or operator room.

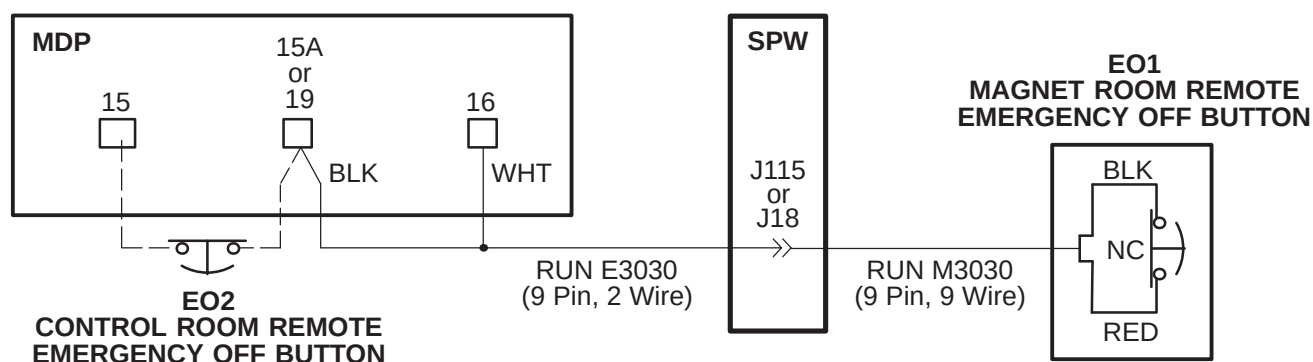
Note: If MR750 system being installed is in an upgraded site, Run 297 in the Magnet Room can be re-used and re-connected to J18/J115.

Refer to applicable site installation drawings for detailed wiring diagram.

1. In the Magnet room, connect Run M3030 to the SPW and route the cable to the remote E-off pushbutton location and trim cable to length. Locate Red and Black pair and prepare ends for applicable terminals.

Note: Run M3030 has nine wires. Make sure to use the black and red pair when connecting to the E-off pushbutton.

2. Connect Run E3030 to the Secondary Pen Wall (SPW) and route the cable to the Main Disconnect Panel (MDP) and trim cable to length. Locate white and black pair of wires and prepare ends for applicable terminals.
3. If the MDP power has been connected, have the proper personnel connect the E-off wiring. See illustration below.



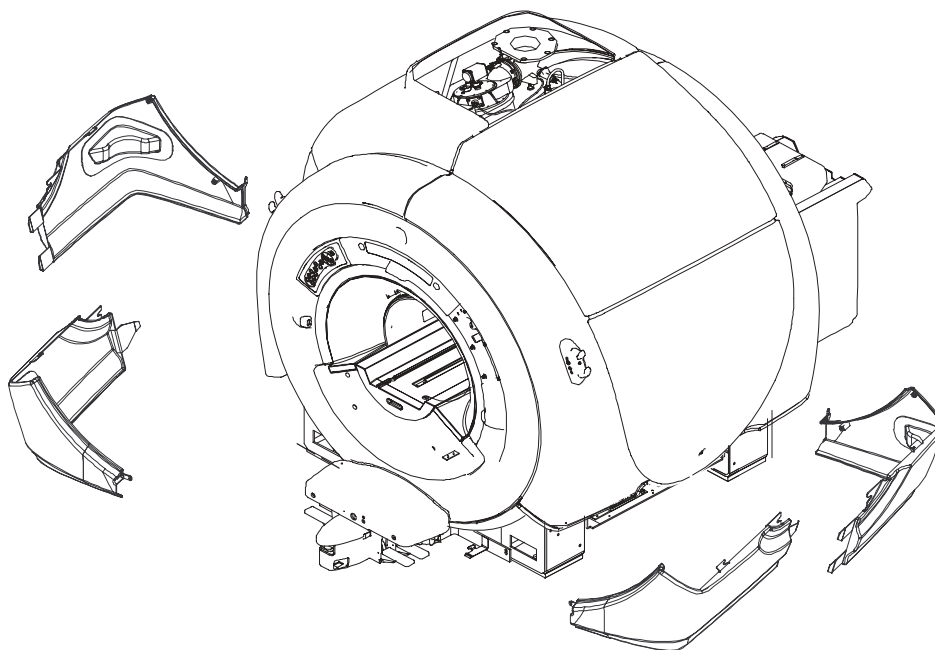
SECTION 5 – FINAL ENCLOSURE INSTALLATION

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5-1 INSTALL SKIRT COVERS

Install the left and right side skirt covers as shown in the following procedures.



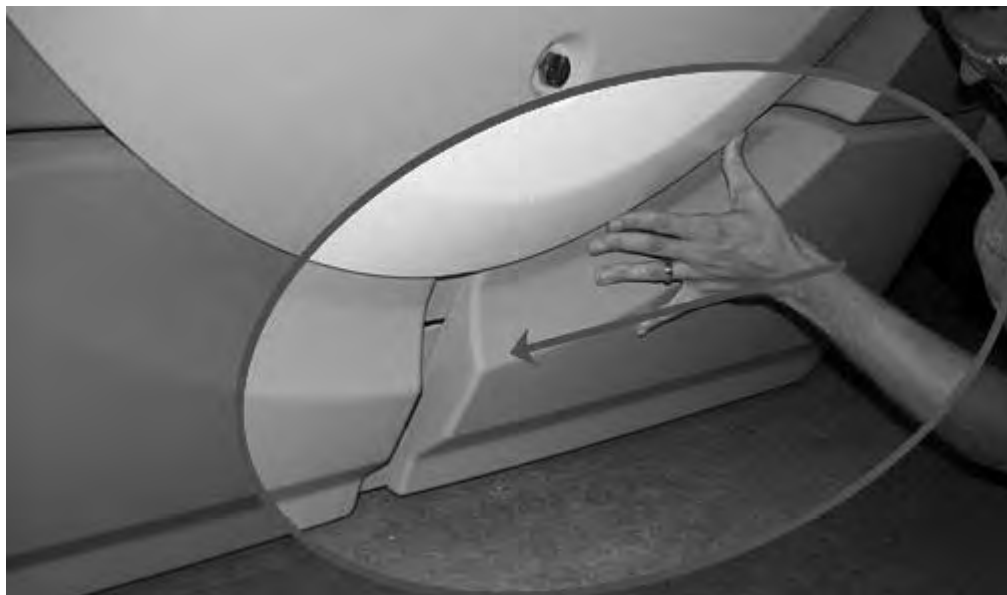
5-1-1 Align Skirt's Hook with Slot

Align fastener on skirt cover to the notch on the arc cover and slide cover towards the center of the magnet.



5-1-2 Align Skirt Tabs

Align tabs as to interlock with tabs on adjoining skirt cover.



5-1-3 Snap Covers in Place

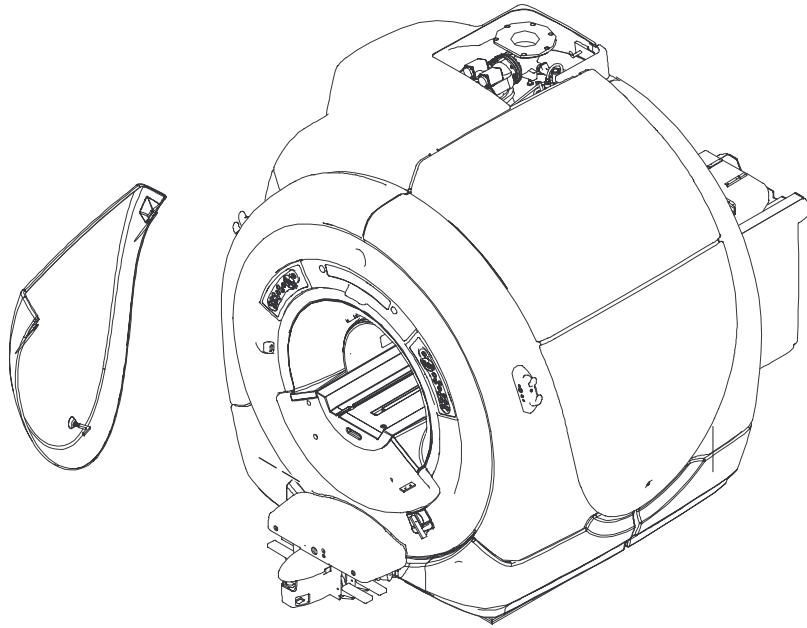
Snap poppers on the front end of arc cover into place.



5-2 INSTALL LEFT SIDE COVERS

Install the left side enclosure covers as shown in the following procedures.

Note: To install the left side covers, the right side cover must be installed



5-2-1 Using the Arc Cover as Guide Rails

Using the arc covers as guide rails, slide the side cover up to meet the right side cover at the top of the magnet.



5-2-2 Align Tabs at Left and Right Covers

Hold the handles away from the enclosure to ensure that the alignment tabs get properly slotted under the right side's slot.

**Note:**

This is not easily accomplished since you cannot see all the alignment tabs while holding on to the cover's handles. Before proceeding to install the lower side cover, go over to the front and rear of the magnet and visually verify that the alignment tabs have been properly inserted and are not protruding over the top of the right side cover.



Make sure tab is properly placed.

5-2-3 Place Arc Cover Foot

Set the round foot on the enclosure side of the cover into the upper oval on the arc cover. Snap into place.



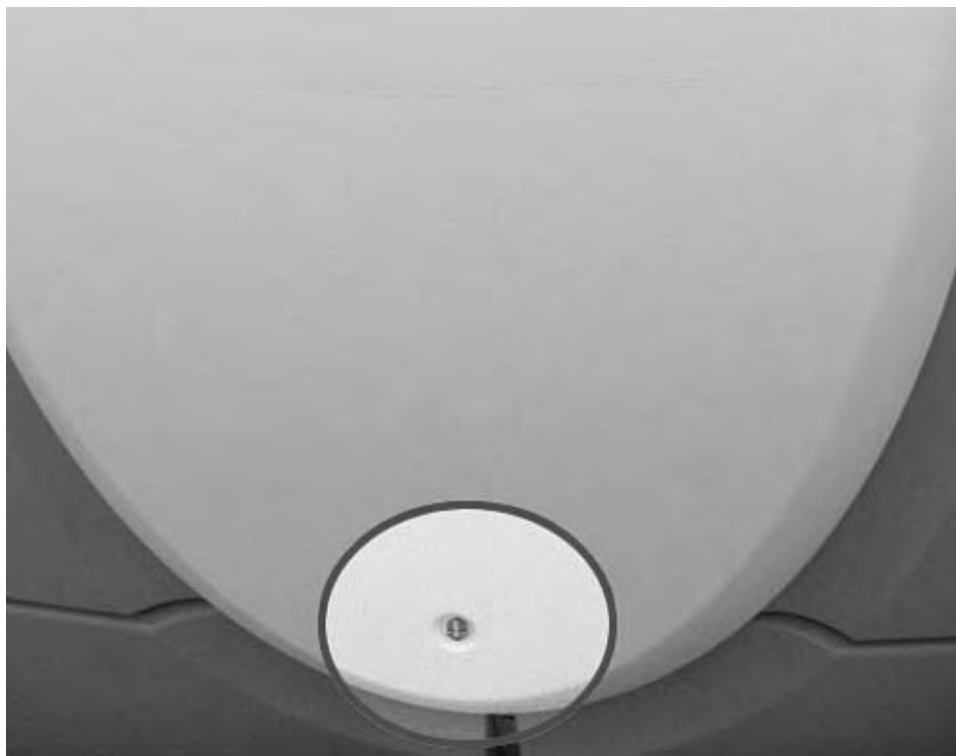
5-2-4 Installing Left Side Cover

Hold the sides of the Lower Cover and align the round feet on the enclosure-side of the lower cover with the arc cover's ovals. Snap into place.



5-2-5 Secure Side Cover

Latch the cover into place by turning the screw clockwise so the latch is secure.



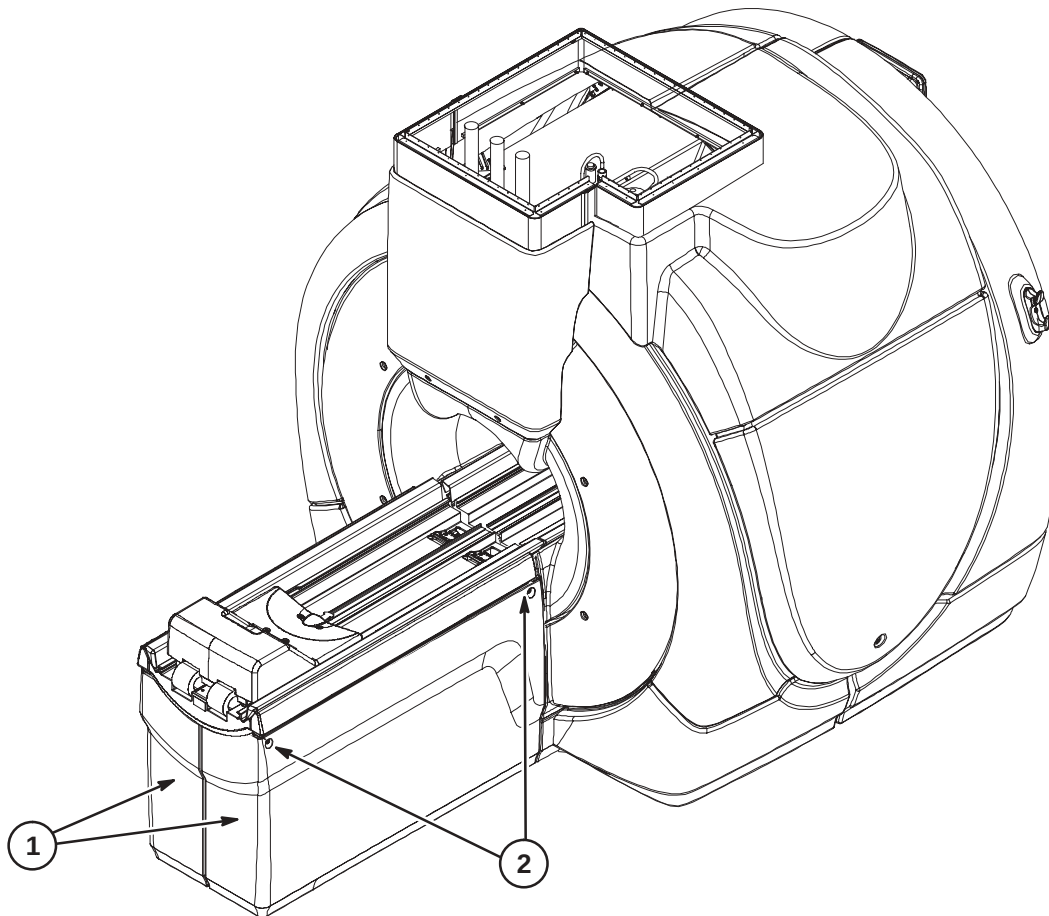
5-3 INSTALL CHIMNEY COVER

Position Chimney cover and secure in place with provided screws.



5-4 INSTALL REAR PEDESTAL COVERS

1. Move Rear Pedestal covers into place.
2. Attach each with two provided screws,

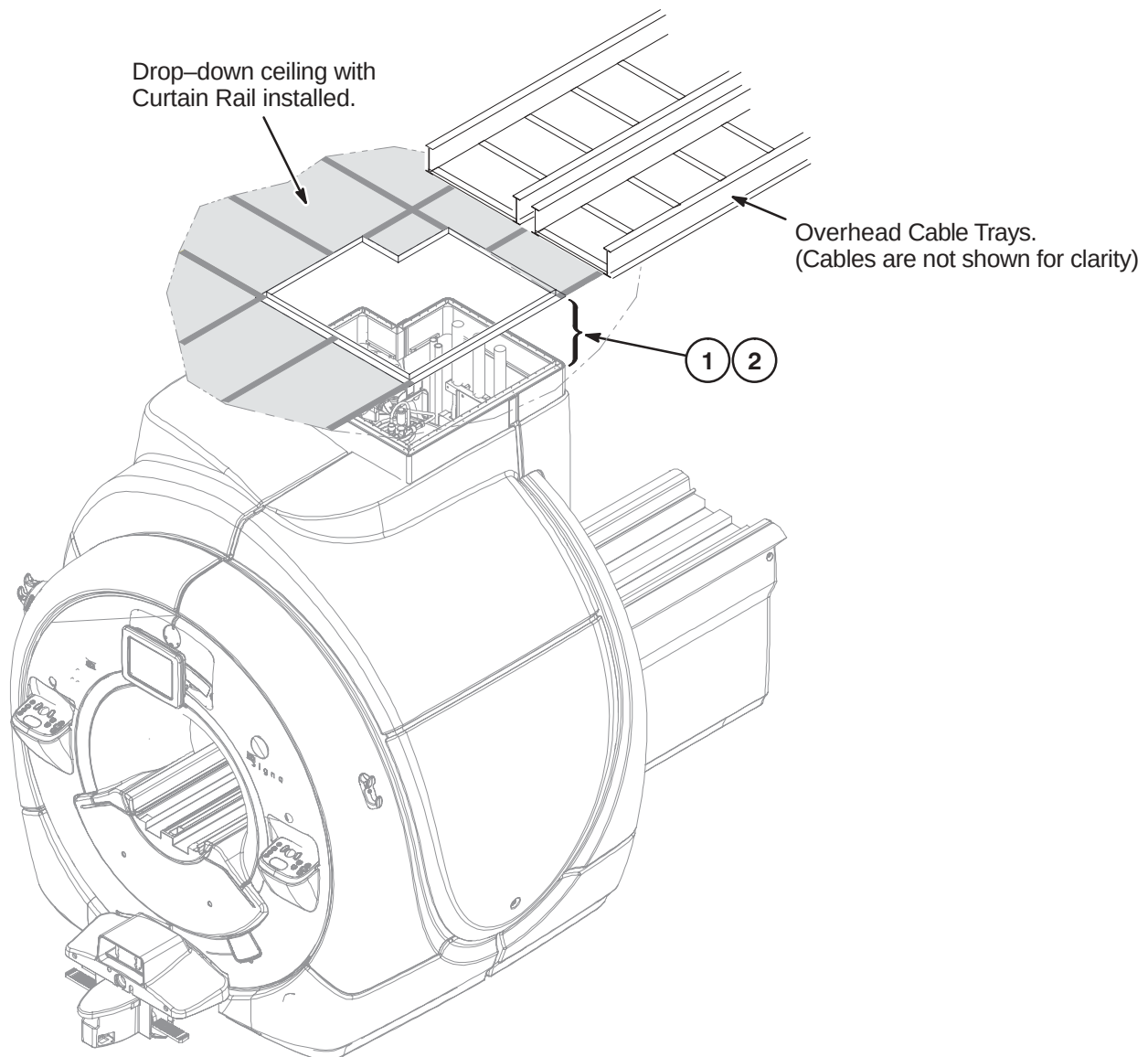


5-5 INSTALL CABLE CURTAIN AT TOP OF MAGNET (Optional)

If the site has designated that the Curtain Cover Kit (5241434) is to be installed, the Curtain Rail (5237762 & 5237763) should have already been positioned and installed in the drop-down ceiling. The following procedure provides instructions for cutting to length the curtain and installing it on the curtain rail. The Curtain and associated installation parts were delivered with the Curtain Cover Kit prior to the delivery of the components that are installed using the following procedure steps.

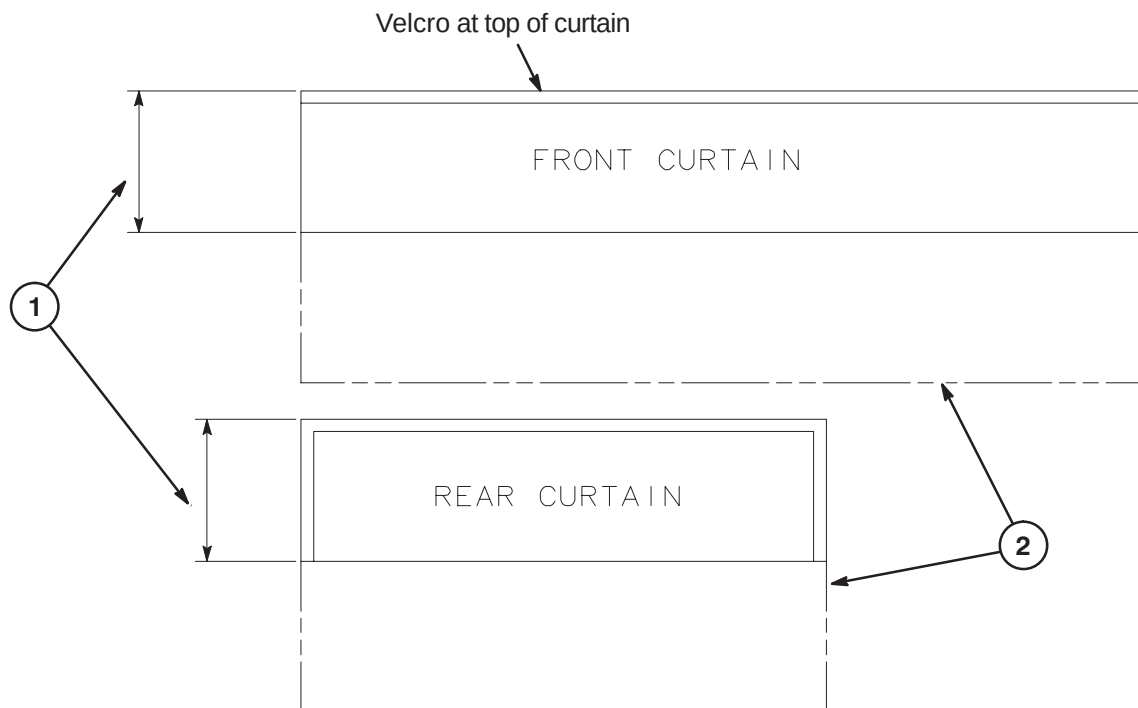
5-5-1 Measure for Length of Curtain

1. Measure the distance from the ceiling to the top covers of the magnet.
2. **Alternative Measuring Method** – Measure the distance from the ceiling to the top of the magnet cylinder and subtract 212mm



5-5-2 Cut Curtain to Length

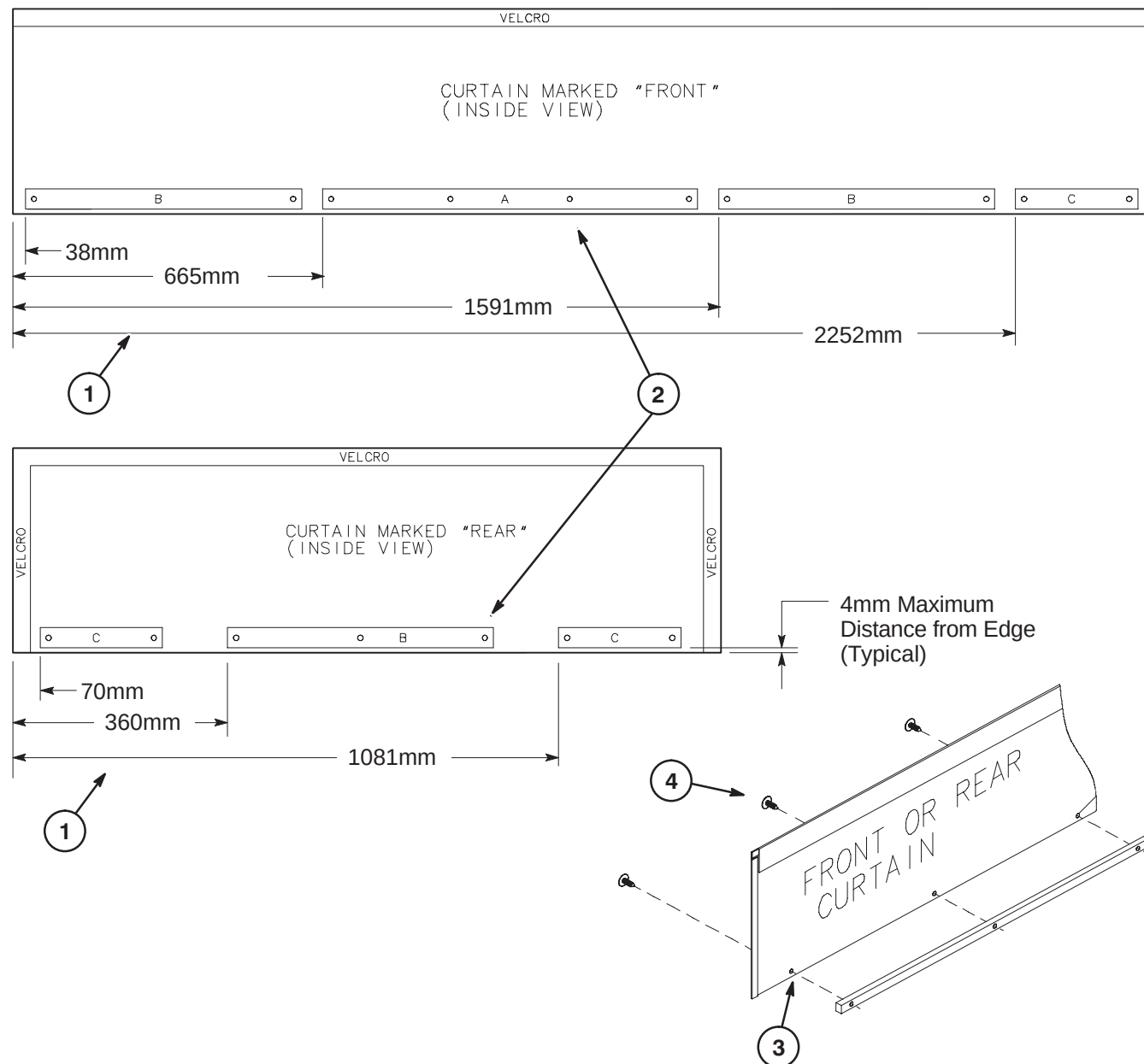
1. Measure the length of curtain to be installed for both Front and Back curtain.
2. Cut curtains to length.



5-5-3 Attach Weight Bars to Curtains

Curtain weight bars are to be installed at the bottom of the Front and Back curtains as indicated in illustration below.

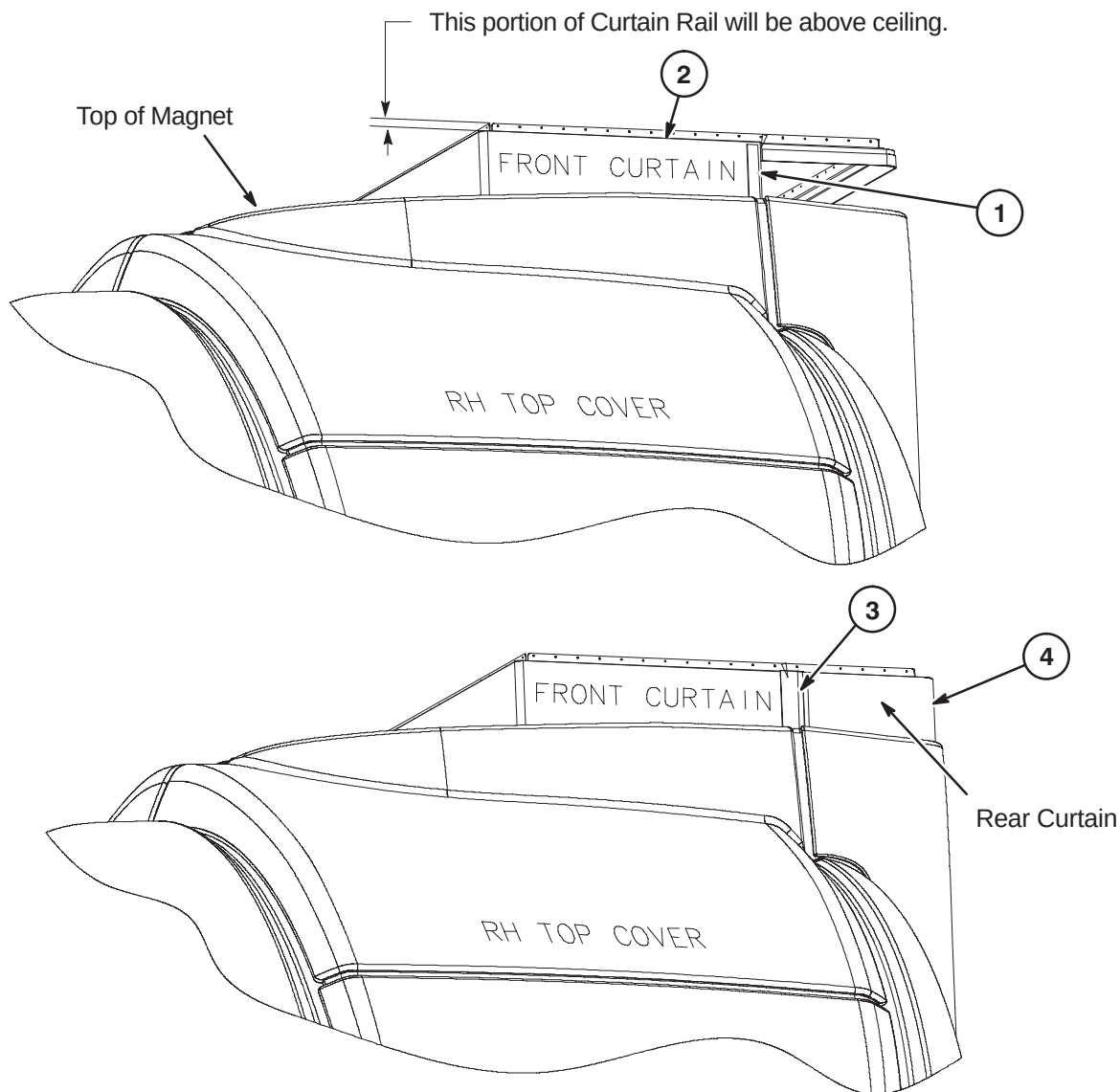
1. Measure and mark the indicated location distances below on the inside of each curtain.
2. Position each of the different size A, B, and C Curtain Weight Bars and mark the mounting holes for each.
3. Create a small hole in the curtain at each mounting hole location.
4. Attach the A, B, and C bars to the curtain by inserting a Plastic Pin Clip (5241226-3) thru the hole in the curtain into the mounting holes of each bar.



5-5-4 Attach Weight Bars to Curtains

Note: The views below show the enclosure side, top, and cable covers installed. It may be preferable to install the curtain covers with these covers removed.

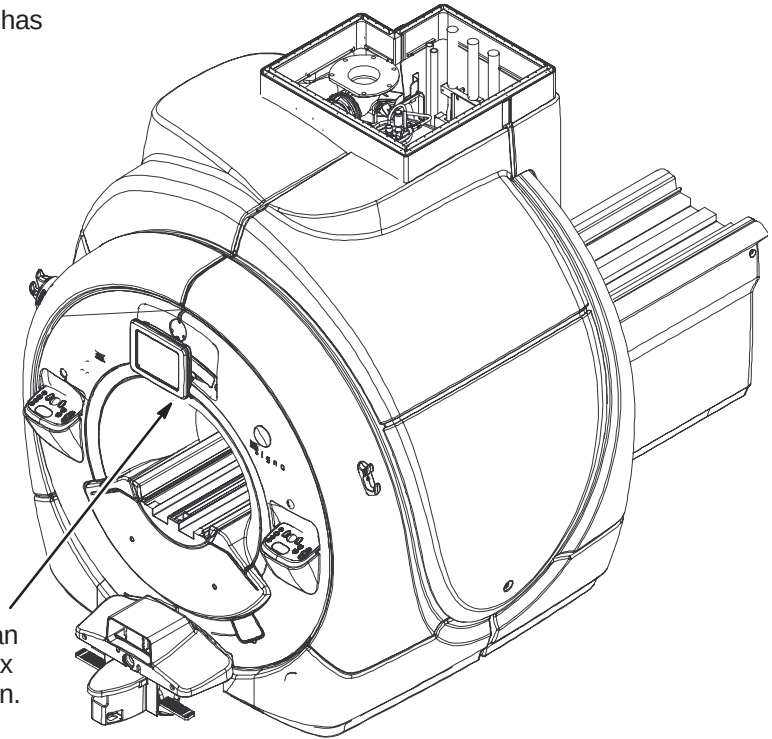
1. Install Front cover first. Align the right hand edge of curtain with gap in ceiling rail velcro.
2. Align top edge of curtain flush with velcro as curtain is being wrapped around to the left side.
3. Install Rear Cover. Start on right hand side with overlap on Front Curtain.
4. Wrap around back to left hand side.



5-6 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below the laser light.



- ① For those sites which require a label other than English, peel off the appropriate label and affix over the English label at the laser light location.



French



German



Portuguese



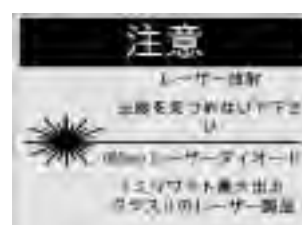
Spanish



Italian



Swedish



Japanese



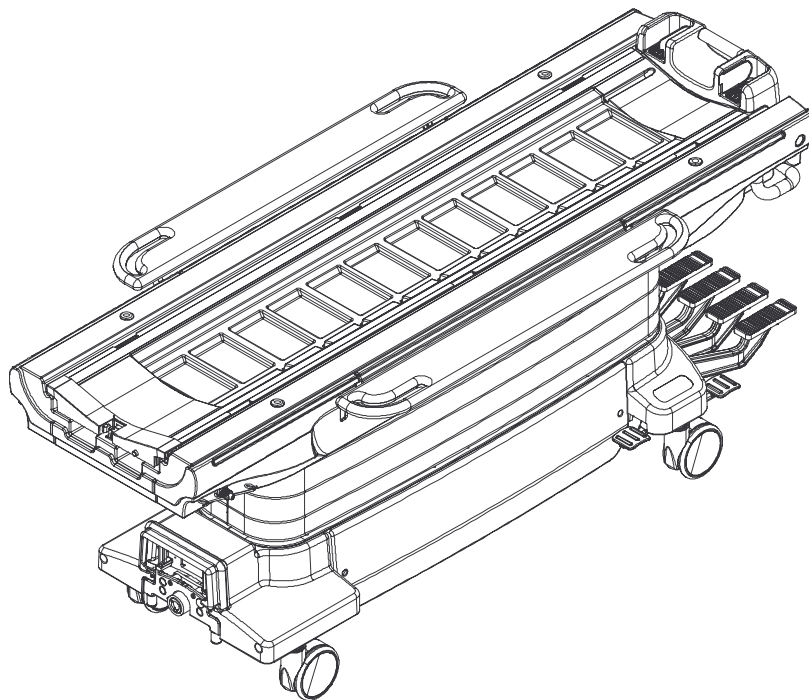
Chinese

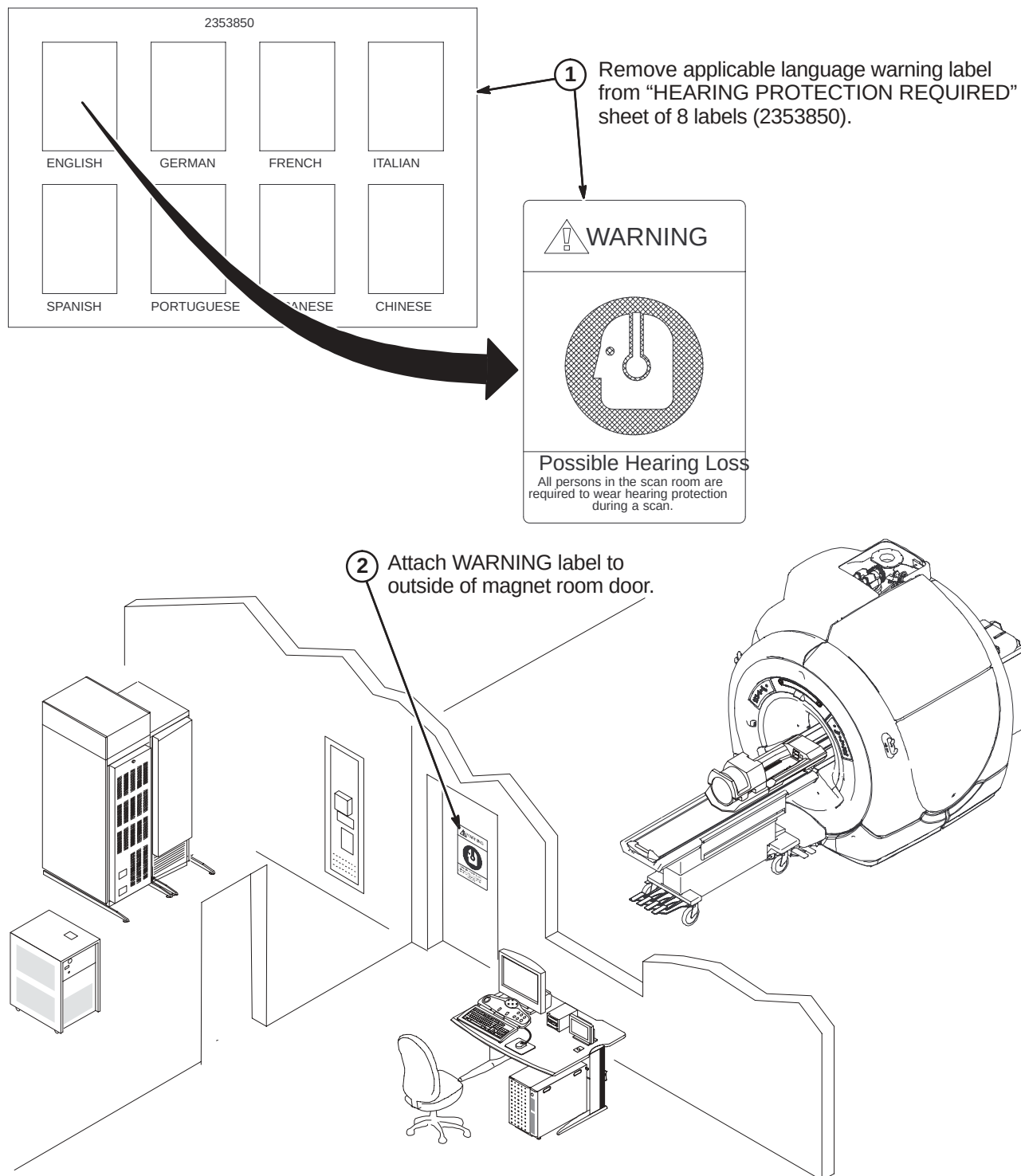
5-7 PATIENT TRANSPORT INSTALLATION

Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanical procedures in MR Service Methods DVD, Set up and Calibration: Patient Transport

Note: If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.



5-8 INSTALLATION OF HEARING LOSS WARNING SIGN

SECTION 6 – OPERATOR WORKSPACE

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6-1 OPERATOR WORKSPACE TABLE INSTALLATION

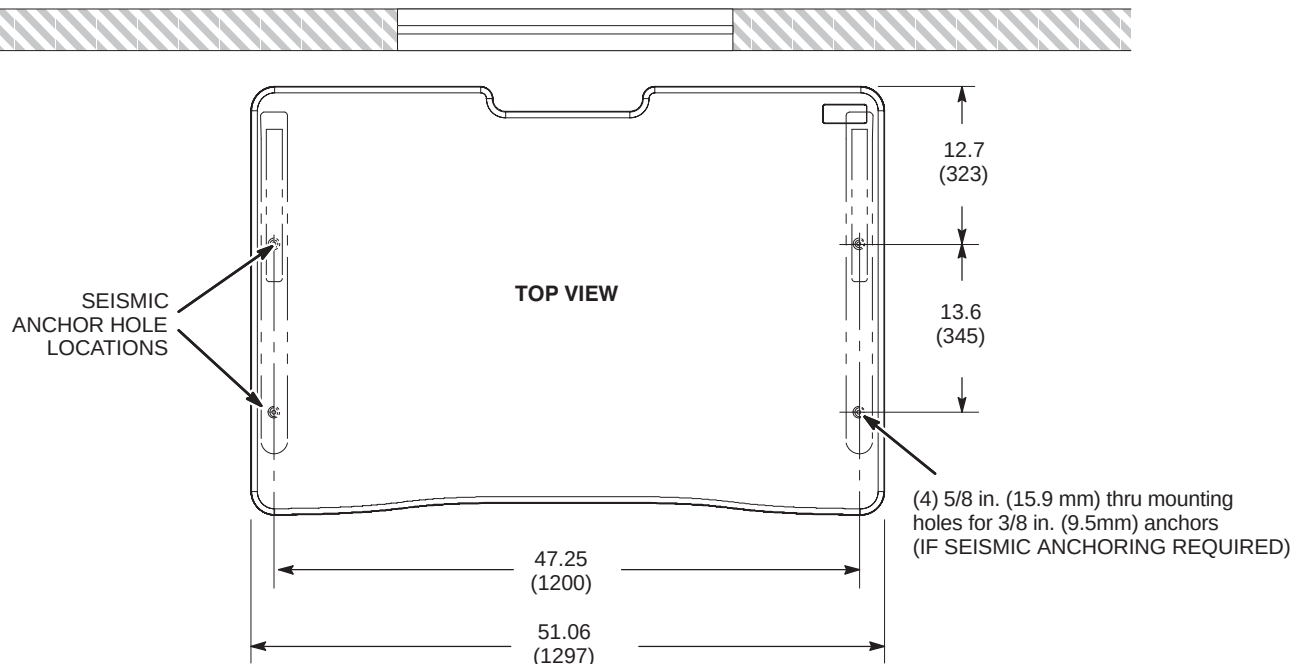
The customer may have the Operator Workspace components installed on a customer designed countertop of other type of table. If the GE supplied Operator Workspace Table is to be installed, refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions* (5189137-50) to assemble the model 5191031 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.

Note

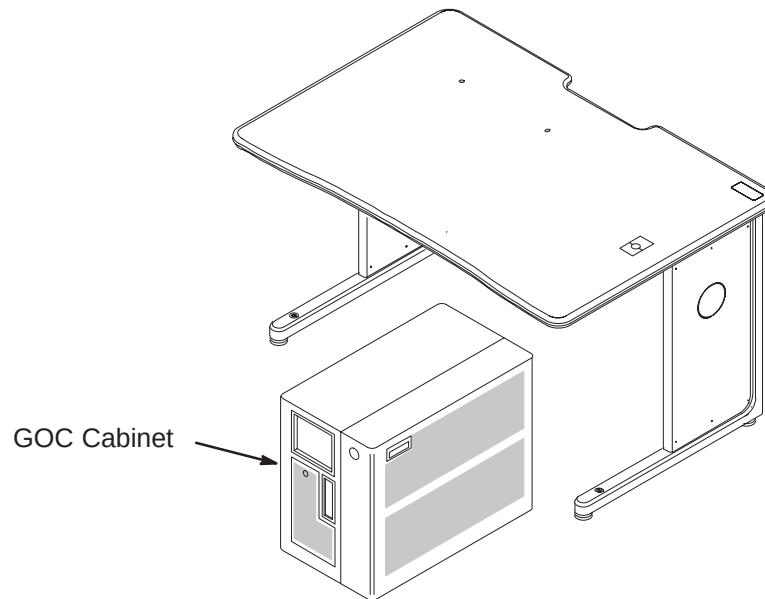
The design of the new model 5191031 table does not require anti-tip anchors. This table meets the ANSA/BIFMA anti-tip standards.

3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.



6-2 POSITION GOC

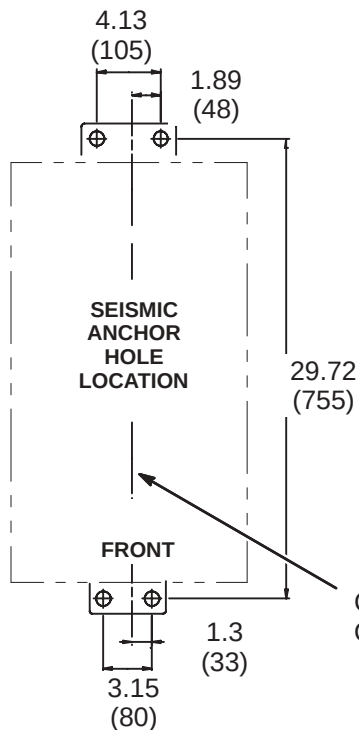
Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



6-3 SEISMIC ANCHORING (IF REQUIRED)

GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

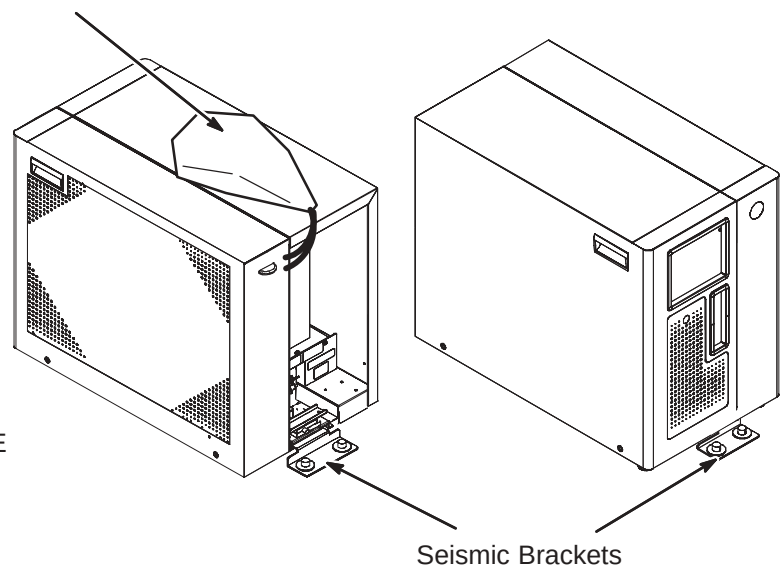
If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Preinstallation manual. Illustration below indicates hole location for installing brackets.



Plastic bag for the cables and GOC Seismic Mount kit

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



6-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC Cabinet is shipped with cables that will be routed and connected to components installed on the top of the OW Table.

- Two power cords
- One video cable
- Keyboard cable
- SCIM cable
- Modem cable (Option)

These cables are already connected to the computer as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will to be routed thru the rear opening.

Note

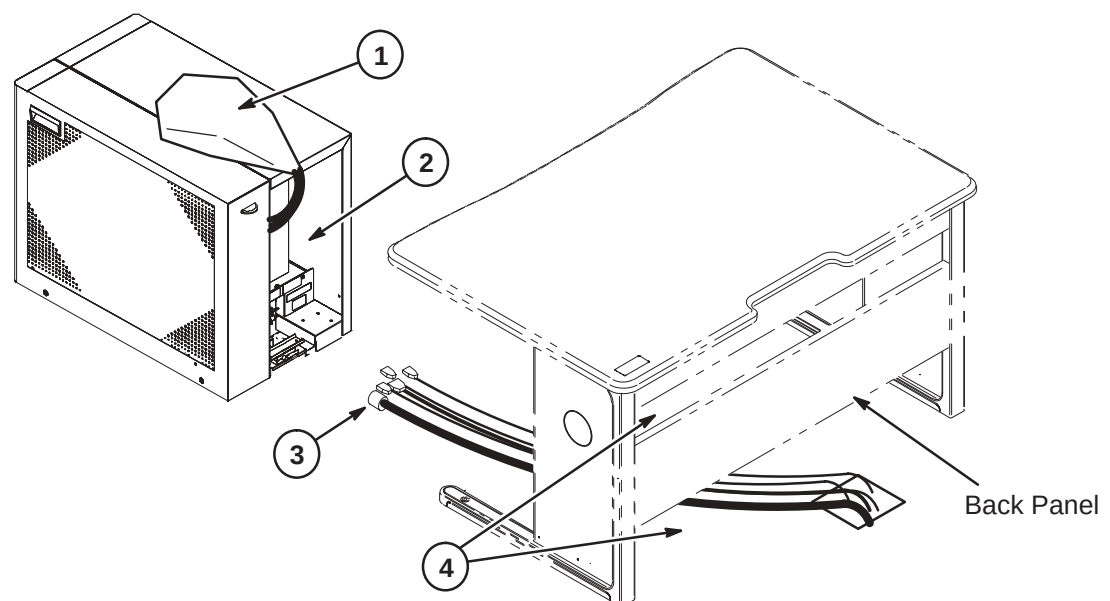
FOR UPGRADES ONLY: SCSI signal cable and SCSI power cable are not connected to the computer. For upgrade site, re-use the signal & power cable for SCSI tower which were used for the system before upgrade. When reusing the SCSI cable, it is required to attach the new label on the old label because the destination (J#) of power cable is different between old GOC and new GOC. New label is included in the upgrade kit for DVMR.

1. Runs E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening. (These cables are inside of plastic bag)
2. Runs E0003, E3011, E3015, E3017, E3018, the power cord for the Patient Alert system, and any other option cables will be routed thru the rear opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.

Note

Make sure a cable service loop of at least 4 feet (1300mm) is provided for future servicing.

4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.
5. Refer to Section 4-17 for connection of cables to GOC.



6-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

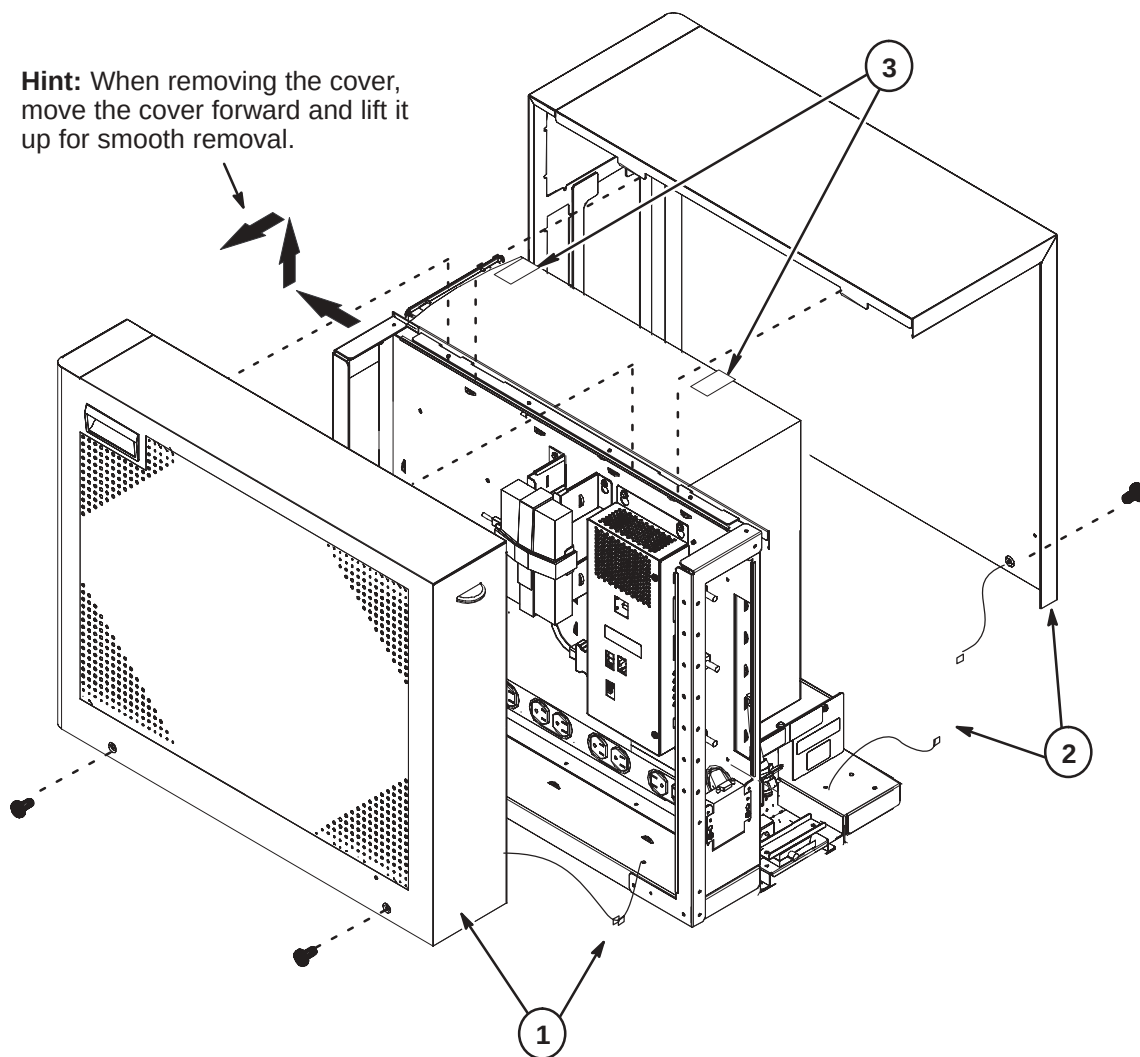
1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

3. Peel the two shipping tapes off from PC.

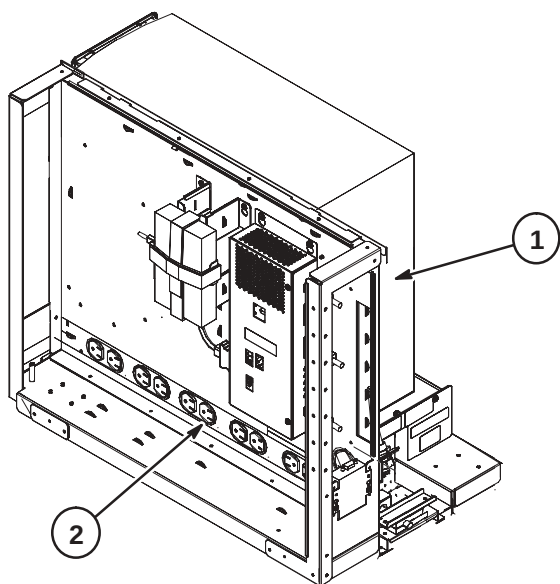
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



6-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

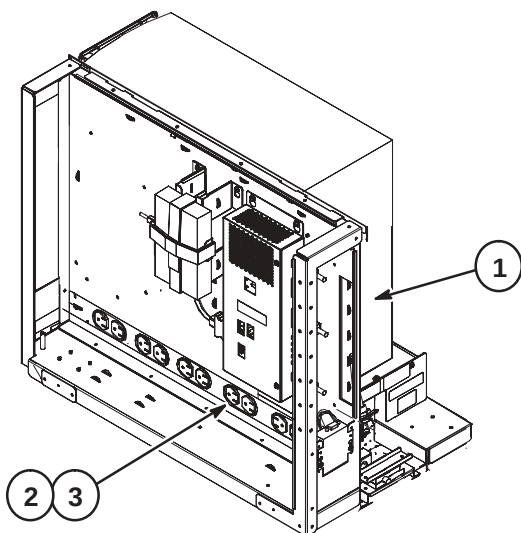
If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.

**6-7 INSTALL SCSI TOWER CABLE TO GOC CABINET (If Applicable)**

For sites being upgraded and have a SCSI Tower, the SCSI Tower will be placed on Operator Table or GOC. Reuse the cables of SCSI tower from old system.

1. Connect SCSI cable (RUN1064) to the Linux PC SCSI 1 port.
2. When reusing the SCSI power cable, attach the new label on the old label because the destination (J#) of power cable is different between old GOC and new GOC. New label is included in the upgrade kit for MR 750.
3. SCSI Tower power cord (RUN1070) must be connected to J6 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



6-8 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

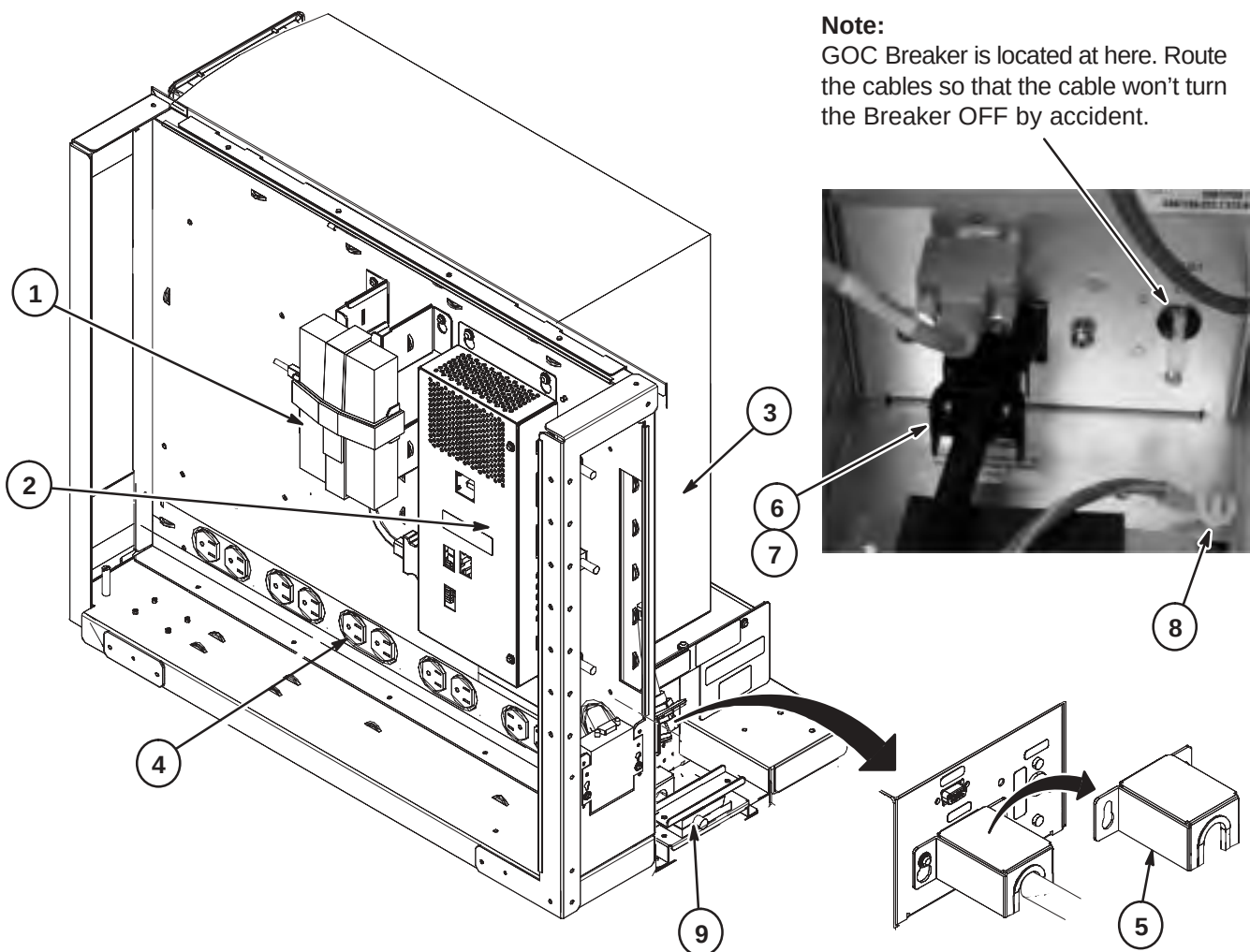
1. Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

2. Connect Run E3031 to J7 on GOC AA.
3. Refer to Section 6-10 for all cables connected to Linux PC computer.
4. Refer to Section 6-9 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
5. Before connecting GOC power cable E0003 to J1, loosen screws and remove cover.
6. Route and connect GOC power cable Run E0003 to J1 on PDU.
7. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5 after connecting cable and tighten screws.
8. Connect Run 4008 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
9. Tighten clamp to secure cables in place.

Note:

GOC Breaker is located at here. Route the cables so that the cable won't turn the Breaker OFF by accident.



6-9 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

- 1. If Modem is installed, route power cable and connect to J7.
- 2. Tie-wrap point for Modem power cable.
- 3. Route power cables connected to J3 thru J7 through access hole.
- 4. route and connect Patient Alert power cable to J3 and tie-wrap at this point.
- 5. Tie-wrap power cables J4 to J7 at this point.

POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Comp
J5	Host LCD
J6	SCSI Tower (upgrade)
J7	Modem (option)
J8	
J9	LAN Hub
J10	
J11	IRD (option)
J12	

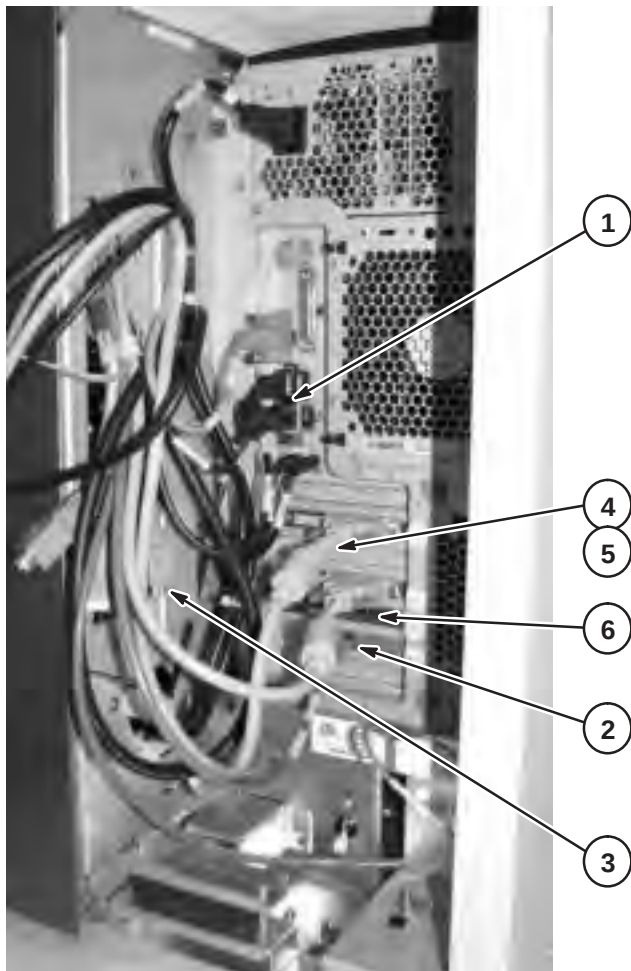
6-10 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the PC are described in the following two sections. Several of the cables shown are delivered with the GOC and are preconnected.

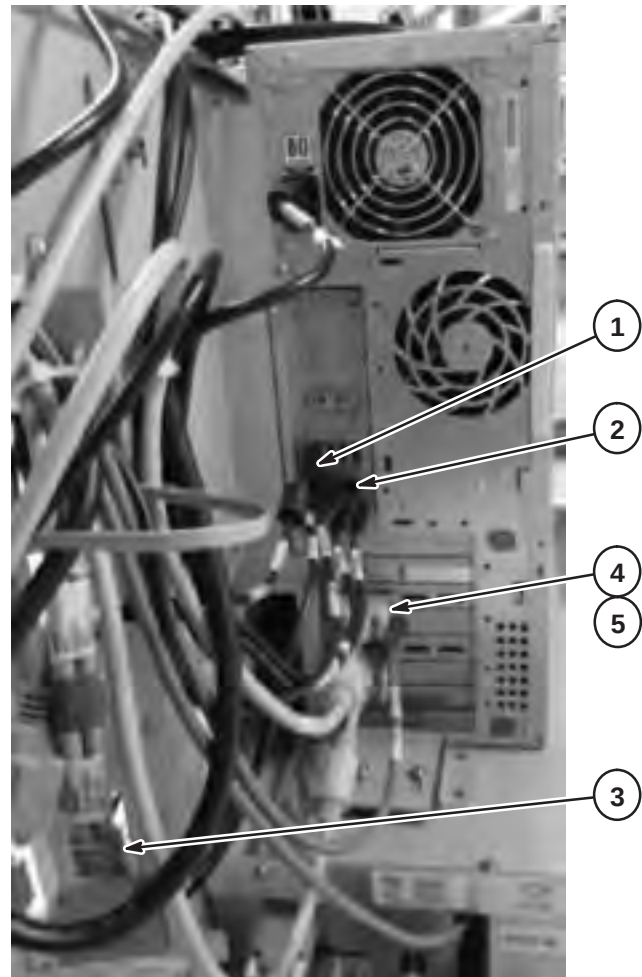
Refer to the applicable PC, XW8400 or Z400, shown below for cable routing and connections.

1. Route and connect the following to the USB ports: Keyboard USB extension cable – Run E3048, Mouse USB cable, Serial Converter USB connector, and IRD USB cable (option)
2. Connect Run E3011 to ethernet connection Port B (XW8400) or upper ethernet port (Z400).
3. Connect Runs E3015, E3017, and E3018 to FJ6, FJ5, and FJ7 as marked.
4. Monitor cable Run E3046 will be connected to MNTR1 port.
5. If the IRD option is installed, connect DVI-D cable to MNTR2 port.
6. Connect Run 1064 to SCSI1 for SCSI Tower if applicable.

NOTE: SCSI Tower is not supported for Z400.



XW8400

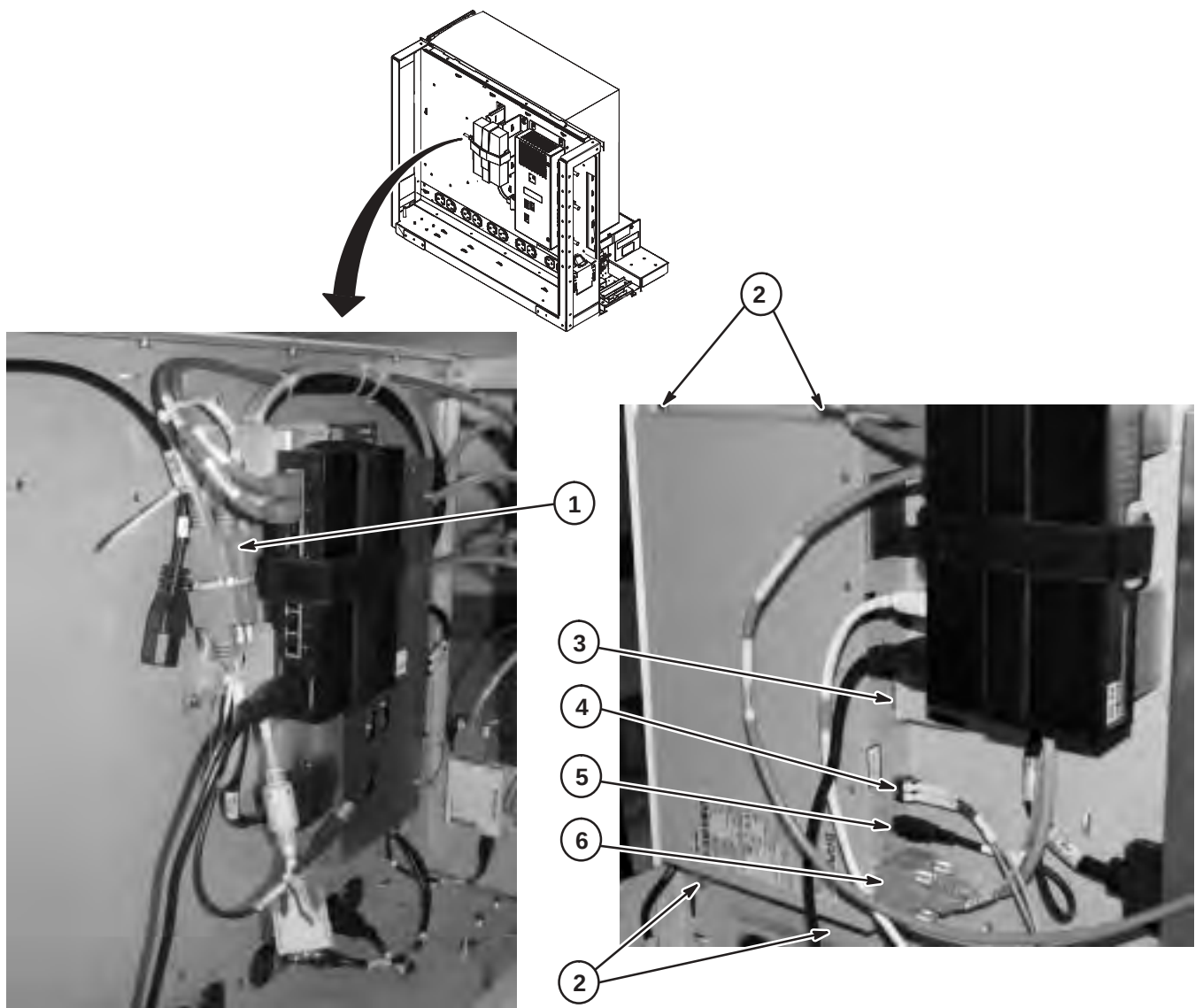


Z400

6-11 IN ROOM DISPLAY (IRD) INSTALLATION (Option)

Complete this procedure if IRD is part of system installation. The GOC is delivered with cables pre-installed for connection to the Fiber Optic Converter Box.

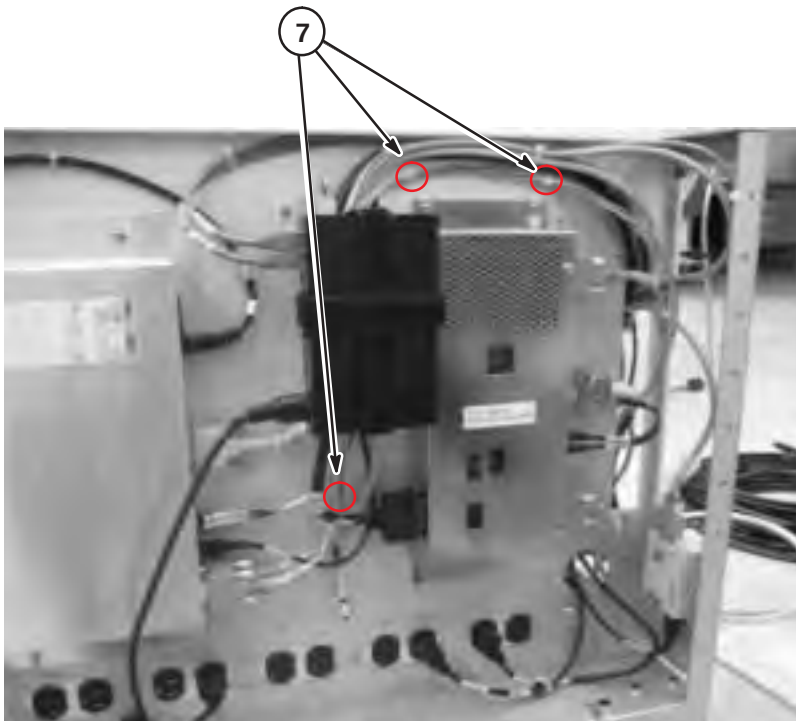
1. Disconnect and discard jumper cable 5267721-2 connected to E3019 and Sensor cable.
2. Install Fiber Optic Converter Box (5133768-2) using the provided four M5 screws.
3. Connect the DVI-D cable from the Host PC to J2.
4. Route fiber optic cable Run P2026 to GOC and connect to J3.
5. Connect the USB cable from the Host PC to J1.
6. Connect Run E3019 to J4 and the Sensor cable to J5.



6-11 IN ROOM DISPLAY (IRD) INSTALLATION (Continued)

7. Secure the optical fiber cable with tie wraps at points indicated below. Tie the tie wrap loosely as to not damage the optical cable.

Notice: Do NOT tighten the the tie wraps firmly. Damage to the optical cable will occur if tie wrap is to tight.



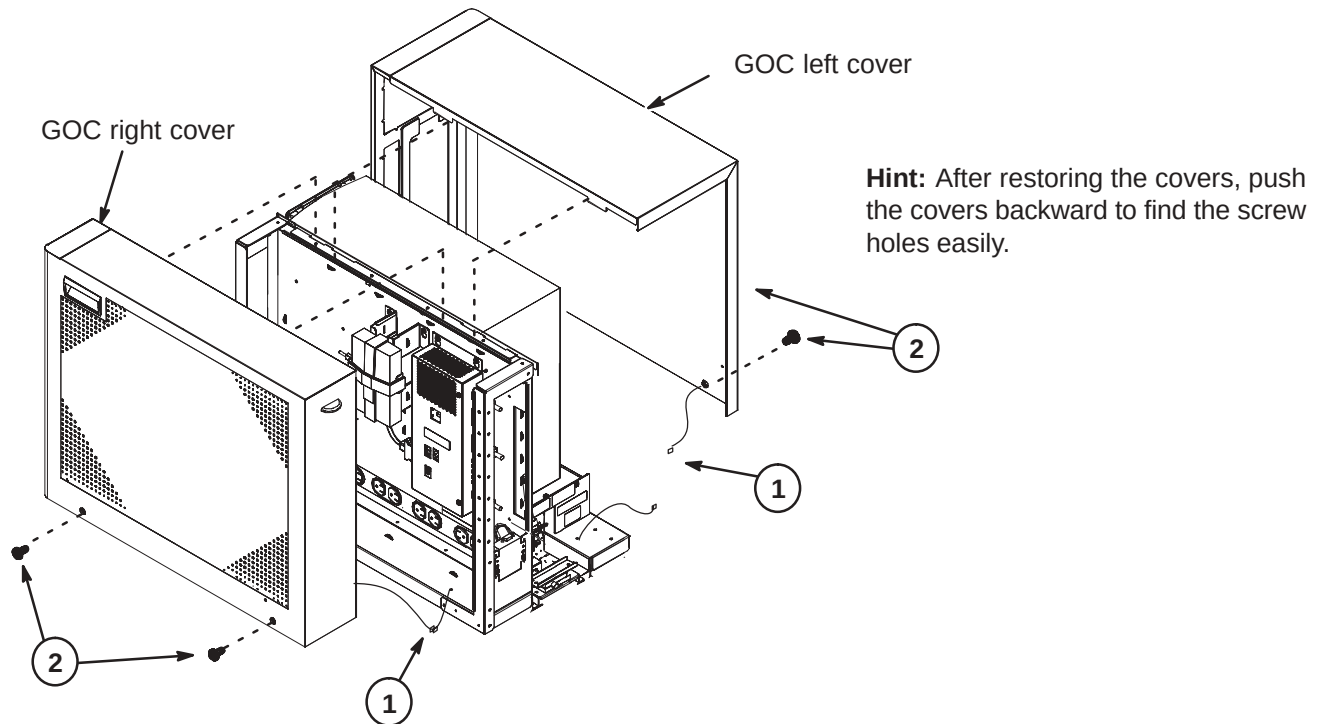
GOC side view



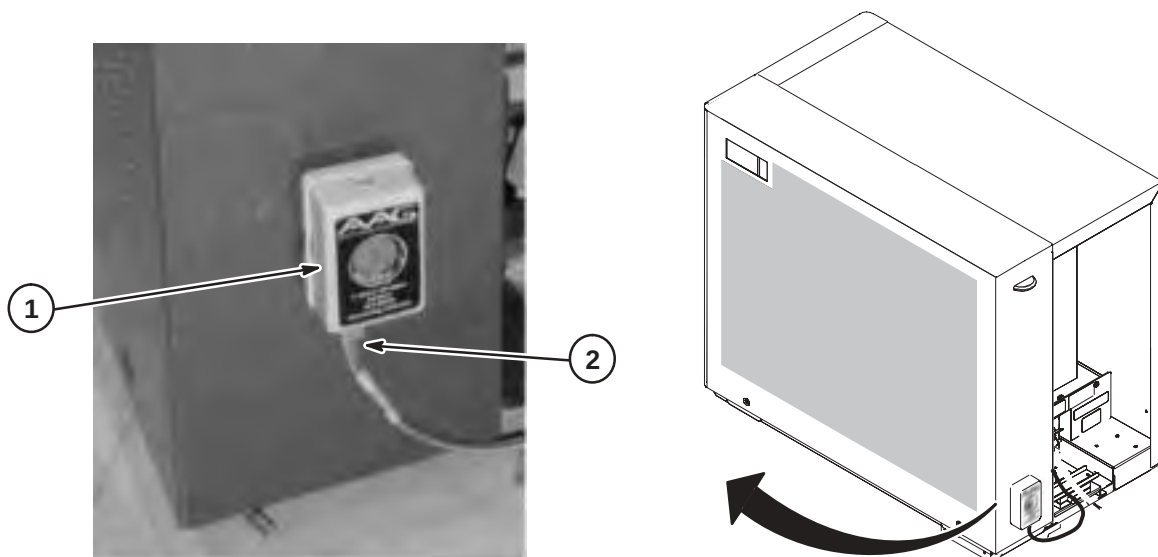
GOC rear view

6-12 RESTORE GOC COVERS

1. Connect Ground cables and install both side covers by inserting tabs at top of covers into slots.
2. Install screws that were removed in earlier procedure

**6-13 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC**

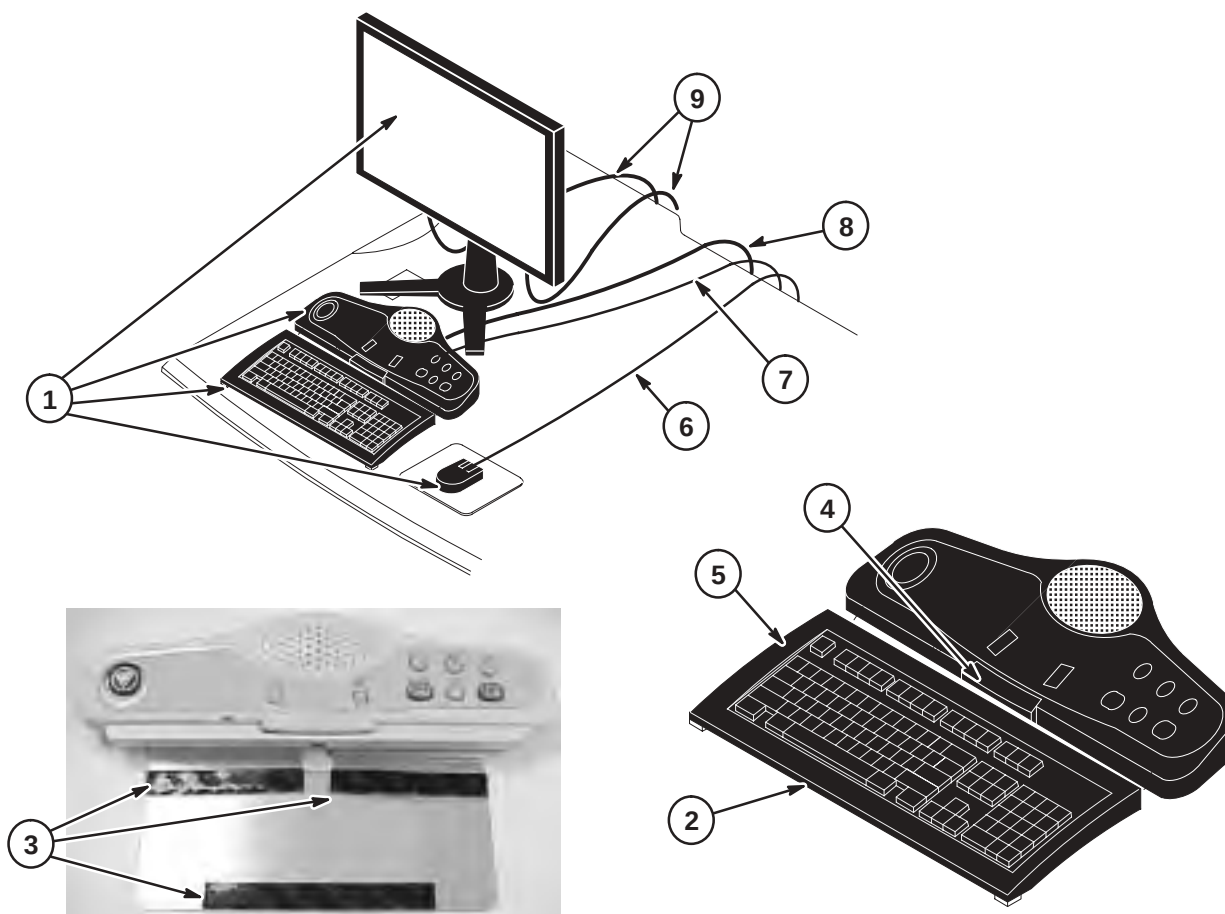
1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Connect cable (5267721) to Sensor.



6-14 INSTALL MONITOR AND SCIM & KEYBOARD

Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

1. Position Host Display Monitor, SCIM, Mouse, and keyboard on table.
2. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
3. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).
4. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
5. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.
6. Connect Mouse cable to USB port at rear of computer in GOC.
7. Connect Run E3048 to USB Keyboard cable. Route and connect other end to computer in GOC.
8. Connect Run E3047 to SCIM. Route and connect other end to computer in GOC.
9. Route power cord Run E0015 and Run E3046 from GOC and connect to Display Monitor.



Note

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

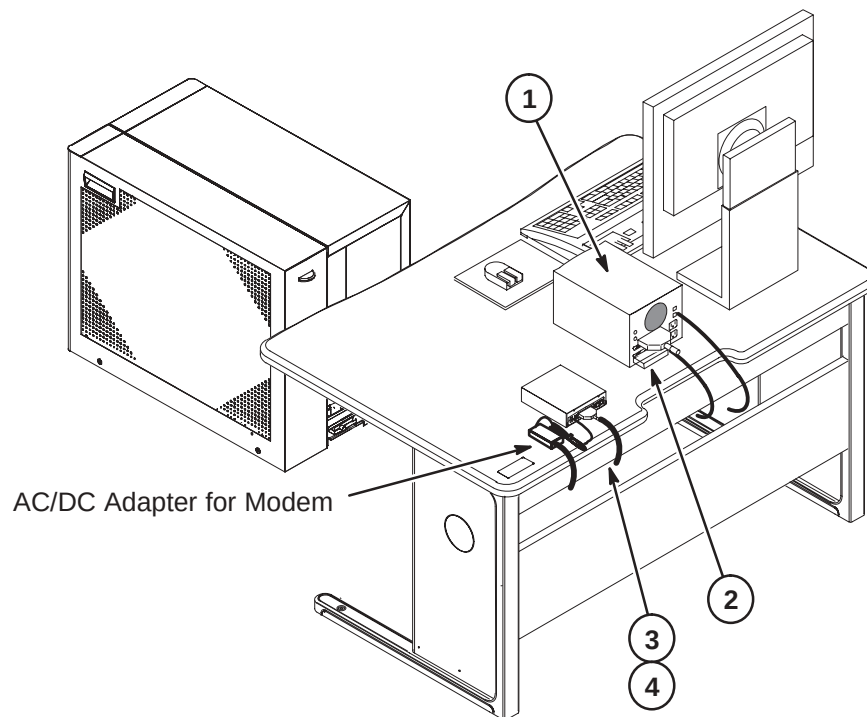
6-15 INSTALL SCSI TOWER (upgrade) AND MODEM (Option)

NOTE: SCSI Tower is only applicable to existing sites that are being upgraded to MR750. SCSI Tower is not supported for Z400.

1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet.
2. Install terminator on back of SCSI Tower.

The following steps are for the Modem option

3. If applicable, Place the Modem on the OW table and connect Runs E3045 and power cable from GOC Cabinet.
4. Connect customer supplied telephone line.



6-16 FINAL POSITIONING OF GOC CABINET

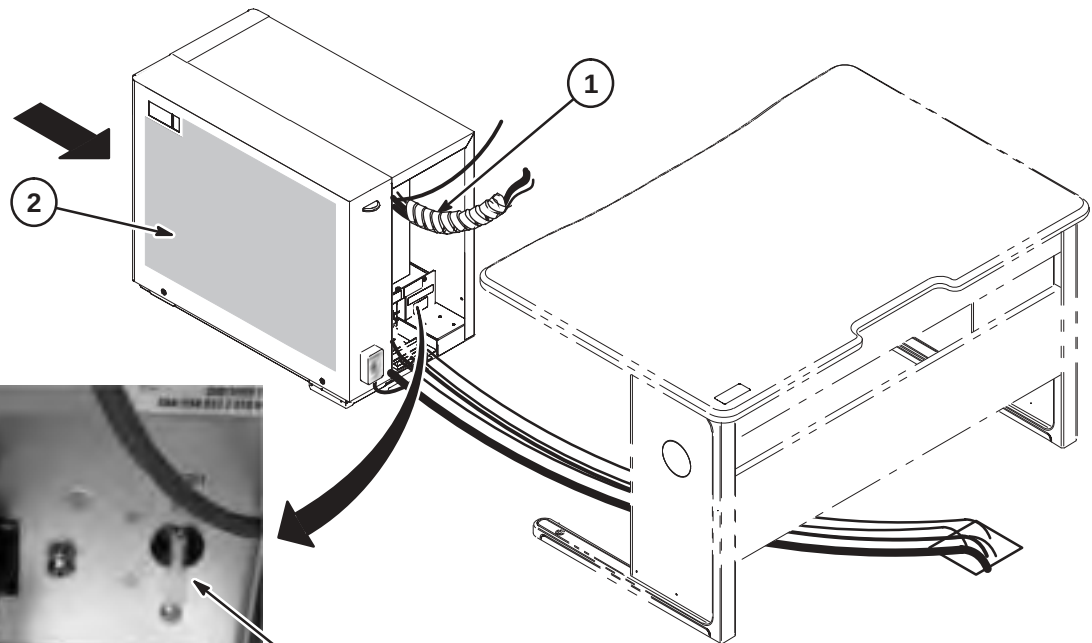
1. Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m) Spiral tube is included in the plastic bag which is shipped with GOC.

Notice

Do Not include Modem power cable in Spiral Tube.

- Three power cords (LCD Monitor ,SCSI Tower(if applicable))
- One video cable
- Keyboard cable
- SCIM cable
- SCSI Tower cable (Upgrades Only)
- Modem cable (if applicable)

2. Place cabinet in final position as determined by customer.



Notice:

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

6-17 POWER ON INFORMATION

When powering up the GOC, make sure the Main breaker located in the rear of the GOC is ON.

SECTION 7 – COMPLETION CHECKLIST

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7-1-1 RETURNING SHIPPING MATERIAL	7-2
7-1-2 FINAL MECHANICAL INSTALLATION STEPS	7-2
7-2 GE FIELD ENGINEER RESPONSIBILITIES	7-2
7-2-1 PHANTOMS AND SERVICE KITS	7-2
7-2-2 FINAL GE FIELD ENGINEER INSTALLATION STEPS	7-3
7-2-3 SUPPLIER PERFORMANCE REPORT	7-5

7-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map
- ☐ Correct for wiring errors if necessary
- ☐ Recheck all coolant hose connections for tightness and that there are no leaks. Correct if necessary.
- ☐ Recheck all air hose connections for tightness and that there are no leaks. Correct if necessary.
- ☐ Complete Section 7-1-1.
- ☐ Complete Section 7-1-2.

7-1-1 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies and carts. These can be included with the "Return pile", Recycling Operation will route them to appropriate location.

7-1-2 FINAL MECHANICAL INSTALLATION STEPS

1. Complete assembly of PPG cable and probe head.
2. Make sure all Cabinet, Magnet Enclosure, and Rear Pedestal covers have been installed.
3. Make sure Patient Transport is in position at front of Magnet.

7-2 GE FIELD ENGINEER RESPONSIBILITIES

7-2-1 PHANTOMS AND SERVICE KITS

1. Locate Phantom Kit and SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. Locate the multilingual (provided in eight languages) phantom attention labels shipped with each of the phantoms.



Three types of multilingual phantom attention labels are provided and each type must be applied to the correct phantom. The directions at the top of each label sheet describe which phantoms that can receive that label.

7–2–1 PHANTOMS AND SERVICE KITS (Continued)

4. Read the directions provided at the top of each of the multilingual attention label sheets and apply the appropriate language attention label for the site to each of the phantoms that should receive that type of label.
 - a. Each phantom will receive the correct type of attention label written in the appropriate language for that site.
 - b. Note that the LVshim phantom assembly will receive 3 labels, one for each phantom component.
5. Place service phantoms into SPT Phantom Cart or service storage cabinet to be organized and ready to be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.
6. Locate Signa Spares Kit. Place kits into service storage cabinet for ready use.

7–2–2 FINAL GE FIELD ENGINEER INSTALLATION STEPS

1. Record and enter applicable data into applicable site configuration files and records.
2. Complete Product Locator information for all installed serialized components, new or updated, via either:
 - **(U.S. Only)** *FE Site Verification Web Site*
or
 - Process and return product locator installation cards for all serialized components to:
Product Locator File, P.O. Box 414, W–523, Milwaukee, WI 53201–0414

Refer to Section 1–7 for details on submitting Product Locator information.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

3. Process the **Supplier Performance Report** located in Common Forms and Section 7–2–3 (reference copy).
4. Store the delivered site's set of service tools and spares kit in service cabinet at site.
5. Locate surface coil ("quick disconnect") adapters and any optional surface coils and positioning accessories and place on Patient Transport.
6. Locate Box "To be opened by Applications Specialist" and set aside for later visit by Applications Specialist.
7. Leave the site's set of Manuals and CD–ROM at site. Install Service and Operator documentation on the host computer. Organize and set up reference cabinet for the Manuals.
8. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
9. Verify that coordination of application support for instruction of site users on the appropriate Software Release has been made before returning system to the users. Turn site over to applications who will instruct users.

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7-2-3 SUPPLIER PERFORMANCE REPORT**GE Medical Systems Field Service**

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: _____ Date: _____

Work Performed / Evaluated: _____

Commodity / Item Purchased: _____

GE Customer Name: _____

FDO Number: _____ Test Equipment Bar-code No; _

GEMS Evaluator Name: _____

Please rate suppliers as 1 to 5 on each of the 15 questions below:

1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

Supplier schedules well and:					
1	starts on time?	1	2	3	4 5
2	finishes on time?	1	2	3	4 5
3	avoids disruptions?	1	2	3	4 5
Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4 5
5	cooperates with changes in schedule?	1	2	3	4 5
6	deals with unexpected problems?	1	2	3	4 5
Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4 5
8	adequate for contracted work?	1	2	3	4 5
9	calibrated and in good working order?	1	2	3	4 5
Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4 5
11	always keep job sites clean and free of debris?	1	2	3	4 5
12	ensure both function and appearance are as expected?	1	2	3	4 5
Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4 5
14	behavior?	1	2	3	4 5
15	attitude?	1	2	3	4 5

Examples / Details: _____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

1. Senior Operations Specialist for your LCT
2. The contractor who performed the work
3. ISS (already in default distribution)
4. Sourcing Supplier Quality (already in default distribution)

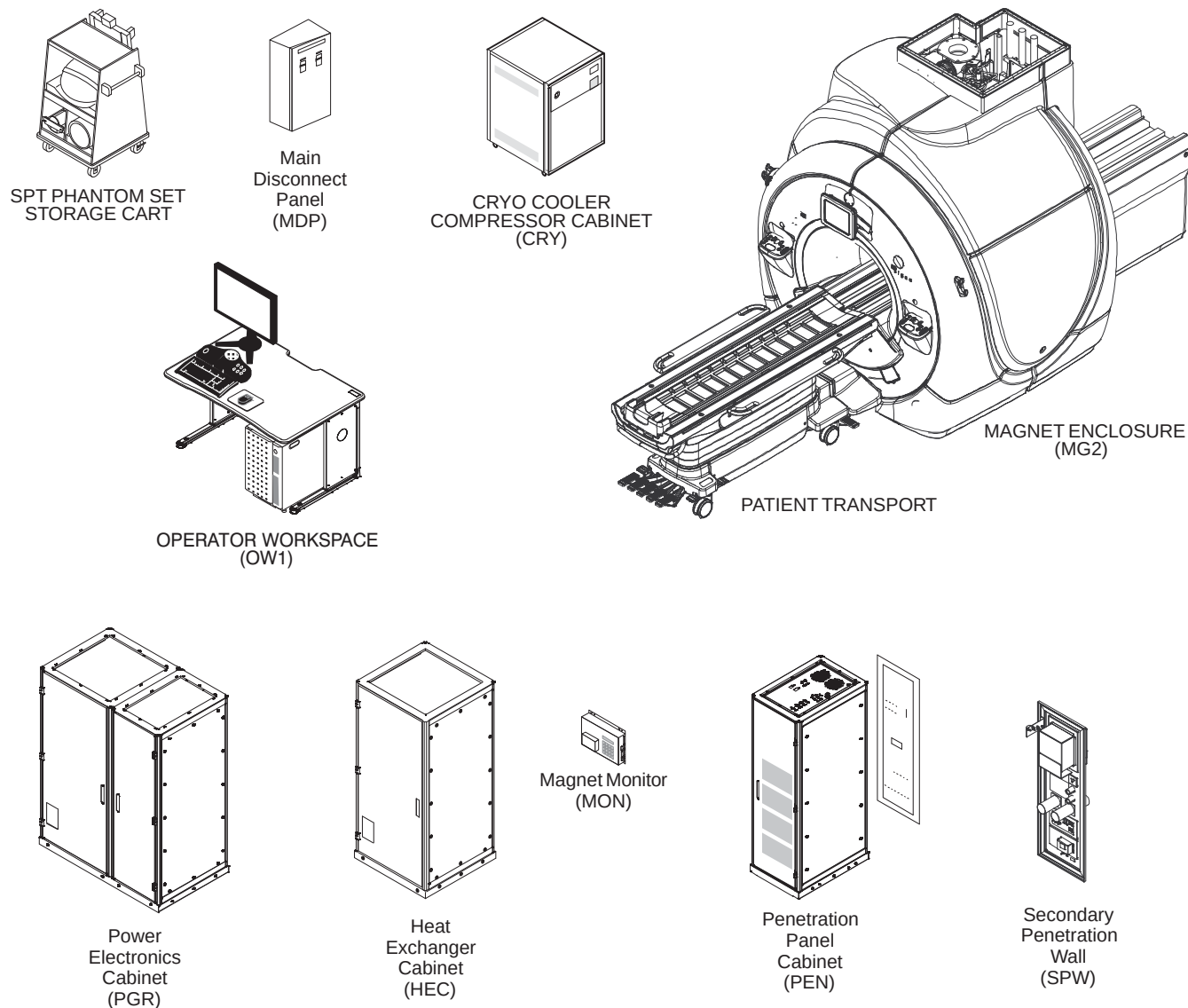
SECTION 8 – APPENDIX

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8-2 MR450 1.5T SYSTEM CONFIGURATION	8-3
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8-1 MR750 3.0T SYSTEM CONFIGURATION

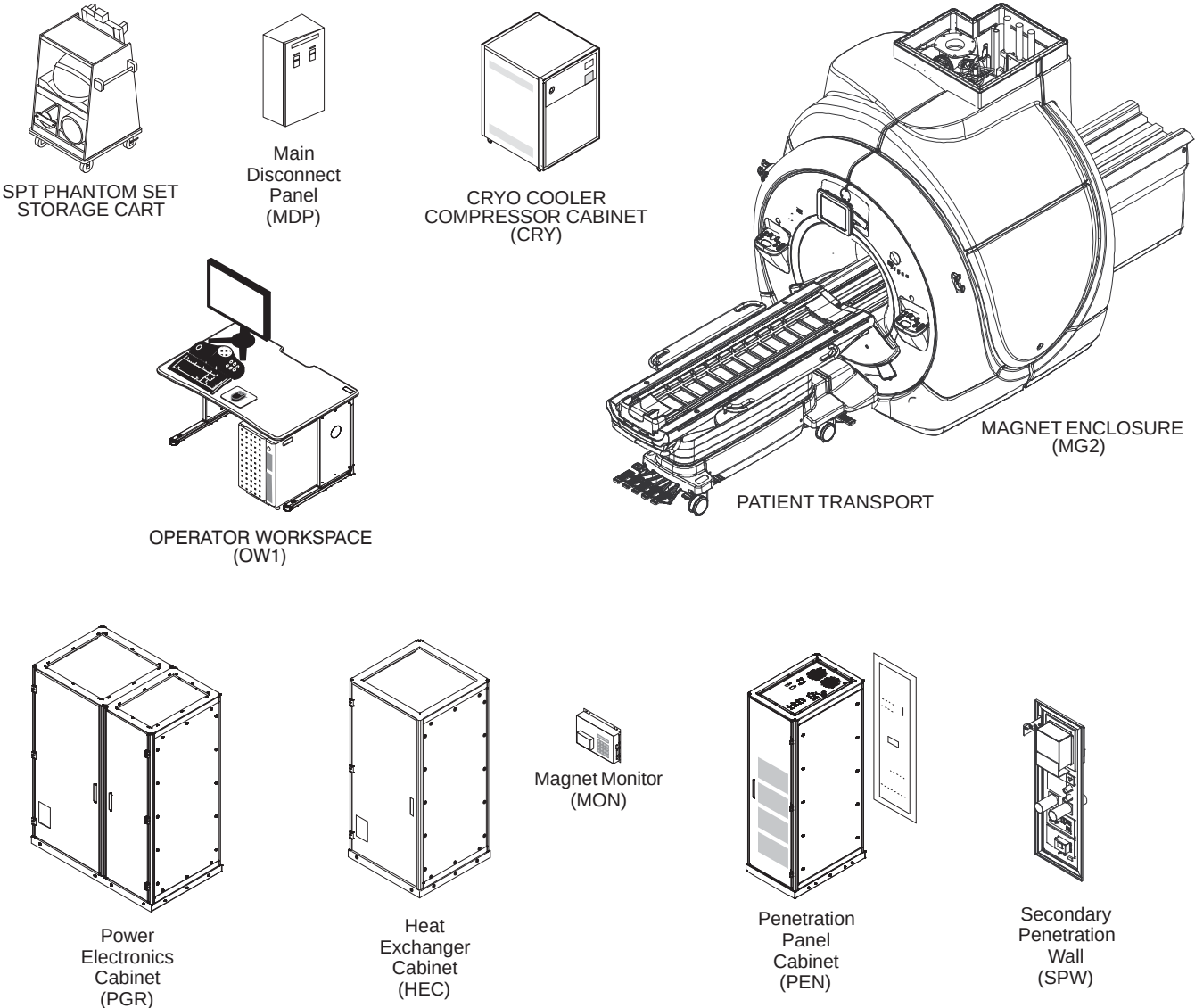
The illustration below shows the hardware equipment which comprise the Discovery MR750 3.0T system. The basic system is configured from several catalogs.



DISCOVERY MR750 3.0T SYSTEM
ILLUSTRATION 8-1

8-2 MR450 1.5T SYSTEM CONFIGURATION

The illustration below shows the hardware equipment which comprise the Discovery MR450 1.5T system. The basic system is configured from several catalogs.



DISCOVERY MR450 1.5T SYSTEM
ILLUSTRATION 8-2

8-3 CABLE SPECIFICATION

Cables delivered to site for installation have specifications dependent on its usage. The following tables contain information on cable runs for this system as it was originally installed.

The table below provides definitions for the following cable tables.

TABLE 8-1
CABLE TABLE DEFINITIONS

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
DESCRIPTION	Includes the Run number, destination labels of the cable, and length	Total length of cable delivered to site, function of cable specifies power from data.
NUMBER OF CONDUCTORS	Wires in cable	
CERTIFICATION	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given.
TEMPERATURE RATING	Maximum temperature insulating material may be used without loss of basic properties	Specific to the core cable manufacturer given.
NOTES	Describe the cable's use in the MR system	

The cables listed in Table 8-2 list all cables routed and connected at time of system installation.

TABLE 8-2
MR450 1.5T and MR750 3.0T CABLES INSTALLED ON SITE

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E0003, PGR-J102 to OW-PDM-J1, Host Power, 30000mm	3	10	AWM 2587, FT4	600	90°	Minimum bend radius is 6 inches (152mm)
Run E0004 PGR-J16 to Pen-J1, Pen Power, 17000mm	5	10	AWM 2587, FT4	600	90°	Minimum bend radius is 6 inches (152mm)
Run E0004, PGR-J16 to PEN-J1, PEN Power, 10000mm	5	10	AWM 2587, FT4	600	90°	Minimum bend radius is 6 inches (152mm)
Run E1001, PGR-J19 to SPW-J106, 3.0T Body, 10327mm ref.	1	N/A	CMR, FT4	6000	85°	LMR-1200 coaxial cable. Minimum bend Radius is 12 inches (305mm)
Run E1001, PGR-J19 to SPW-J106, 3.0T Body, 16523mm ref.	1	N/A	CMR, FT4	6000	85°	LMR-1200 coaxial cable. Minimum bend Radius is 12 inches (305mm)
Run E1002 SPW-J128 to PGR-J18, RF Transmit, 10218mm	1	N/A	CMR, FT4	5000	85°	LMR-900 coaxial cable. Minimum bend radius is 6 inches (152mm)
Run E1002 SPW-J128 to PGR-J18, RF Transmit, 16348mm	1	N/A	CMR, FT4	5000	85°	LMR-900 coaxial cable. Minimum bend radius is 6 inches (152mm)
Run E1003 PGR-J12 to PEN-J5, Body TR, 17000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable. Minimum bend Radius is 2.5 inches (64mm)
Run E1003, PGR-J12 to PEN-J5, Body TR, 10000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable. Minimum bend Radius is 2.5 inches (64mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E1004 PGR-J13 to Pen-J6, Head TR, 10000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable. Minimum bend radius is 6 inches (152mm)
Run E1004, PGR-J13 to PEN-J6, Head TR, 17000mm	1	N/A	CMR, FT4	1500	85°	LMR-240 coaxial cable. Minimum bend radius is 6 inches (152mm)
Run E1005, PGR-J11 to PEN-J15, Tx Output, 3T 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run E1005, PGR-J11 to PEN-J15, Tx Output, 3T 17000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run E1006, PGR-J9 to SPW-J107, Dummy Load SAR Monitoring, 3T 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run E1006, PGR-J9 to SPW-J107, Dummy Load SAR Monitoring, 3T 17000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run E1015, PGR-J10 to SPW-J108, Dummy Load SAR Monitoring, 3T 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run E1015, PGR-J10 to SPW-J108, Dummy Load SAR Monitoring, 3T 17000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run E2002 PGR-J1 to PEN-J19, 10000mm	2	N/A	OFNR	N/A	85°	Optical fibers. 2 simplex connectors bundled on each end. Minimum bend radius is 4.8 inches (122mm).
Run E2002 PGR-J1 to PEN-J19, 17000mm	2	N/A	OFNR	N/A	85°	Optical fibers. 2 simplex connectors bundled on each end. Minimum bend radius is 4.8 inches (122mm).
Run E2020 PGR-J22 to PEN-J17, 10000mm	6	N/A	LFNC	N/A	85°	Optical fibers in conduit, 3 connectors bundled on each end. Minimum bend radius is 7 inches (178mm).
Run E2020 PGR-J22 to PEN-J17, 17000mm	6	N/A	LFNC	N/A	85°	Optical fibers in conduit, 3 connectors bundled on each end. Minimum bend radius is 7 inches (178mm).
Run E2023 PGR-J21 to PEN-J18, 10000mm	2	N/A	OFNR	N/A	85°	Optical fibers. 2 simplex connectors bundled on each end. Minimum bend radius is 4.8 inches (122mm).
Run E2023 PGR-J21 to PEN-J18, 17000mm	2	N/A	OFNR	N/A	85°	Optical fibers. 2 simplex connectors bundled on each end. Minimum bend radius is 4.8 inches (122mm).
Run E3002 PGR-J103 to Pen-J11, 17000 mm	15	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3002, PGR-J103 to PEN-J11, E-Stop, 10000mm	15	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E3003, PGR-J7 to MDP-TB1-14-17, E-Off,10000mm	2	22	CMG, FT4	300	105°	Minimum bend Radius is 6 inches (152mm)
Run E3006 PGR-J5 to Pen-J14,17000 mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run E3006, PGR-J5 to PEN-J14, HART,10000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run E3008, PGR-J4 to PEN-J9, CAN,10000mm	9	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3008, PGR-J9 to PEN-J9, CAN,17000mm	9	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3009 PGR-J8 to Pen-J3,10000 mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3009 PGR-J8 to Pen-J3,17000 mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3011, PGR-J2 to OW1-Host, Ethernet, 30000mm	8	26	CMP	300	75°	Minimum bend Radius is 6 inches (152mm)
Run E3013, PGR-J6 to HEC-CAT 5, Cooling Control Ethernet,10000mm	8	26	CMP	300	75°	Minimum bend Radius is 6 inches (152mm)
Run E3014, PGR-J20 to RF Door SW, E-Stop, 25000mm	2	22	CMG, FT4	300	105°	Minimum bend Radius is 6 inches (152mm)
Run E3015, PGR-J17 to OW-FJ6, E-stop, 30000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run E3017, PEN-J10 to OW-FJ5, E-stop, 30000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run E3018, PEN-J12 to OW-FJ7, HART-RS422, 30000mm	15	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3020 Pen-J13 to Mon-J10,17000mm	15	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3022,MON-J7 to SPW-J116, Instrumentation Box, 17000mm	6	22	CMG, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3023 SPW-J117 to MON-J8,17000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3024, MON-J9 to MON-FJ4 and MON-FJ3, Flow Monitoring, 400mm	14	22	CMG, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3025, CRY to MON-FJ4, Flow Monitor-Sumitomo, 10000mm	10	22	CMG, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3026, HEC-FL-Temp to MON-FJ3, Compressor Interface, 2000mm	4	22	CMG, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3027, HEC-MDP to MDP-TB1-17-18, E-Off,10000mm	2	22	CMG, FT4	300	105°	Minimum bend Radius is 6 inches (152mm)
Run E3028, HEC-PLC-RS232 to PGR-J3, Cooling Control Communication, 10000mm	8	26	CM, FT4	300	75°	Minimum bend Radius is 6 inches (152mm)
Run E3030 E-Off Switch to SPW-J115,17000mm	2	22	CMG, FT4	300	105°	Minimum bend Radius is 6 inches (152mm)
Run E3030, SPW-J115 to E-Off SW, E-off,10000mm	2	22	CMG, FT4	300	105°	Minimum bend Radius is 6 inches (152mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run E3031, SPW-J118 to OW-J7, Patient Intercom, 30000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run E3034 MON-J5 to OW-Port 7 or Network, 30000mm	8	26	CMP	300	75°	Minimum bend Radius is 6 inches (152mm)
Run E3317 Gradient Cable X, Equip Room, Long, 17000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run E3317 Gradient Cable X, Equip Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run E3318 Gradient Cable Y, Equip Room, Long, 17000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run E3318 Gradient Cable Y, Equip Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run E3319 Gradient Cable Z, Equip Room, Long, 17000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run E3319 Gradient Cable Z, Equip Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run E4002, PEN-GND to PGR-GND, Ground Cable, 10000mm	1	10	CMG	300	80°	Single conductor ground cable. Minimum bend Radius is 5 inches (127mm)
Run E4002, PEN-GND to PGR-GND, Ground Cable, 17000mm	1	10	CMG	300	80°	Single conductor ground cable. Minimum bend Radius is 5 inches (127mm)
Run E4005 RF COM GND STUD to PGR-GND STUD, 10000mm	1	1/0	AWM 1284, TEW	600	105°	Single conductor ground cable. Minimum bend Radius is 7 inches (178mm)
Run E4005 RF COM GND STUD to PGR-GND STUD, 17000mm	1	1/0	AWM 1284, TEW	600	105°	Single conductor ground cable. Minimum bend Radius is 7 inches (178mm)
Run E4007, HEC-GND to PGR-GND, Ground Cable, 10000mm	1	10	CMG	300	80°	Single conductor ground cable. Minimum bend Radius is 5 inches (127mm)
Run E4008, OW-GOC-GND to PGR-GND, Ground Cable, 30000mm	1	10	CMG	300	80°	Single conductor ground cable. Minimum bend Radius is 5 inches (127mm)
Run E4009 RF COM GND STUD to SPW-GND STUD, 2800mm	1	1/0	AWM 1284, TEW	600	105°	Single conductor ground cable. Minimum bend Radius is 7 inches (178mm)
Run E4010 RF COM GND STUD to PEN-GND STUD, 2800mm	1	1/0	AWM 1284, TEW	600	105°	Single conductor ground cable. Minimum bend Radius is 7 inches (178mm)
Run M1001, RUN M1319 to SPW-J106, 1.5T RF Transmit, 10327mm ref.	1	N/A	CMR, FT4	5000	85°	LMR-1200 coaxial cable. Minimum bend Radius is 12 inches (305mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M1001, RUN M1319 to SPW-J106, 1.5T RF Transmit, 14457mm ref.	1	N/A	CMR, FT4	5000	85°	LMR-1200 coaxial cable. Minimum bend Radius is 12 inches (305mm)
Run M1001, RUN M1319 to SPW-J106, 3.0T Body, 10327mm ref.	1	N/A	CMR, FT4	6000	85°	LMR-1200 coaxial cable. Minimum bend Radius is 9 inches (229mm).
Run M1001, RUN M1319 to SPW-J106, 3.0T Body, 14457mm ref.	1	N/A	CMR, FT4	6000	85°	LMR-1200 coaxial cable. Minimum bend Radius is 9 inches (229mm).
Run M1002, RF Switch Cable,PEN PANEL J128 to RF-SW1-J121, 990.5mm	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable. Minimum bend Radius is 9 inches (229mm).
Run M1006, SPW-J107 to PED-DUM-J13, Dummyload, 3T, 13500mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run M1006, SPW-J107 to PED-DUM-J13, Dummyload, 3T, 17500mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run M1015, SPW-J108 to PED-DUM-J14, Dummyload, 3T, 13500mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run M1015, SPW-J108 to PED-DUM-J14, Dummyload, 3T, 17500mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend radius is 2 inches (51mm)
Run M1302, PEN-J29 to MAG-TDM-J11, Receiver Clock, Twinax, 10000mm	2	20	CL2, FT4	300	75°	twinaxial cable. Minimum bend Radius is 5 inches (127mm)
Run M1302, PEN-J29 to MAG-TDM-J11, Receiver Clock, Twinax, 14000mm	2	20	CL2, FT4	300	75°	twinaxial cable. Minimum bend Radius is 5 inches (127mm)
Run M1303, PEN-J32 to MAG-J13 or PED-HUB Slot 11-J16, NB Exciter, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend Radius is 2 inches (51mm)
Run M1303, PEN-J32 to MAG-J13 or PED-HUB Slot 11-J16, NB Exciter, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial cable. Minimum bend Radius is 2 inches (51mm)
Run M1304, PEN-J30 to MAG-J8 or PED-HUB Slot 11-J17, Rx LO - Exciter Reference Clock, Twinax, 10000mm	2	20	CL2, FT4	300	75°	twinaxial cable. Minimum bend Radius is 5 inches (127mm)
Run M1304, PEN-J30 to MAG-J8 or PED-HUB Slot 11-J17, Rx LO - Exciter Reference Clock, Twinax, 14000mm	2	20	CL2, FT4	300	75°	twinaxial cable. Minimum bend Radius is 5 inches (127mm)
Run M1305, SPW-J122 to MAG-J14, P1 RF Transmit, 10218mm	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable. Minimum bend Radius is 9 inches (229mm).
Run M1305, SPW-J122 to MAG-J14, P1 Transmit, 14305mm	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable. Minimum bend Radius is 9 inches (229mm).

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M1306, SPW-J120 to MAG-J12, A Transmit, 10218mm	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable. Minimum bend Radius is 9 inches (229mm).
Run M1306, SPW-J120 to MAG-J12, A Transmit, 14305mm	1	N/A	CMR, FT4	4000	85°	LMR-600 coaxial cable. Minimum bend Radius is 9 inches (229mm).
Run M1310 PEN-J72 to MAG-J9, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1310 PEN-J72 to MAG-J9, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1311 PEN-J76 to MAG-J10, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1311 PEN-J76 to MAG-J10, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1312 PEN-J74 to MAG-HYB-J7, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1312 PEN-J74 to MAG-HYB-J7, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1324, MAG-HYB-J4 to PED-DUM-J12, 3.0T, 5986mm	1	N/A	CMR, FT4	1000	100°	LMR-400 coaxial cable
Run M1341 PEN-J73 to MAG-J18, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1341 PEN-J73 to MAG-J18, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1342 PEN-J75 to MAG-J19, 10000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M1342 PEN-J75 to MAG-J19, 14000mm	1	N/A	CMR, FT4	1000	100°	LMR-195 coaxial. Minimum bend Radius is 2 inches (51mm)
Run M2025 Mag to Pen , 12500mm	8	N/A	LFNC	N/A	85°	optical fibers in conduit. 4 duplex connectors bundled. Minimum bend radius is 7 inches (178mm).
Run M2025 Mag to Pen, 16500mm	8	N/A	LFNC	N/A	85°	optical fibers in conduit. 4 duplex connectors bundled. Minimum bend radius is 7 inches (178mm).
Run M3022, SPW-J116 to MAG-P403, 14000 mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run M3023 SPW-J117 to Mag-FJ2, 14000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M3030, E-OFF, 30000mm	9	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3031, SPW-J118 to MAG-J1, Console Intercom, 10000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3031, SPW-J118 to MAG-J1, Console Intercom, 14000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3300,PEN-J51 to MAG-TDM-J9, TDM Power, 24W7, Low-Mag, 10000mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	Minimum bend Radius is 8.3 inches (211mm)
Run M3300,PEN-J51 to MAG-TDM-J9, TDM Power, 24W7, Low-Mag, 14000mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	Minimum bend Radius is 8.3 inches (211mm)
Run M3301, PEN-J52 to MAG-TDM-J10, TDM Power, 24W7, Low-Mag, 10000mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	Minimum bend Radius is 8.3 inches (211mm)
Run M3301,PEN-J52 to MAG-TDM-J10, TDM Power, 24W7, Low-Mag, 14000mm	24	7/C 12AWG + 17/C 22AWG	CL2, FT4	300	105°	Minimum bend Radius is 8.3 inches (211mm)
Run M3302 PEN-J40 to MAG-J4,SRI Power, 10000mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 13 inches (330mm)
Run M3302 PEN-J40 to MAG-J4,SRI Power, 14000mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 13 inches (330mm)
Run M3303 PEN-J39 to MAG-J3, MCPDB Interface, 10000mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 13 inches (330mm)
Run M3303 PEN-J39 to MAG-J3, MCPDB Interface, 14000mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 13 inches (330mm)
Run M3304 PEN-J48 to PED-HUB-SLOT12-J2, RFHub Power, 13W6, 17500mm	12	6/C 10AWG + 6/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 8 inches (203mm)
Run M3304,PEN-J48 to PED-HUB-SLOT12-J2, RFHub Power, 13W6, 13500mm	12	6/C 10AWG + 6/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 8 inches (203mm)
Run M3306, PEN-J46 to PED-HUB-SLOT12-J1, RFHub Power, 17C5, 13500mm	12	6/C 10AWG + 6/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 8 inches (203mm)
Run M3306, PEN-J46 to PED-HUB-SLOT12-J1, RFHub Power, 17C5, 17500mm	12	6/C 10AWG + 6/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 8 inches (203mm)
Run M3307, PEN-J50 to PED-HUB-SLOT12-J5, RFHub Power, 21WA4, 13500mm	8	4/C 10AWG + 4/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 8 inches (203mm)
Run M3307, PEN-J50 to PED-HUB-SLOT12-J5, RFHub Power, 21WA4, 17500mm	8	4/C 10AWG + 4/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 8 inches (203mm)
Run M3308, PEN-J47 to PED-HUB-SLOT12-J6, RFHub Power, 9C4, 13500mm	8	4/C 10AWG + 4/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 7.3 inches (185mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M3308, PEN-J47 to PED-HUB-SLOT12-J6, RFHub Power, 9W4, 17500mm	8	4/C 10AWG + 4/C 22AWG	AWM 2464, FT4	300	105°	Minimum bend Radius is 7.3 inches (185mm)
Run M3309, SPW-J123 to MAG-J15, Tx Transmit, 14000mm	7	22	CMG, FT4	300	80°	Minimum bend Radius is 6.8 inches (173mm)
Run M3310 PEN-J45 to PED-HUB-SLOT12-J4, RF Power, 13500mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6.8 inches (173mm)
Run M3310 PEN-J45 to PED-HUB-SLOT12-J4, RF Power, 17500mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6.8 inches (173mm)
Run M3311, Mag-J16 or PED-MTR-J1 to Pen-J36, Long Drive Power, 10000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3311, Pen-J36 or PED-MTR-J1 to Mag-J16, Long Drive Power, 14000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3312 PEN-J41 to MAG-J6 or PED-HUB Slot 11-J9, PAC Power, 10000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3312 PEN-J41 to MAG-J6 or PED-HUB Slot 11-J9, PAC Power, 14000mm	25	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3313 PEN-J35 to MAG-J5, Dock Power, 10000mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3313 PEN-J35 to MAG-J5, Dock Power, 14000mm	37	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3314 PEN-J42 to MAG-J2, LCD Power, 10000mm	15	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3314 Pen-J42 to Mag-J2, 14000 mm	15	22	CL3, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3315 PEN-J54 to MAG-J17 or PED-TIF-J4, 14000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run M3315, PEN-J54 to MAG-J17 or PED-TIF-J4, Hall's Signal, 10000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run M3309, SPW-J123 to MAG-J15, Tx Transmit, 10000mm	7	22	CMG, FT4	300	80°	Minimum bend Radius is 6 inches (152mm)
Run M3316 Pen-J34 to Mag-J7 or PED-HUB Slot 11-J10, 14000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run M3316, PEN-J34 to MAG-J7 or PED-HUB Slot 11-J10, TRDD Unblank, 10000mm	9	22	CL3, FT4	300	90°	Minimum bend Radius is 6 inches (152mm)
Run M3317 Gradient Cable X, Scan Room, Long, 14000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run M3317 Gradient Cable X, Scan Room, Short, 10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run M3318 Gradient Cable Y, Scan Room, Long, 14000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)

8-3 CABLE SPECIFICATION (Continued)

Description	Number of Conductors	AWG	Certification	Voltage Rating (volts)	Temp Rating (Celsius)	Notes
Run M3318 Gradient Cable Y, Scan Room, Short,10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run M3319 Gradient Cable Z, Scan Room, Long,14000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run M3319 Gradient Cable Z, Scan Room, Short,10000mm	5	4/C 2/0 AWG + 1/C 6 AWG	TC-ER, FT4	600	90°	Minimum bend Radius is 13 inches (330mm)
Run M4005 RF COM GND STUD to MAG-GND STUD, 13500mm	1	1/0	AWM 1284, TEW	600	105°	Single conductor ground cable. Minimum bend Radius is 7 inches (178mm)
Run M4005 RF COM GND STUD to MAG-GND STUD, 17500mm	1	1/0	AWM 1284, TEW	600	105°	Single conductor ground cable. Minimum bend Radius is 7 inches (178mm)
Run P2003 PGR-J15 to Mag-J20,31000mm	6	N/A	OFNR	N/A	85°	optical fibers, 3 duplex connectors bundled on each end. Minimum bend radius is 5.7 inches (145mm).
Run P2004 PGR-J14 to Mag-J21,20000mm	6	N/A	OFNR	N/A	85°	optical fibers, 3 duplex connectors bundled on each end. Minimum bend radius is 5.7 inches (145mm).
Run P2004 PGR-J14 to Mag-J21,31000mm	6	N/A	OFNR	N/A	85°	optical fibers, 3 duplex connectors bundled on each end. Minimum bend radius is 5.7 inches (145mm).
Run P2026 OW to Mag-J11,Scan Room Display,40000mm	4	N/A	OFNR	N/A	85°	optical fibers. 2 duplex connectors bundled on each end. Minimum bend radius is 4.8 inches (122mm).
Run P2026 OW to Mag-J11,Scan Room Display,44000mm	4	N/A	OFNR	N/A	85°	optical fibers. 2 duplex connectors bundled on each end. Minimum bend radius is 4.8 inches (122mm).